M.Sc. In Electrical Energy Conversion and Power Systems



Unit 4.- Three Phase Converters

4.1 DC-AC conversion. The single-phase inverter

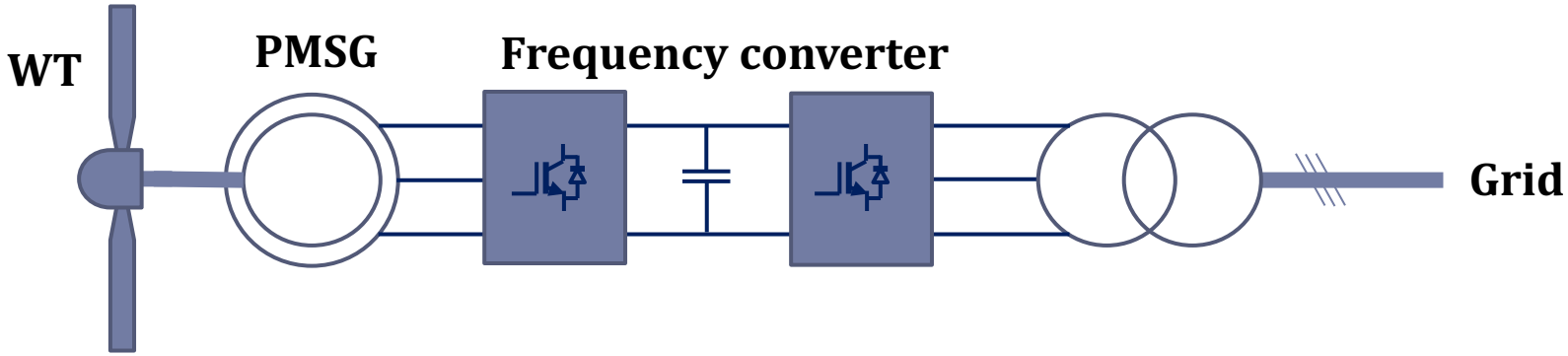
Semester 2 – Industrial electronics in renewable energy generation systems

Lecturer: Jorge García, garciajorge@uniovi.es

Unit 4.- Three-phase converters	
4.1	DC-AC conversion. The single-phase inverter
	General concepts in inverters. Characteristics and types Half-bridge square wave single-phase inverter Full-bridge square wave single-phase inverter Sine-waveform PWM modulation Output LC filter and harmonics analysis Input filter and DC-link current
4.2	DC-AC conversion. The three-phase inverter
	General concepts in inverters. 6-pulses inverter. Sine-waveform PWM modulation Output LC filter and harmonics analysis Input filter and DC-link current Effect of the dead-times. Limitations. Alternative modulations, 3rd harmonic injection, SVPWM.
4.3	AC-DC conversion. The three-phase controlled rectifier
	Reverse operation of the three-phase inverter Equivalent single-phase circuit Control of the filter current Main electrical magnitudes relations Evolution of waveforms

Introduction and General Concepts

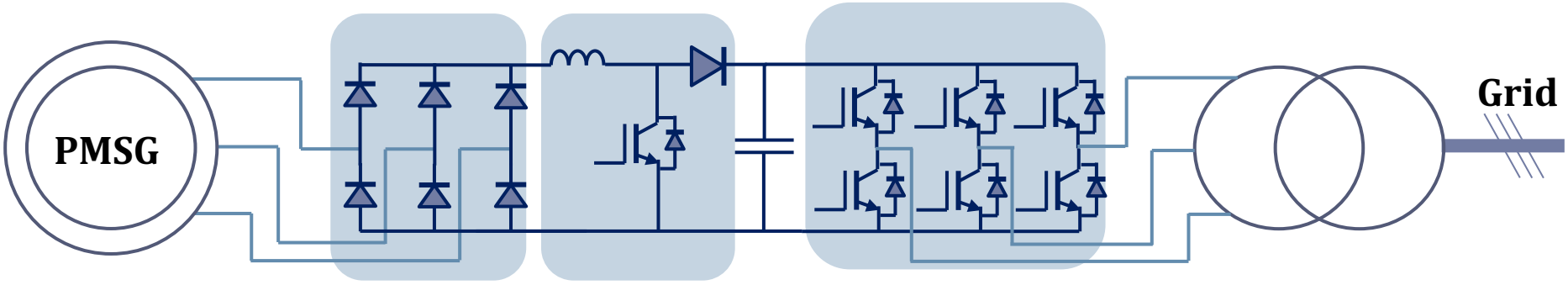
Basic Power Schemes



Permanent Magnet Synchronous Generator

- Pitch control
- PC withstands full power
- variable speed
- No gearbox (multipole generator)
- Up to ? MW

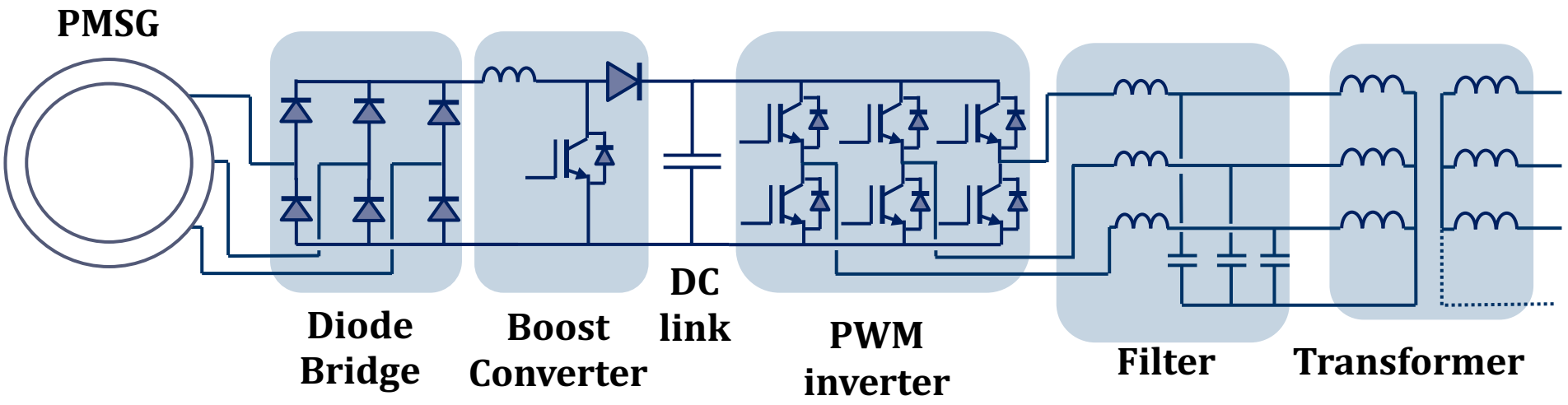
Basic Power Schemes



Frequency converter
(diode bridge+boost converter+ PWM inverter)

Filter + Transformer

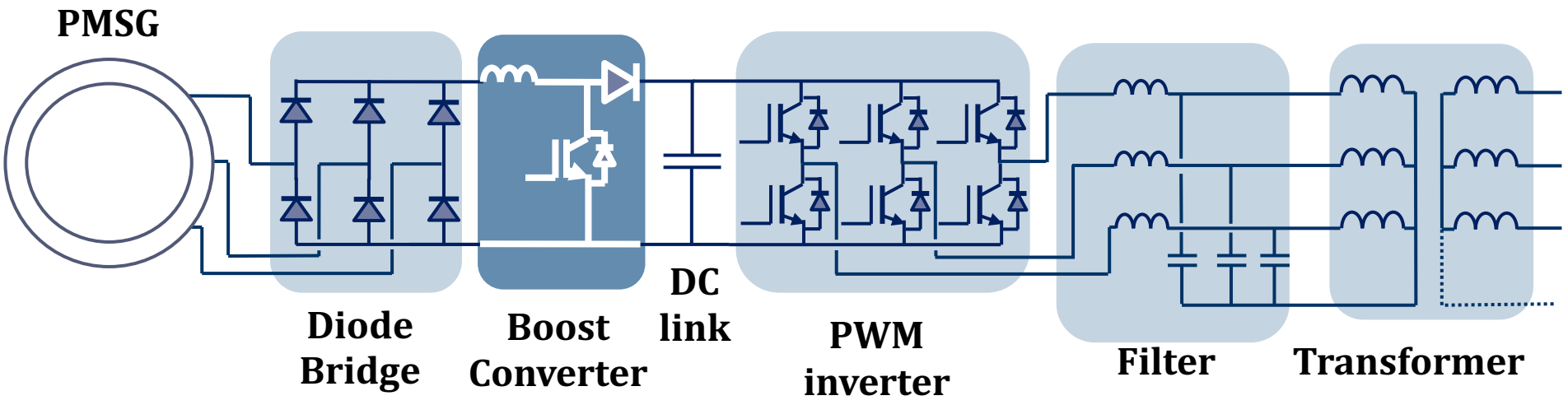
Basic Power Schemes



TARGETS:

- Operation of the different power electronics circuits
- Voltage and current waveforms at each converter
- Power losses estimation
- Design of the reactive components
- Size of the semiconductor switches
- Large and small signal average models (dynamic of converters)
- Simulation of the topologies
- Control of the converters and feedback loop design
- Simulation of the full system
- Practical implementation, laboratory setup, experimental verification

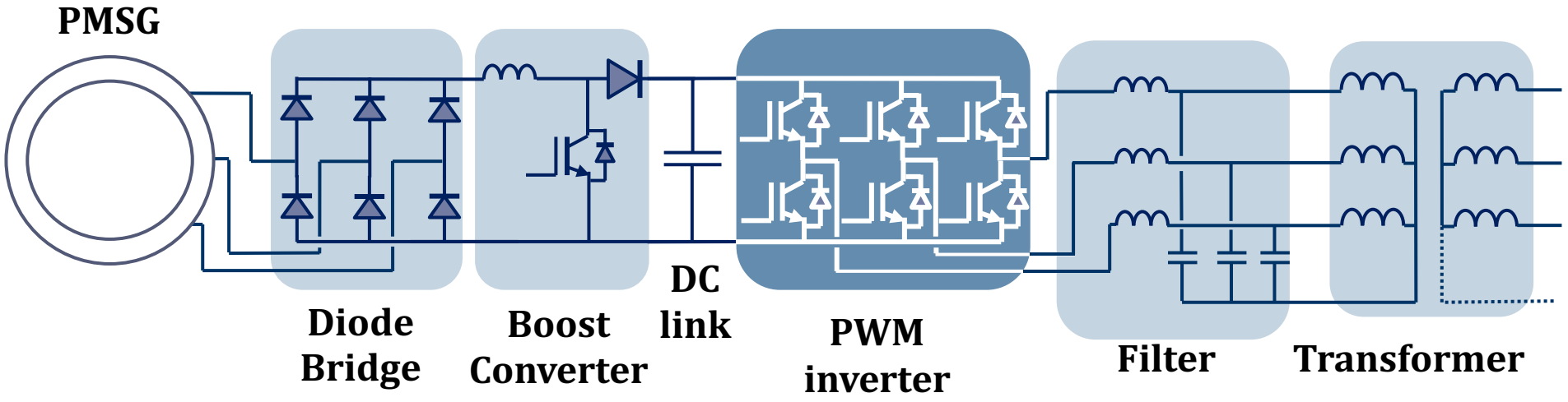
Basic Power Schemes: The boost converter



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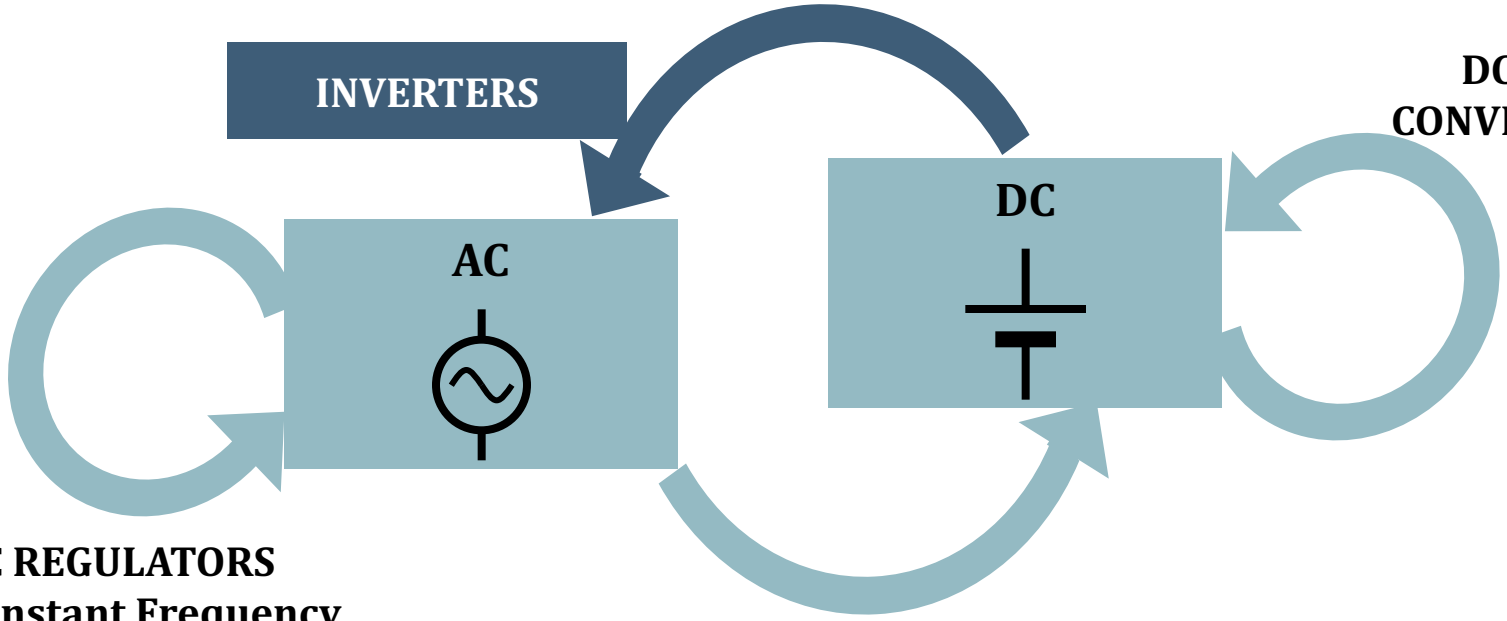
Basic Power Schemes: The PWM inverter



TARGETS:

- Operation of the different power electronics circuits
- Voltage and current waveforms at each converter
- Power losses estimation
- Design of the reactive components
- Size of the semiconductor switches
- Large and small signal average models (dynamic of converters)
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Overview: DC/AC energy conversion



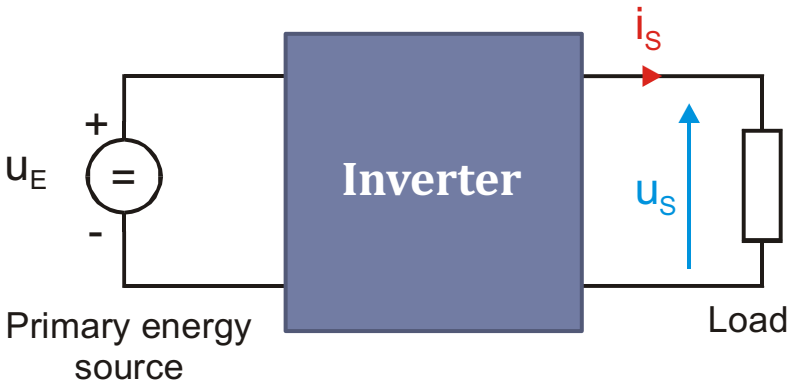
AC REGULATORS
Constant Frequency

CYCLOCONVERTERS
Variable Frequency

RECTIFYERS
- *Controlled (thyristors)*
- *Uncontrolled (diodes)*
- *Partially Controlled (both)*
- *PWM Rectifyers (Transistors)*

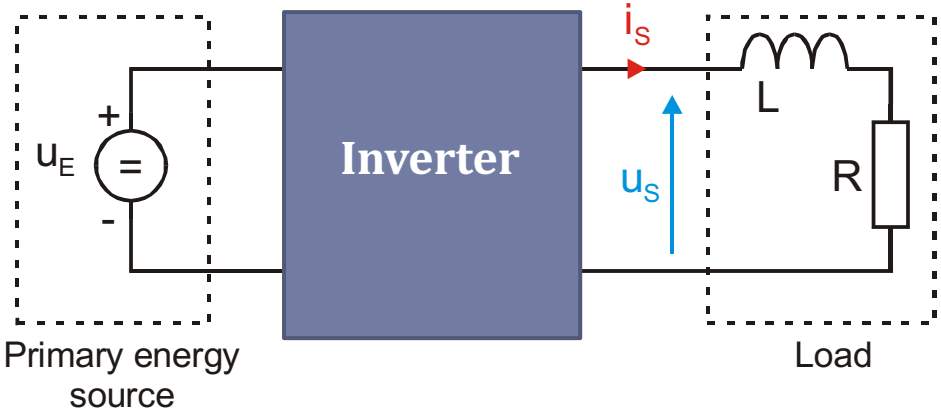
STATIC SWITCHES
- *DC / AC*

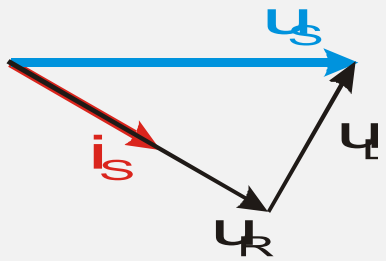
Overview: DC/AC energy conversion



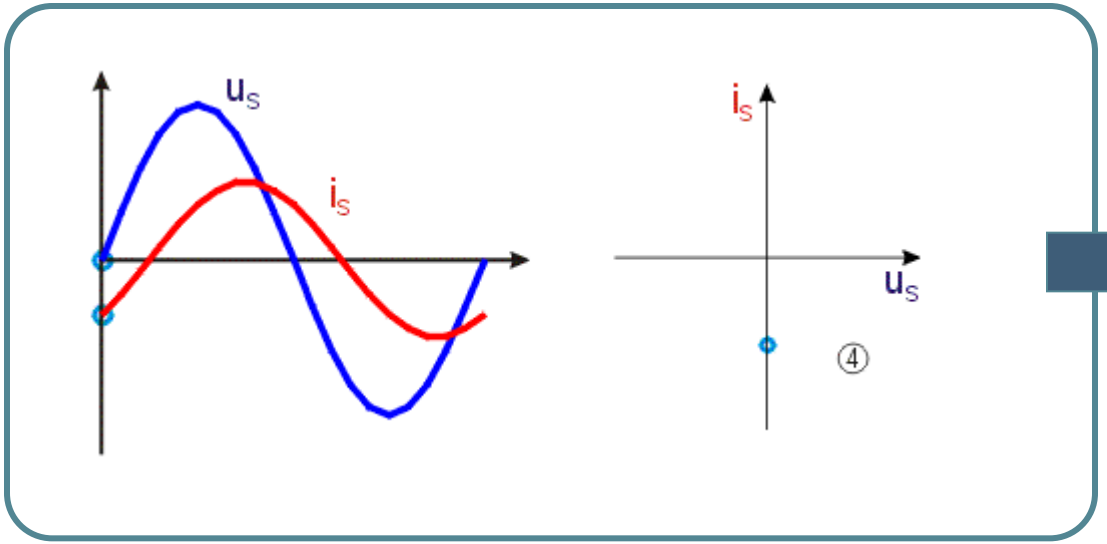
- Inverters: DC/AC conversion
- Many applications require a near-sinusoidal output signal
- Applications:
  - Induction motor supply
  - Uninterruptible power supplies (UPS)
  - Grid injection
  - …

Overview: energy flow in DC/AC conversion



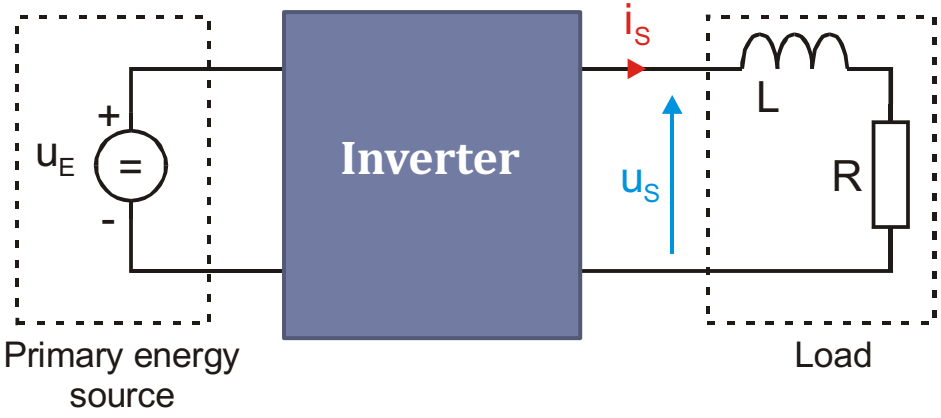


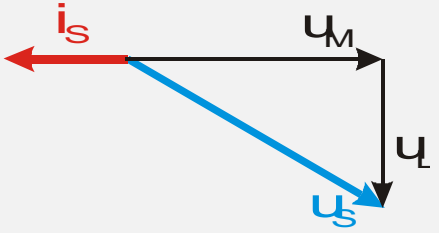
Inductive load

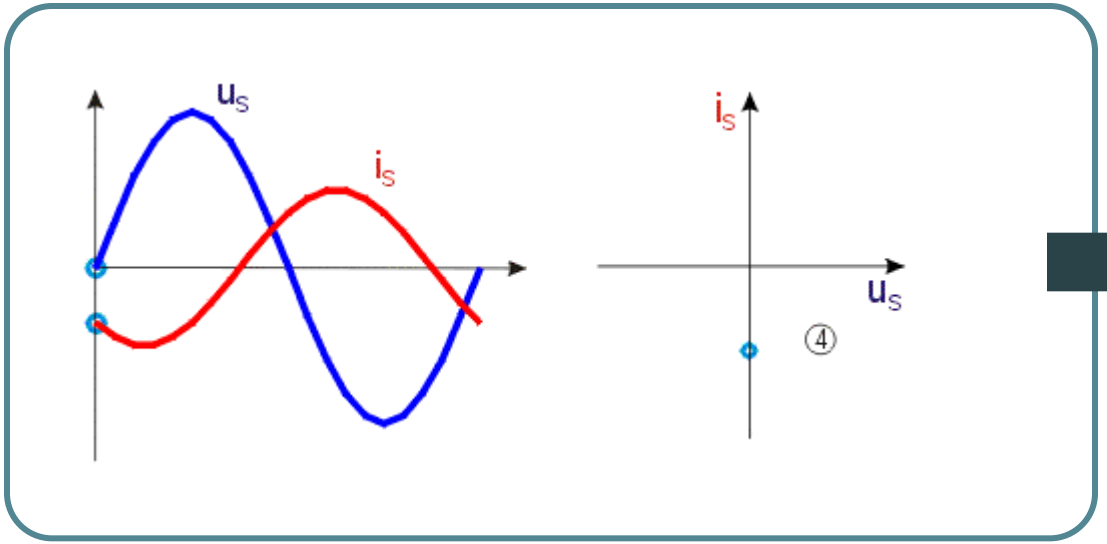


An inverter must be able to work in all four quadrants

Overview: energy flow in DC/AC conversion







Rectifier operation

Characteristic parameters of an inverter:

Basic specifications of an inverter:

- Output voltage and current regulation
- Controllability of the output frequency
- Harmonic content

Other specifications:

- Galvanic isolation
- Size and weight of the converter
- Simplicity of the control circuit: self-oscillating inverters
- Minimize switching losses: resonant inverters

Characteristic parameters of an inverter:

- The ideal output would be a pure sinusoidal signal
- Distortion of the nth harmonic:

$$D_n = \frac{v_n}{v_1}$$

Where v_n is the nth harmonic amplitude

- Total harmonic distortion:

$$THD = \frac{\sqrt{v_2^{\,2} + v_3^{\,2} + \ldots + v_n^{\,2} + \ldots}}{v_1} \cdot 100$$

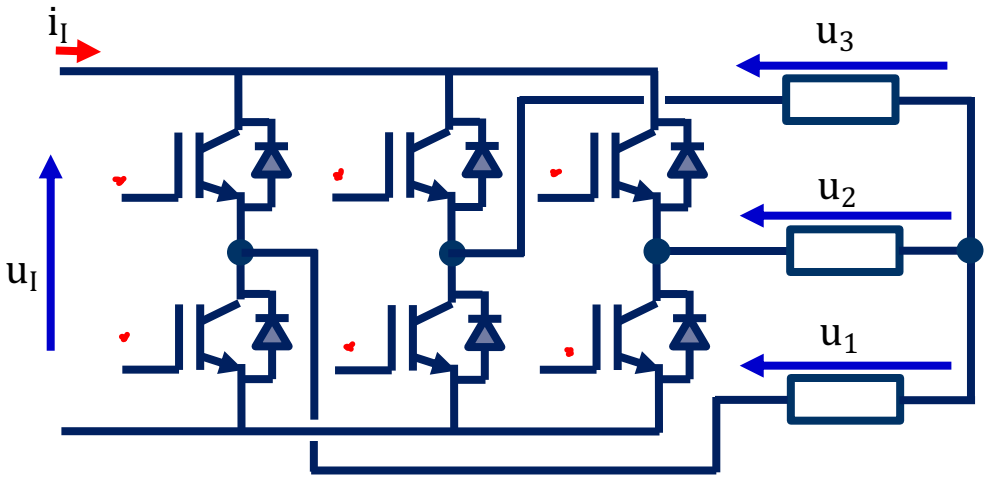
Inverter types:

- Input DC source
  - VSI-Voltage fed inverter
  - CSI-Current fed inverter
- AC output voltage
  - Single phase
  - Three phase
- Power topology
  - Half bridge
  - Full bridge
  - Push-pull
- Control scheme
  - Square wave
  - PWM - Pulse width modulation

Inverter types:

Input DC source	**VSI-Voltage fed inverter** CSI-Current fed inverter
AC output voltage	**Single phase** **Three phase**
Power topology	**Half bridge** **Full bridge** Push-pull
Control scheme	**Square wave** **PWM - Pulse width modulation**

The PWM inverter

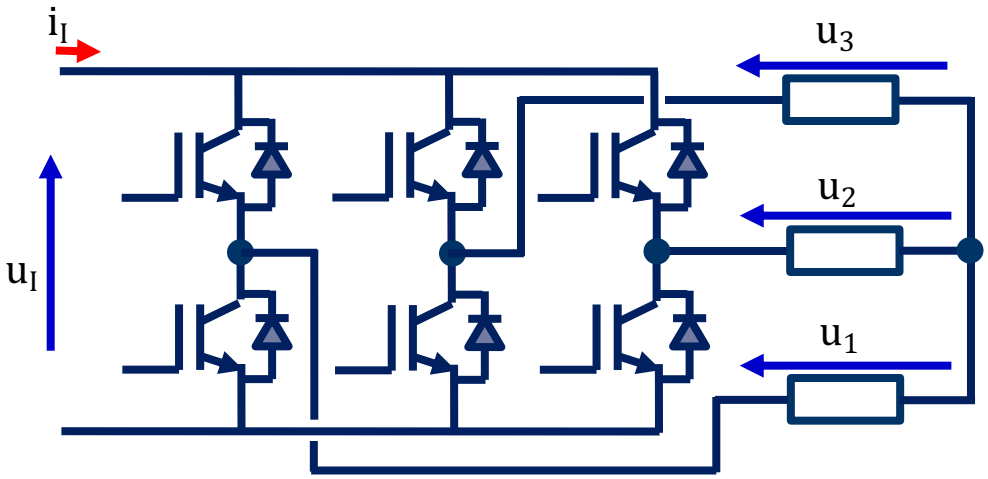


¿How to obtain a sinusoidal line frequency voltage waveform from a DC voltage?

The amplitude, phase and frequency must be easily controlled.

First, the study of a single leg will be carried out: the three-phase inverter will be constructed from single-phase inverters.

The PWM inverter



The switching patterns of each leg determine the phase voltages

$$\left.\begin{matrix} d_{1H}(t), d_{2H}(t), d_{3H}(t) \\ d_{1L}(t), d_{2L}(t), d_{3L}(t) \end{matrix}\right\} \; u_I(t)$$

$$\left.\begin{matrix} u_1(t), u_2(t), u_3(t) \\ Z_1, Z_2, Z_3 \end{matrix}\right\} \rightarrow i_1(t), i_2(t), i_3(t) \rightarrow i_I(t)$$

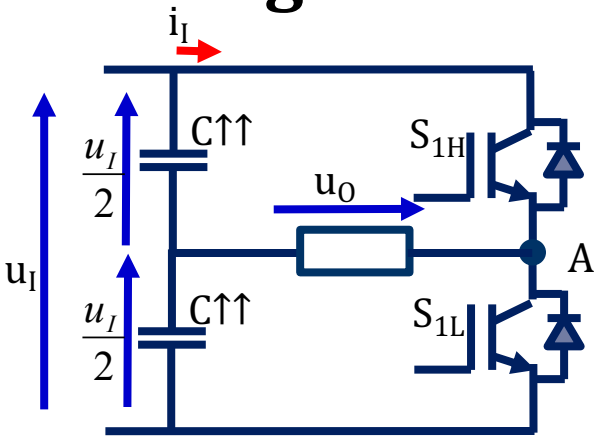
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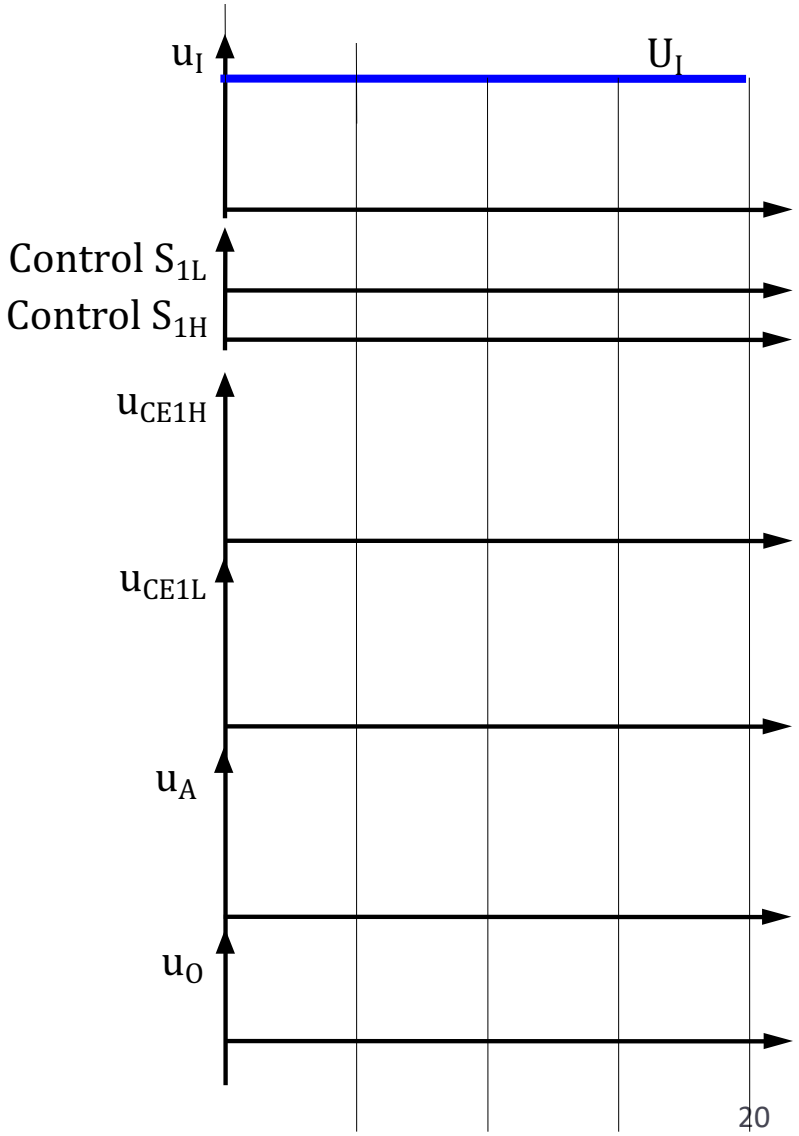
Single-Phase, Square Waveform, Half-Bridge Inverter

Single-Phase Half-Bridge Inverter

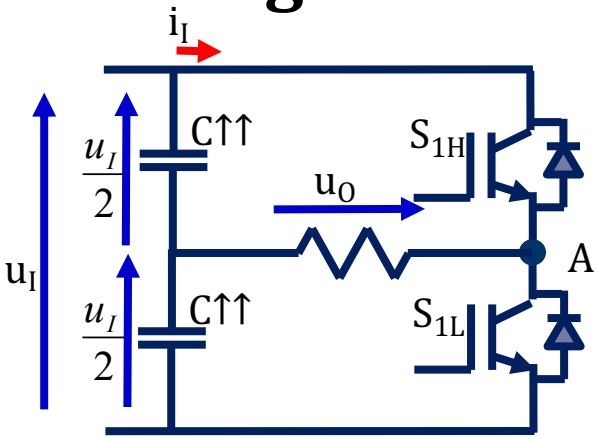


Analysis considerations:

- Ideal switches (no losses)
- Equal voltage at both capacitors

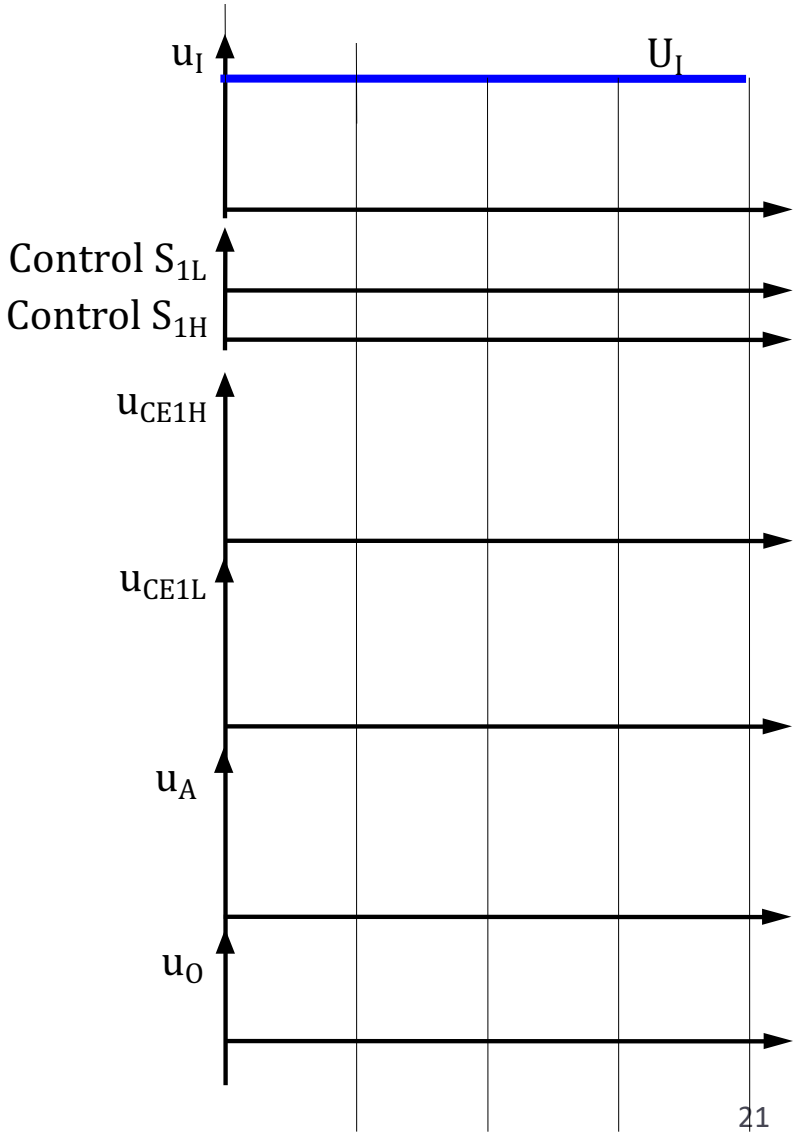


Single-Phase Half-Bridge Inverter

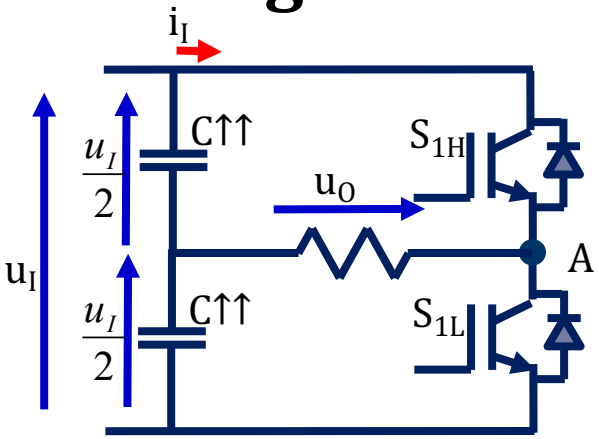


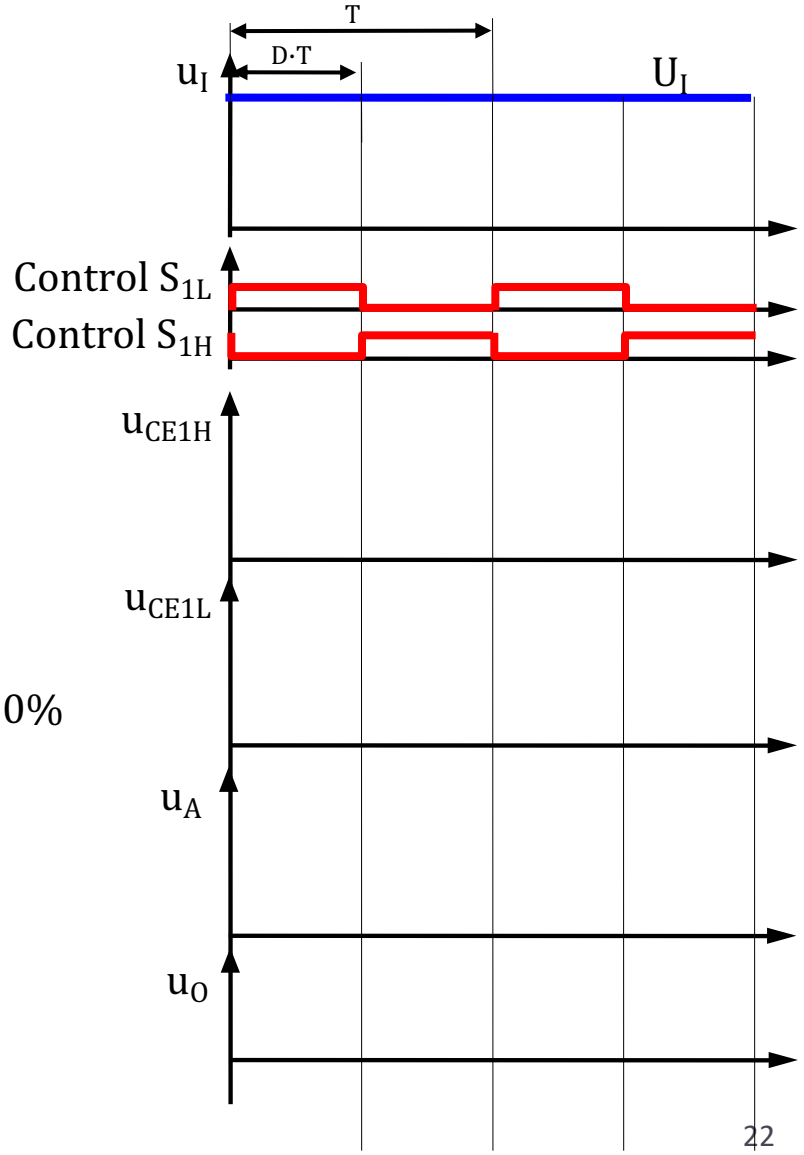
Analysis considerations:

- Ideal switches (no losses)
- Equal voltage at both capacitors
- Resistive load



Single-Phase Half-Bridge Inverter

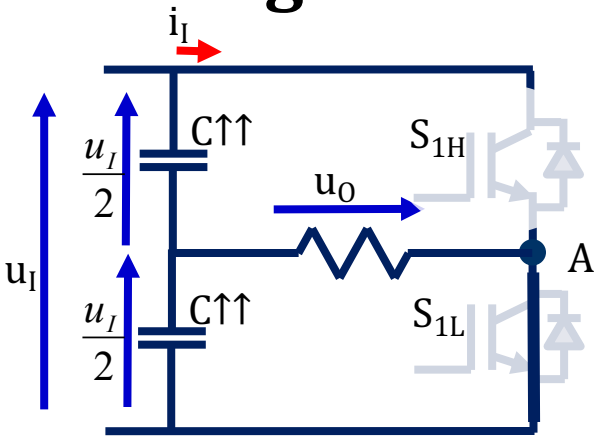


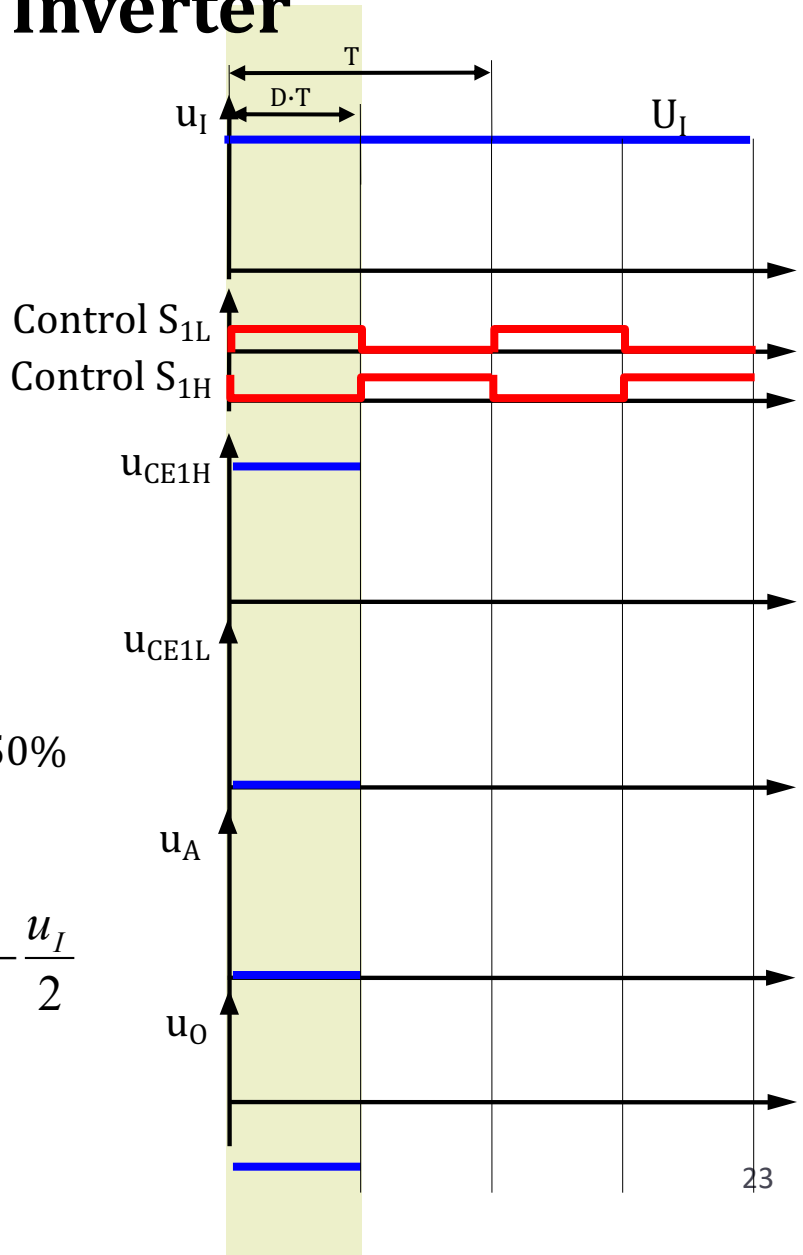


Analysis considerations:

- Ideal switches (no losses)
- Equal voltage at both capacitors
- Resistive load
- Gate pulses for S_{1L} and S_{1H} are complementary, d=50%

Single-Phase Half-Bridge Inverter





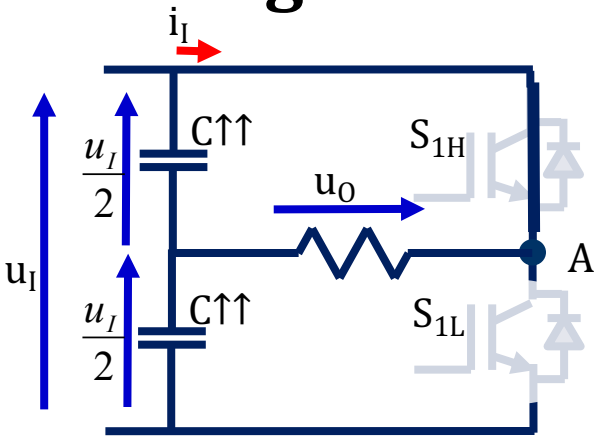
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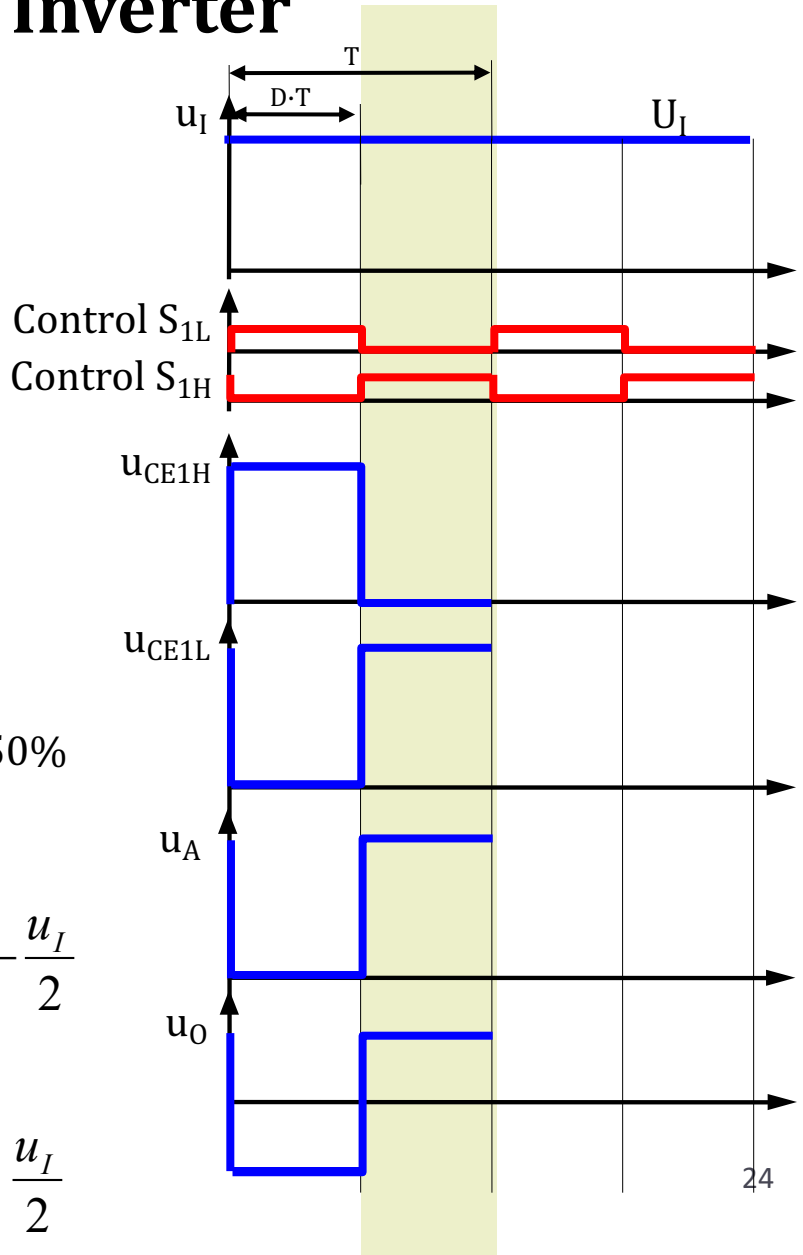
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- Gate pulses for S_{1L} and S_{1H} are complementary, D=50%

When Switch S_{1L} is ON, and S_{1H} is OFF

$$u_{CE1L}(t)=0 \quad u_{CE1H}(t)=u_I \quad u_A(t)=0 \quad u_O(t)=-\frac{u_I}{2}$$

Single-Phase Half-Bridge Inverter





Analysis considerations:

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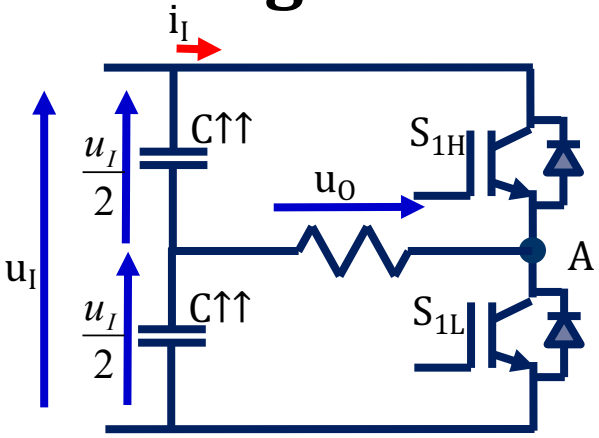
When Switch S_{1L} is ON, and S_{1H} is OFF

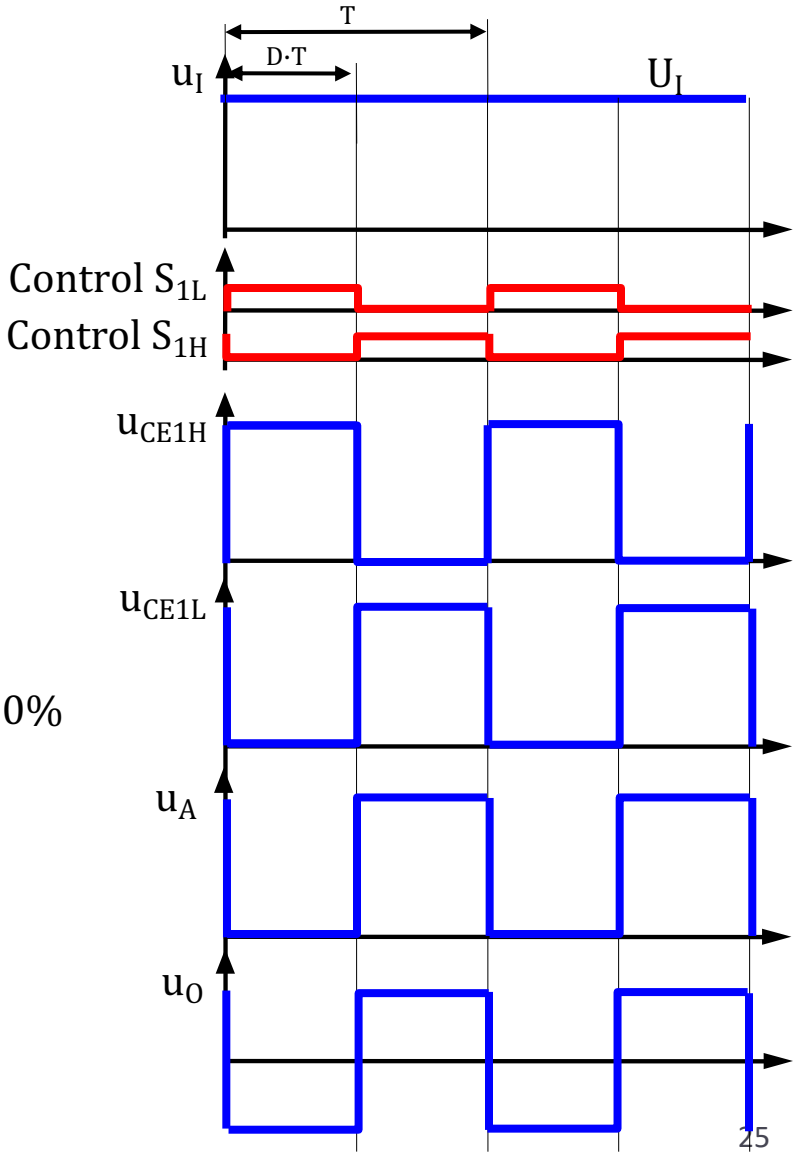
$$u_{CE1L}(t)=0 \quad u_{CE1H}(t)=u_I \quad u_A(t)=0 \quad u_O(t)=-\frac{u_I}{2}$$

When Switch S_{1L} is OFF, and S_{1H} is ON

$$u_{CE1L}(t)=u_I \quad u_{CE1H}(t)=0 \quad u_A(t)=u_I \quad u_O(t)=\frac{u_I}{2}$$

Single-Phase Half-Bridge SW Inverter



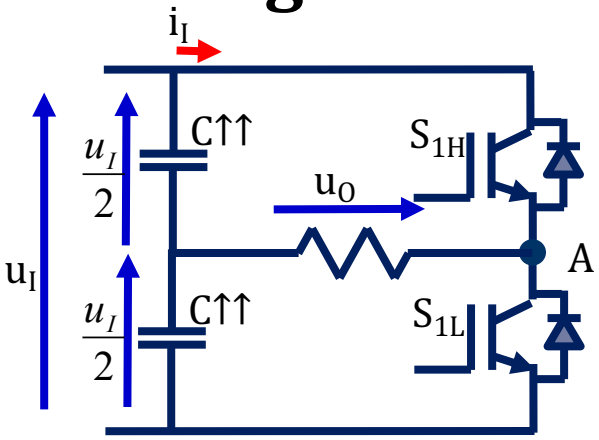


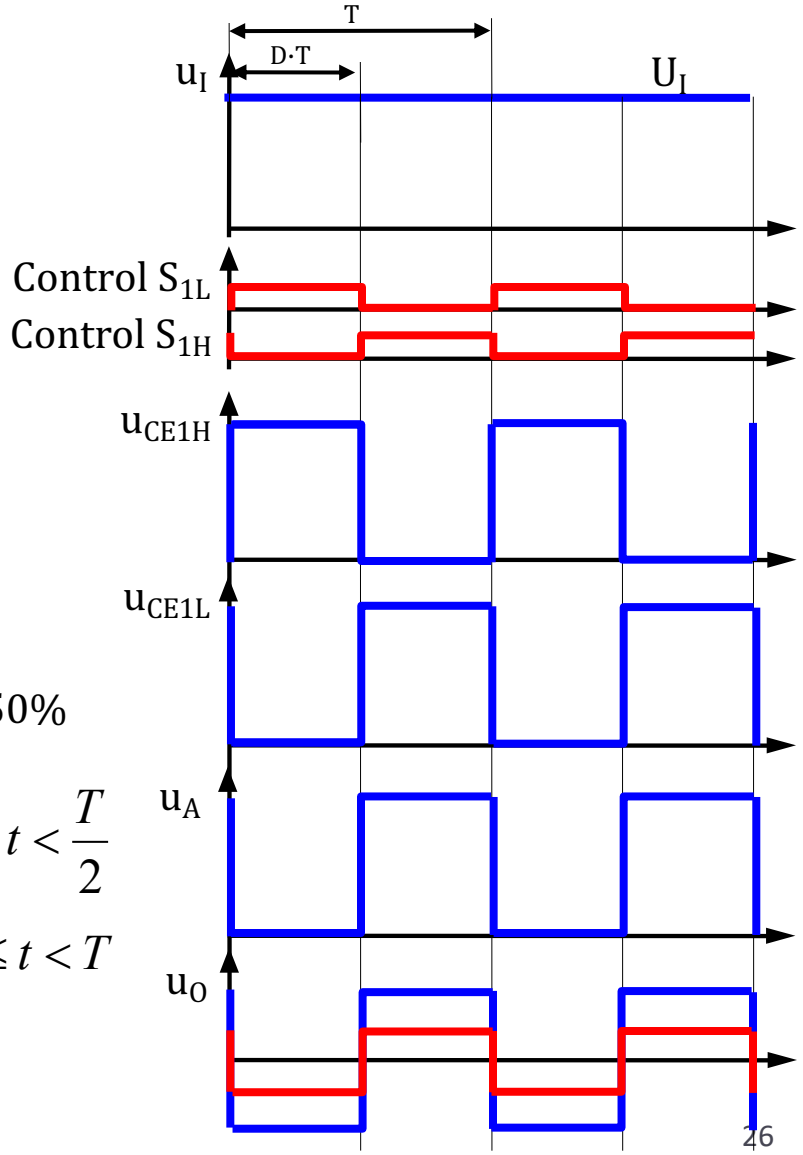
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$$u_O(t) = \begin{cases} -\dfrac{u_I}{2} & 0 \le t < \dfrac{T}{2} \\ \dfrac{u_I}{2} & \dfrac{T}{2} \le t < T \end{cases}$$

Single-Phase Half-Bridge SW Inverter



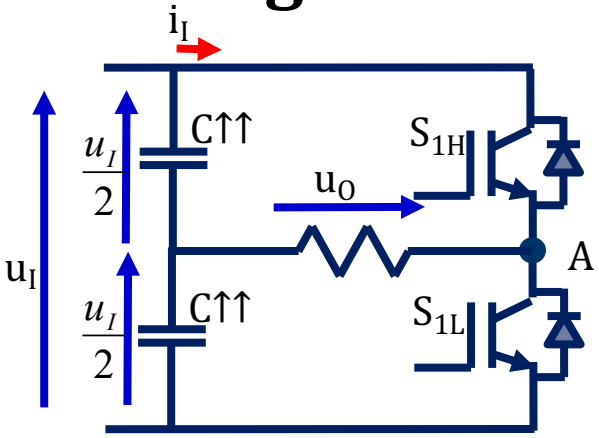


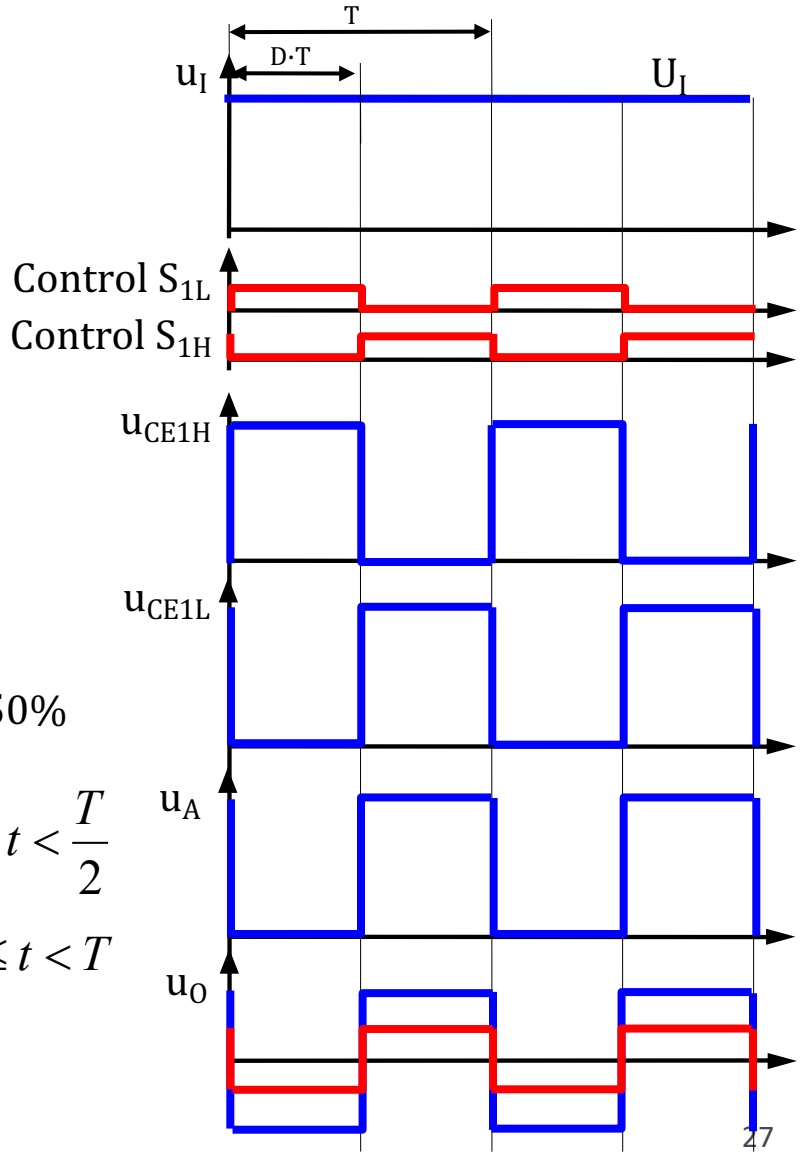
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Single-Phase Half-Bridge SW Inverter





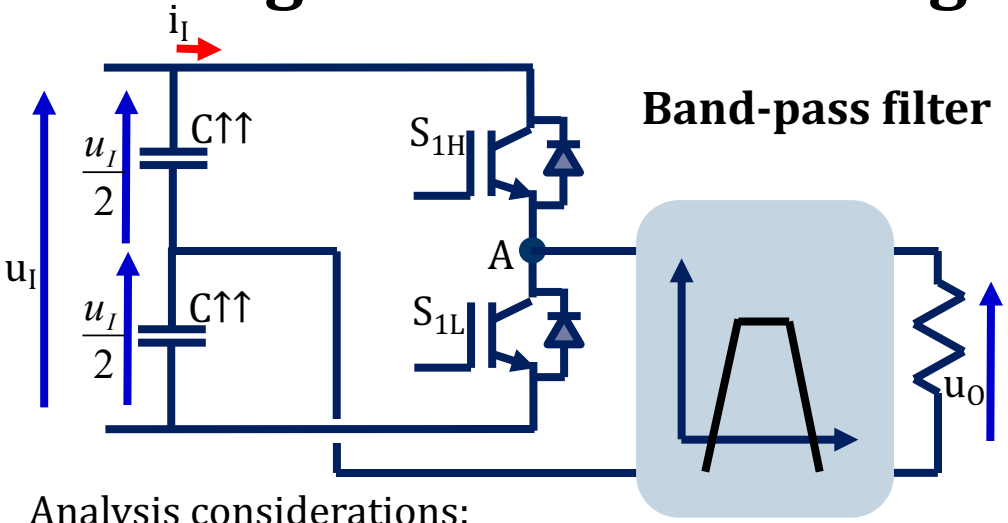
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Obviously, the output waveform is NOT sinusoidal.

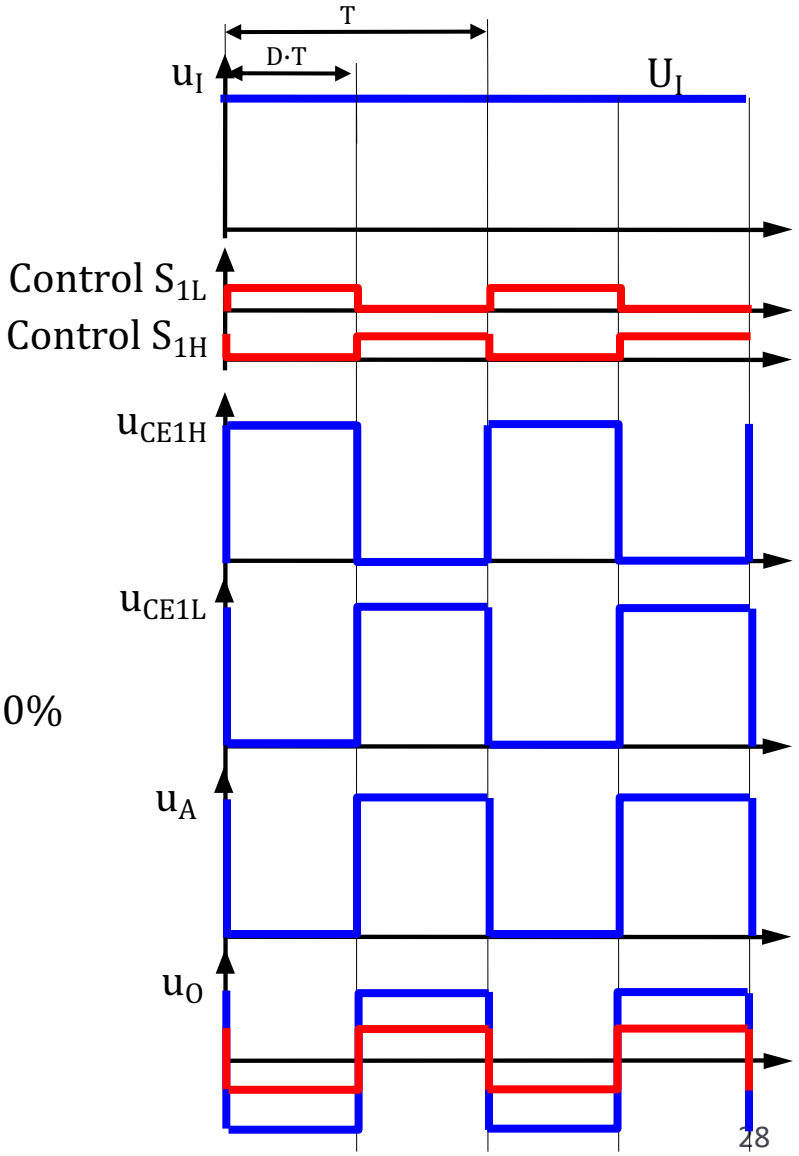
Single-Phase Half-Bridge SW Inverter



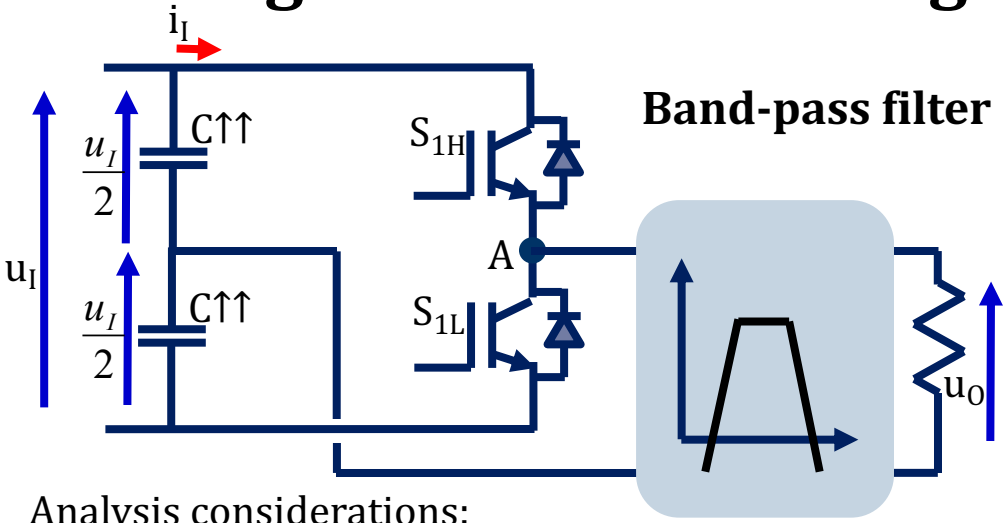
Band-pass filter

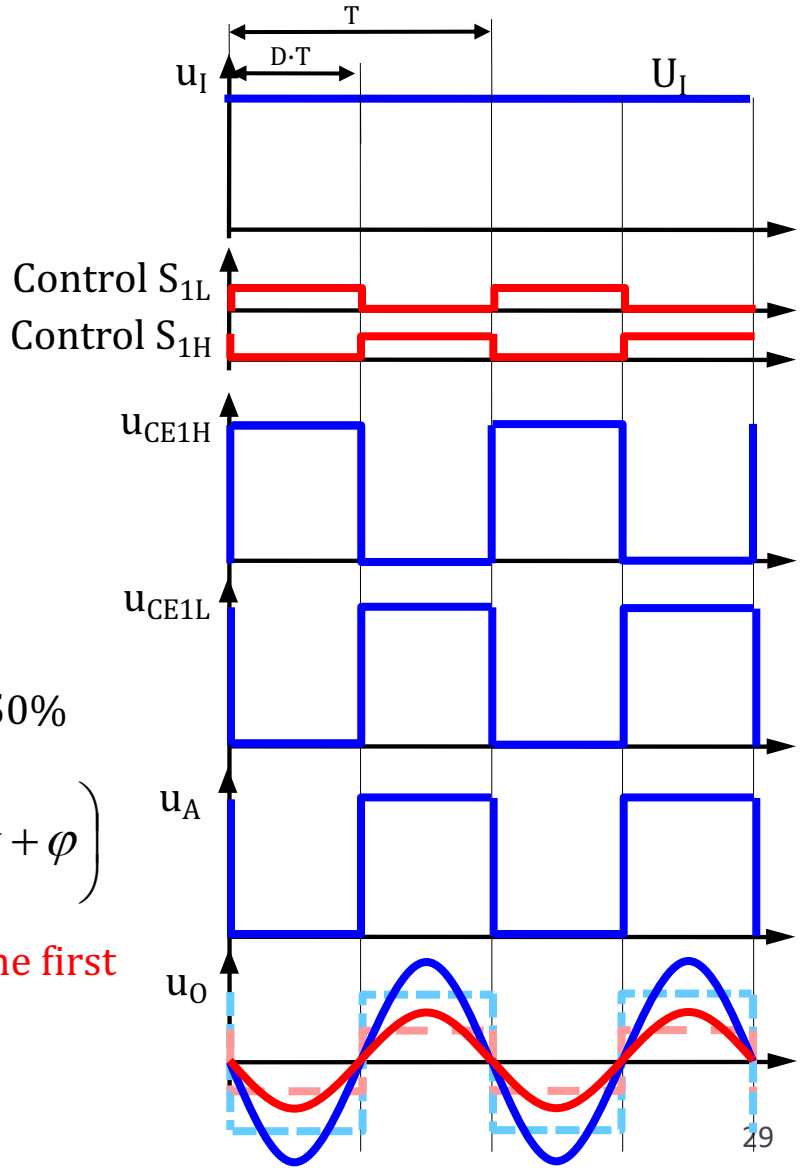
Analysis considerations:

- Ideal switches (no losses)
- Equal voltage at both capacitors
- Resistive load, using a Band Pass Filter (BPF)
- Gate pulses for S_{1L} and S_{1H} are complementary, D=50%



Single-Phase Half-Bridge SW Inverter





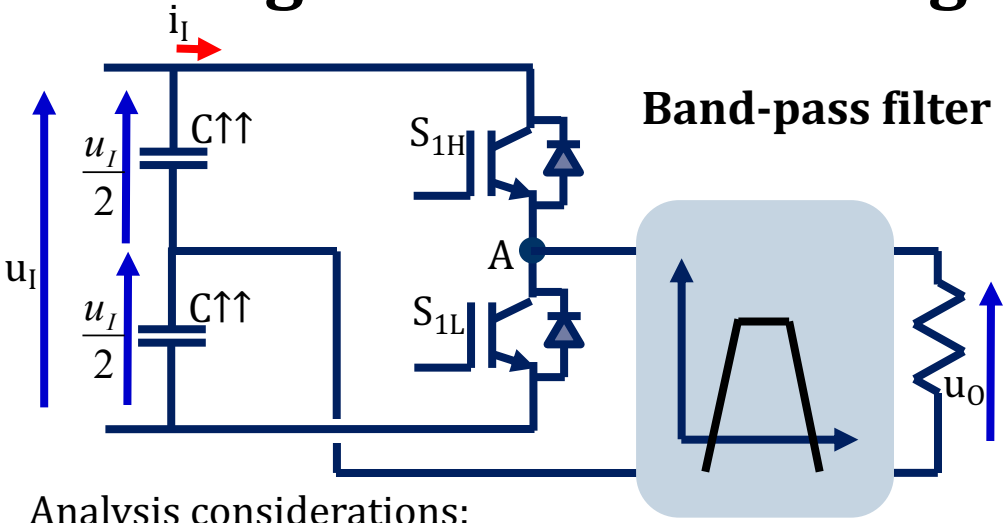
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$$u_O(t) = \frac{2 \cdot u_I}{\pi} \cdot \sin\left(\frac{2\pi}{T} t + \varphi\right) \qquad i_O(t) = \frac{2 \cdot u_I}{\pi \cdot R} \cdot \sin\left(\frac{2\pi}{T} t + \varphi\right)$$

Exercise#3.1: Obtain these expressions (calculating the first order harmonic of a square waveform).

Single-Phase Half-Bridge SW Inverter

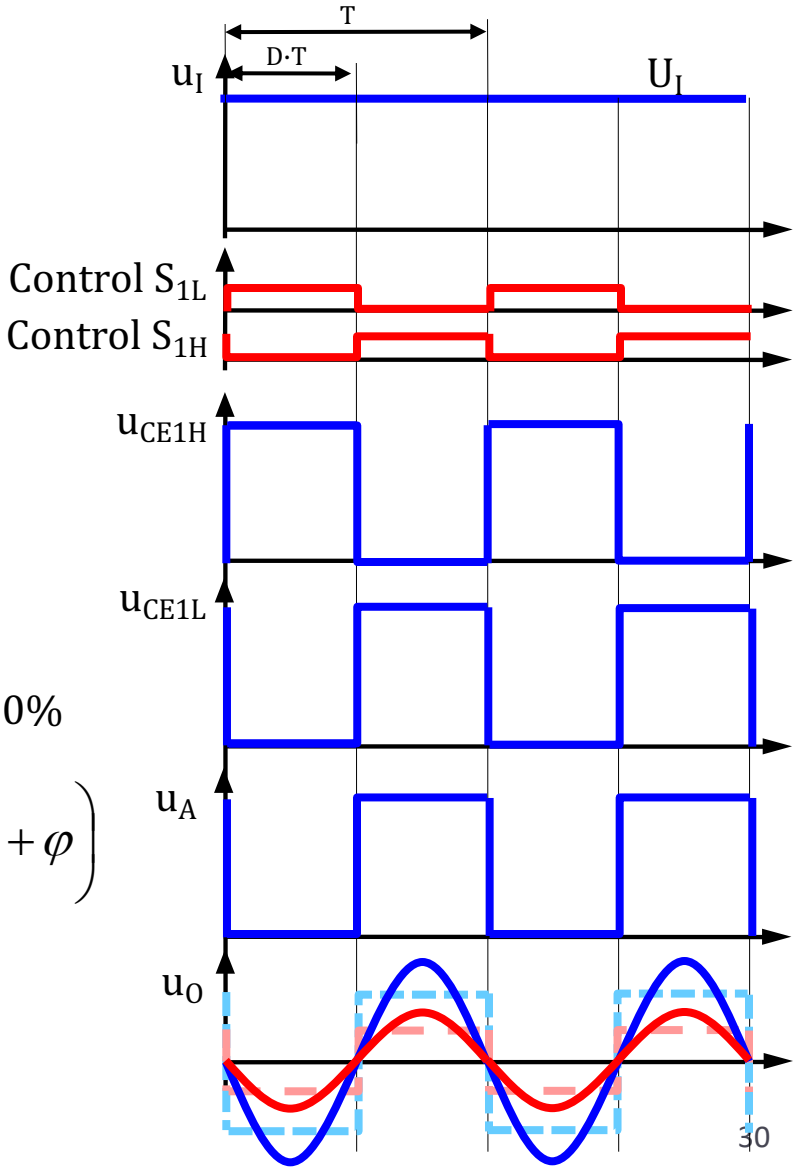


Band-pass filter

Analysis considerations:

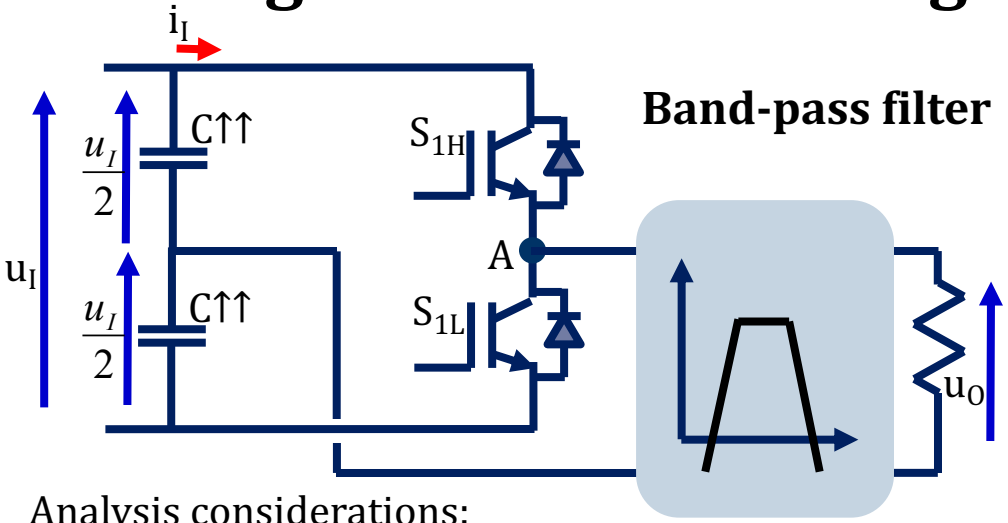
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The frequency can be changed by varying the switching period, T.

Single-Phase Half-Bridge SW Inverter

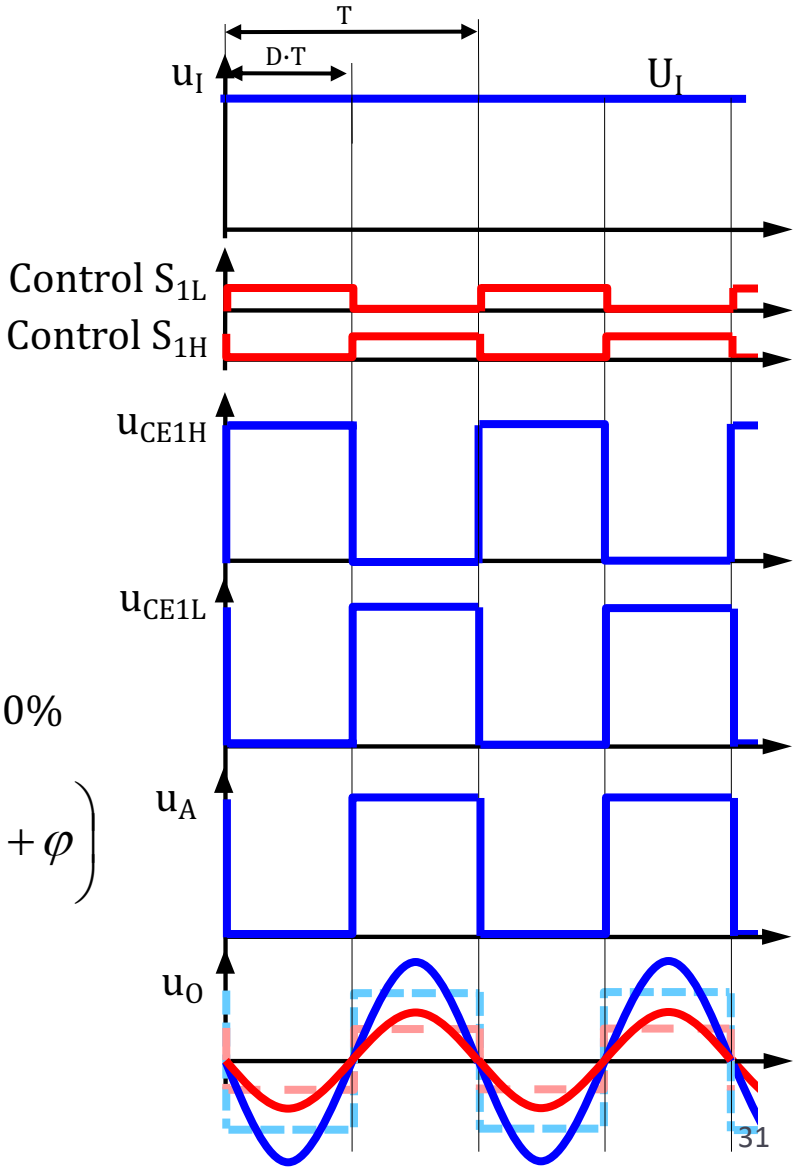


Band-pass filter

Analysis considerations:

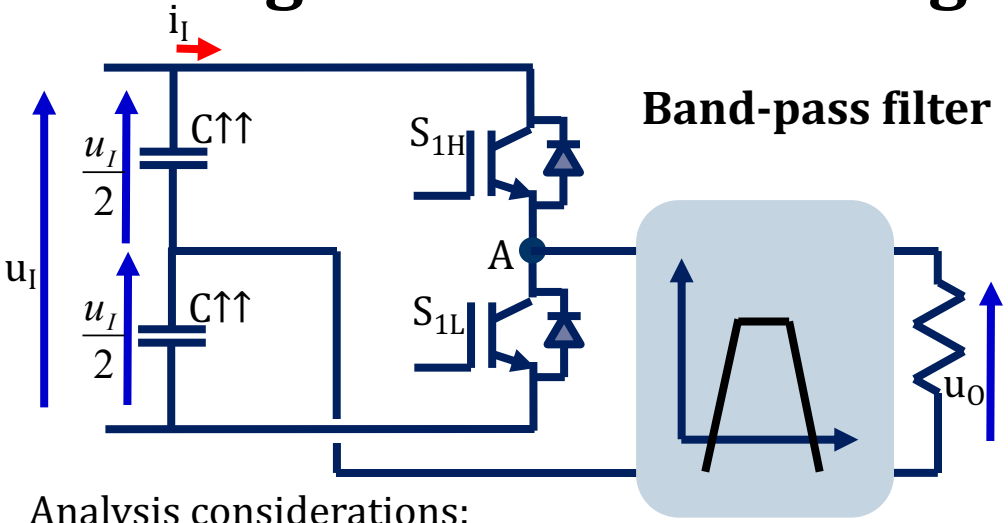
- Ideal switches (no losses)
- Equal voltage at both capacitors
- Resistive load, using a Band Pass Filter (BPF)
- Gate pulses for S_{1L} and S_{1H} are complementary, D=50%

$$u_O(t) = \frac{2 \cdot u_I}{\pi} \cdot \sin\left(\frac{2\pi}{T} t + \varphi\right) \qquad i_O(t) = \frac{2 \cdot u_I}{\pi \cdot R} \cdot \sin\left(\frac{2\pi}{T} t + \varphi\right)$$



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Single-Phase Half-Bridge SW Inverter

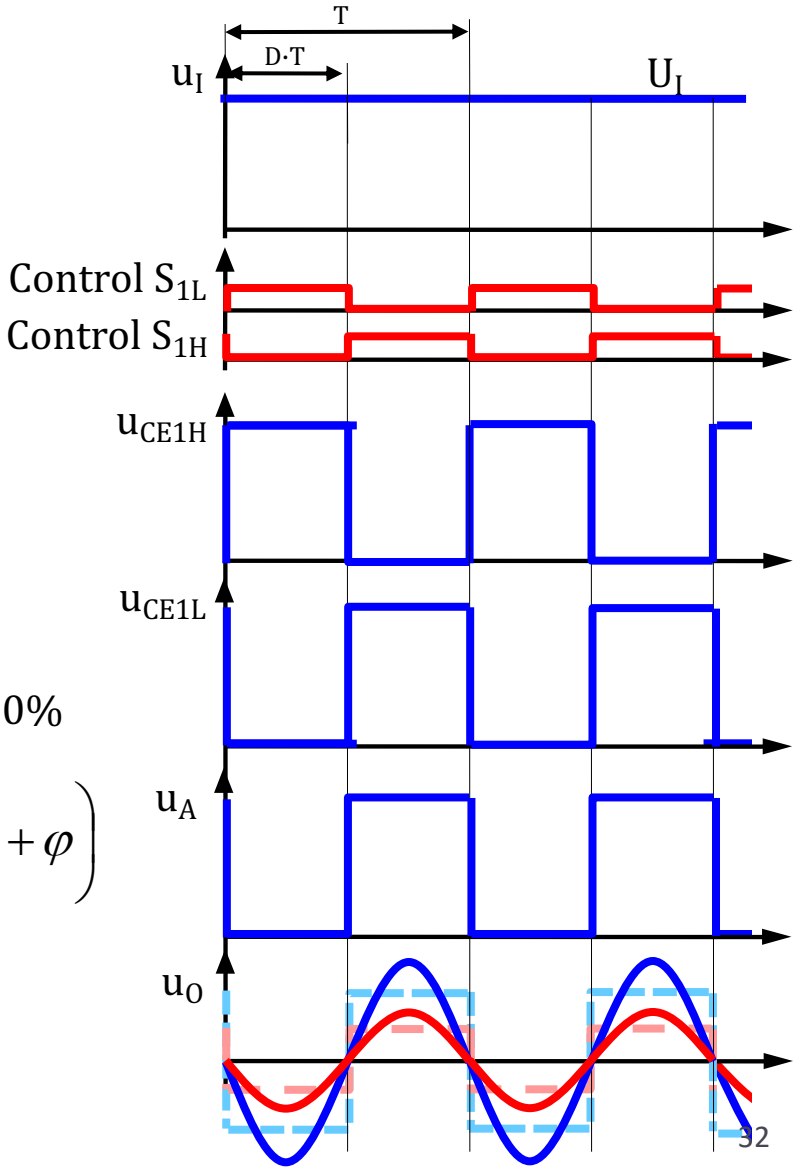


Band-pass filter

Analysis considerations:

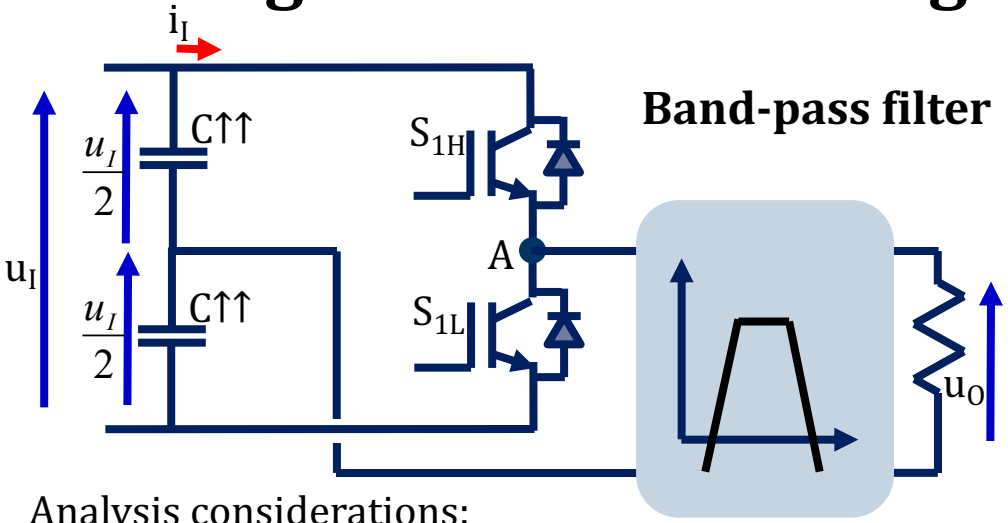
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Single-Phase Half-Bridge SW Inverter

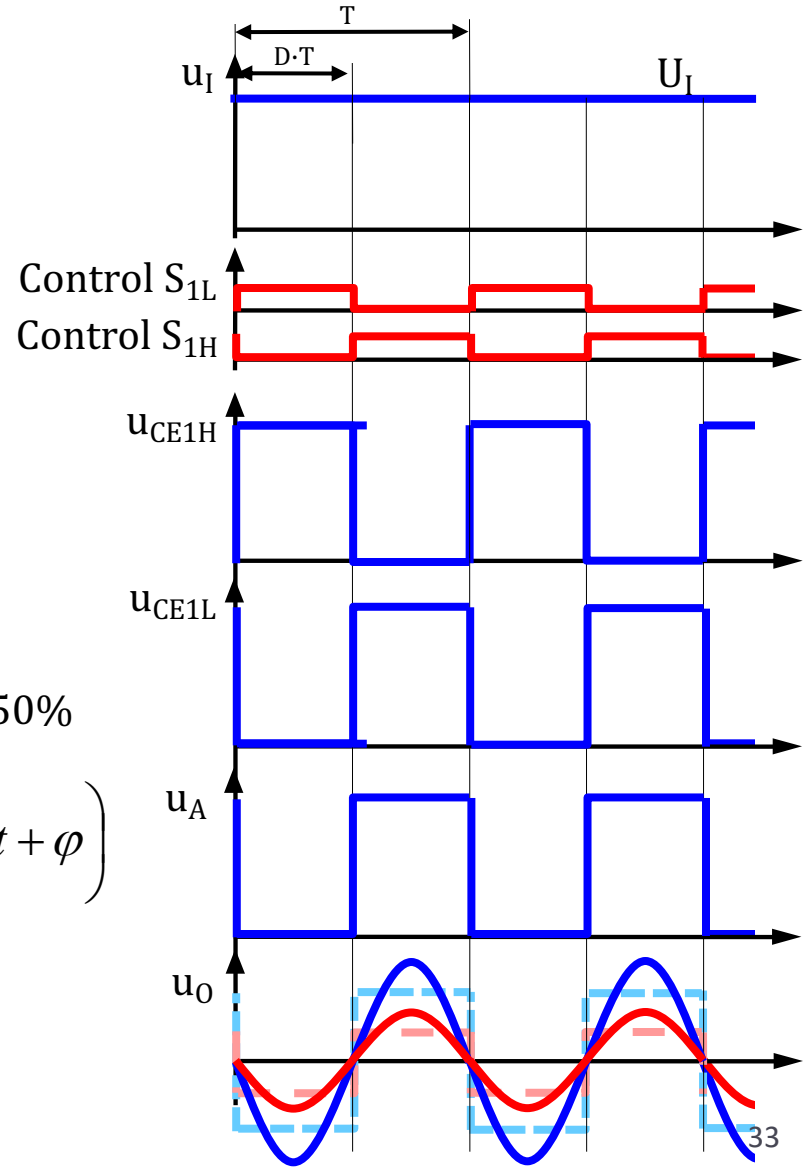


Band-pass filter

Analysis considerations:

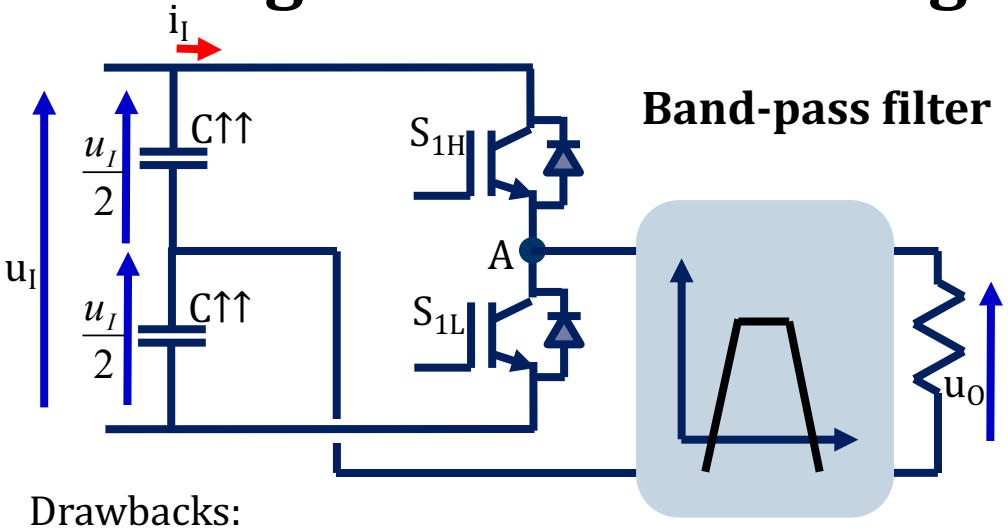
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The frequency can be changed by varying the switching period, T.

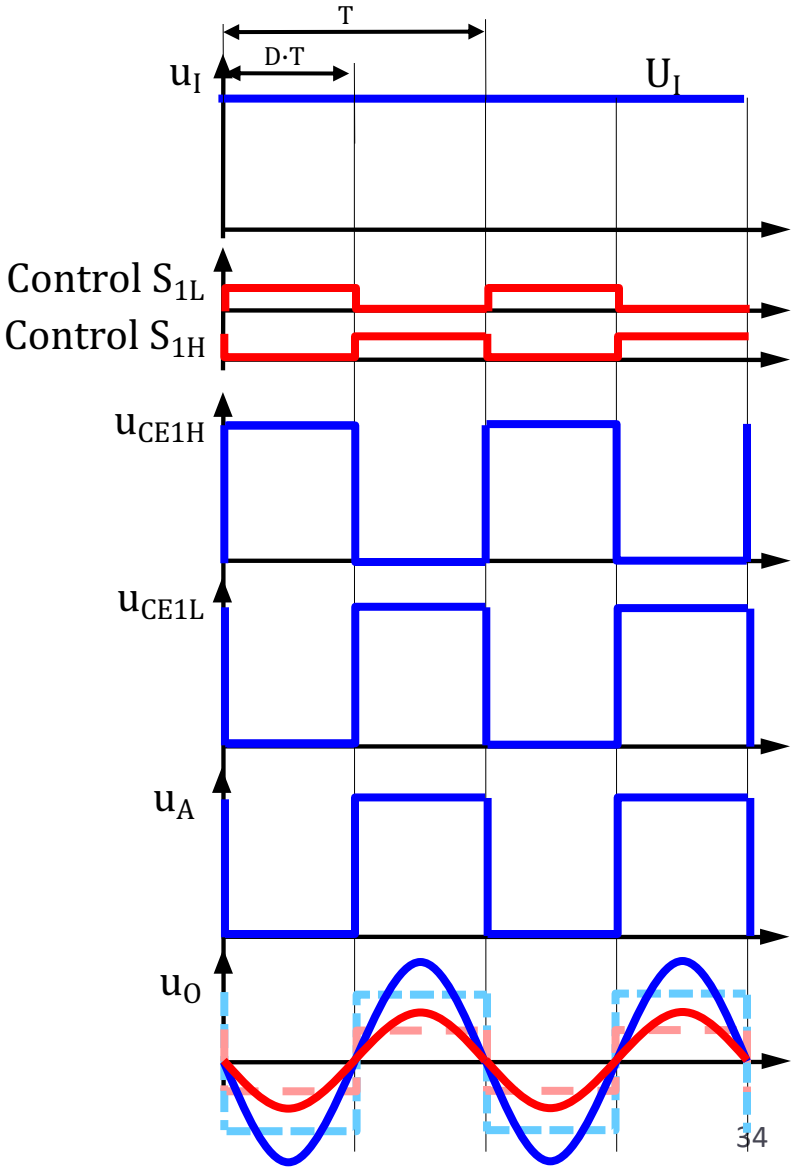
Single-Phase Half-Bridge SW Inverter



Band-pass filter

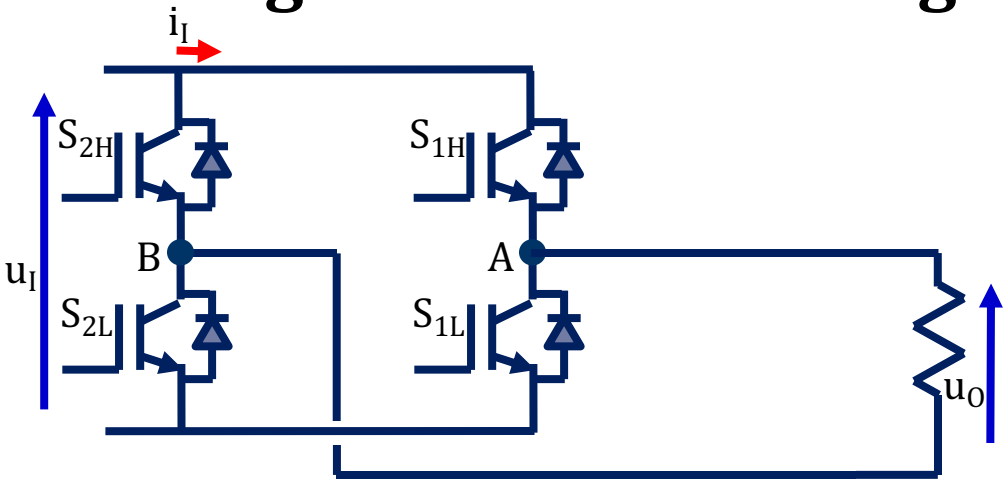
Drawbacks:

- The amplitude can be changed without varying u_I?
- Bulky half-bridge capacitors
- How to equalize the capacitor voltages?
- Is the filter feasible? (later)



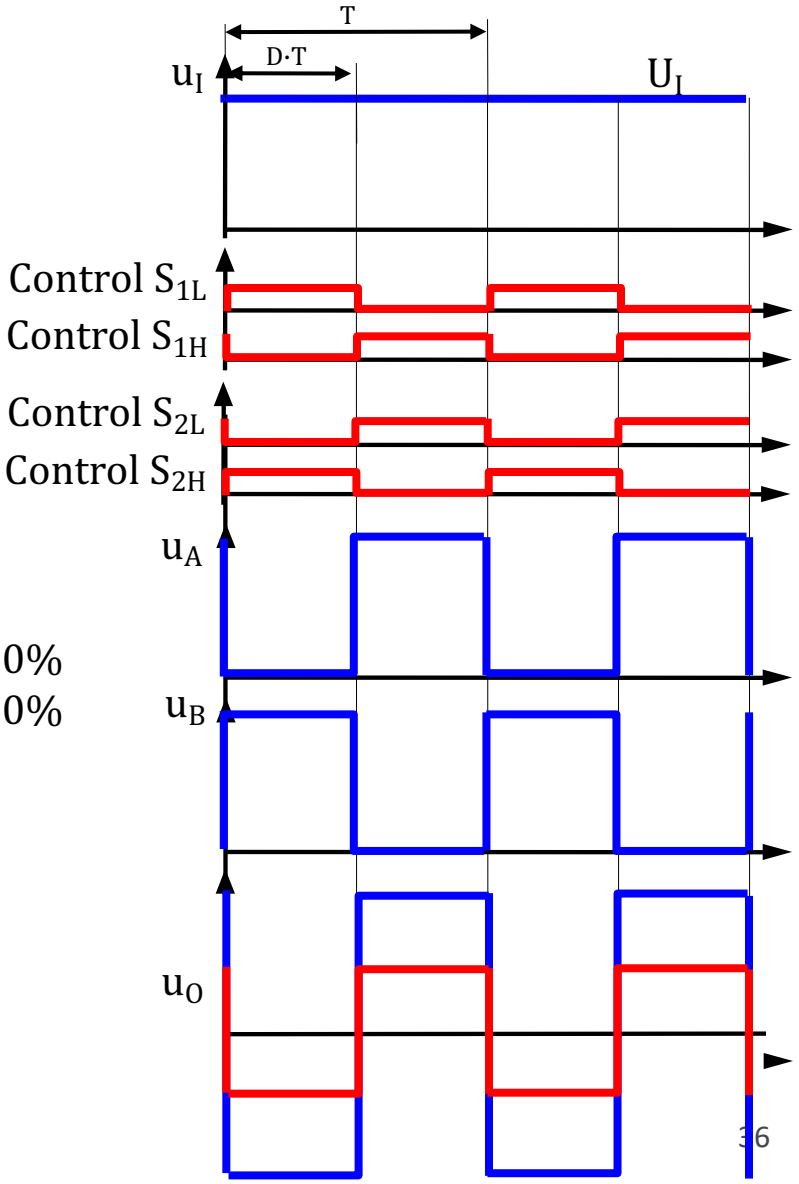
Single-Phase, Square Waveform, Full-Bridge Inverter

Single-Phase Full-Bridge SW Inverter

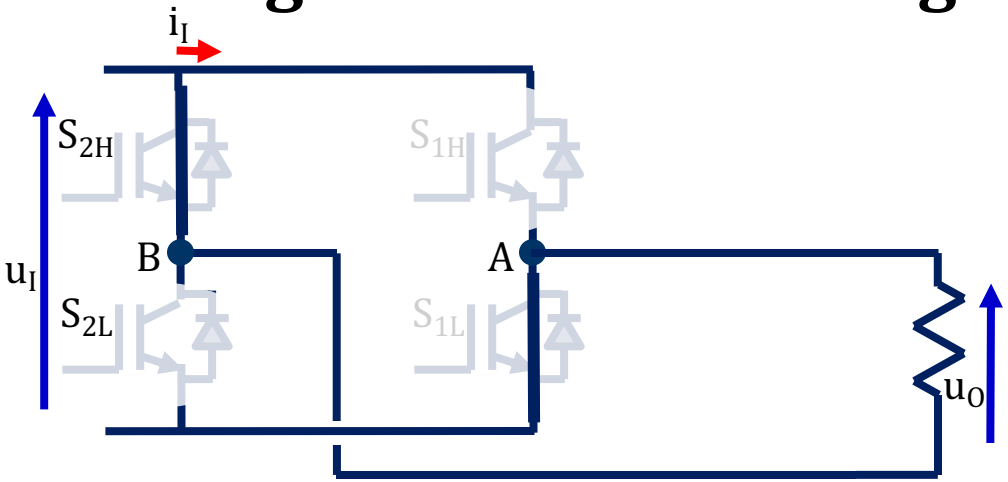


Analysis considerations:

- Ideal switches (no losses)
- Resistive load
- Gate pulses for S_{1L} and S_{1H} are complementary, D=50%
- Gate pulses for S_{2L} and S_{2H} are complementary, D=50%

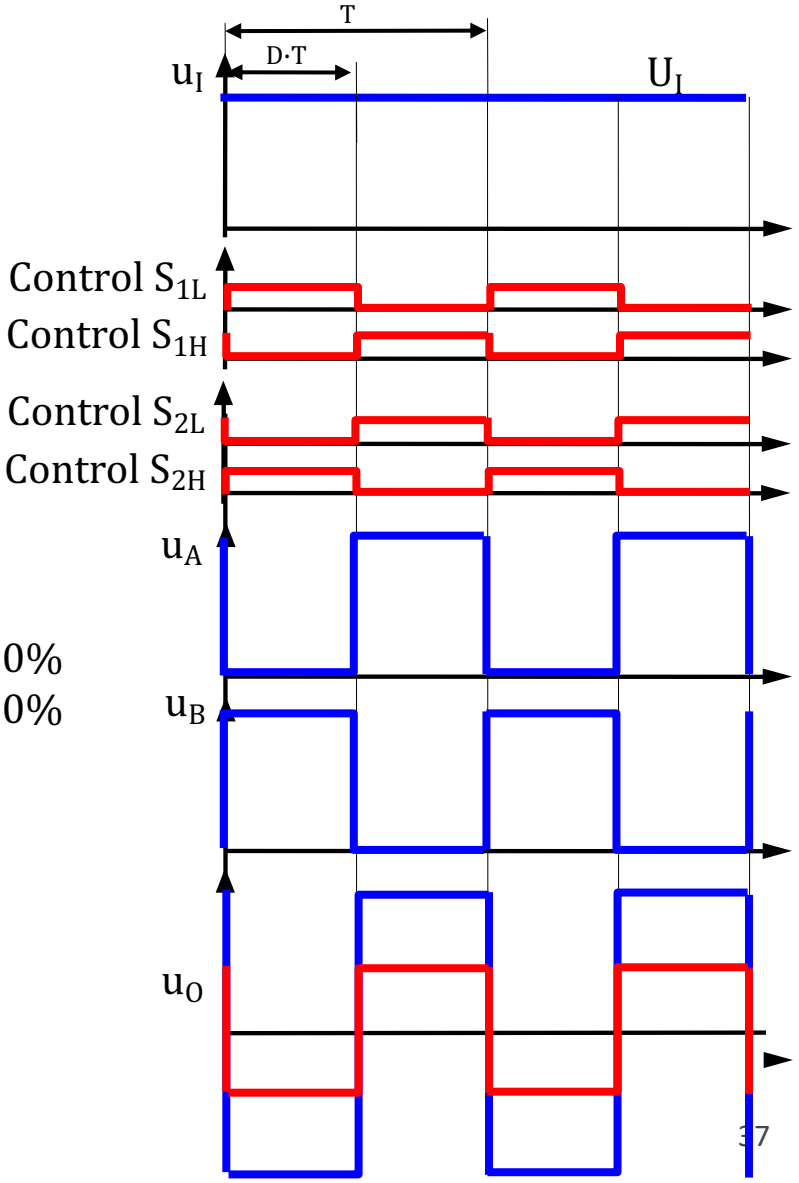


Single-Phase Full-Bridge SW Inverter

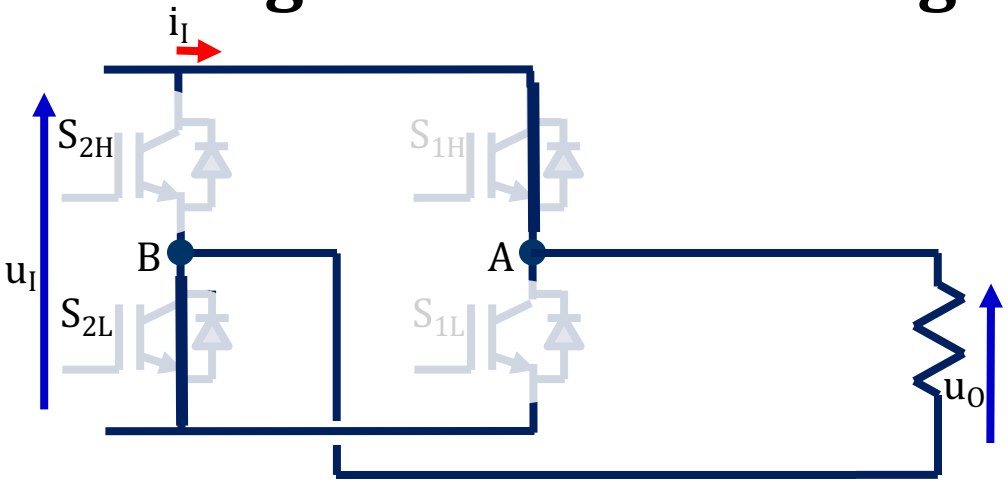


Analysis considerations:

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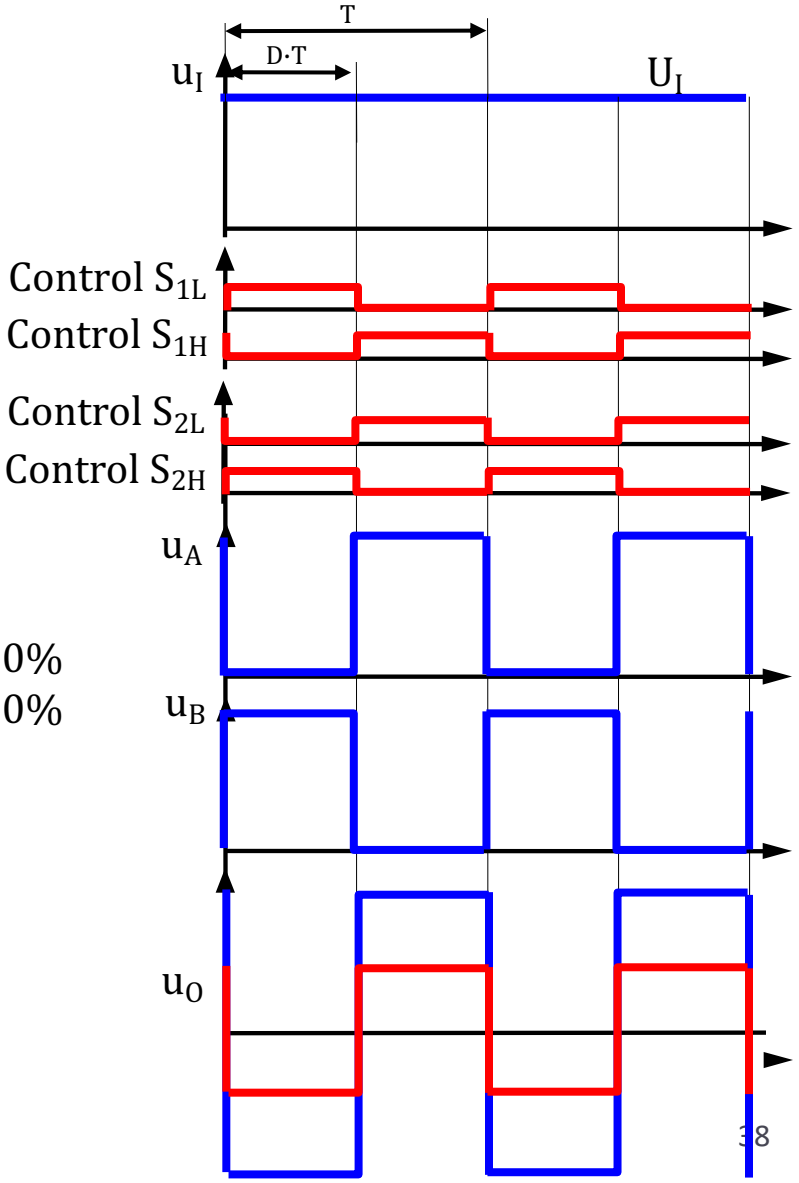


Single-Phase Full-Bridge SW Inverter

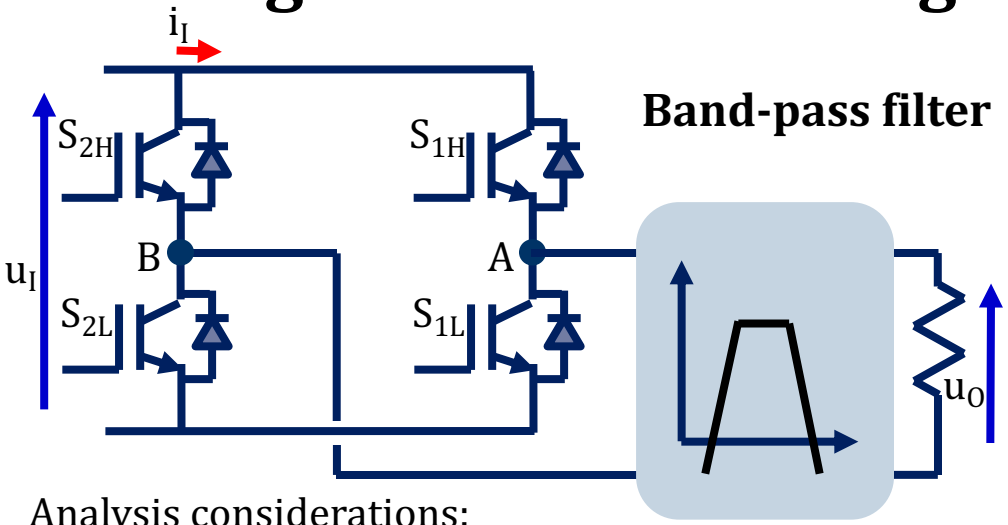


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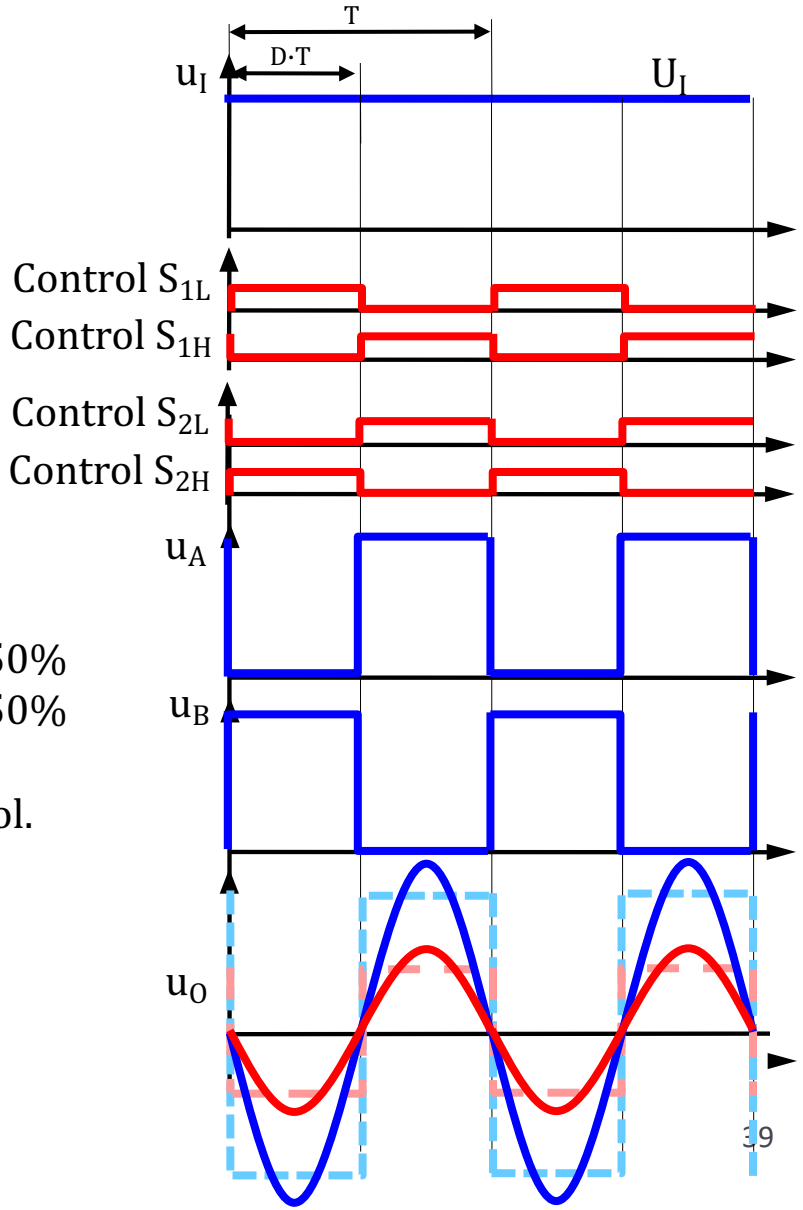
Single-Phase Full-Bridge SW Inverter



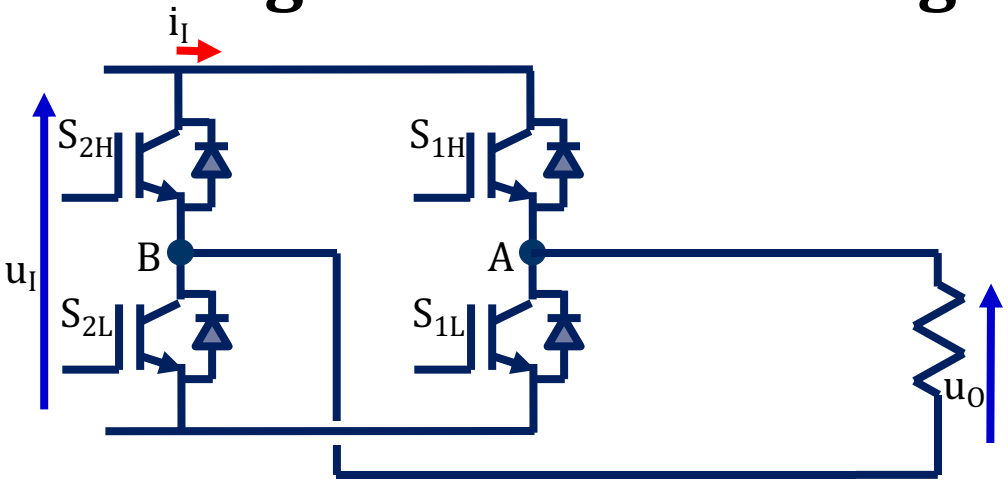
Band-pass filter

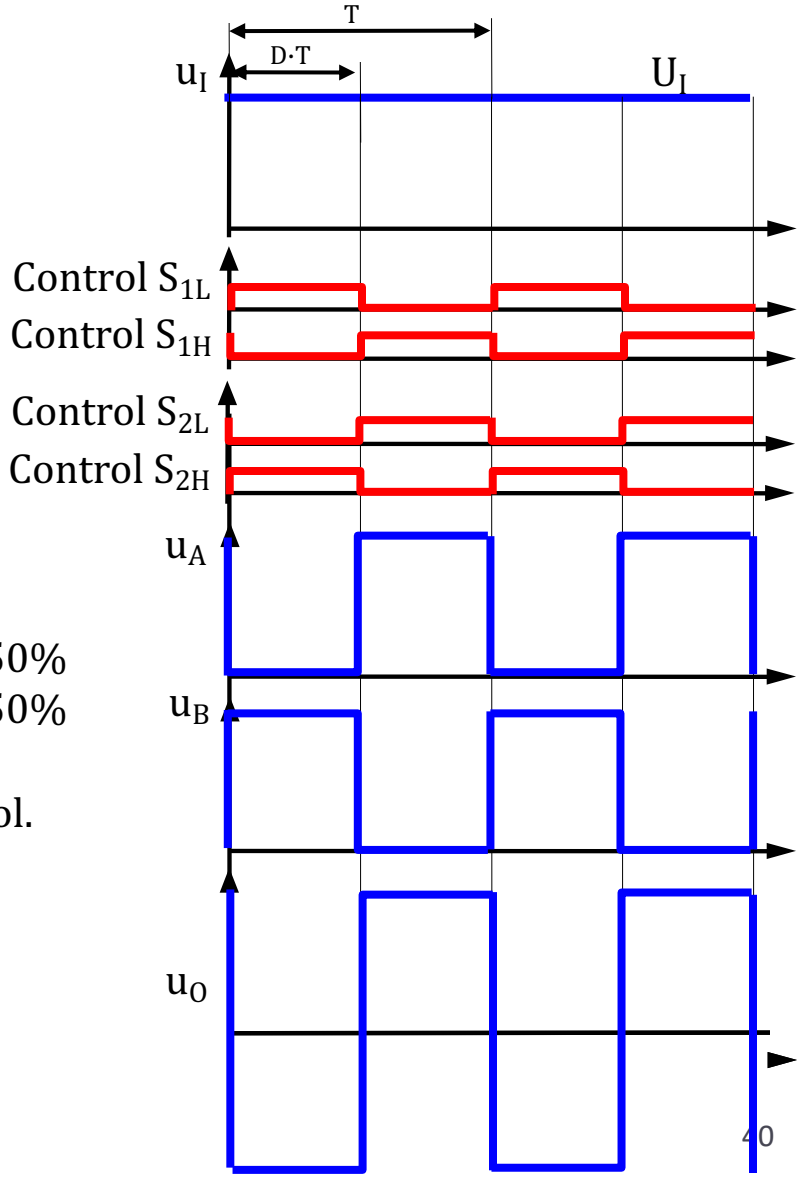
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- The output could be filtered, and thus a sinusoidal waveform could be obtained, with frequency control.
- What about the amplitude?



Single-Phase Full-Bridge SW Inverter

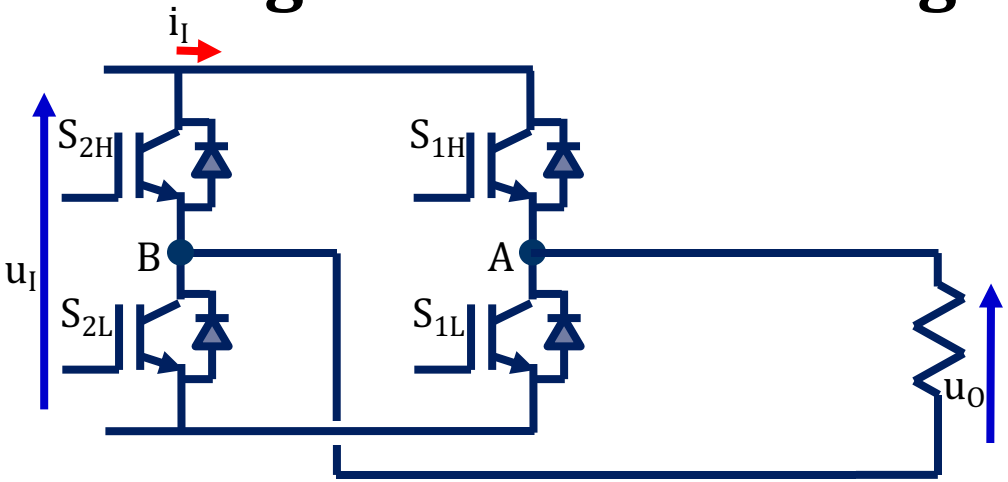




Analysis considerations:

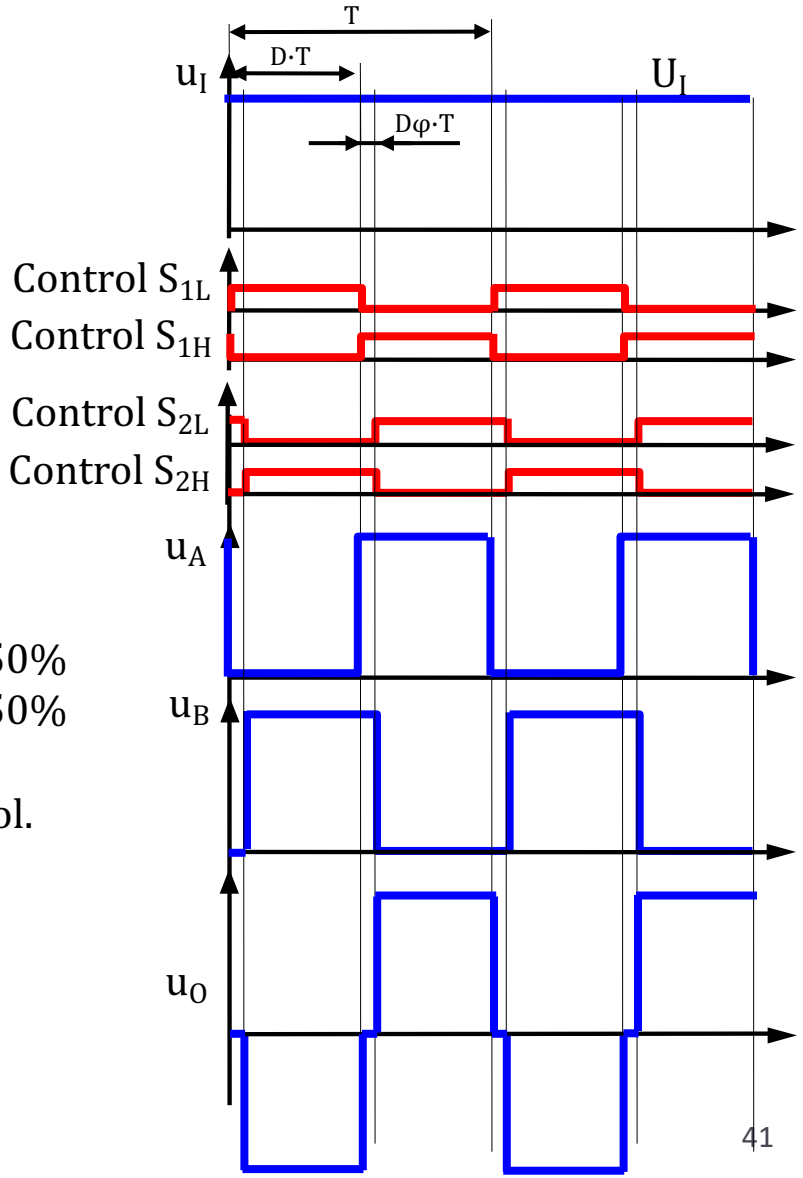
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Single-Phase Full-Bridge SW Inverter

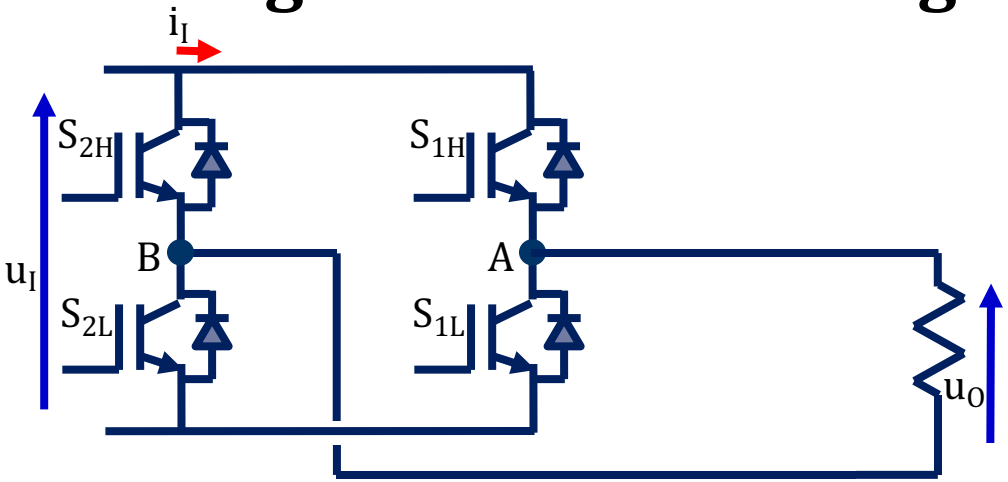


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- What about the amplitude?: **Phase shift**

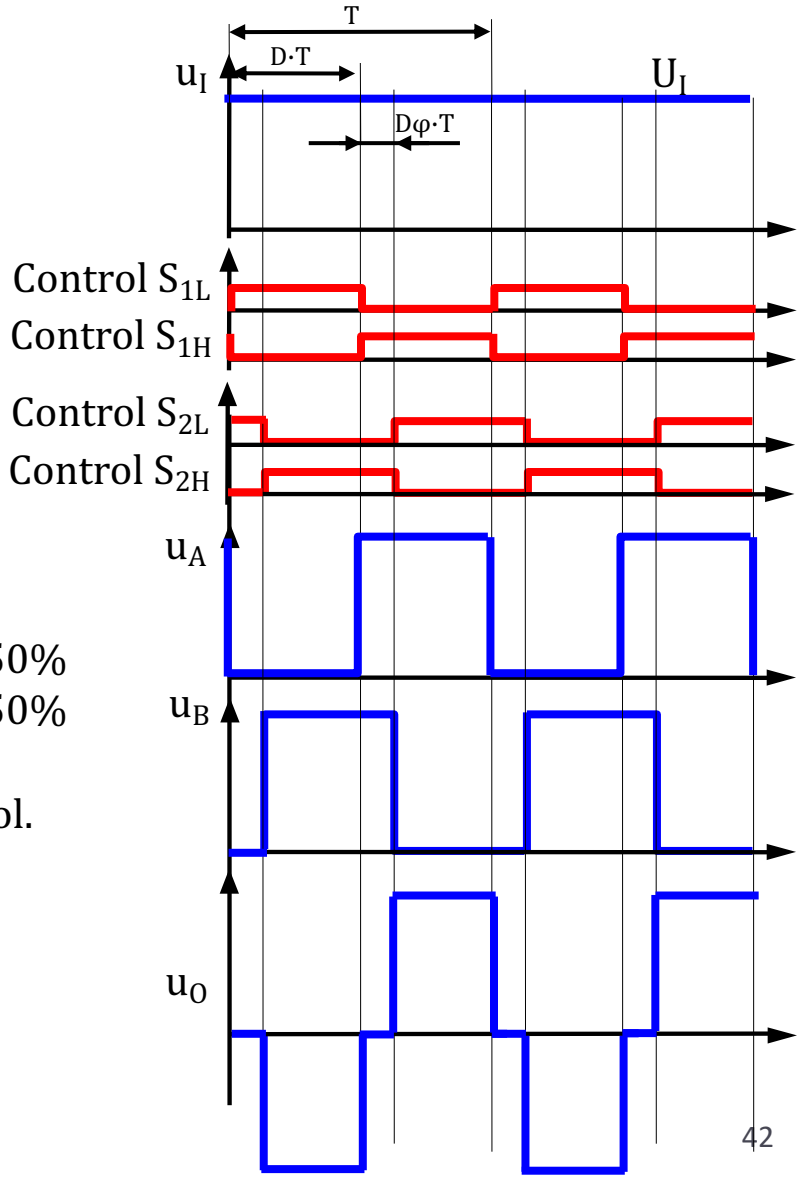


Single-Phase Full-Bridge SW Inverter

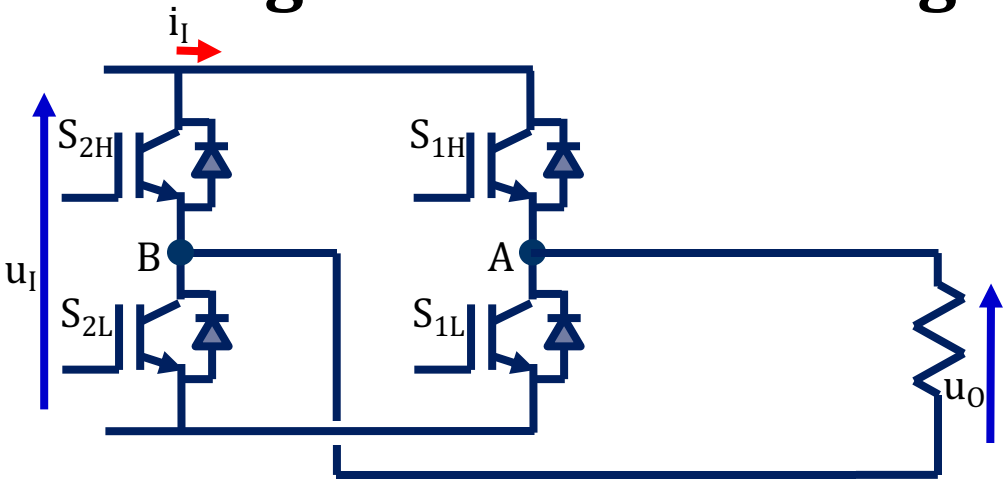


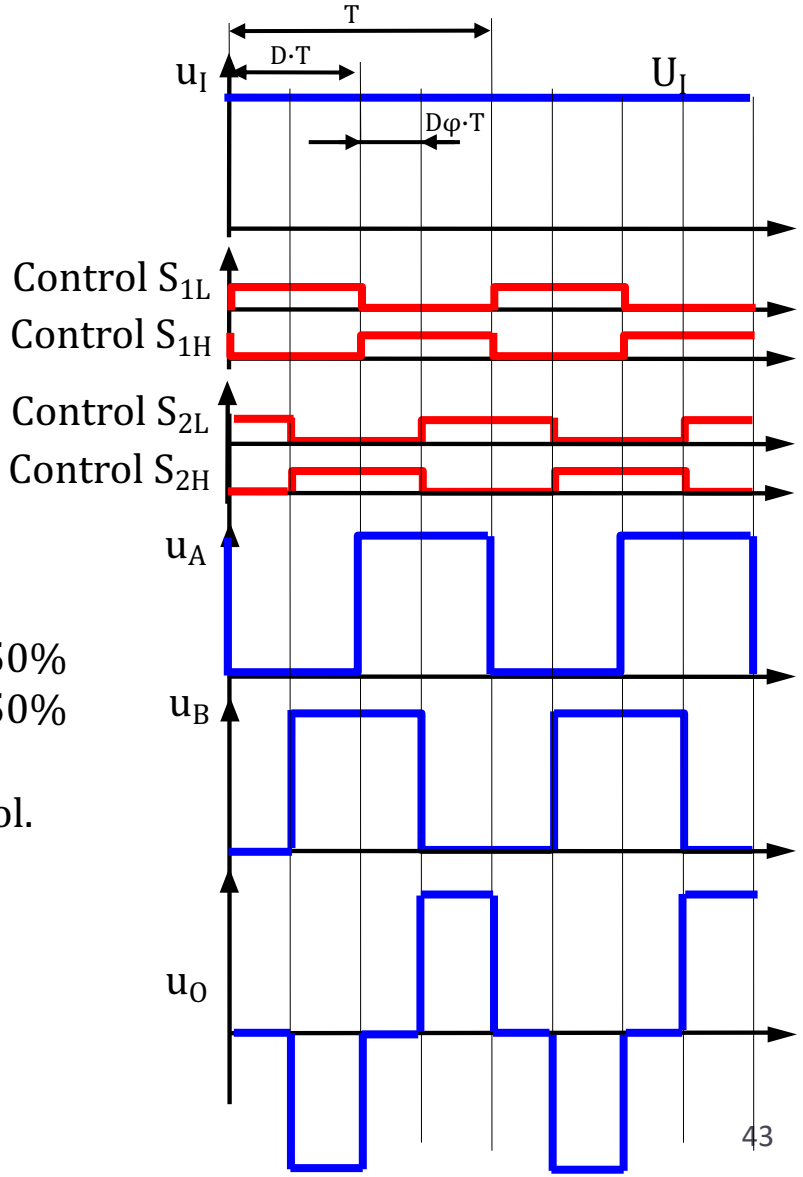
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Single-Phase Full-Bridge SW Inverter

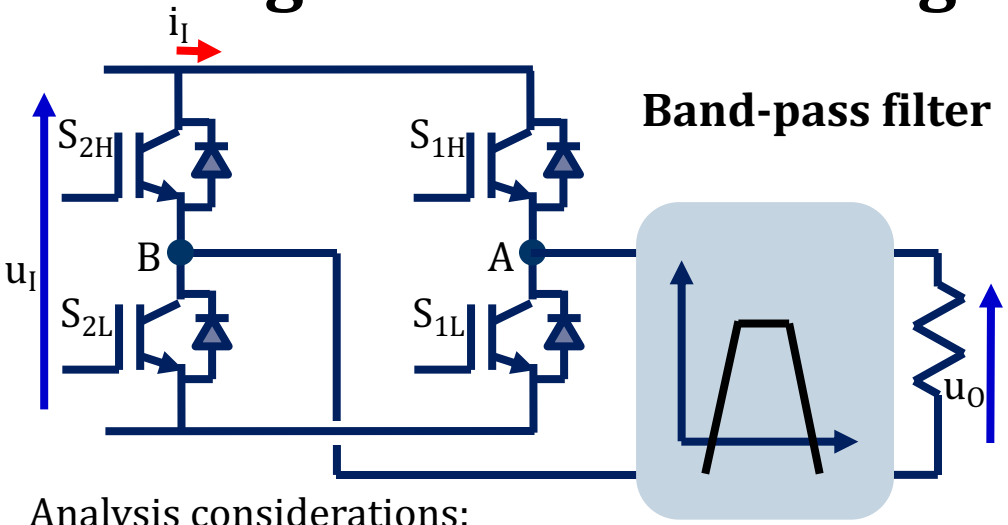




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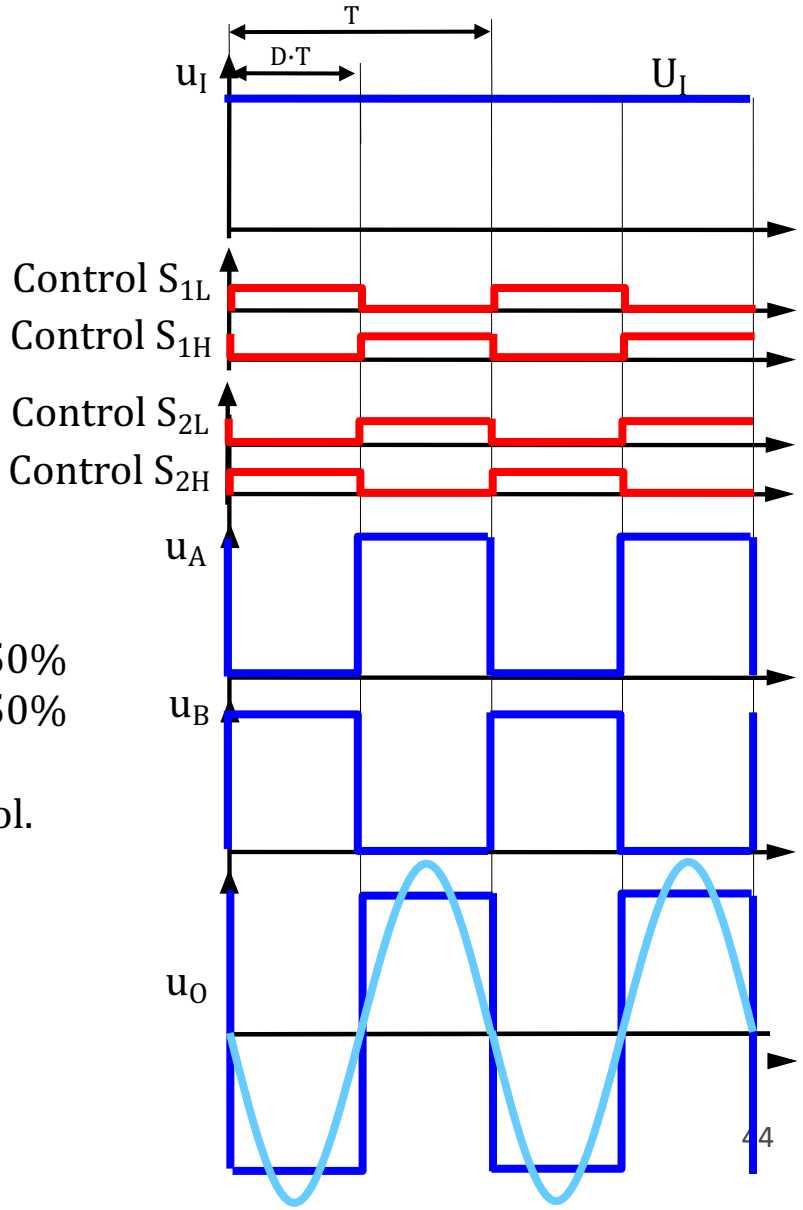
Single-Phase Full-Bridge SW Inverter



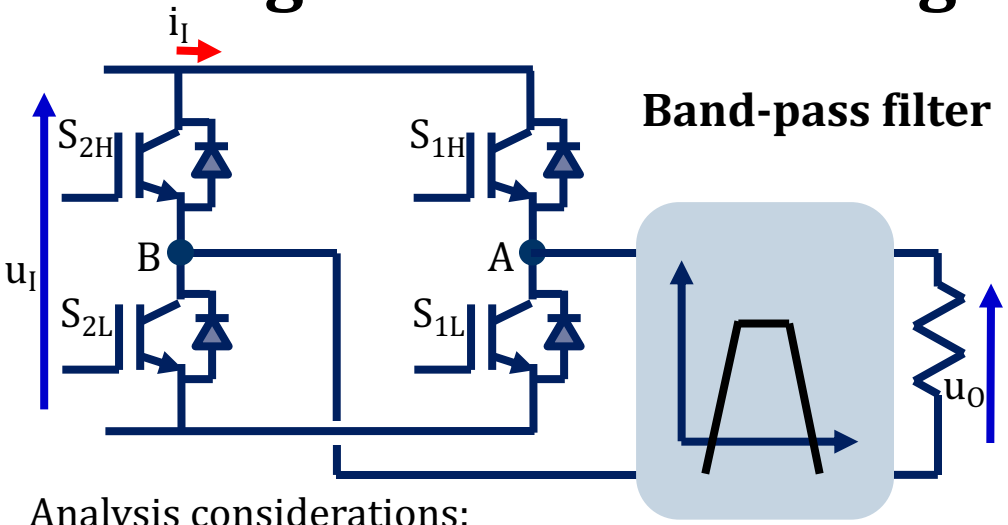
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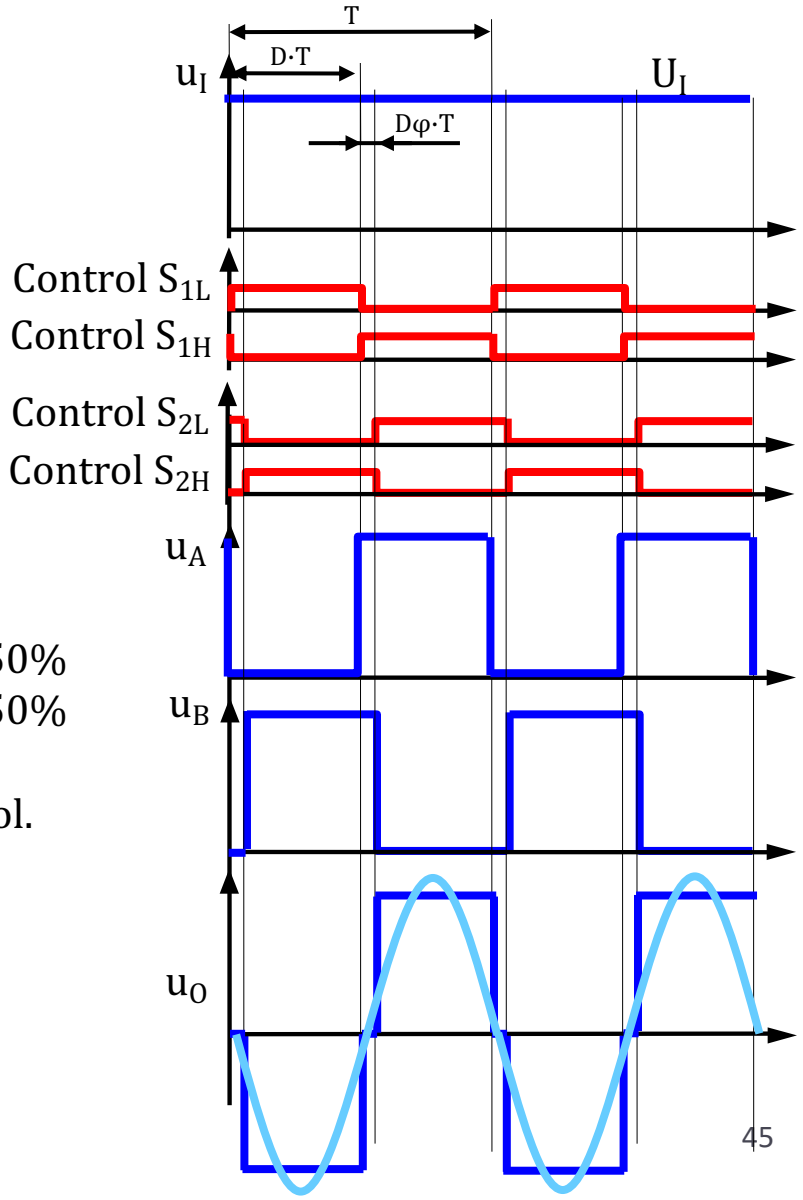
Single-Phase Full-Bridge SW Inverter



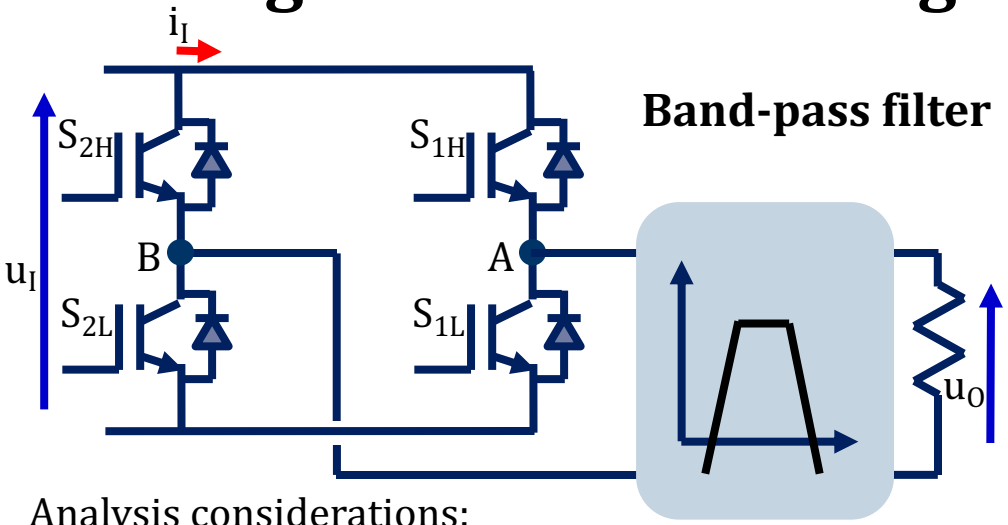
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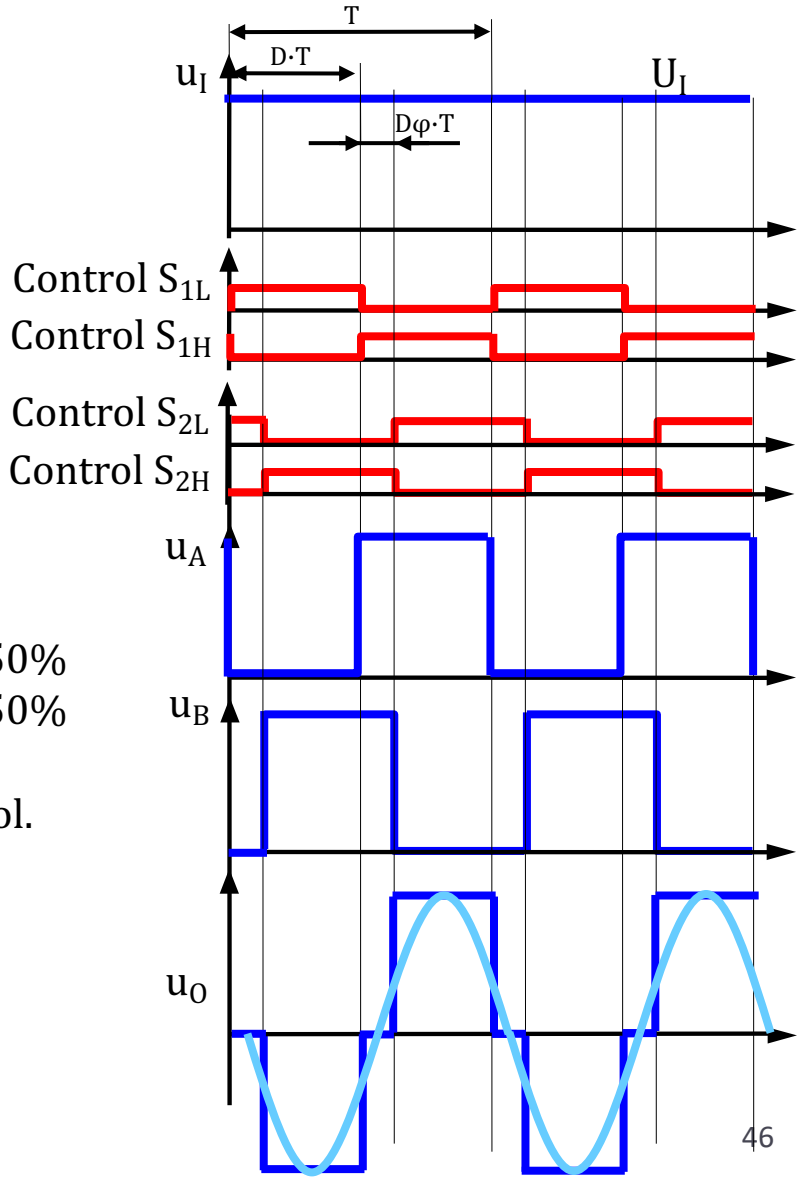
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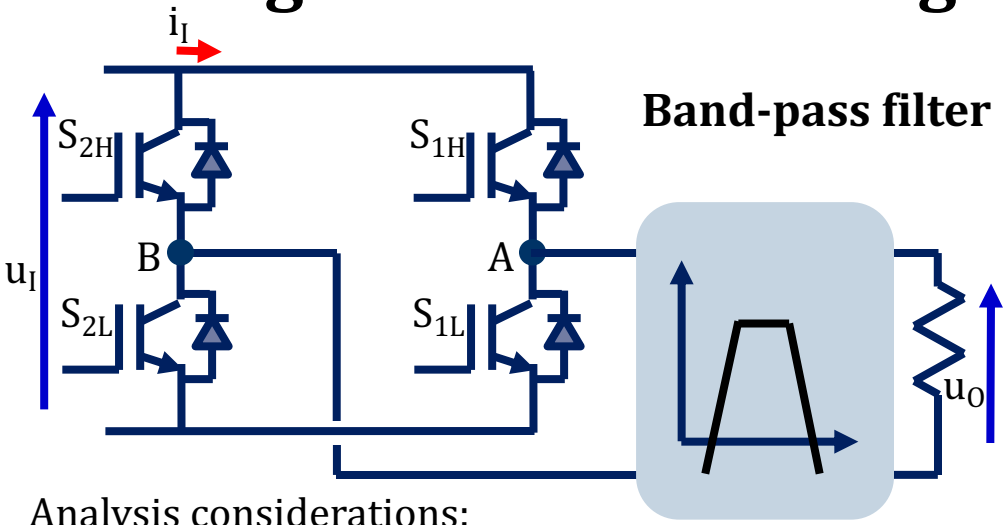
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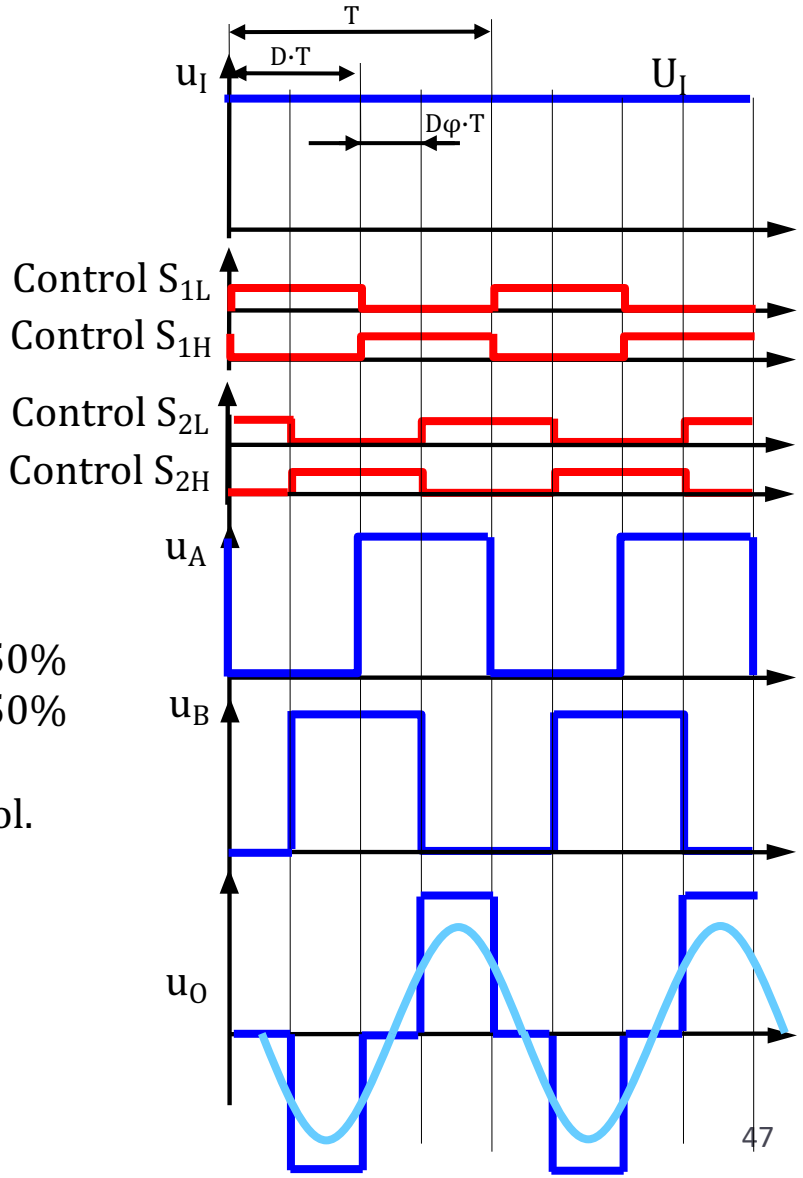
Single-Phase Full-Bridge SW Inverter



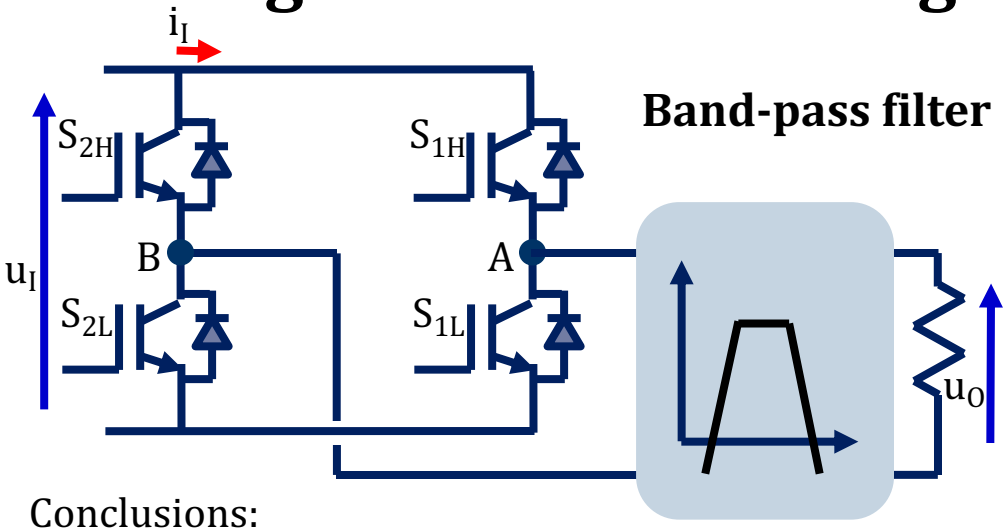
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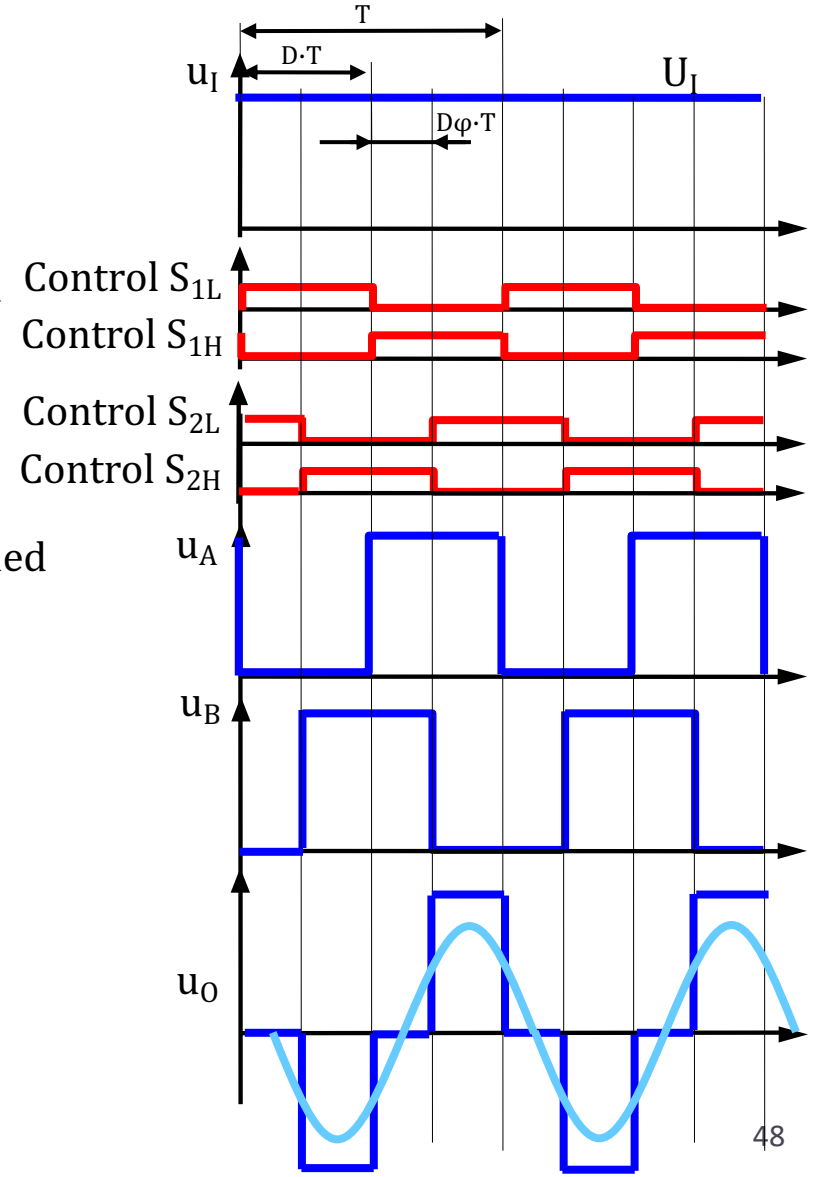
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Single-Phase Full-Bridge SW Inverter



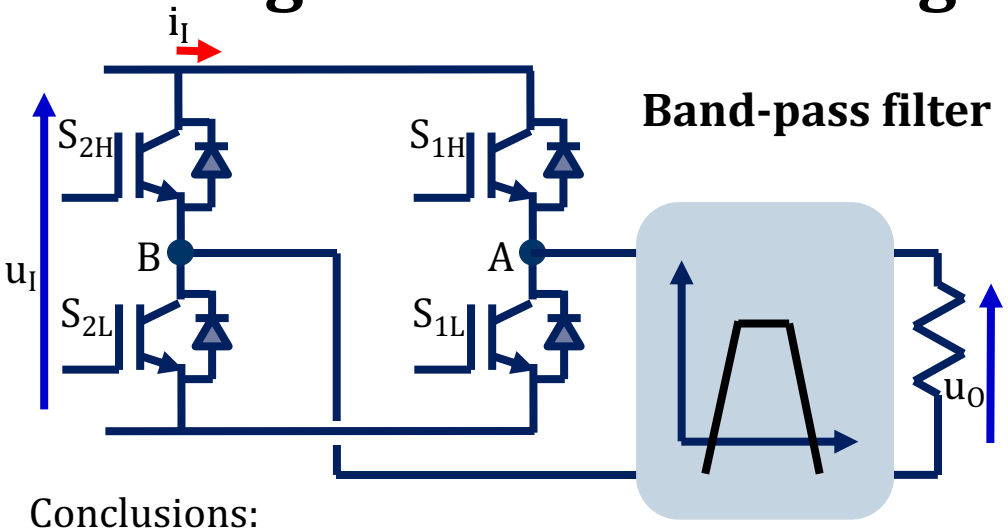
Band-pass filter



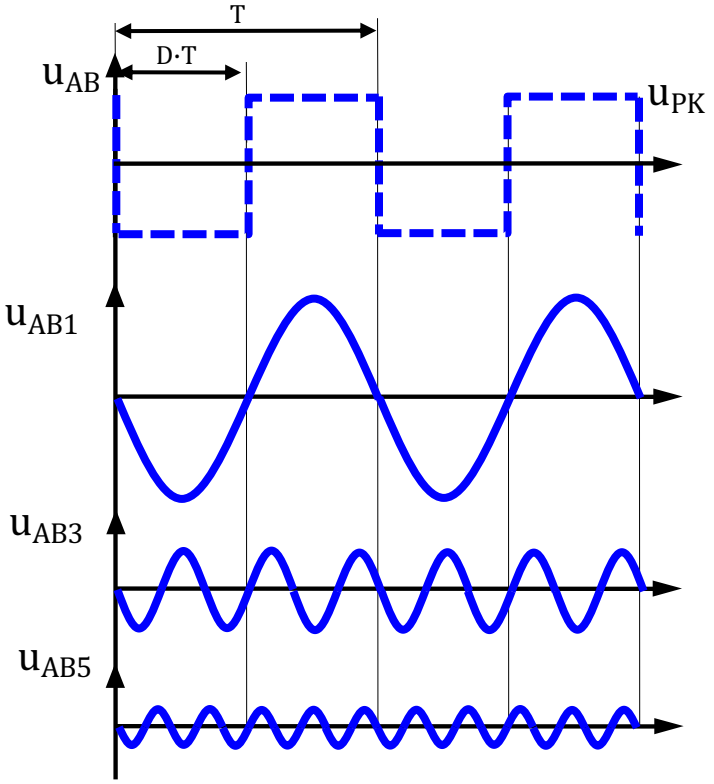
Conclusions:

- Finally, we have an output waveform, with controlled amplitude, phase and frequency.
- But is the waveform sinusoidal?

Single-Phase Full-Bridge SW Inverter



Band-pass filter

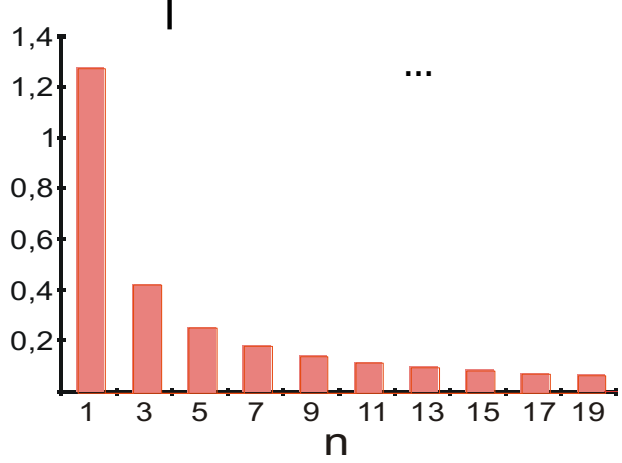


Conclusions:

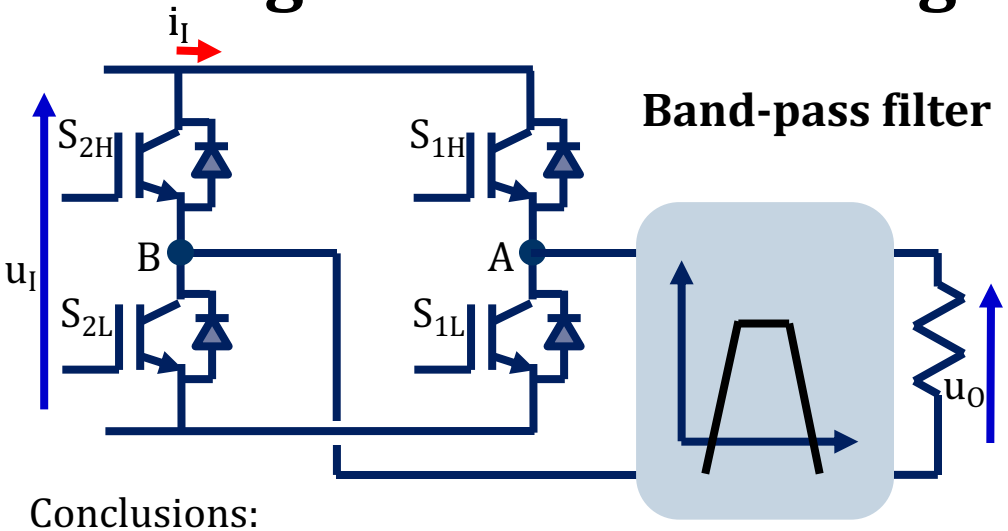
- Finally, we have an output waveform, with controlled amplitude, phase and frequency.
- But is the waveform sinusoidal?

$$u_{ABn\langle PEAK\rangle} = \frac{2}{\pi}\cdot\int_0^{\pi} u_{PK}\cdot\sin(n\cdot\theta)\cdot d\theta$$

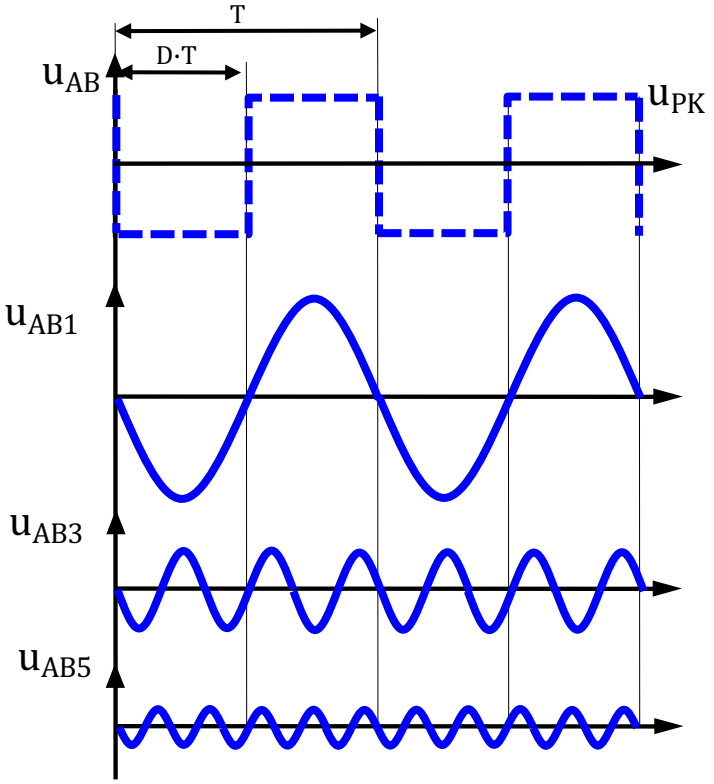
$$u_{ABn\langle PEAK\rangle} = \frac{2\cdot u_{PK}}{n\cdot\pi}\cdot(1-\cos(n\cdot\pi))$$



Single-Phase Full-Bridge SW Inverter



Band-pass filter

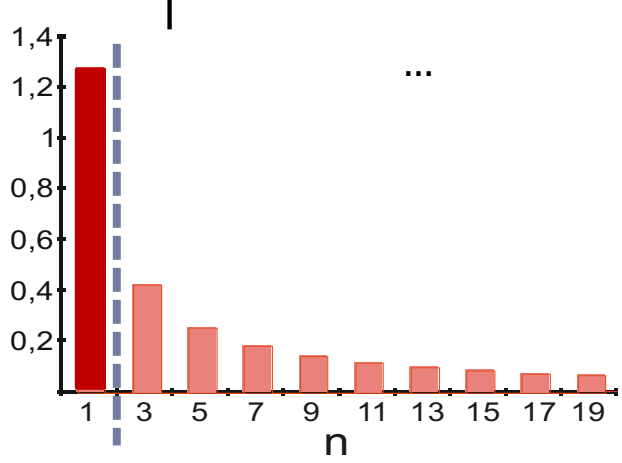


Conclusions:

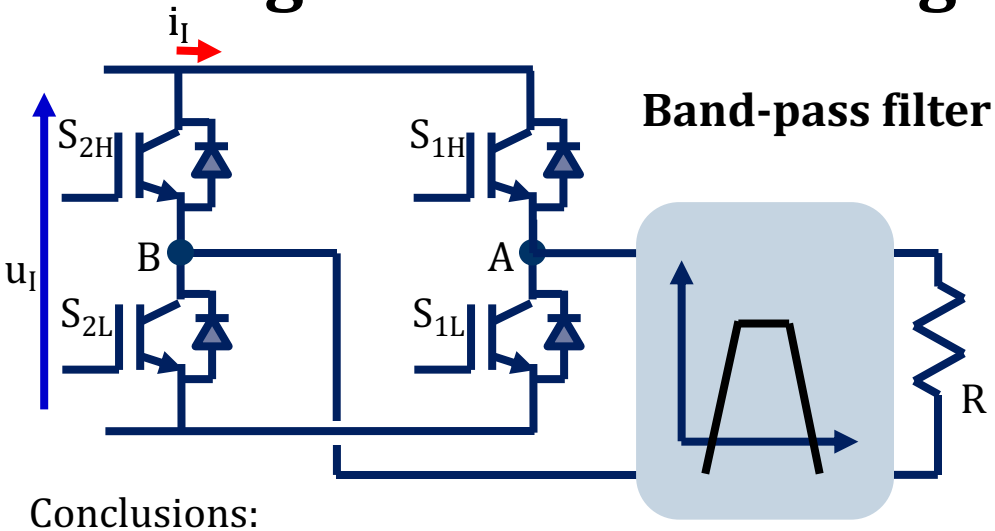
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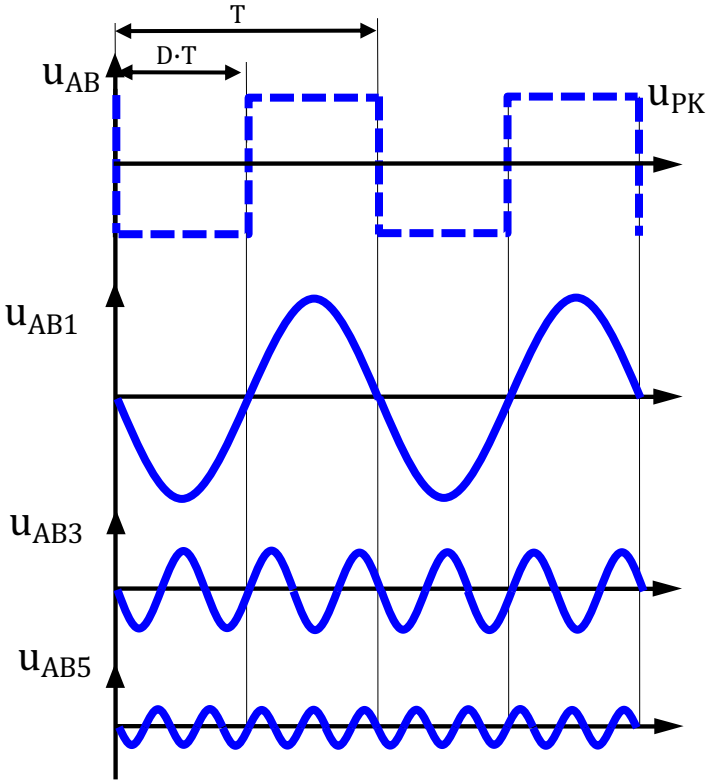
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Single-Phase Full-Bridge SW Inverter

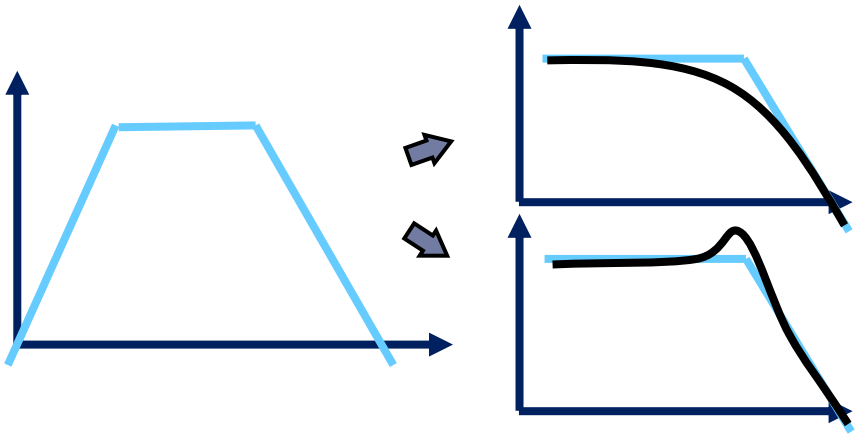


Band-pass filter

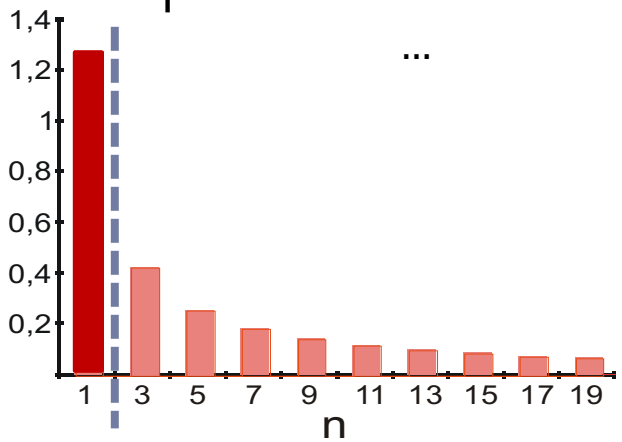


Conclusions:

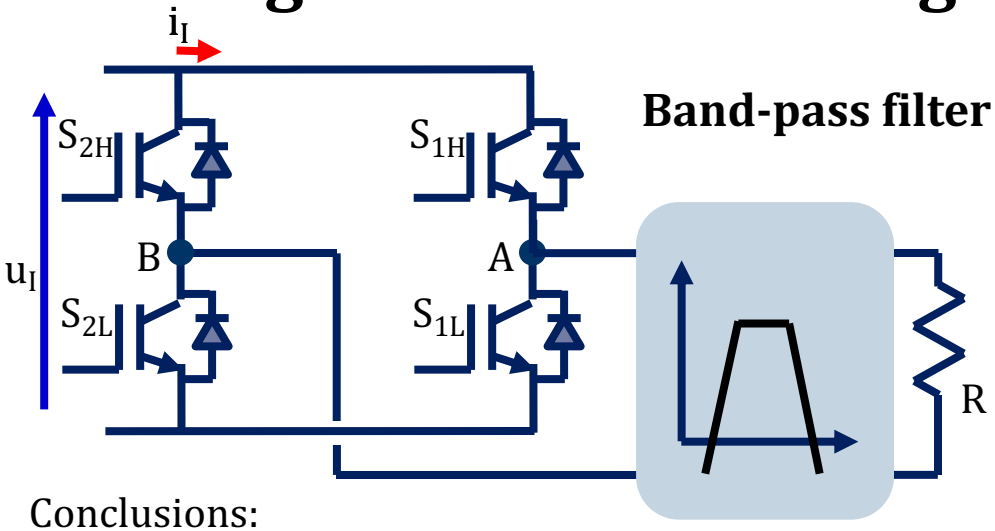
- Finally, we have an output waveform, with controlled amplitude, phase and frequency.
- But is the waveform sinusoidal?



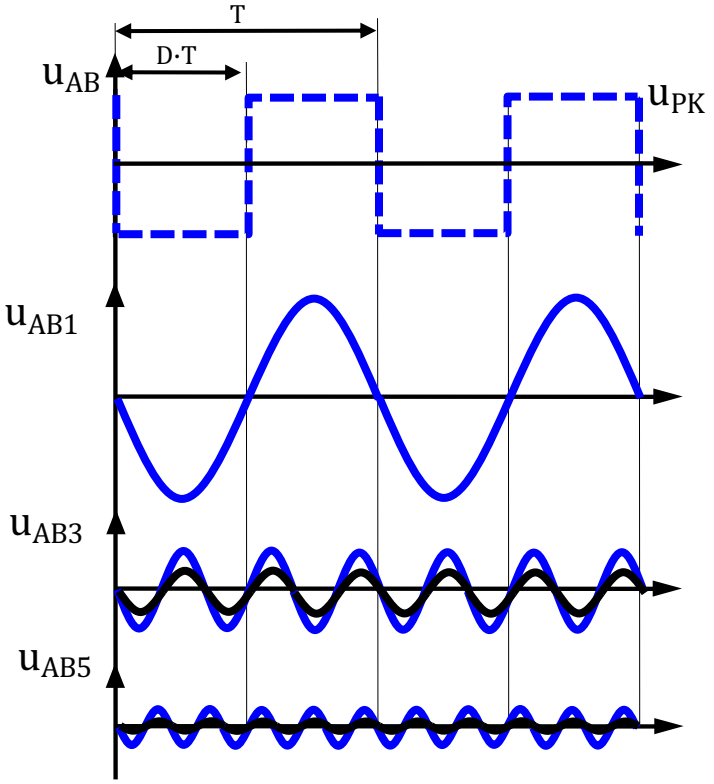
$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$



Single-Phase Full-Bridge SW Inverter

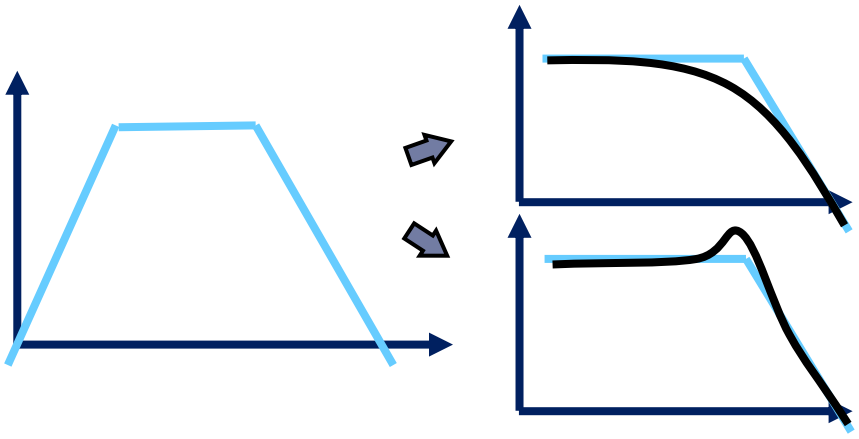


Band-pass filter

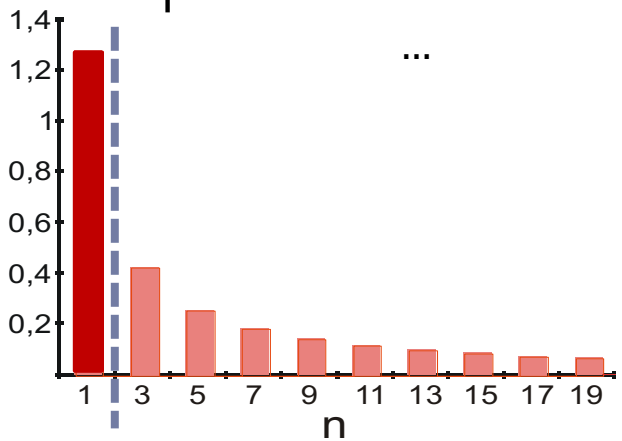


Conclusions:

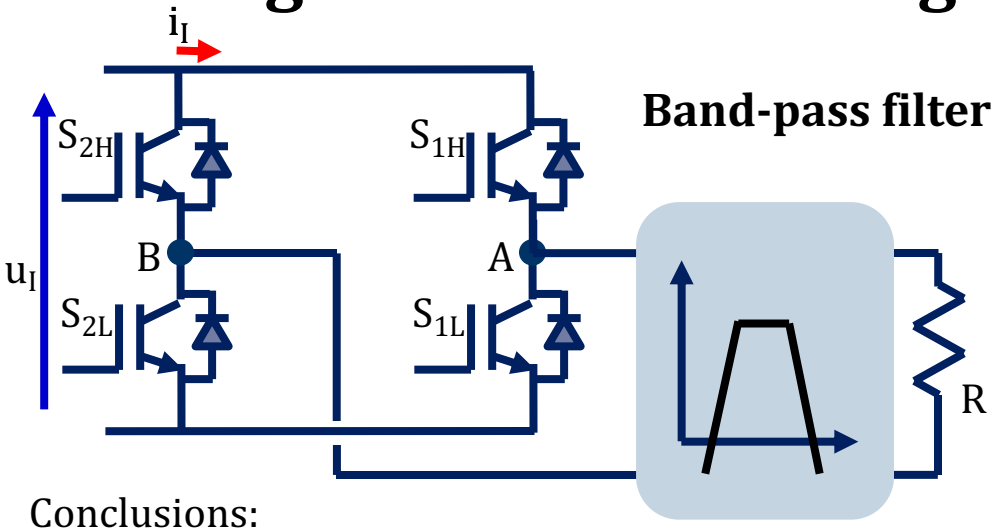
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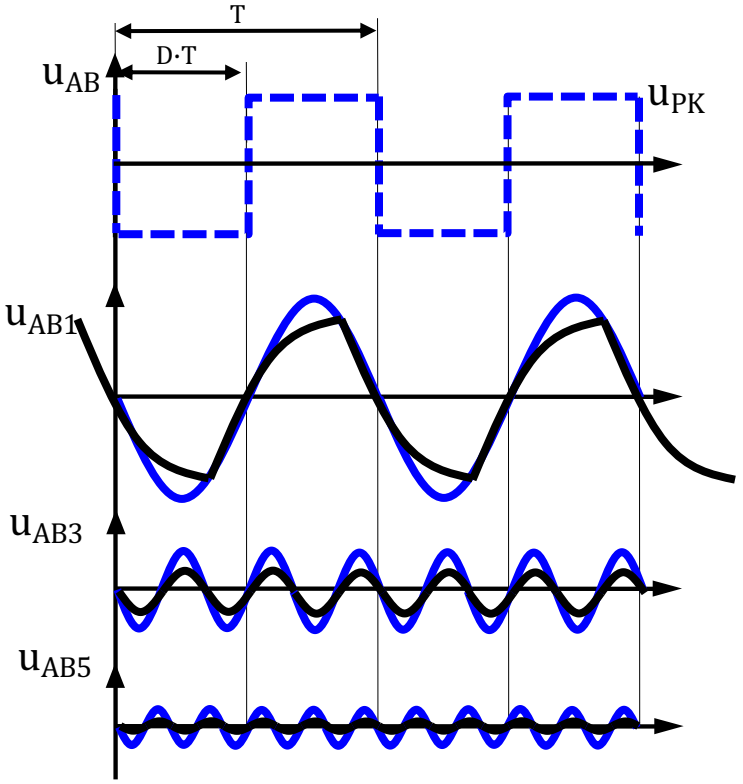
$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$



Single-Phase Full-Bridge SW Inverter

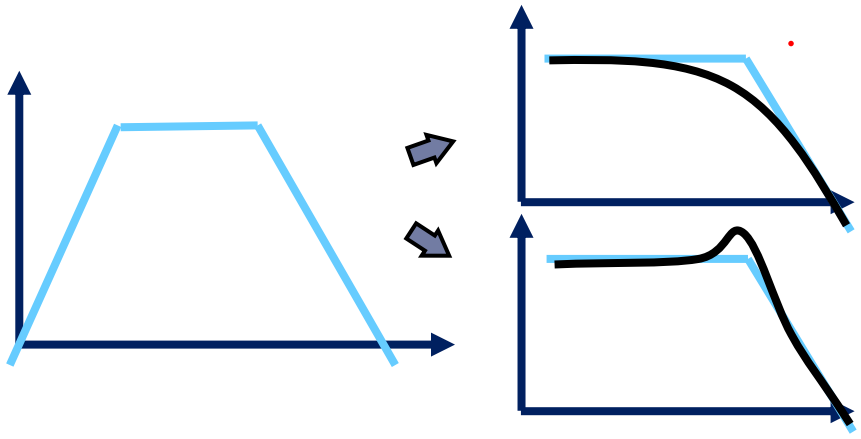


Band-pass filter

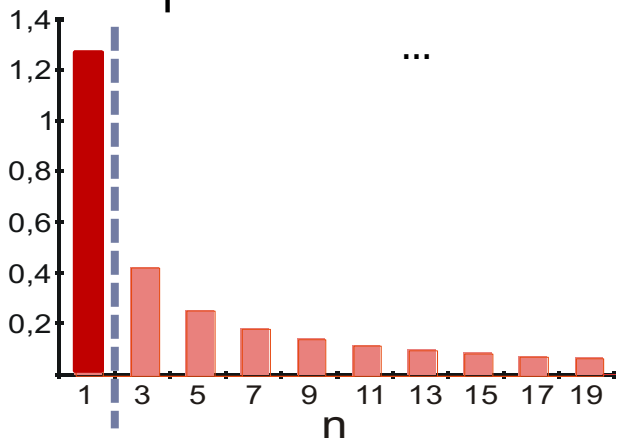


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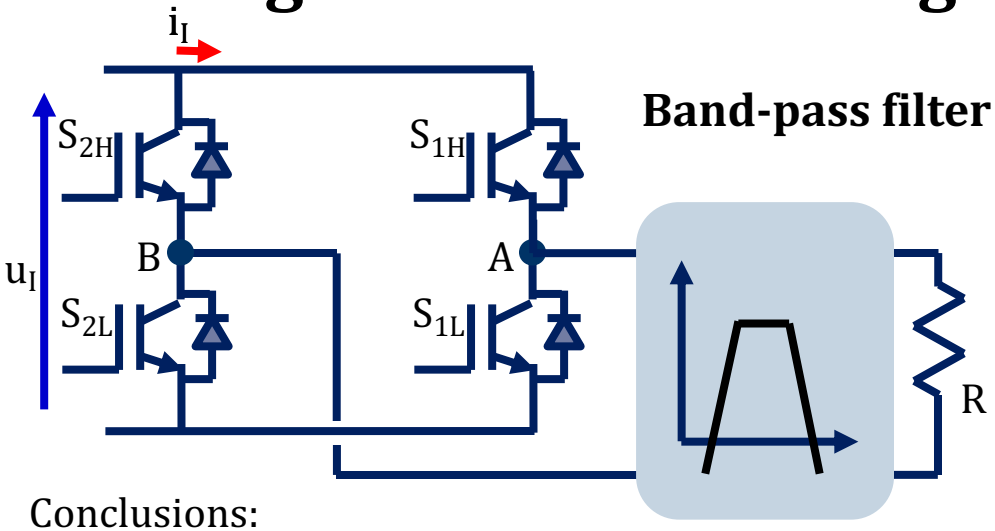


$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$

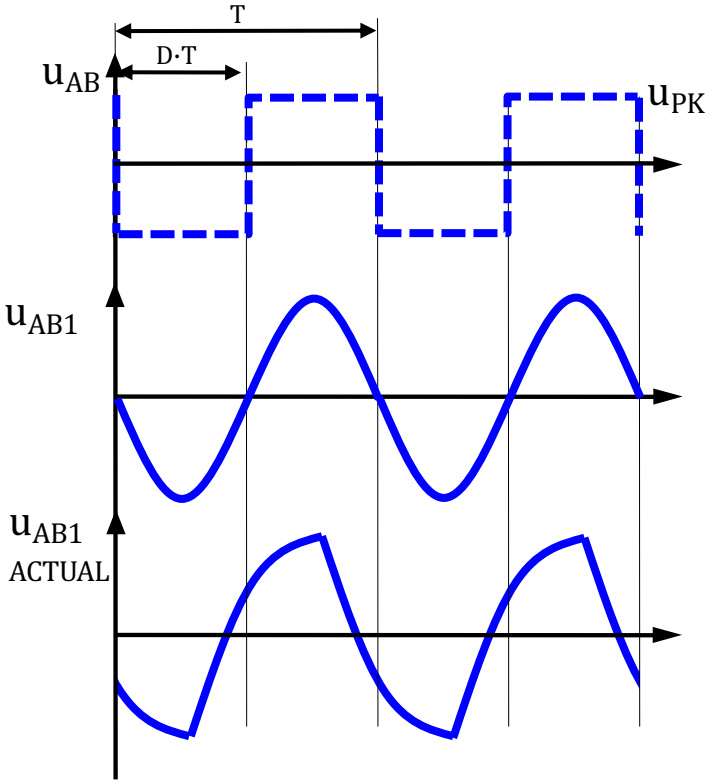


...

Single-Phase Full-Bridge SW Inverter

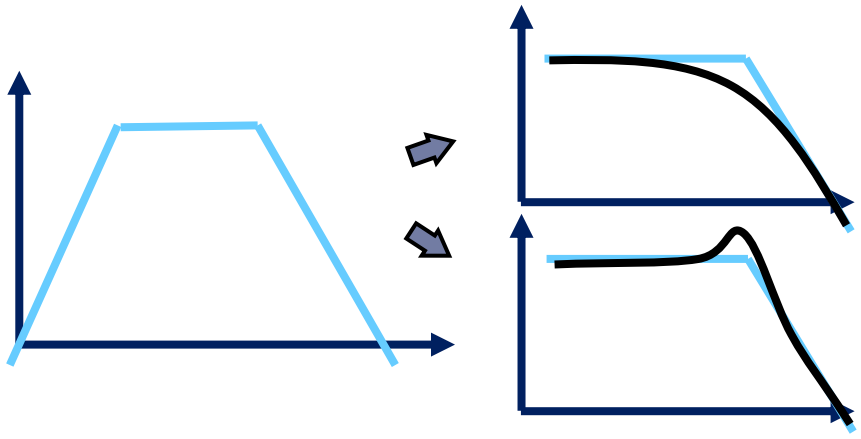


Band-pass filter

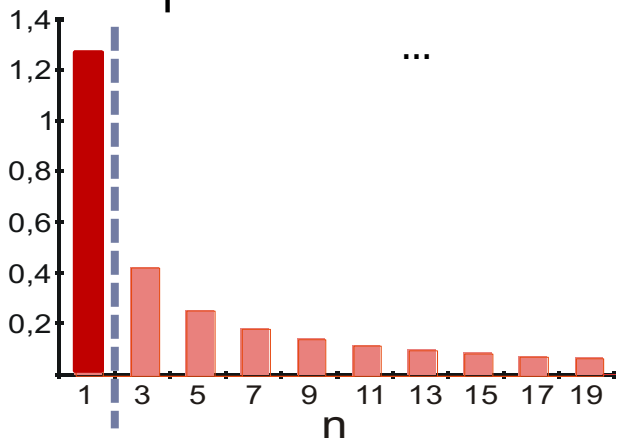


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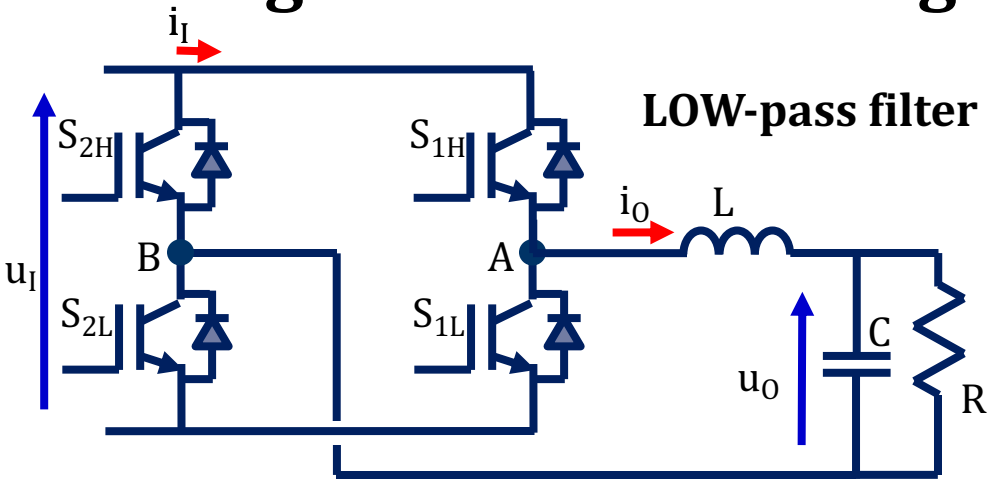
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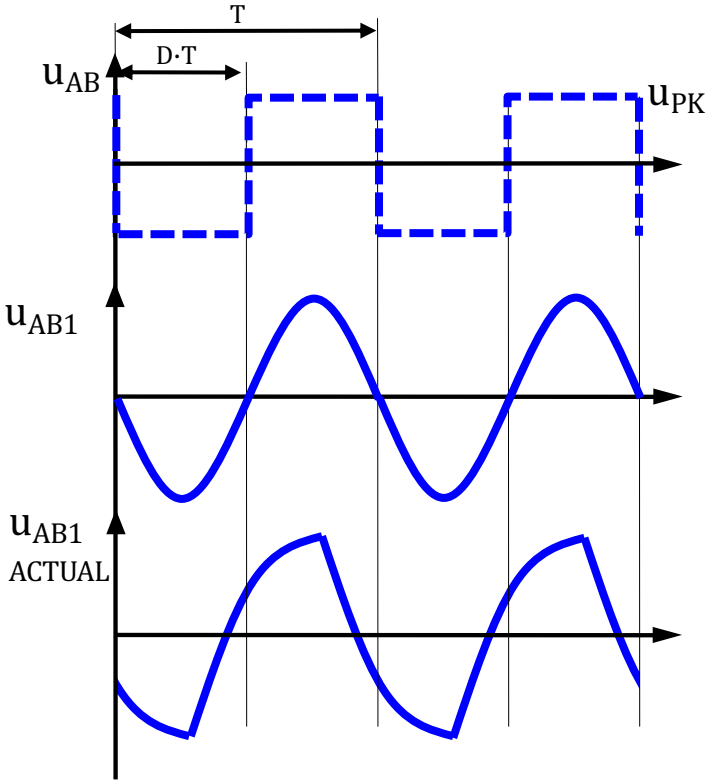


$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$



Single-Phase Full-Bridge SW Inverter

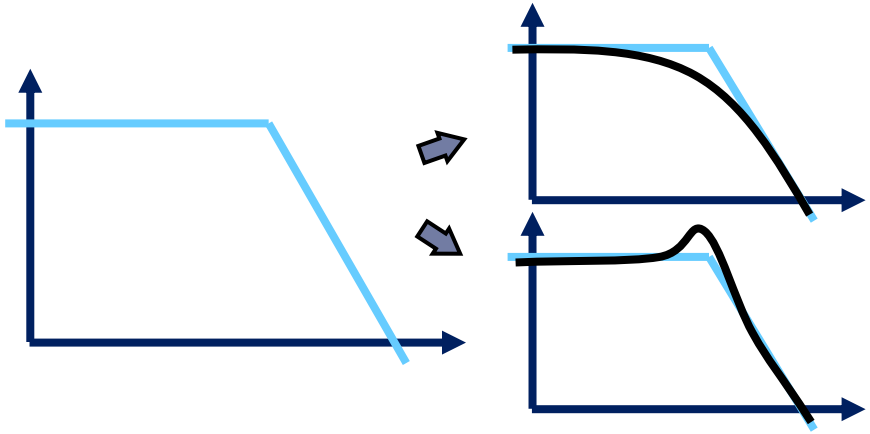




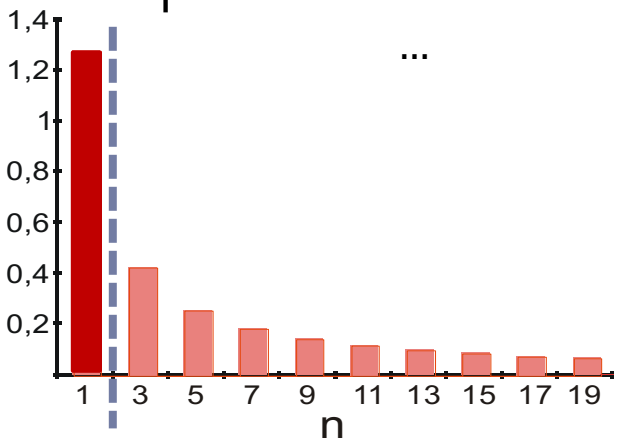
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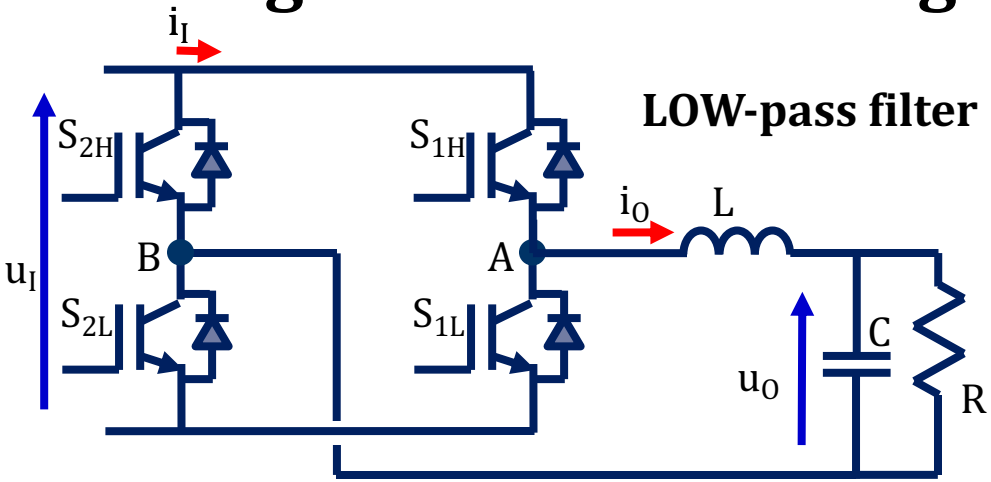
Preferred filter: 2nd order (No DC!!)

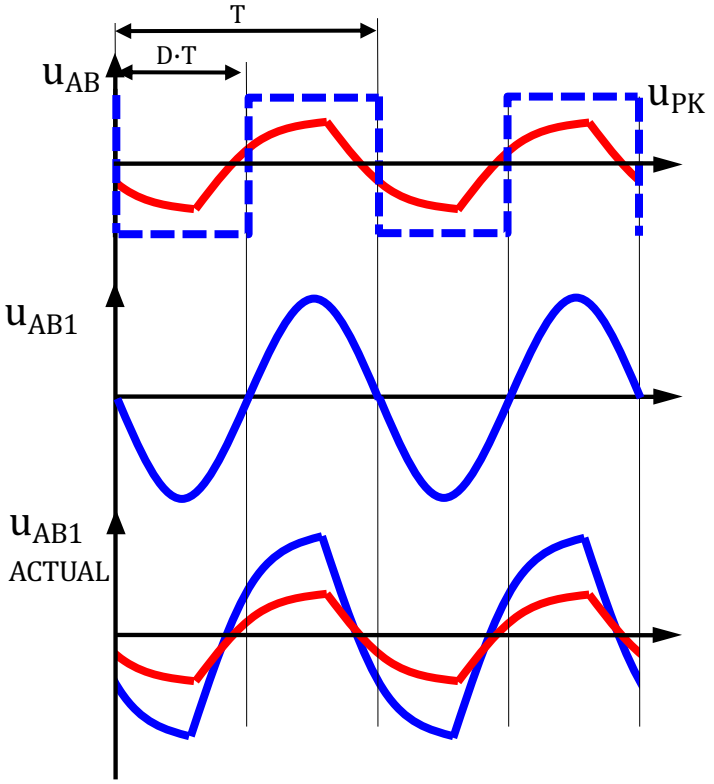


$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$



Single-Phase Full-Bridge SW Inverter

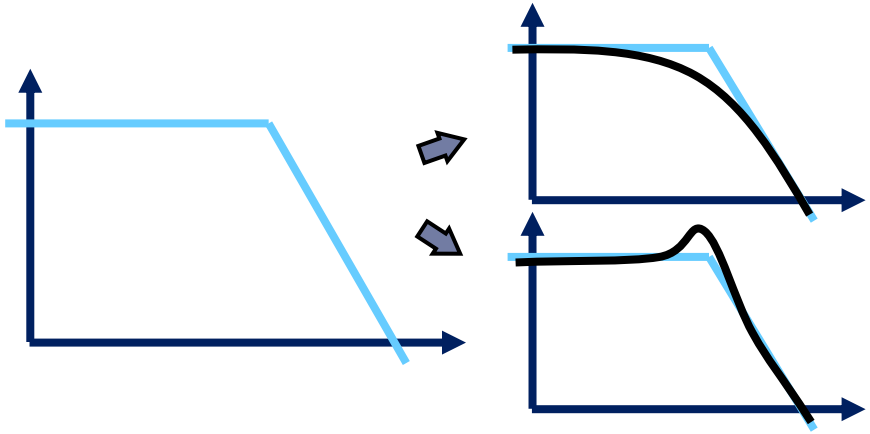




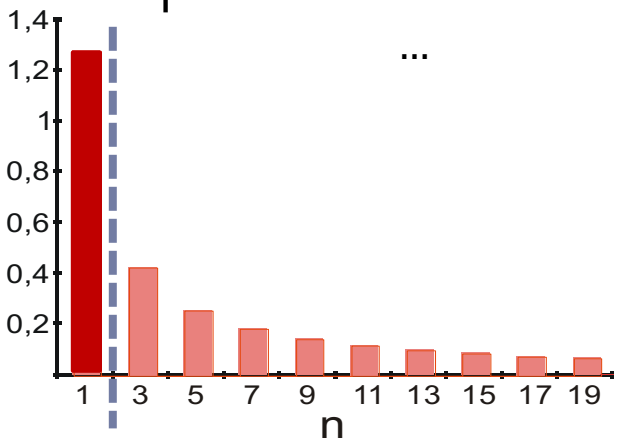
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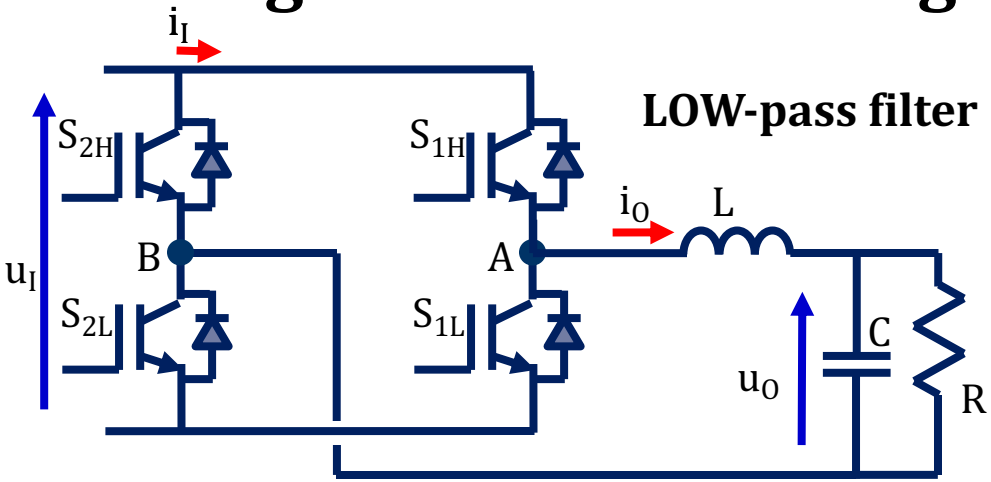
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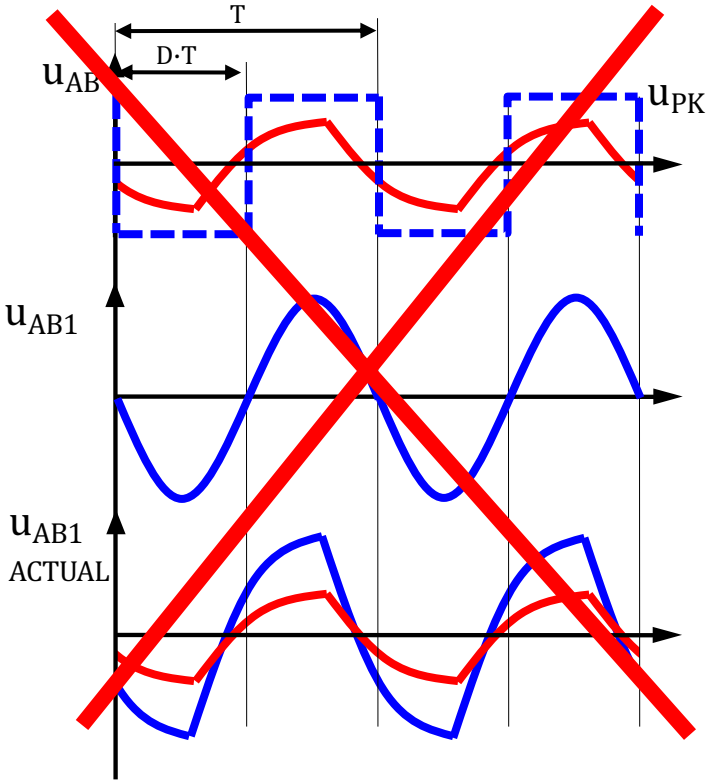


$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$



Single-Phase Full-Bridge SW Inverter

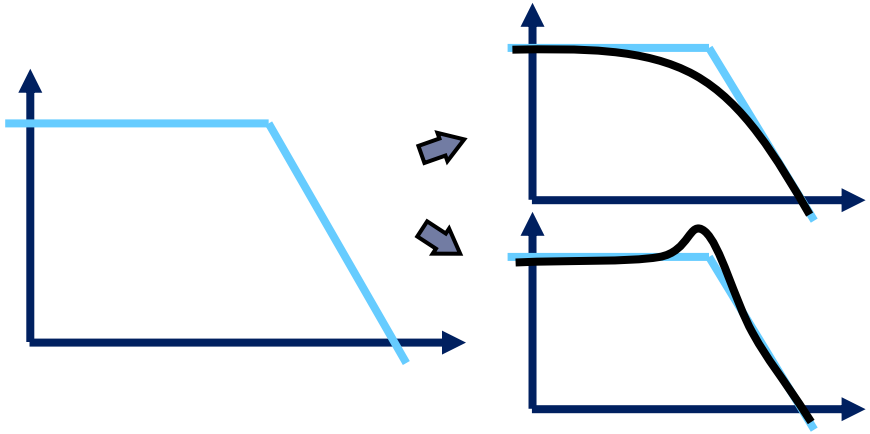




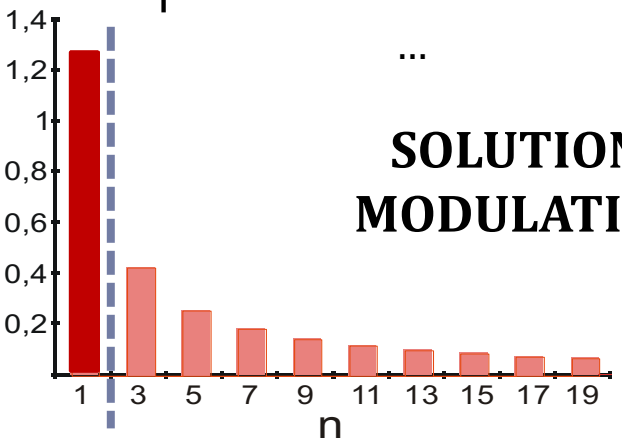
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$$\frac{u_{ABn\langle PEAK\rangle}}{u_{PK}}$$

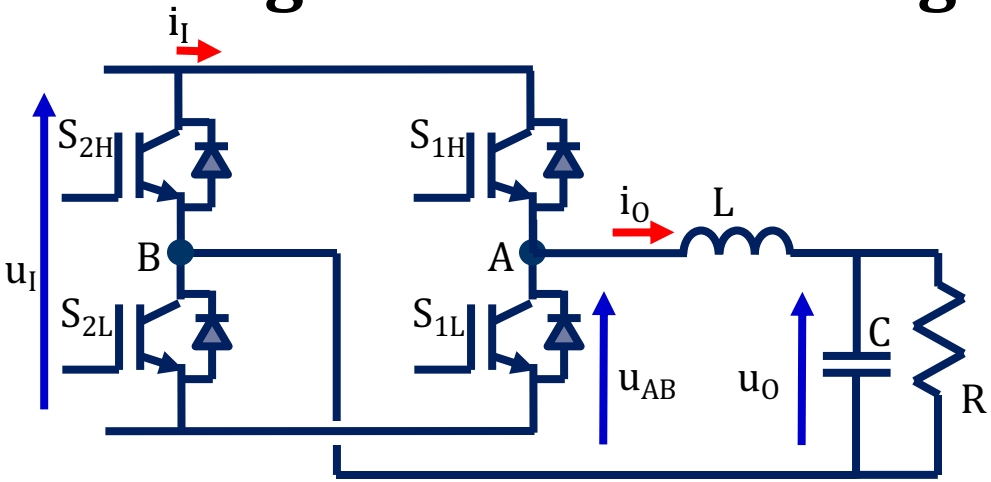


...

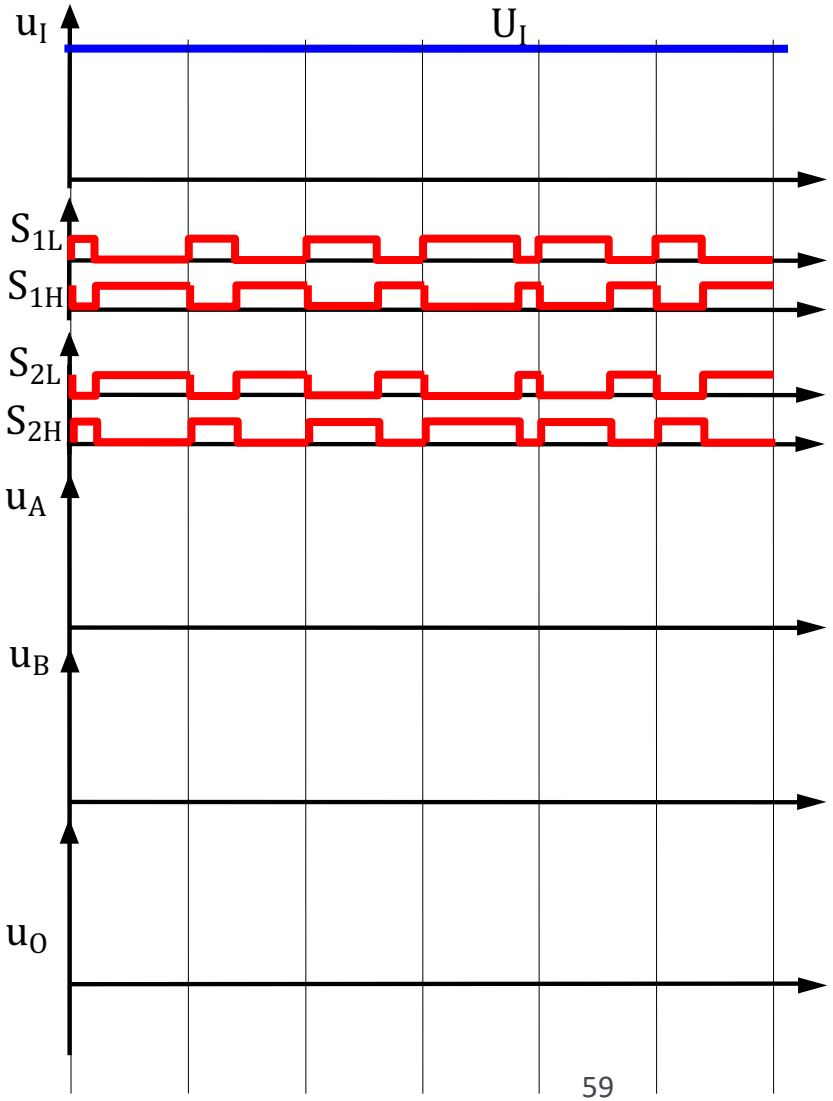
SOLUTION: MODULATION

Single-Phase, Full-Bridge PWM Inverter

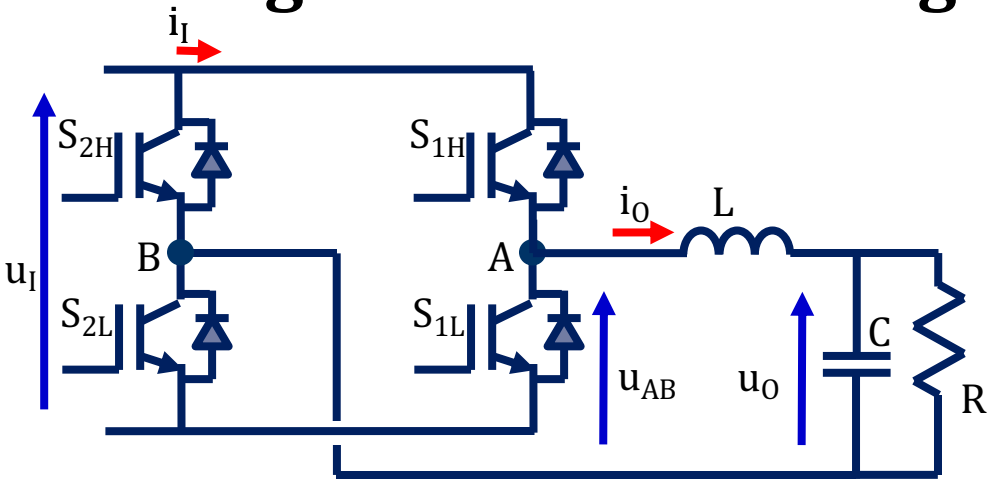
Single-Phase Full-Bridge PWM Inverter



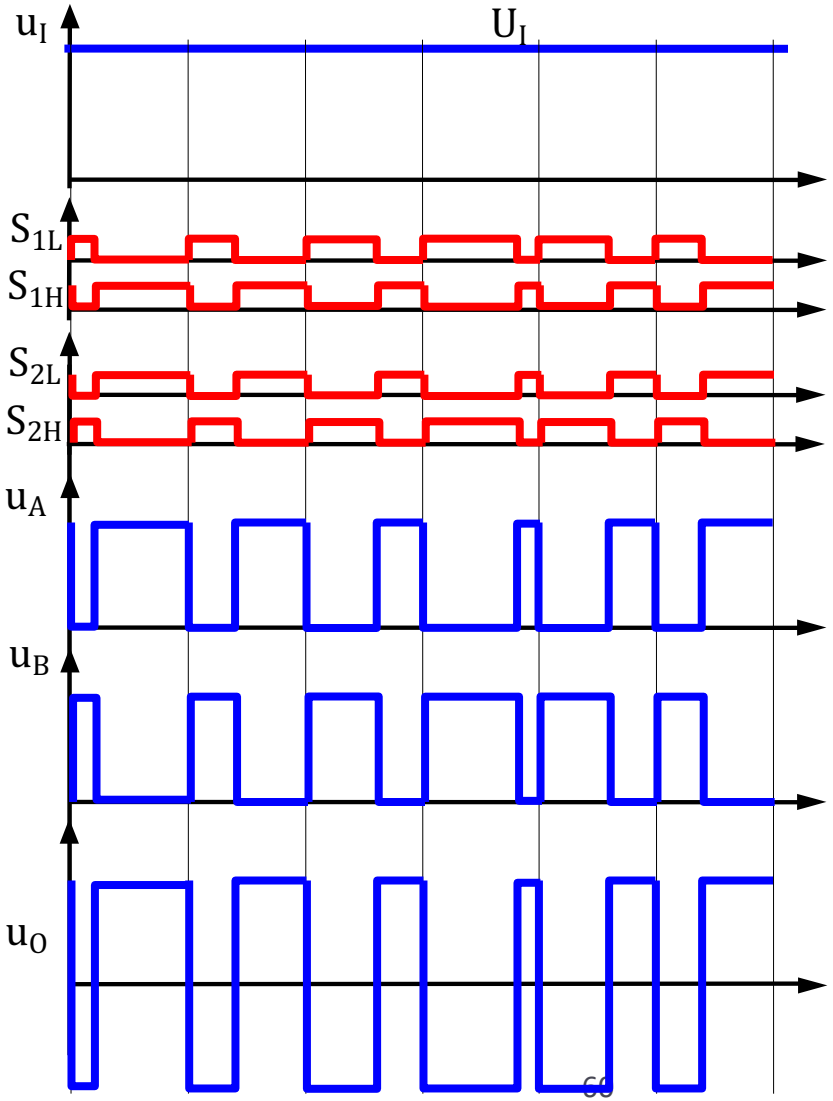
Keeping the hardware scheme, but switching with a different pattern:



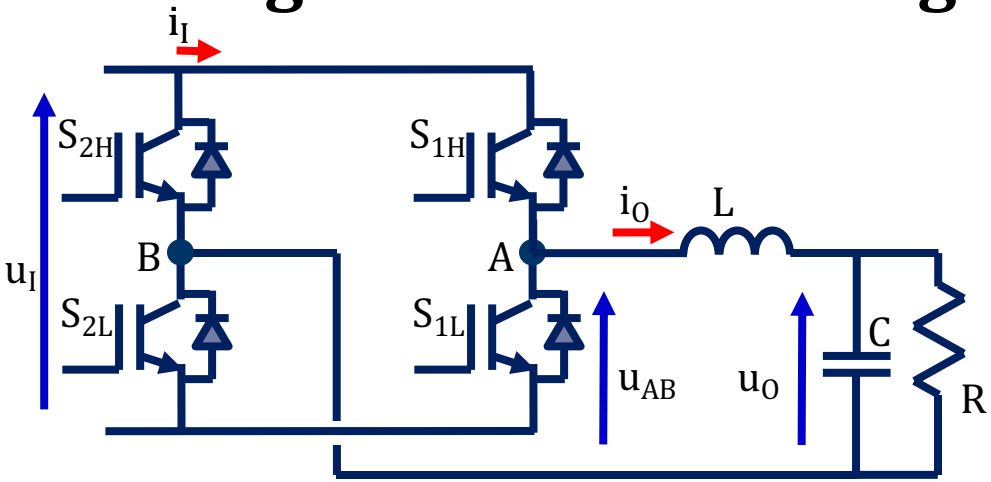
Single-Phase Full-Bridge PWM Inverter

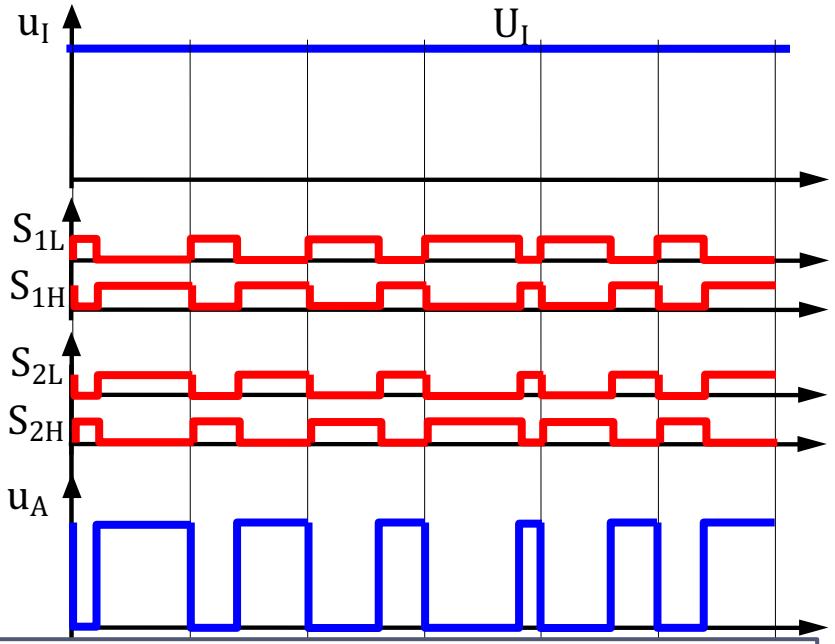


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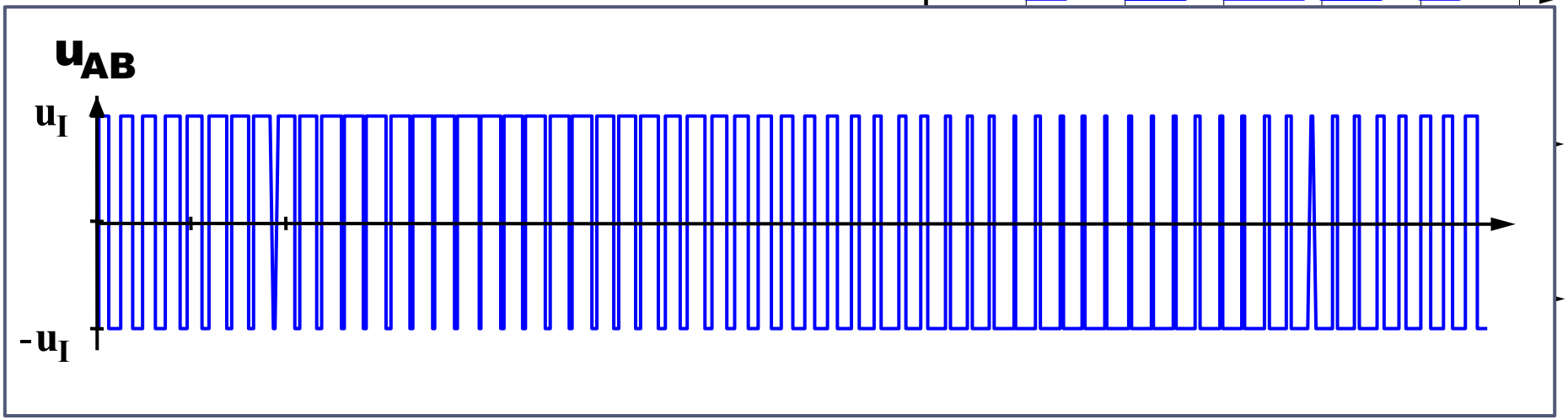


Single-Phase Full-Bridge PWM Inverter

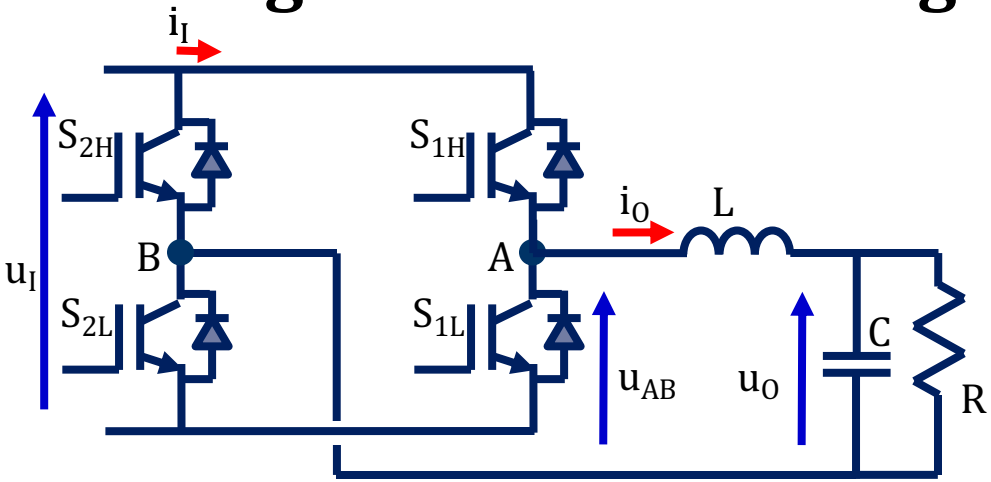


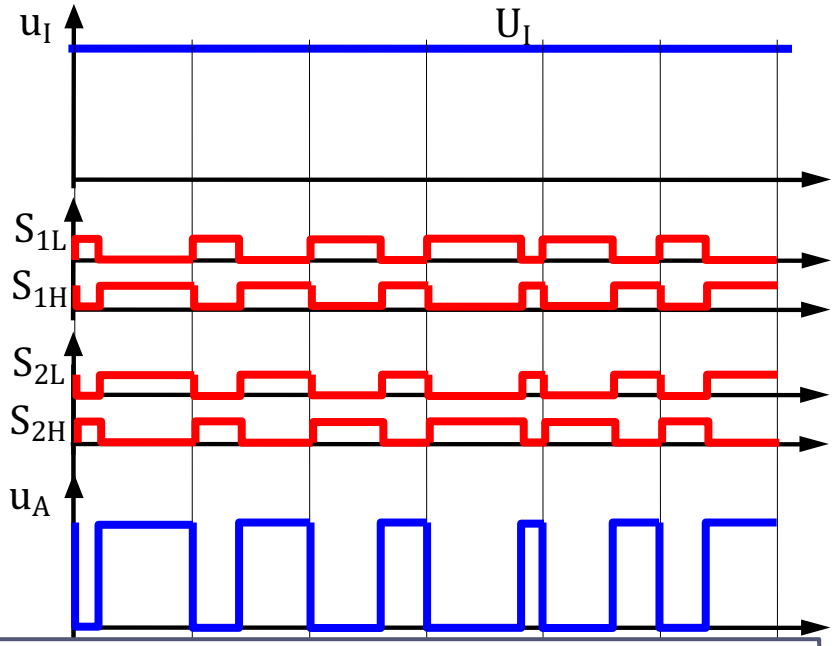


Keeping the hardware scheme, but switching with a different pattern:

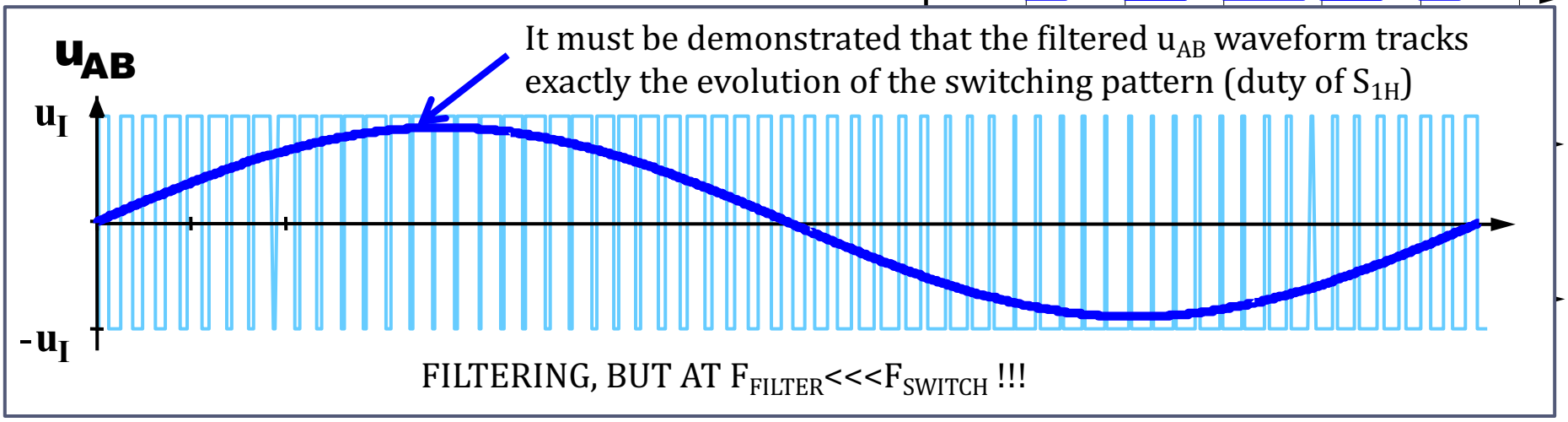


Single-Phase Full-Bridge PWM Inverter

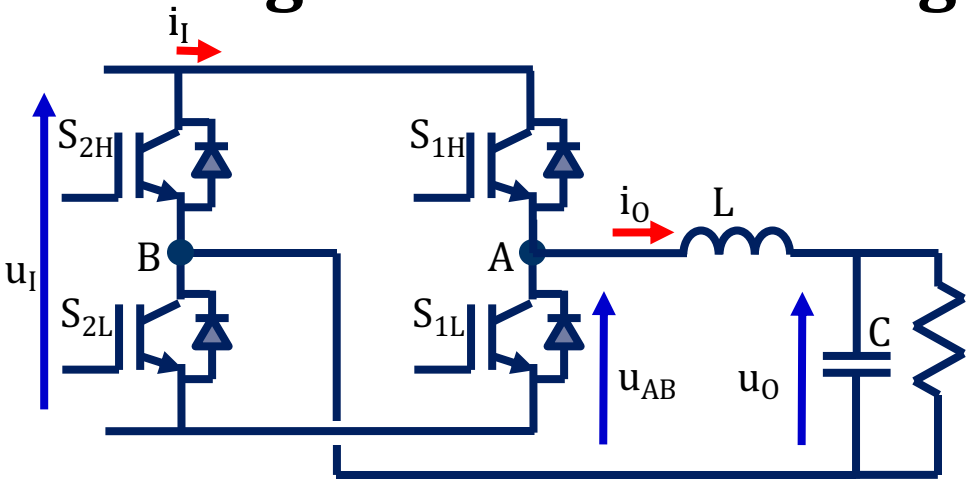


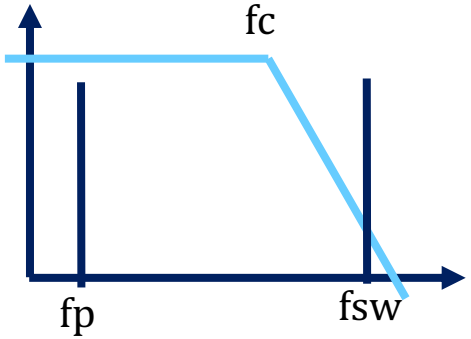


Keeping the hardware scheme, but switching with a different pattern:



Single-Phase Full-Bridge PWM Inverter





Filter design:

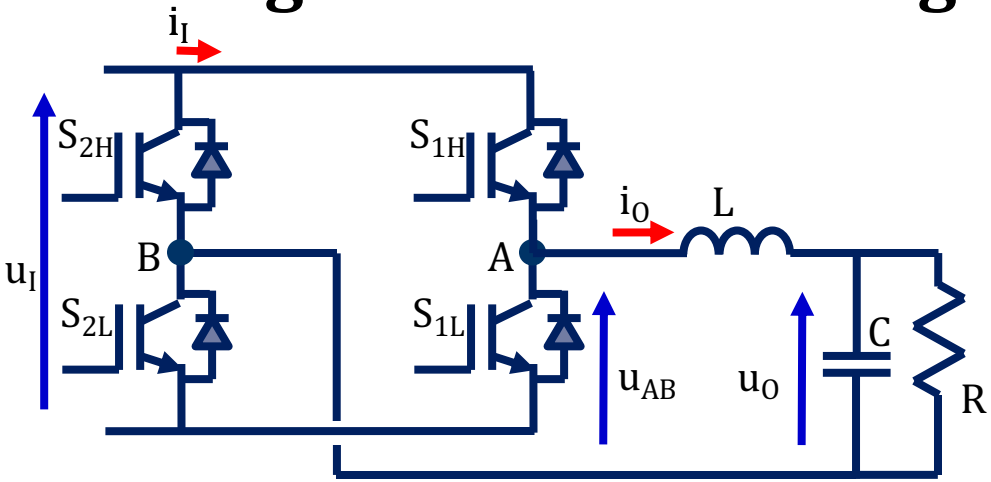
Eq (1): $$fc = \frac{1}{2 \cdot \pi \cdot \sqrt{L \cdot C}}$$

Eq (2):

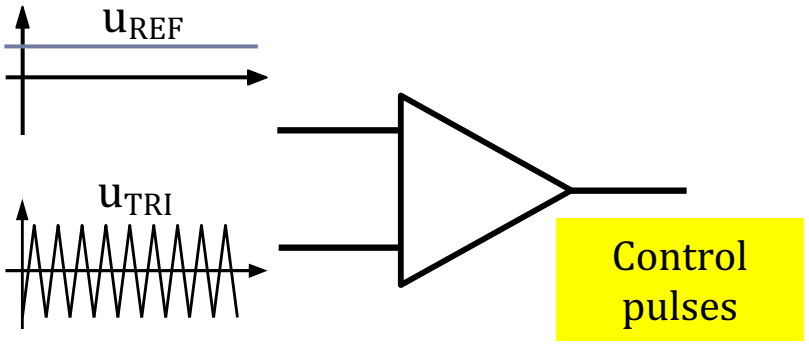
current ripple: (e.g. 20%)
$$\Delta iL = \frac{u_I}{4 \cdot L \cdot fsw}$$

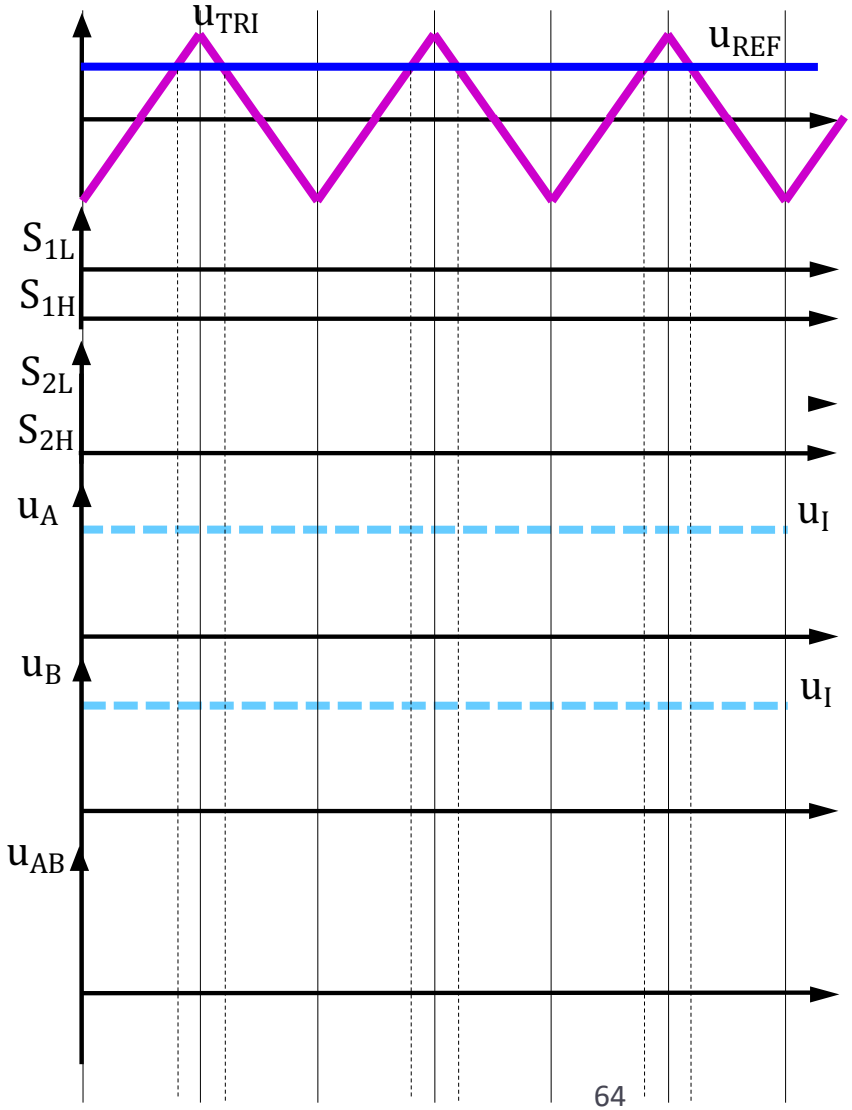
Q factor: (e.g. 3-8)
$$Q = R \cdot \sqrt{\frac{C}{L}}$$

Single-Phase Full-Bridge PWM Inverter: BIPOLAR

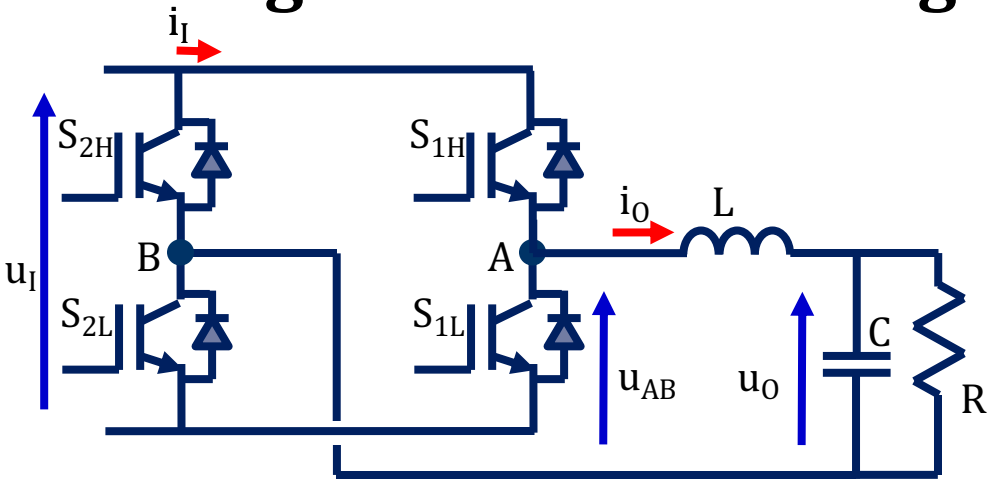


This pattern can be obtaining calculating with a triangle waveform, u_{TRI}

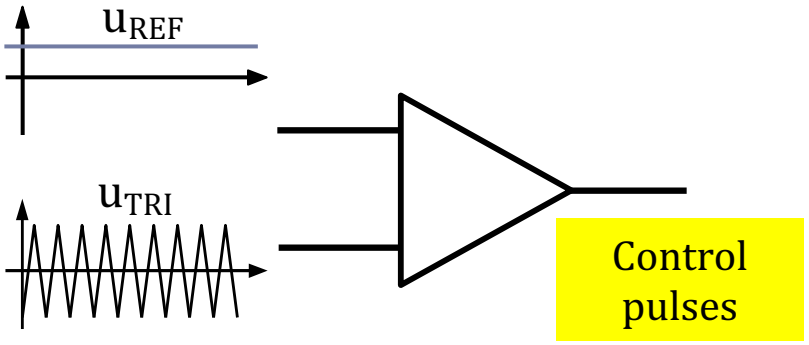


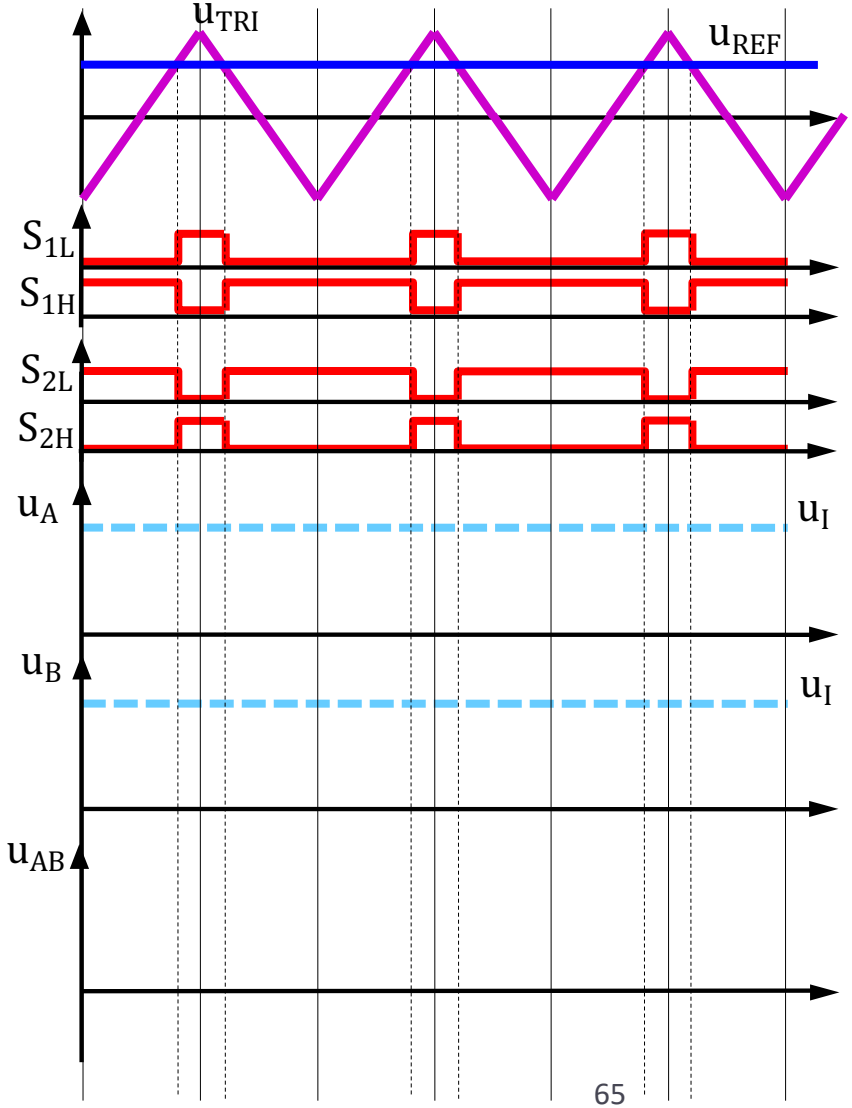


Single-Phase Full-Bridge PWM Inverter: BIPOLAR

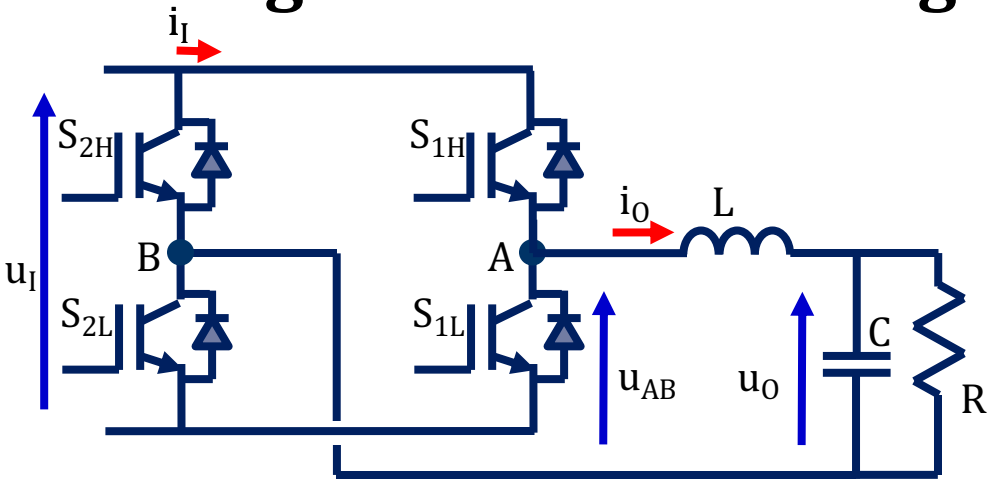


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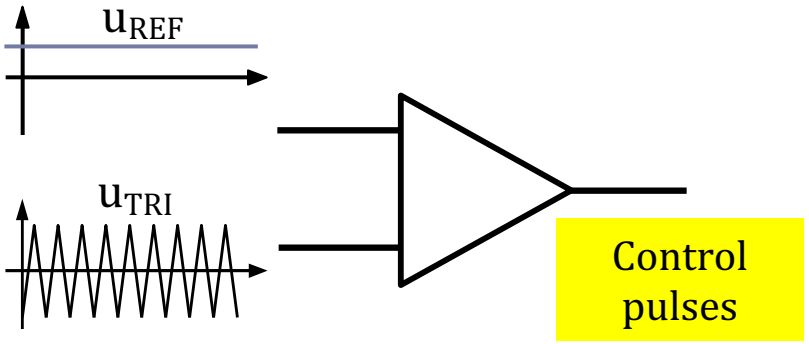


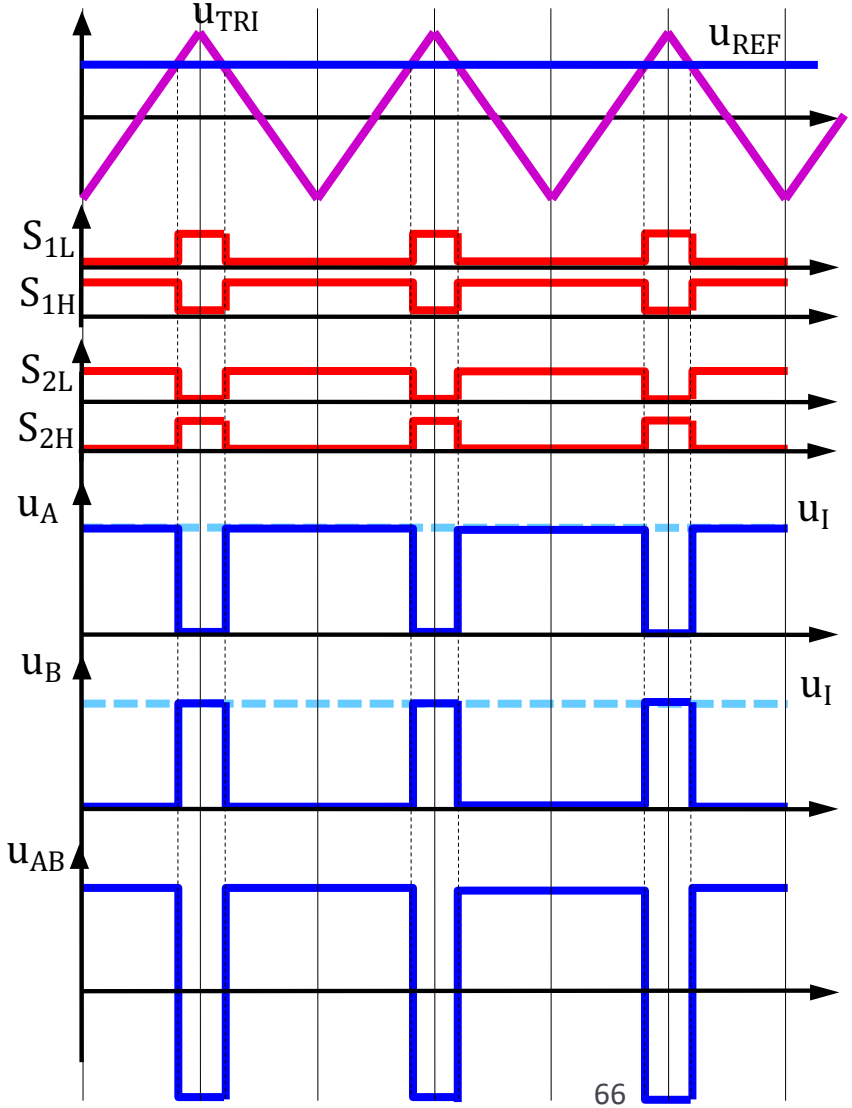


Single-Phase Full-Bridge PWM Inverter: BIPOLAR

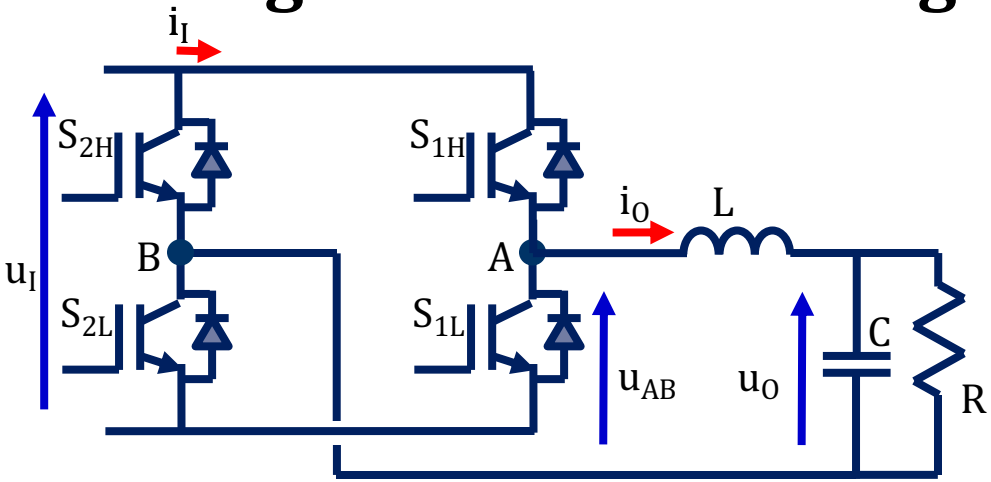


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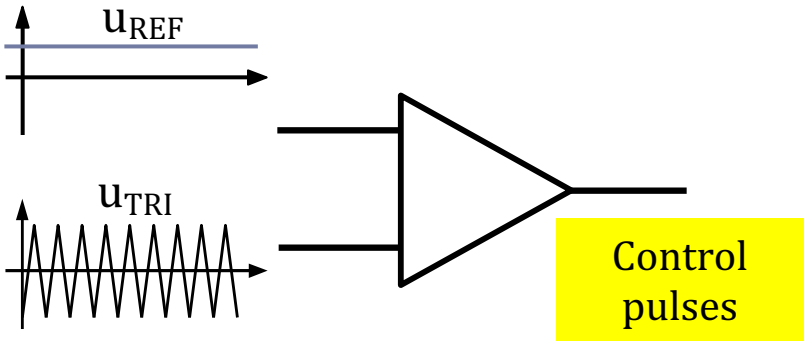


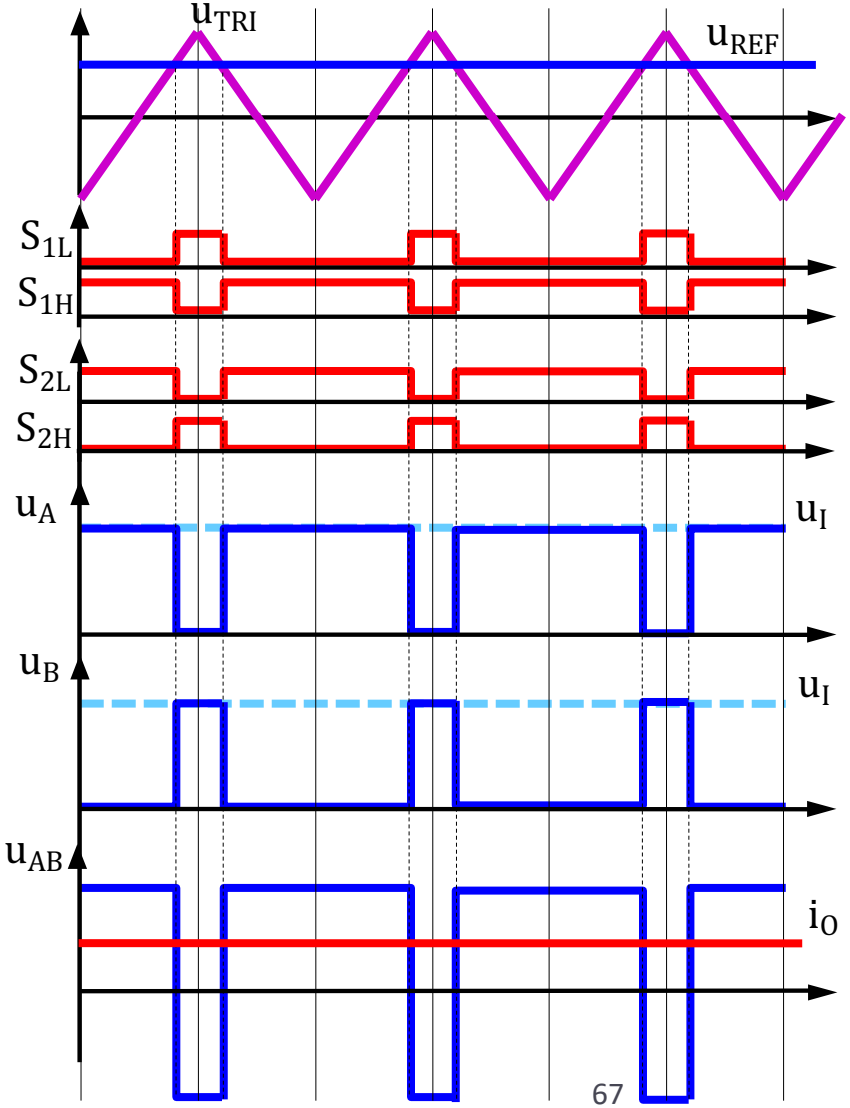


Single-Phase Full-Bridge PWM Inverter: BIPOLAR



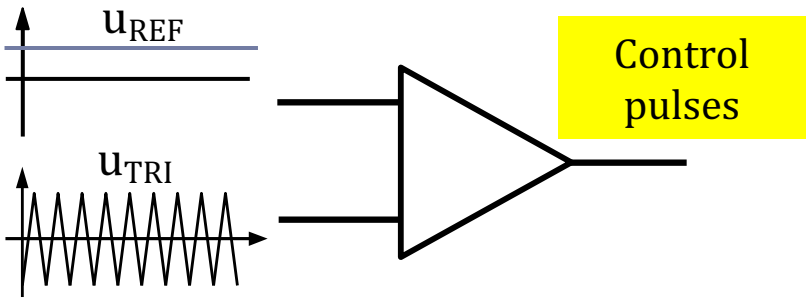
This pattern can be obtaining calculating with a triangle waveform, u_{TRI}





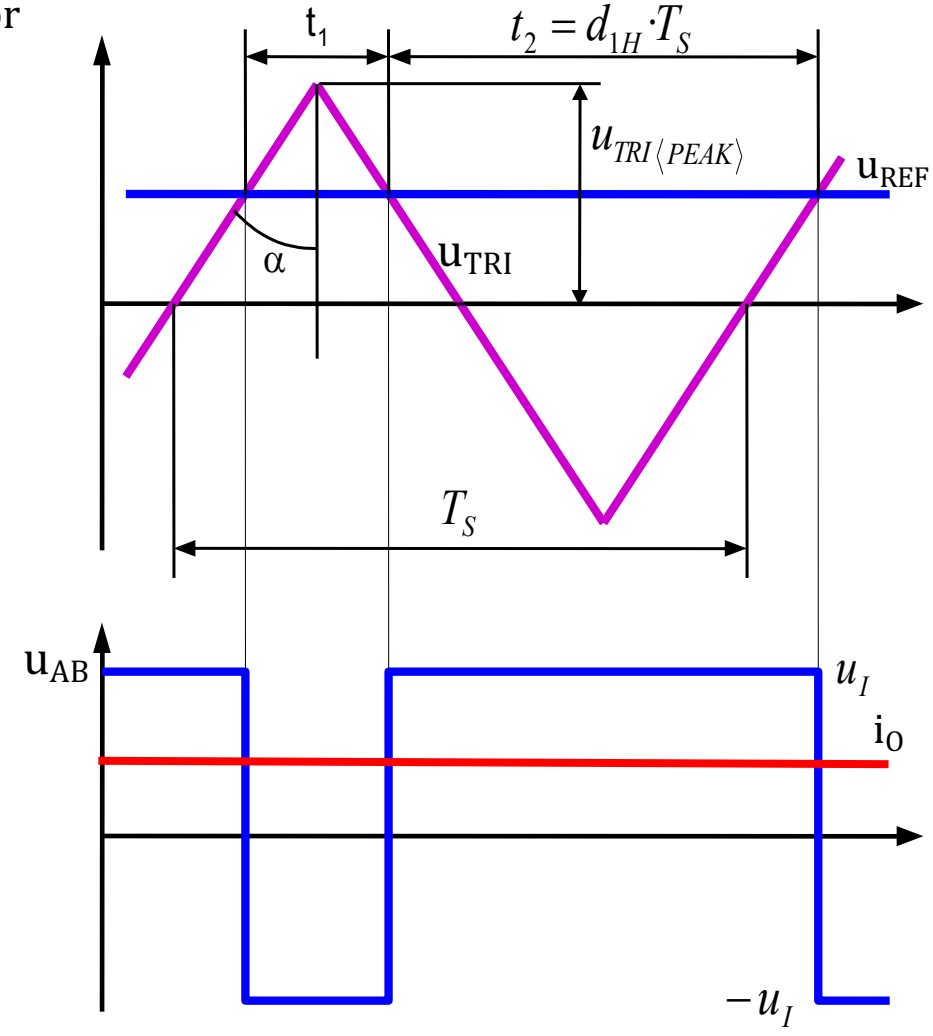
Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



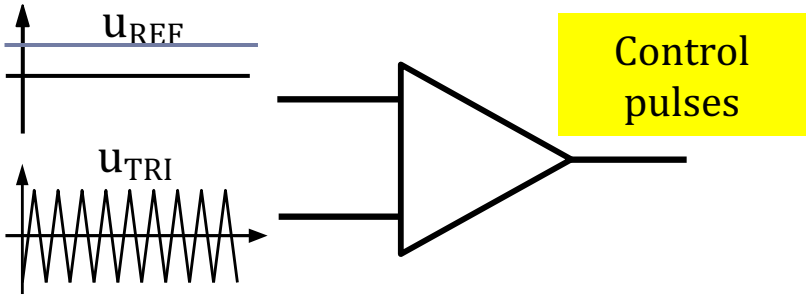
$$u_{REF}(t) = U_{REF}$$

$$u_{TRI}(t) = triangle\left(u_{TRI\langle PEAK\rangle}, T_S\right)$$



Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



$$t_1 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} - U_{REF}\right)$$

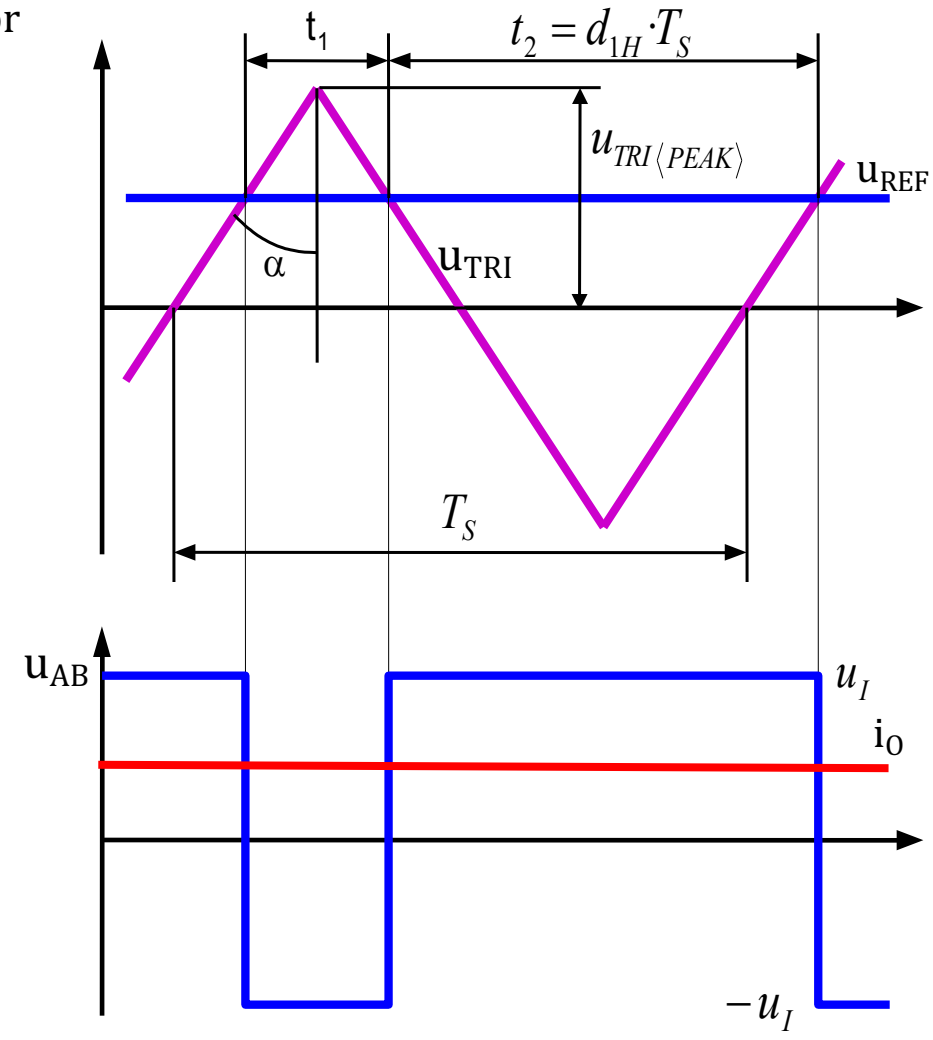
$$t_2 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} + U_{REF}\right)$$

$$tg\alpha = \frac{T_S}{4 \cdot u_{TRI\langle PEAK\rangle}}$$

$$t_2 - t_1 = 4 \cdot tg\alpha \cdot U_{REF}$$

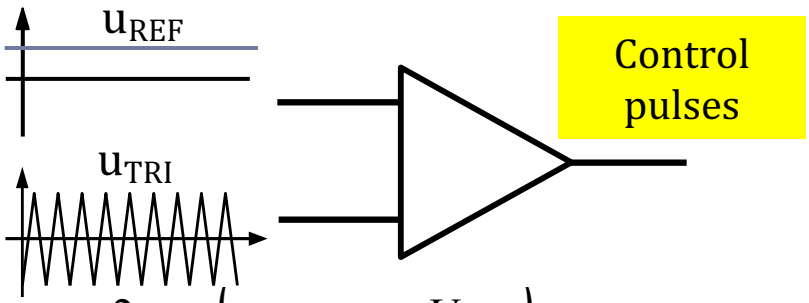
$$t_2 + t_1 = 4 \cdot tg\alpha \cdot u_{TRI\langle PEAK\rangle} = T_S$$

$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot \frac{t_2 - t_1}{T_S} = u_I \cdot (2d_{1H} - 1)$$



Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



$$t_1 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} - U_{REF}\right)$$

$$t_2 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} + U_{REF}\right)$$

$$tg\alpha = \frac{T_S}{4 \cdot u_{TRI\langle PEAK\rangle}}$$

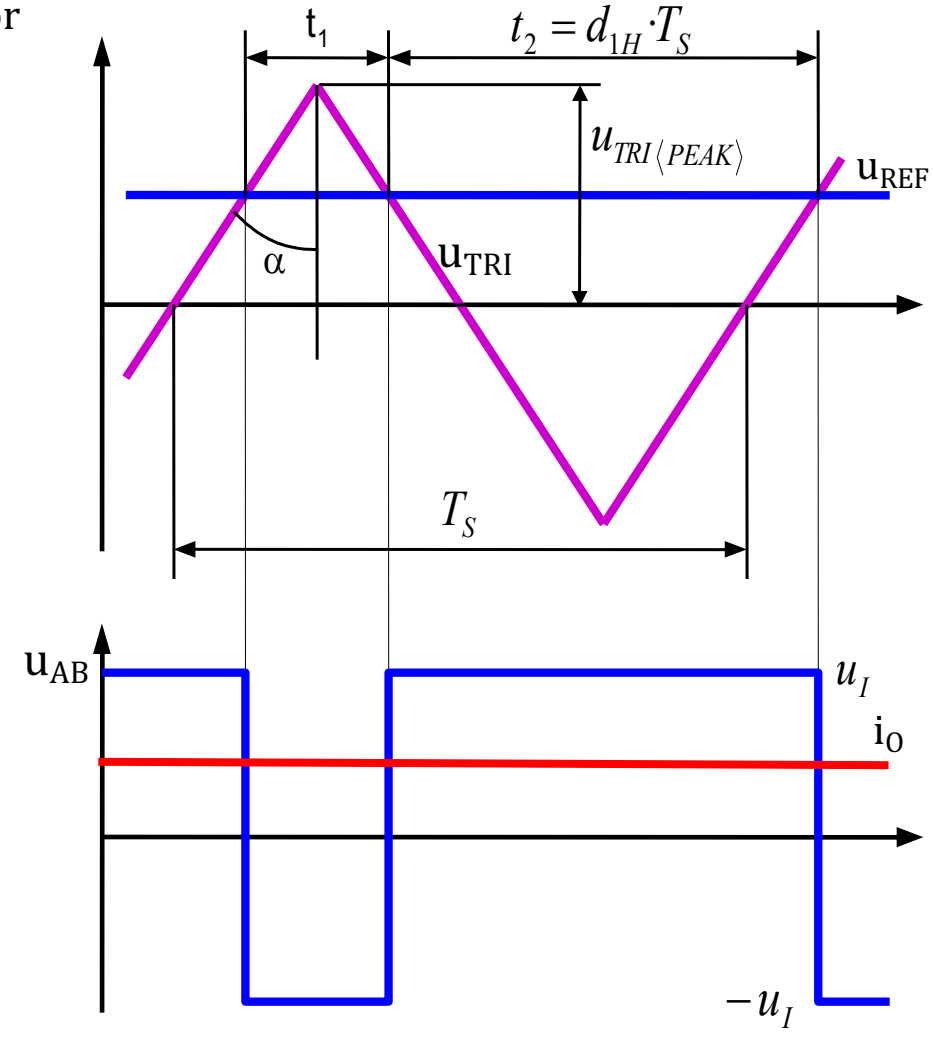
$$t_2 - t_1 = 4 \cdot tg\alpha \cdot U_{REF}$$

$$t_2 + t_1 = 4 \cdot tg\alpha \cdot u_{TRI\langle PEAK\rangle} = T_S$$

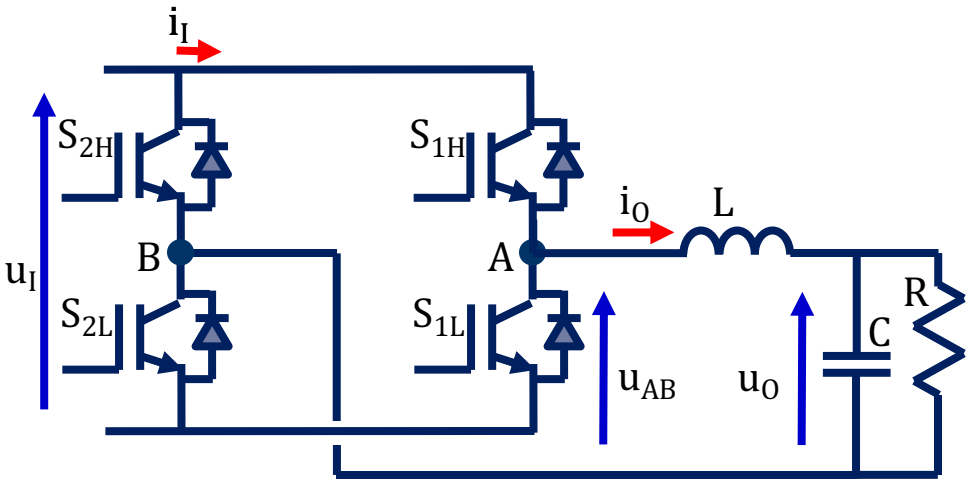
$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot \frac{t_2 - t_1}{T_S} = \ldots = u_I \cdot \frac{U_{REF}}{u_{TRI\langle PEAK\rangle}} =$$

$$= \frac{u_I \cdot U_{REF}}{u_{TRI\langle PEAK\rangle}}$$

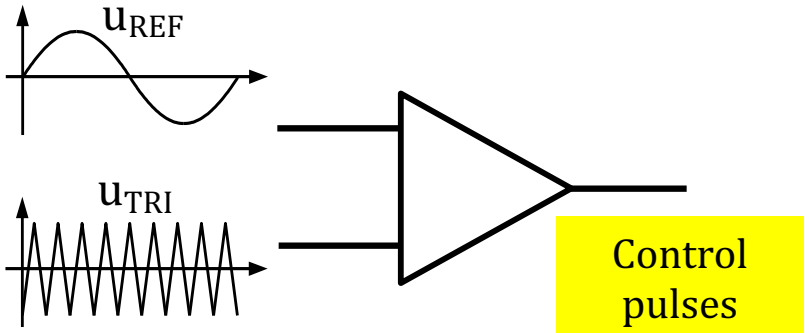
$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot \frac{U_{REF}}{u_{TRI\langle PEAK\rangle}}$$

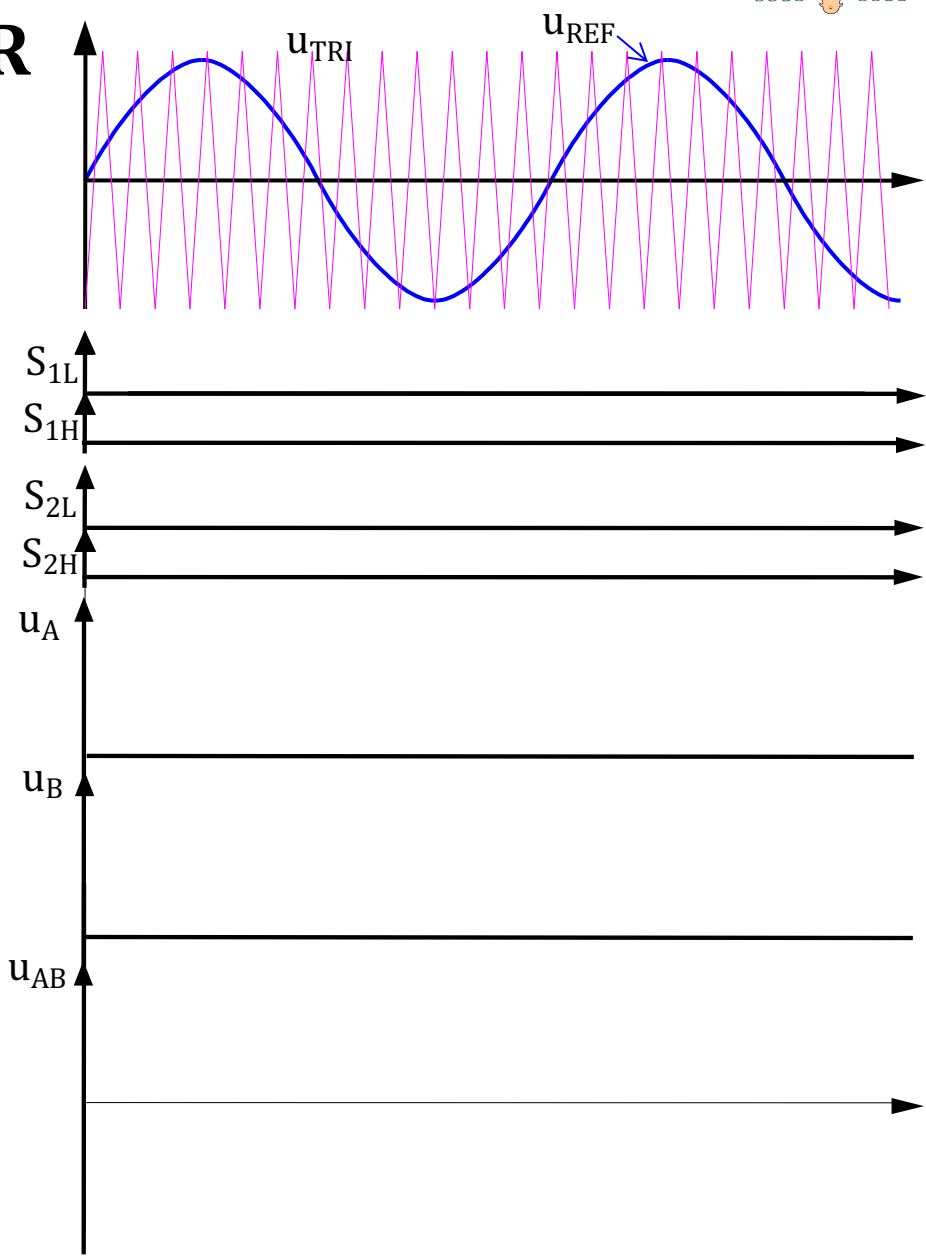


1-Φ FB PWM Inv: BIPOLAR

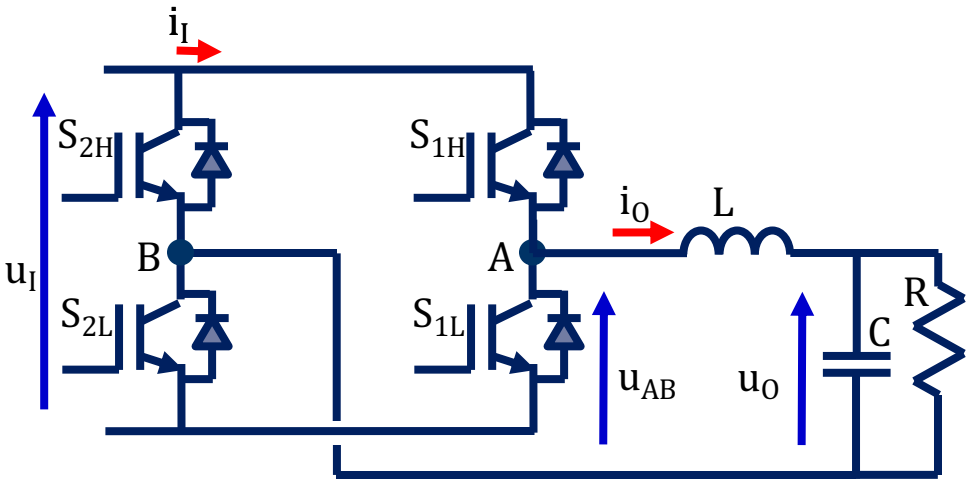


This pattern can be obtaining calculating with a triangle waveform, u_{TRI}

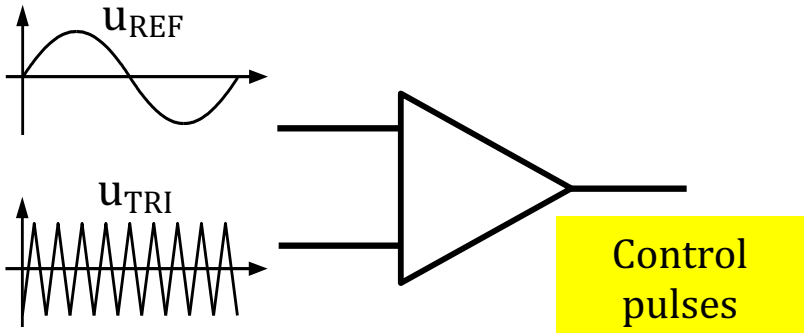


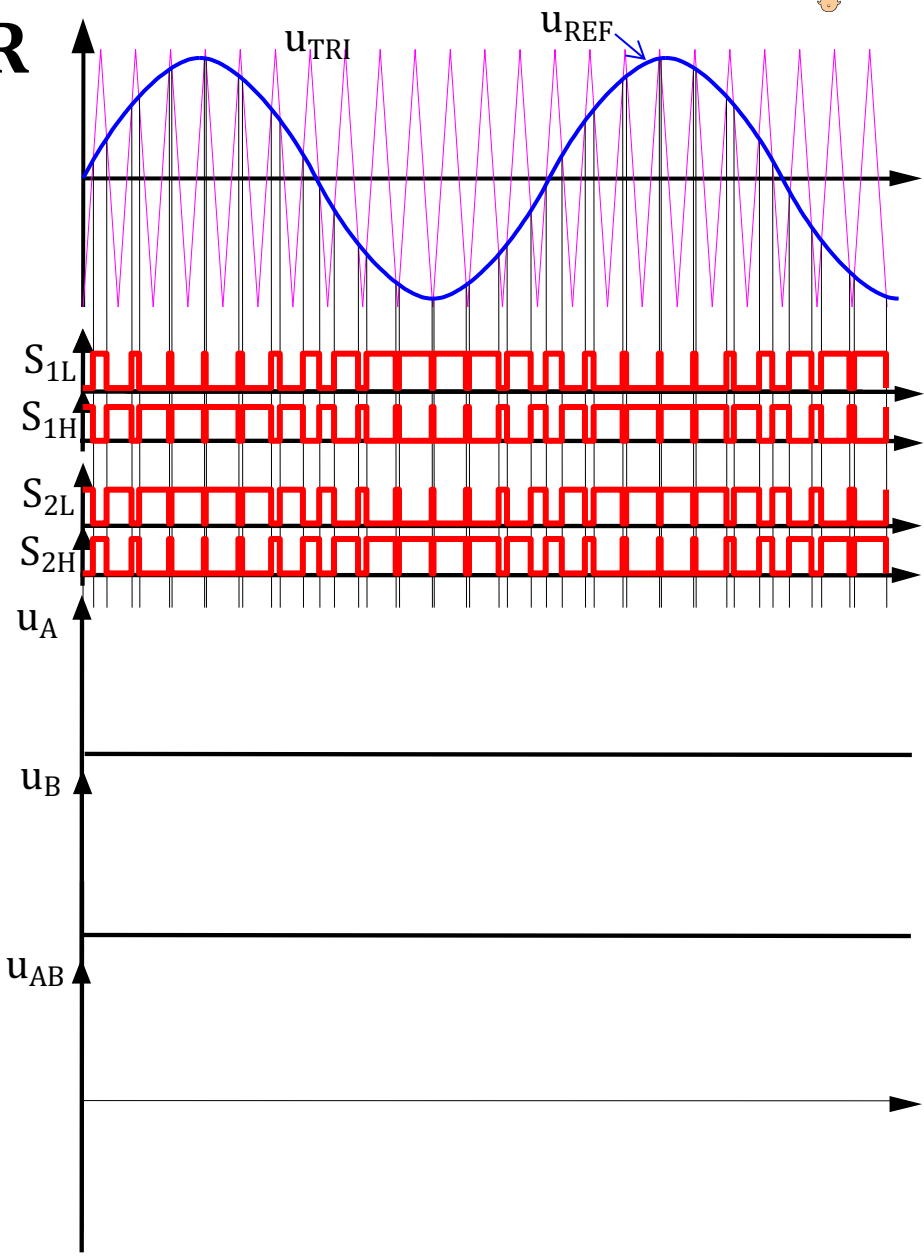


1-Φ FB PWM Inv: BIPOLAR

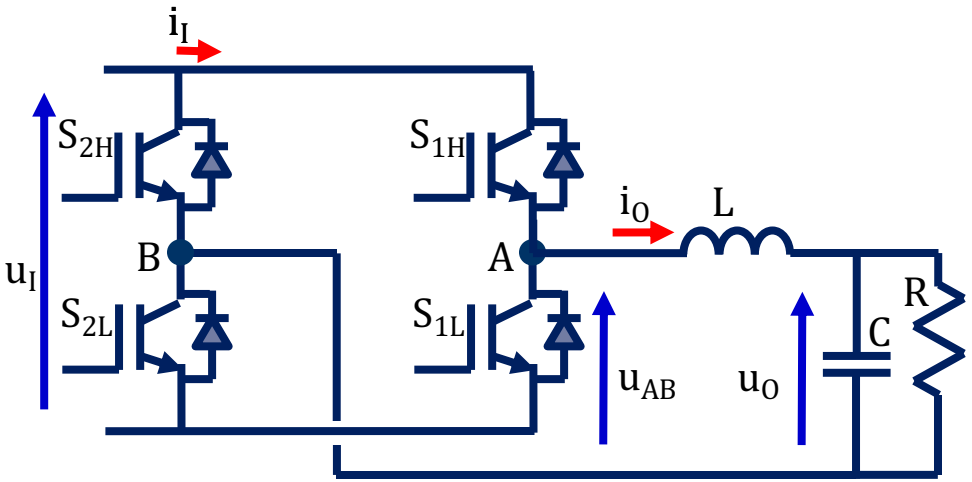


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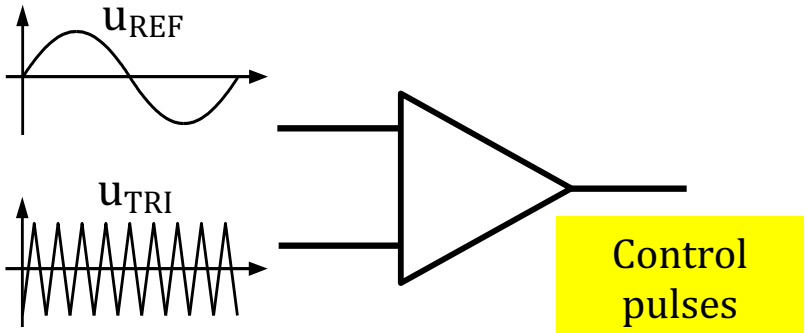


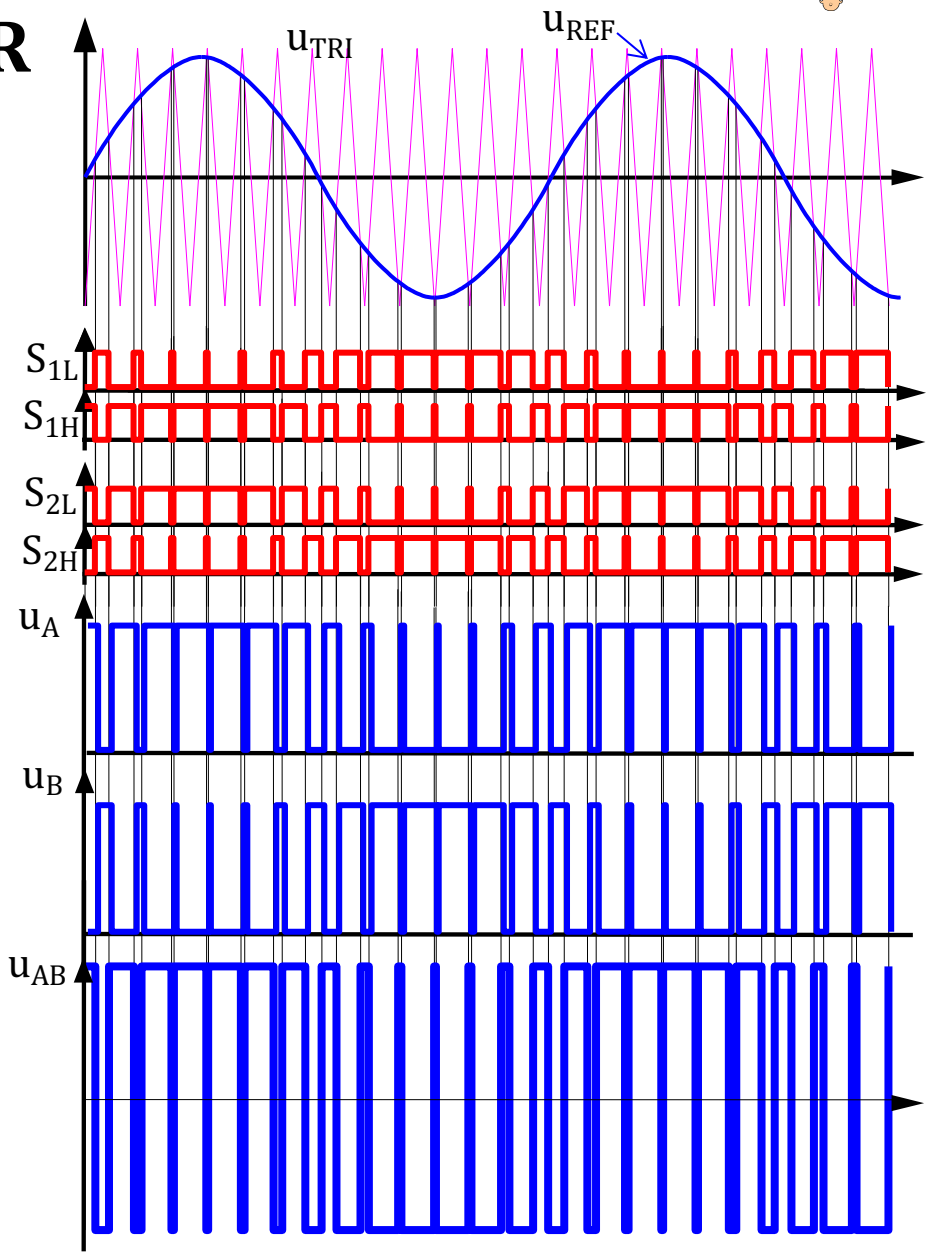


1-Φ FB PWM Inv: BIPOLAR

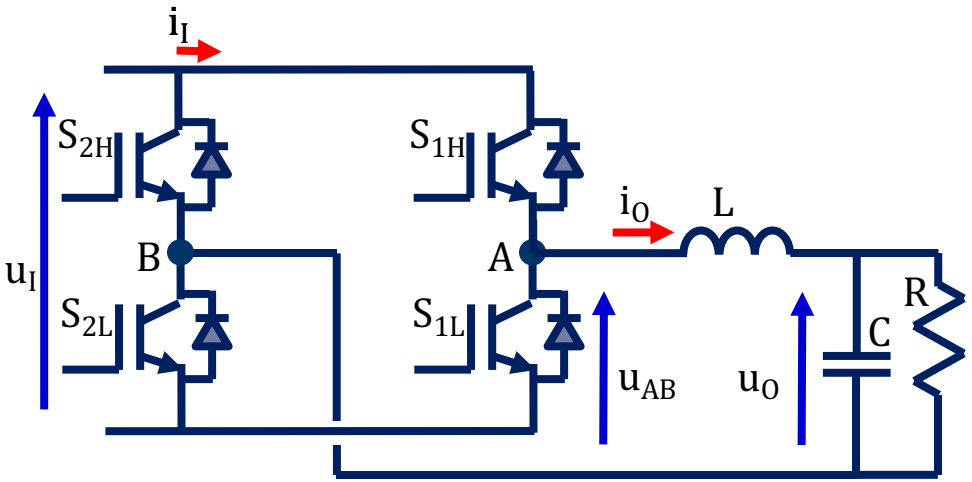


This pattern can be obtaining calculating with a triangle waveform, u_{TRI}

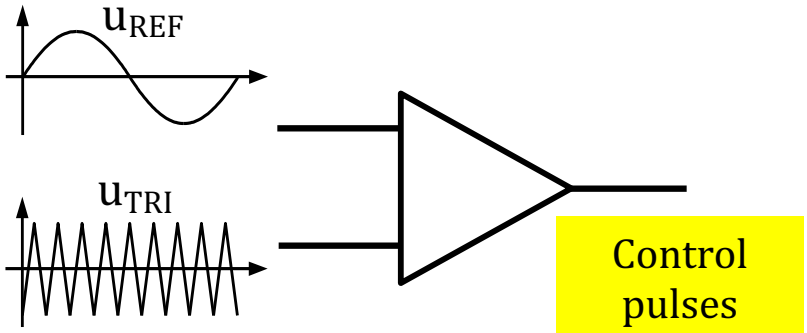




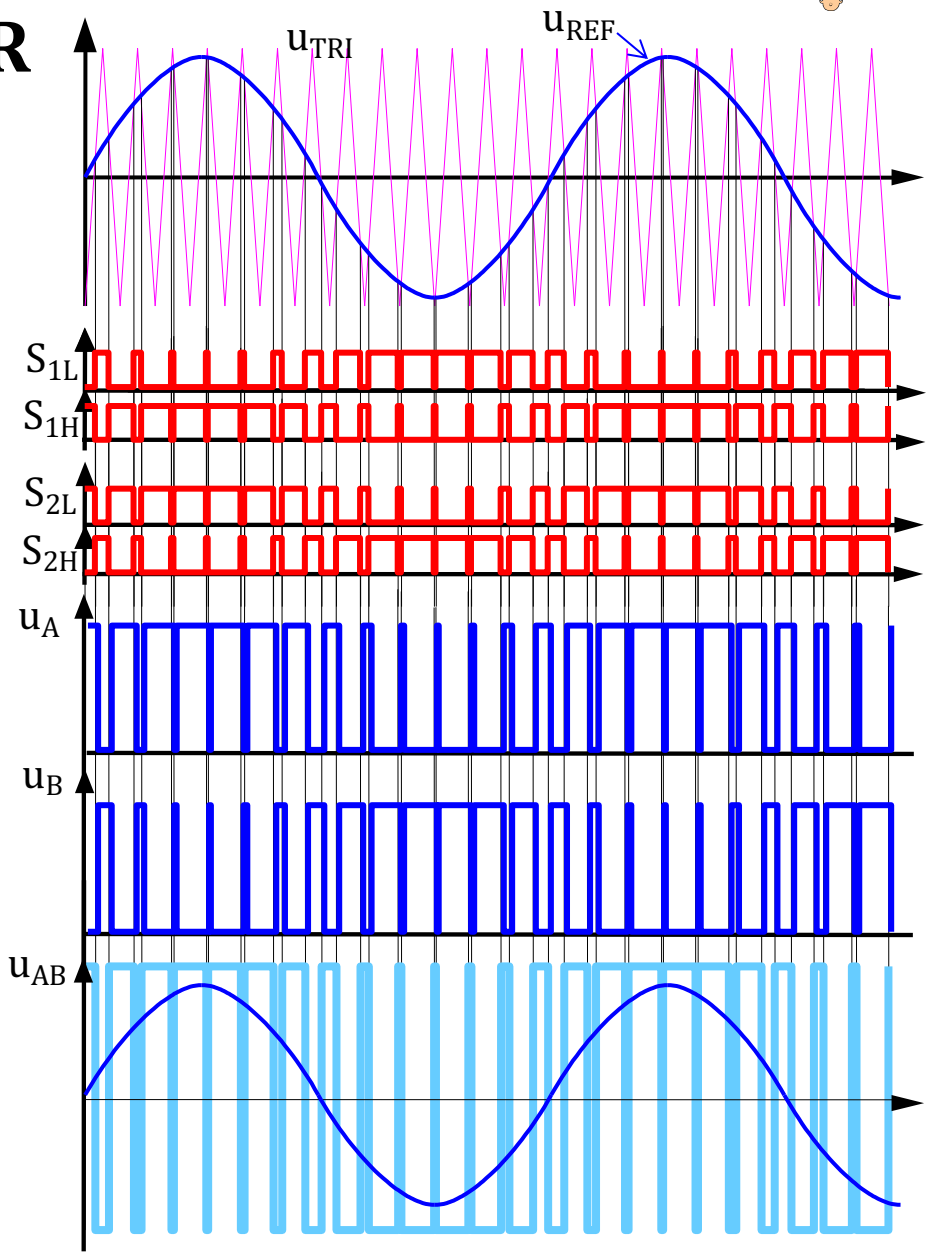
1-Φ FB PWM Inv: BIPOLAR



This pattern can be obtaining calculating with a triangle waveform, u_{TRI}

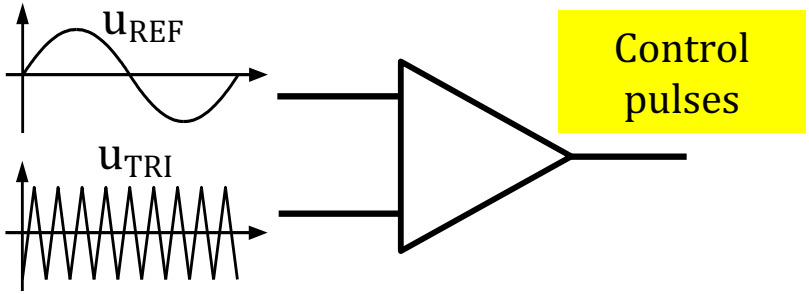


BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



$$u_{REF}(t) = u_{REF\langle PEAK\rangle} \cdot \sin(2 \cdot \pi \cdot f_R \cdot t)$$

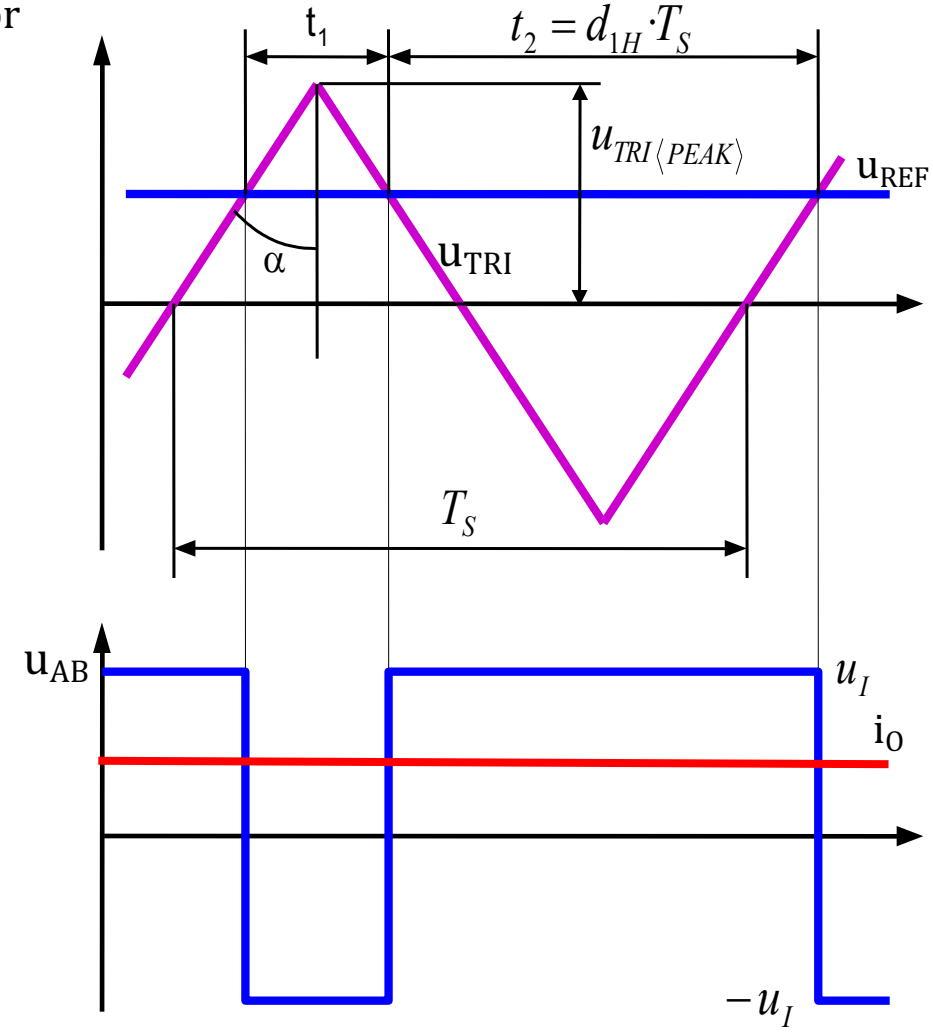
$$u_{TRI}(t) = triangle\left(u_{TRI\langle PEAK\rangle}, T_S\right)$$

We define the **frequency modulation ratio:**

$$m_f = \frac{f_S}{f_R} = \frac{T_R}{T_S}$$

Also, the **amplitude modulation ratio** can be defined:

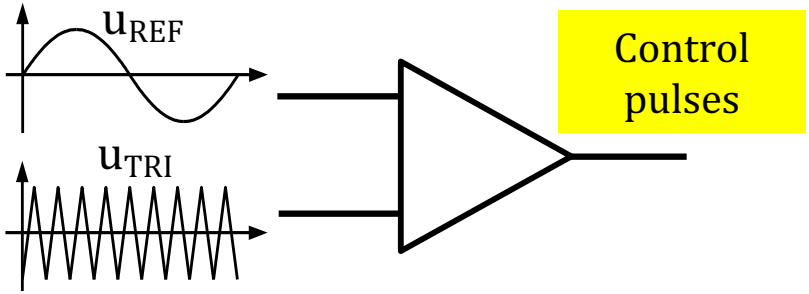
$$m_a = \frac{u_{REF\langle PEAK\rangle}}{u_{TRI\langle PEAK\rangle}}$$



If m_a<1: linear modulation.
If m_a>1: Overmodulation

Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



$$t_1 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} - u_{REF}(t)\right)$$

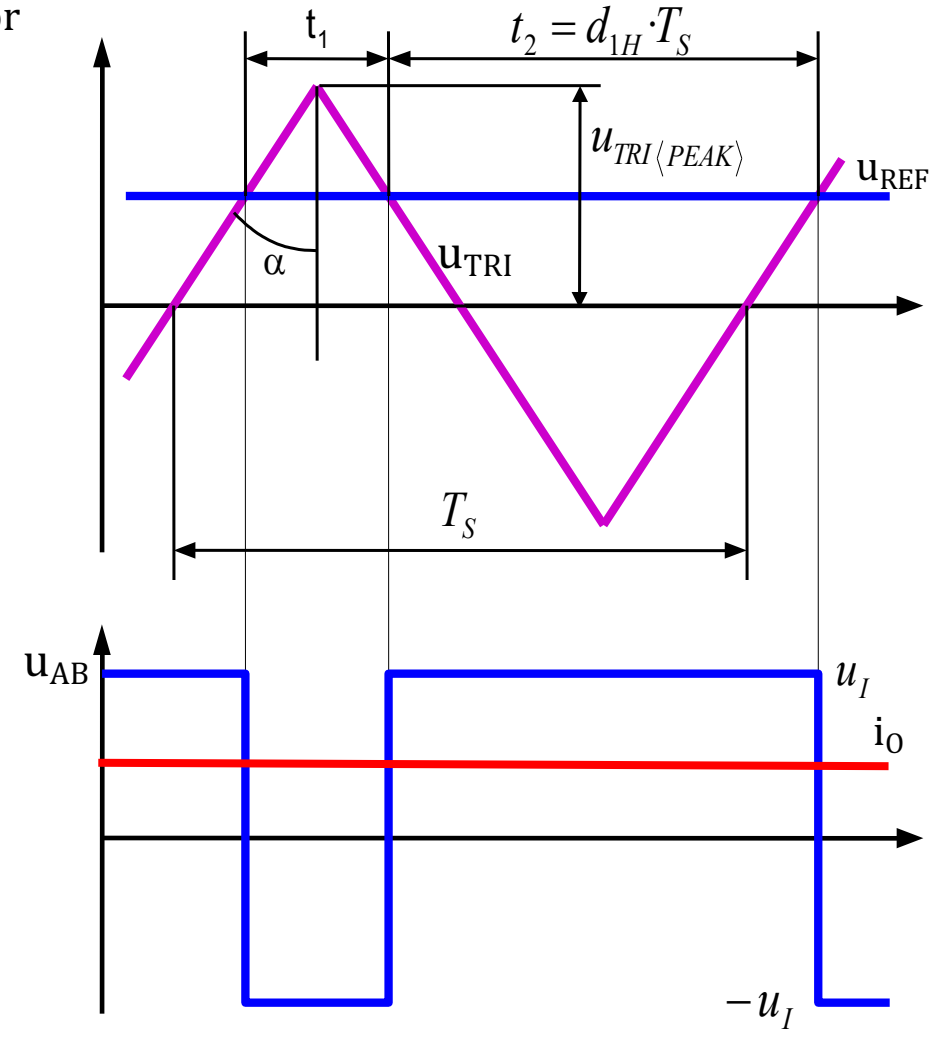
$$t_2 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} + u_{REF}(t)\right)$$

$$tg\alpha = \frac{T_S}{4 \cdot u_{TRI\langle PEAK\rangle}}$$

$$t_2 - t_1 = 4 \cdot tg\alpha \cdot u_{REF}(t)$$

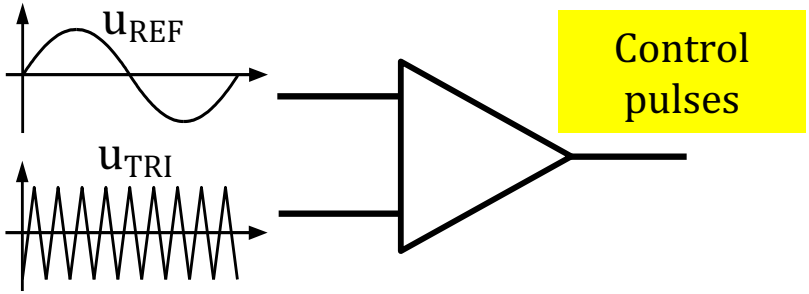
$$t_2 + t_1 = 4 \cdot tg\alpha \cdot u_{TRI\langle PEAK\rangle} = T_S$$

$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot \frac{t_2 - t_1}{T_S} = u_I \cdot (2d_{1H} - 1)$$



Single-Phase Full-Bridge PWM Inverter: BIPOLAR

EQUATIONS FOR BIPOLAR CONTROL (u_{AB} is $+u_I$ or $-u_I$, but never 0)



$$t_1 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} - u_{REF}(t)\right)$$

$$t_2 = 2 \cdot tg\alpha \cdot \left(u_{TRI\langle PEAK\rangle} + u_{REF}(t)\right)$$

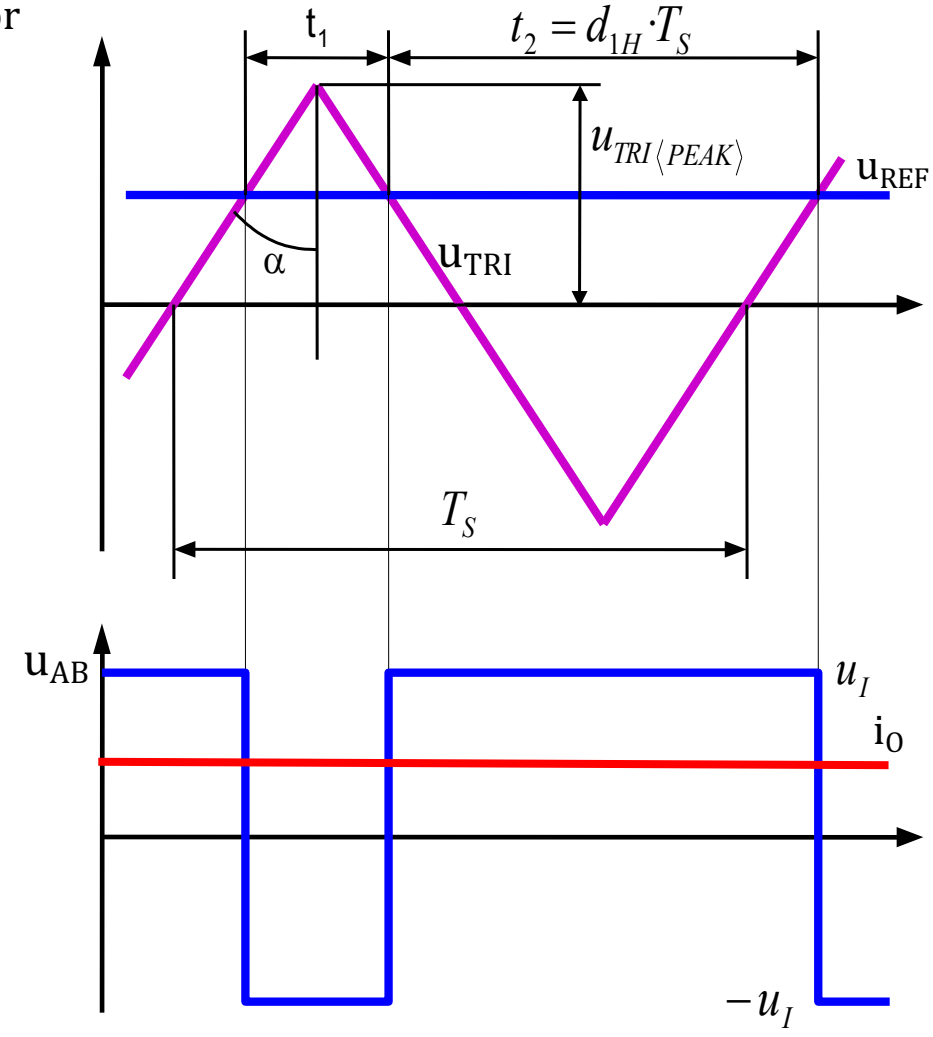
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$$t_2 + t_1 = 4 \cdot tg\alpha \cdot u_{TRI\langle PEAK\rangle} = T_S$$

$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot \frac{t_2 - t_1}{T_S} = \ldots = u_I \cdot \frac{u_{REF}(t)}{u_{TRI\langle PEAK\rangle}} =$$

$$= \frac{u_I \cdot u_{REF\langle PEAK\rangle} \cdot \sin(2 \cdot \pi \cdot f_R \cdot t)}{u_{TRI\langle PEAK\rangle}} = u_I \cdot m_a \cdot sen(2 \cdot \pi \cdot f_R \cdot t)$$



$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot m_a \cdot sen(2 \cdot \pi \cdot f_R \cdot t)$$

VALID ONLY IF m_a<1

Single-Phase Full-Bridge PWM Inverter: BIPOLAR

STUDY OF HARMONICS FOR BIPOLAR CONTROL

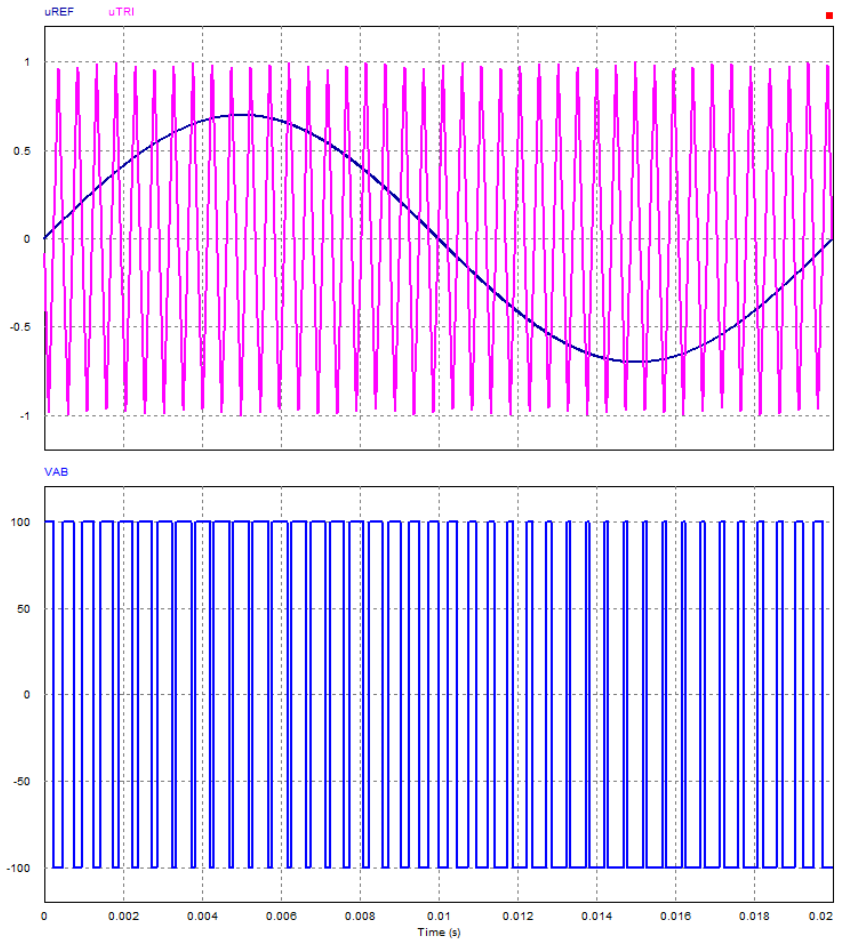
(u_{AB} is $+u_I$ or $-u_I$, but never 0)

$$m_a = \frac{u_{REF\langle PEAK\rangle}}{u_{TRI\langle PEAK\rangle}} = 0.7 \qquad m_f = \frac{f_S}{f_R} = 41$$

$$\langle u_{AB}(t)\rangle_{T_S} = u_I \cdot m_a \cdot sen(2\cdot\pi\cdot f_R \cdot t)$$

Harmonics will be minimized if:

- m_f is an **odd** number
- m_f is a **large** enough.
- For relatively small m_f values, (<21), the $u_{REF}(t)$ and $u_{TRI}(t)$ waveforms must be **synchronized** (same zero-crossing instant)
- Also, at zero crossing, the slopes of $u_{REF}(t)$ and $u_{TRI}(t)$, must present **opposite slopes**.



Single-Phase Full-Bridge PWM Inverter: BIPOLAR

STUDY OF HARMONICS FOR BIPOLAR CONTROL

(u_{AB} is $+u_I$ or $-u_I$, but never 0)

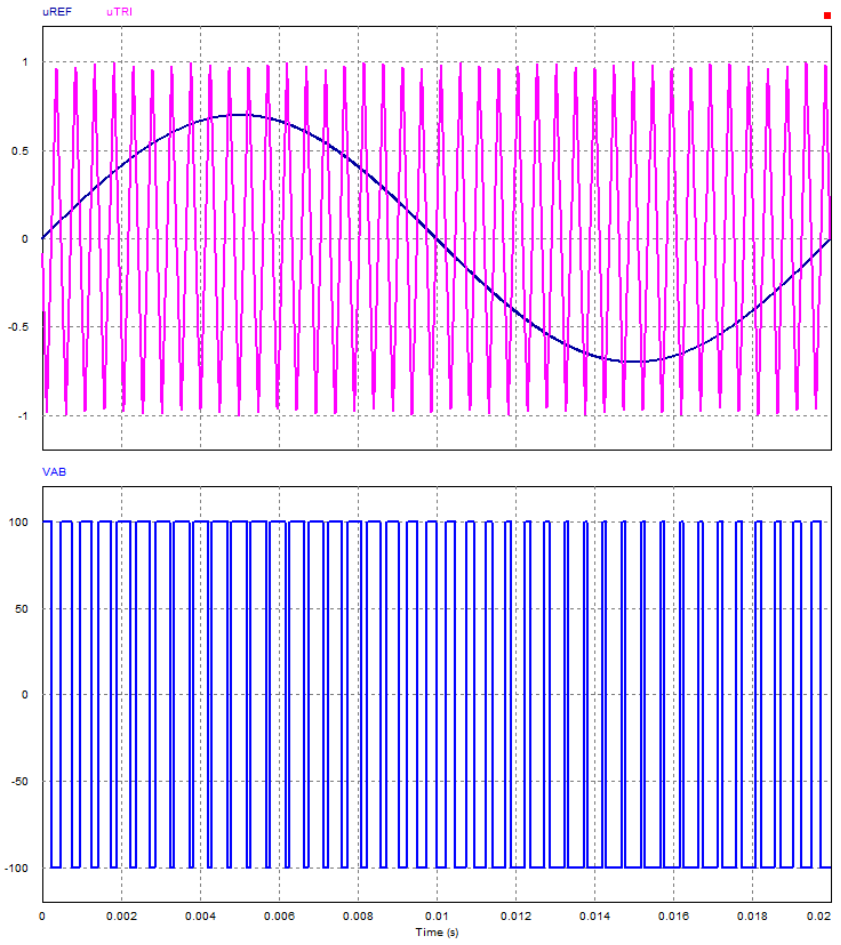
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Table 8-1 Generalized Harmonics of v_{Ao} for a Large m_f.

h \ m_a	0.2	0.4	0.6	0.8	1.0
1 *Fundamental*	0.2	0.4	0.6	0.8	1.0
m_f	1.242	1.15	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
$m_f \pm 4$					0.018
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$		0.024	0.071	0.139	0.212
$2m_f \pm 5$				0.013	0.033
$3m_f$	0.335	0.123	0.083	0.171	0.113
$3m_f \pm 2$	0.044	0.139	0.203	0.176	0.062
$3m_f \pm 4$		0.012	0.047	0.104	0.157
$3m_f \pm 6$				0.016	0.044
$4m_f \pm 1$	0.163	0.157	0.008	0.105	0.068
$4m_f \pm 3$	0.012	0.070	0.132	0.115	0.009
$4m_f \pm 5$			0.034	0.084	0.119
$4m_f \pm 7$	AMPLITUDES			0.017	0.050

Note: $(\hat{V}_{Ao})_h/\frac{1}{2}V_d$ [= $(\hat{V}_{AN})_h/\frac{1}{2}V_d$] is tabulated as a function of m_a.



"Power Electronics Handbook, Second Edition: Devices, Circuits and Applications" *Editor: Muhammad H. Rashid, Ed. Elsevier Inc., 2007.*

Single-Phase Full-Bridge PWM Inverter: BIPOLAR

STUDY OF HARMONICS FOR BIPOLAR CONTROL

(u_{AB} is $+u_I$ or $-u_I$, but never 0)

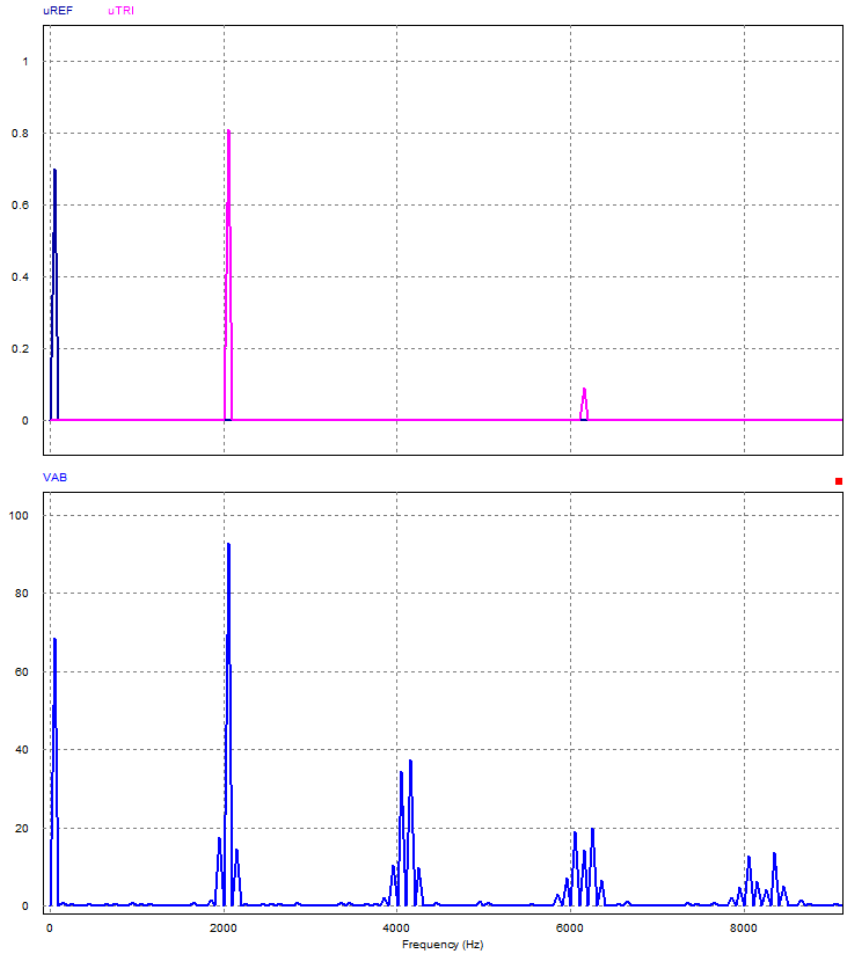
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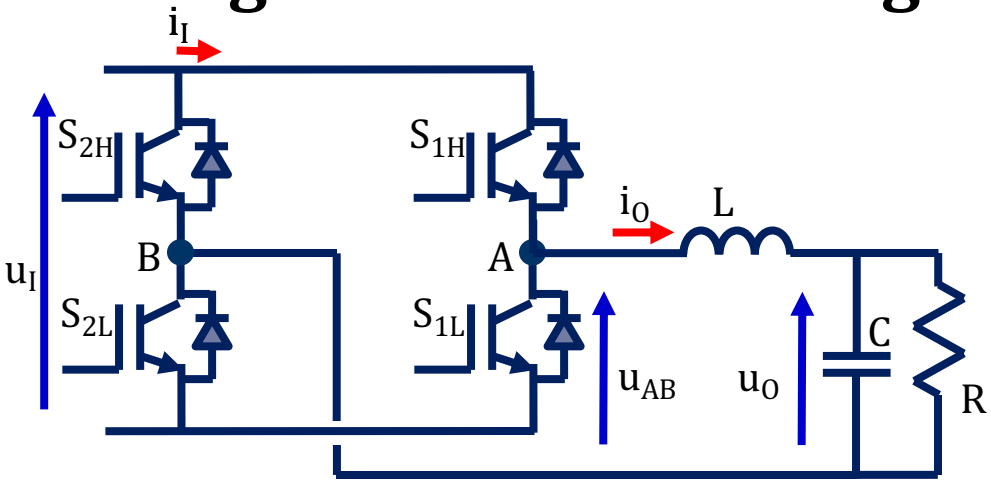
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1 *Fundamental*	0.2	0.4	0.6	0.8	1.0
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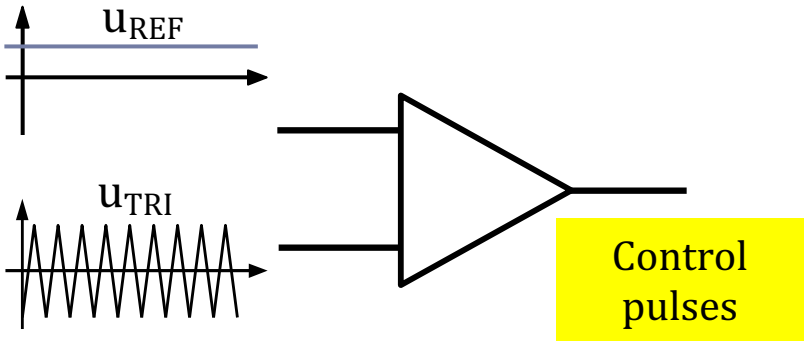


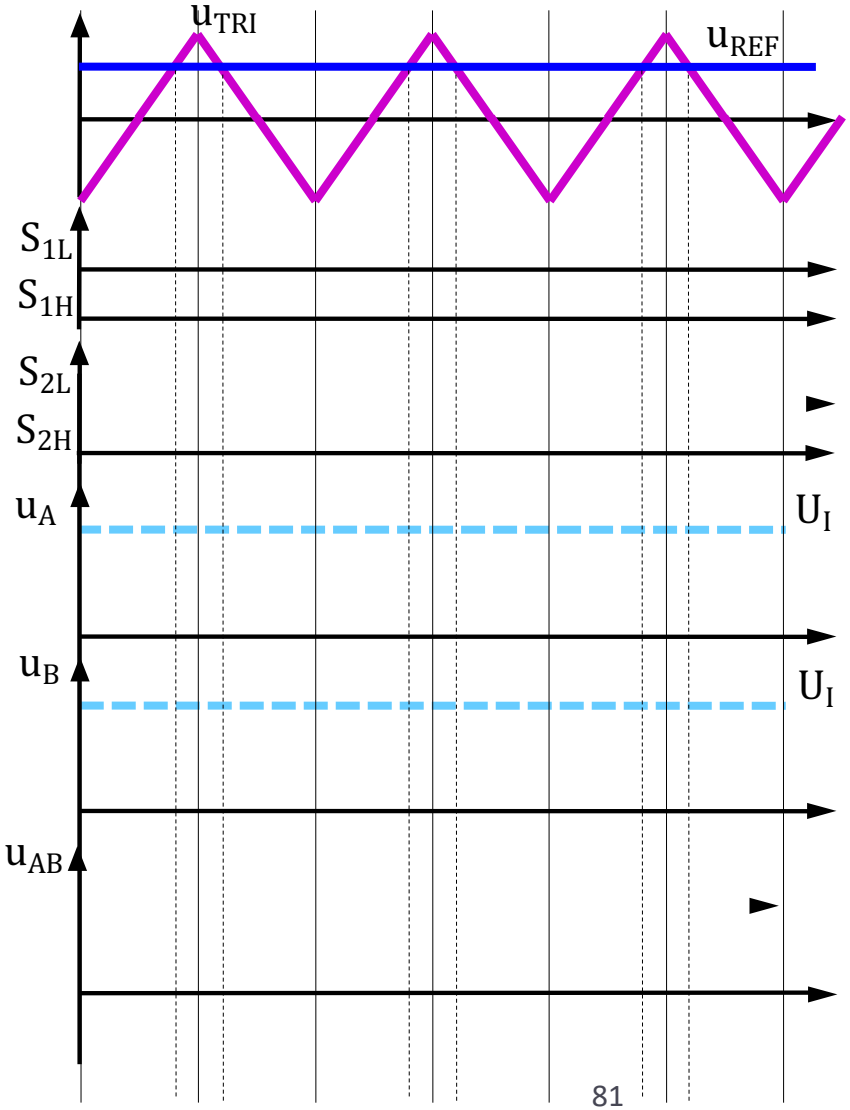
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Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

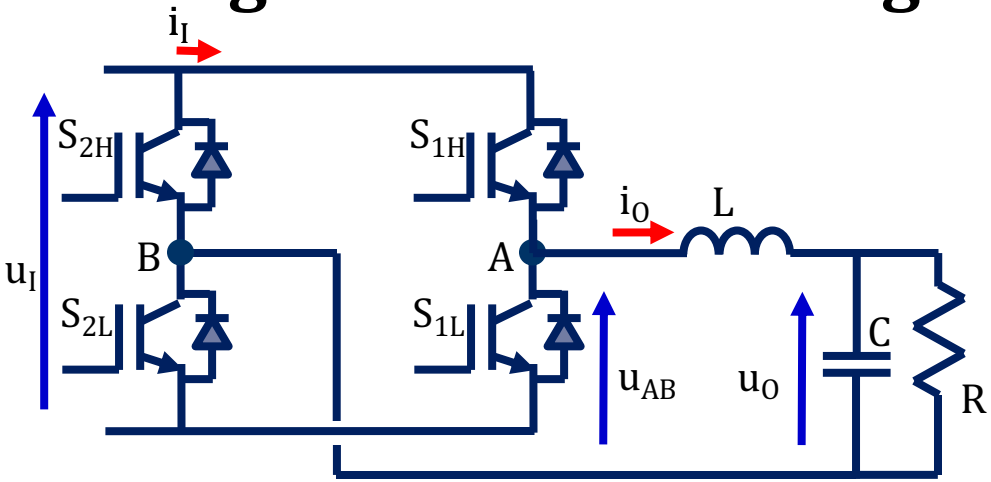


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

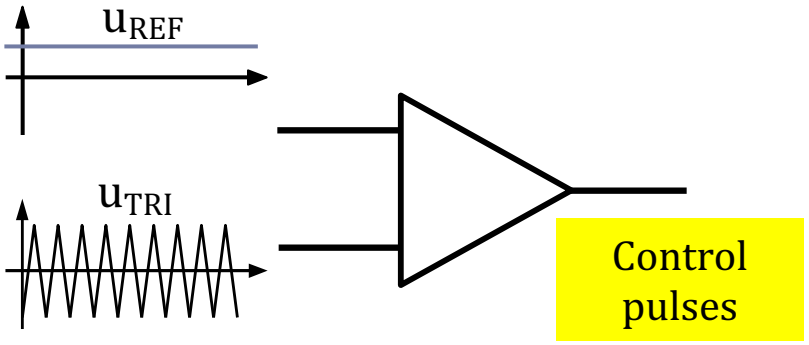


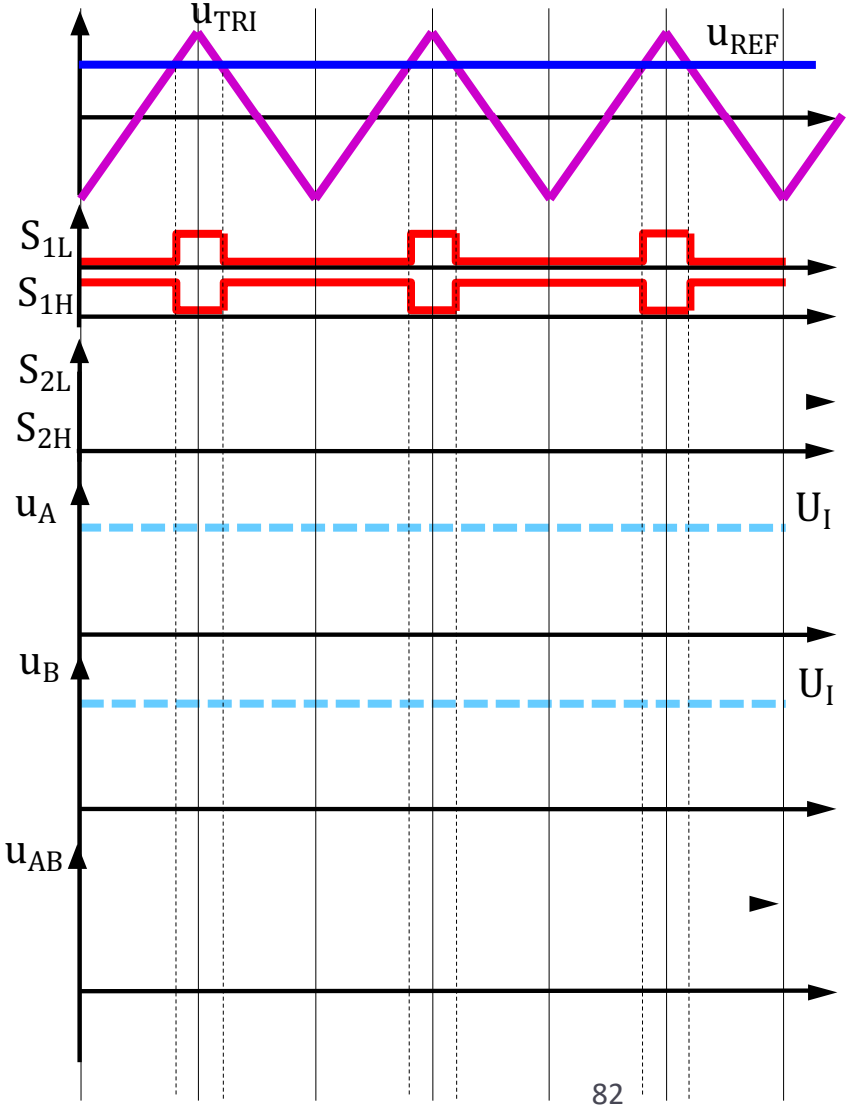


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

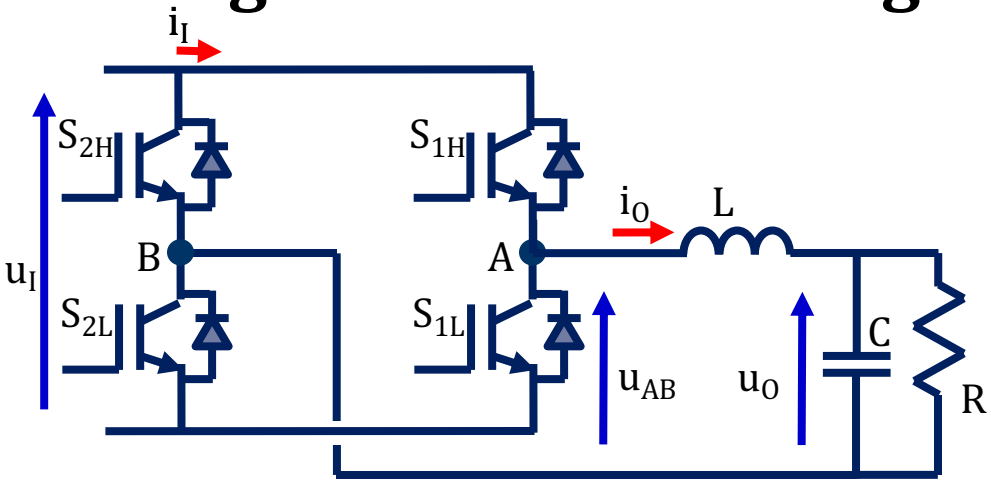


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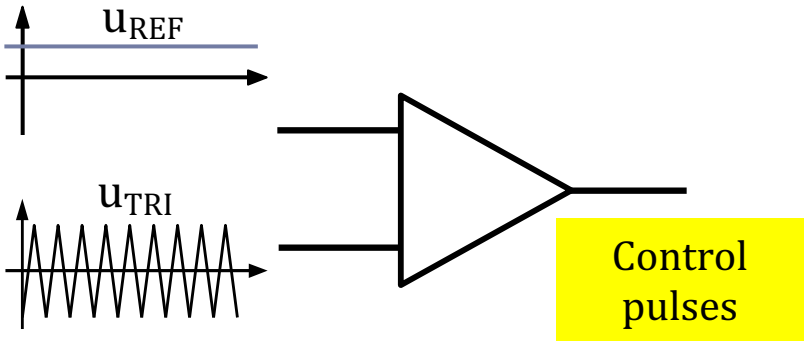


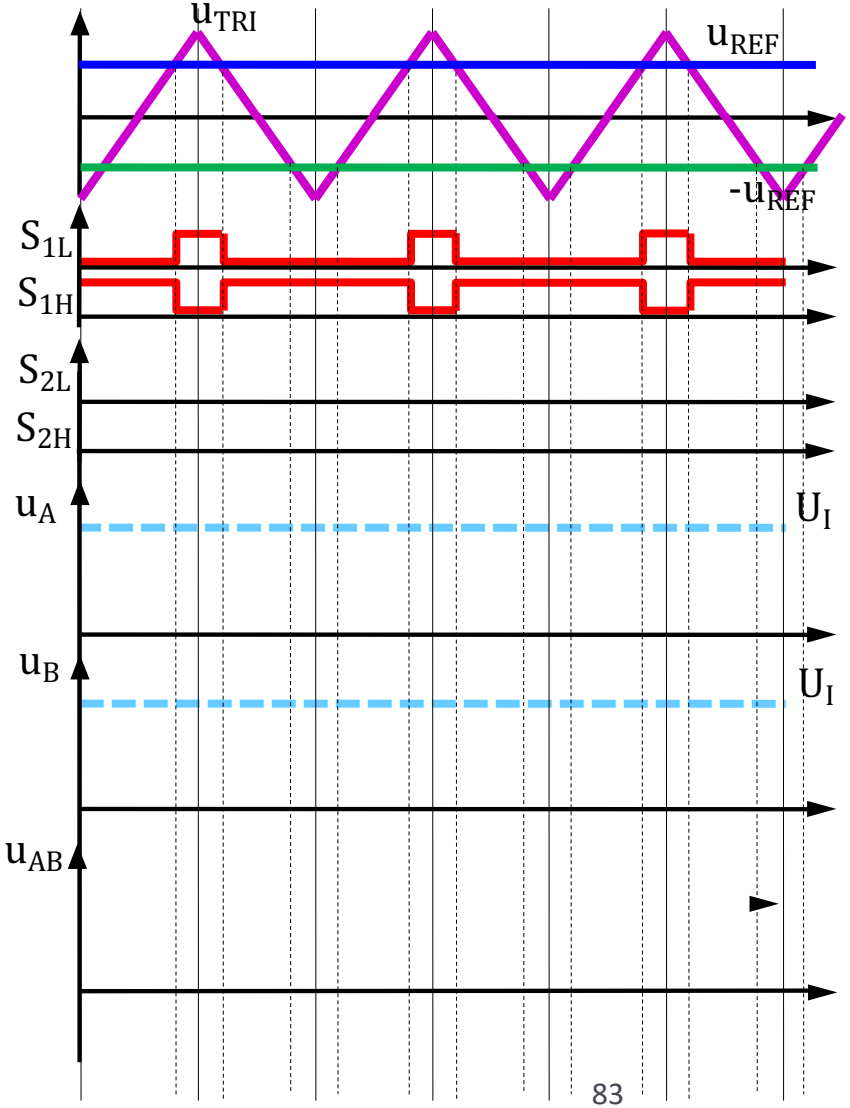


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

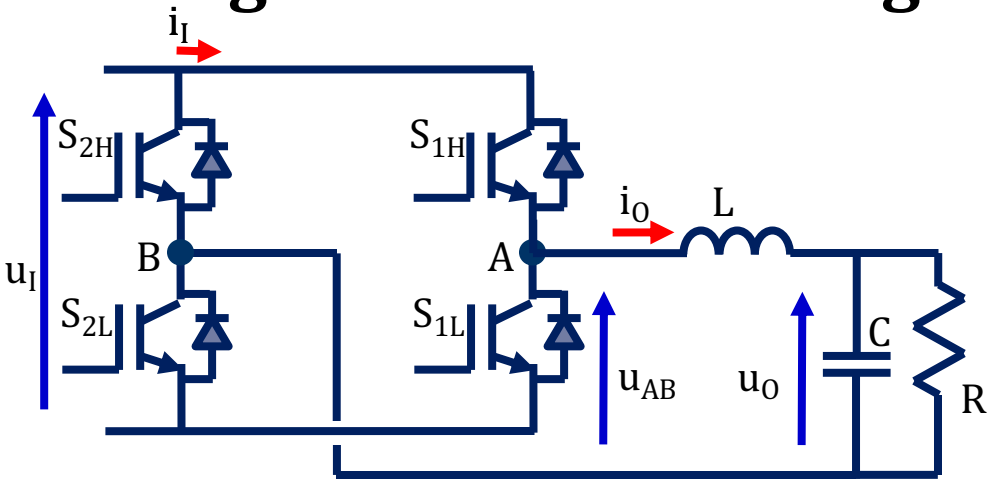


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

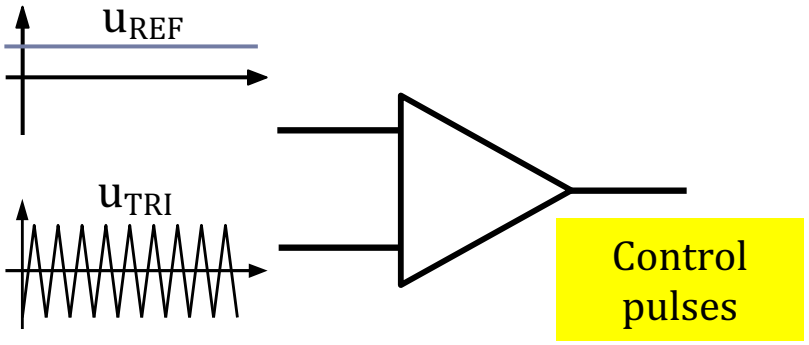


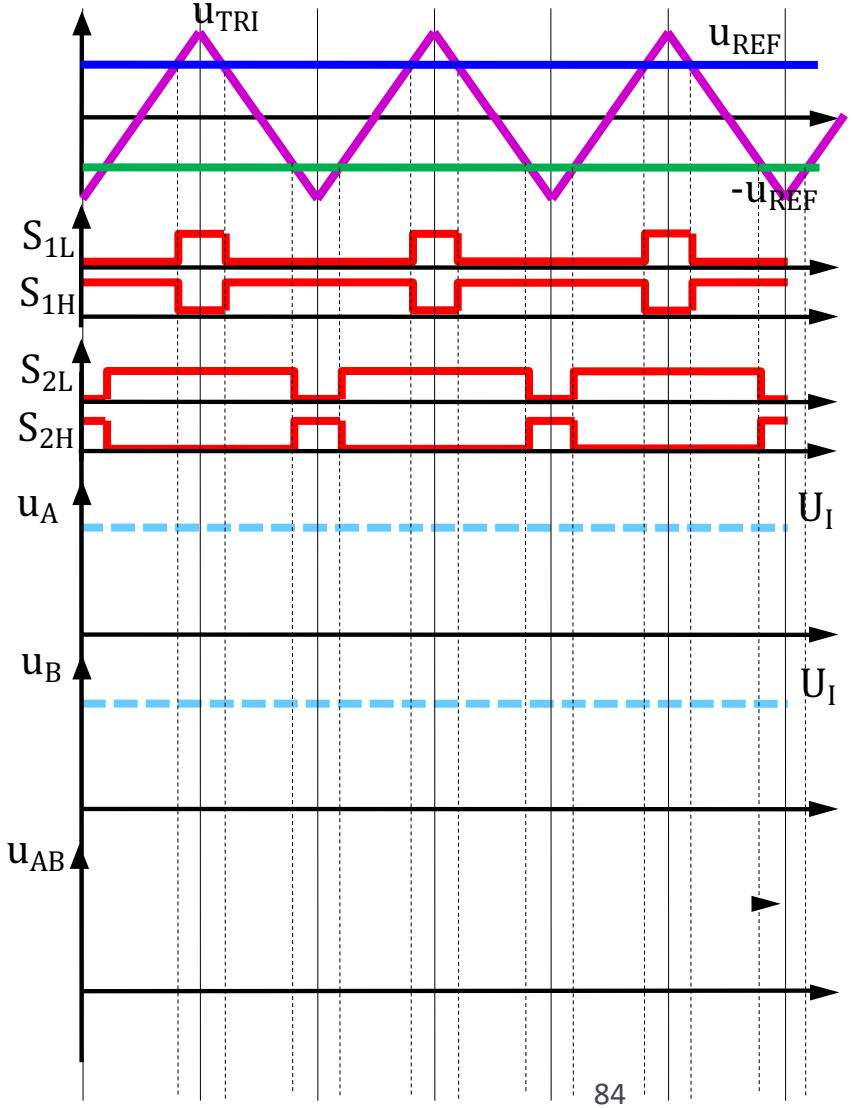


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

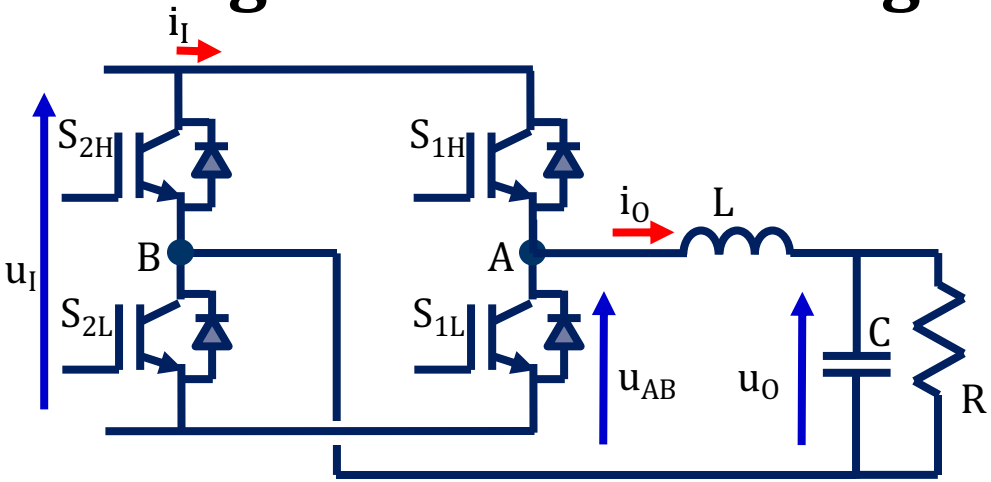


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

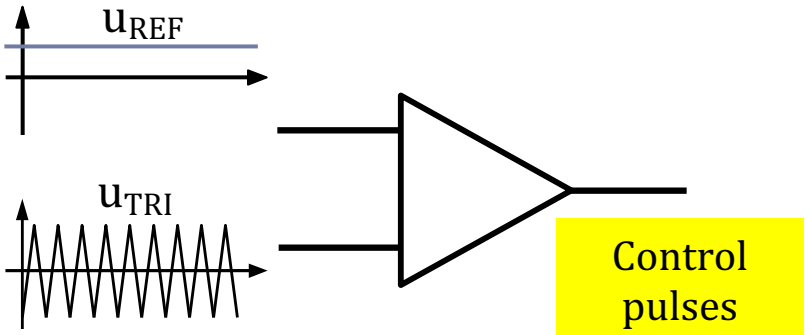


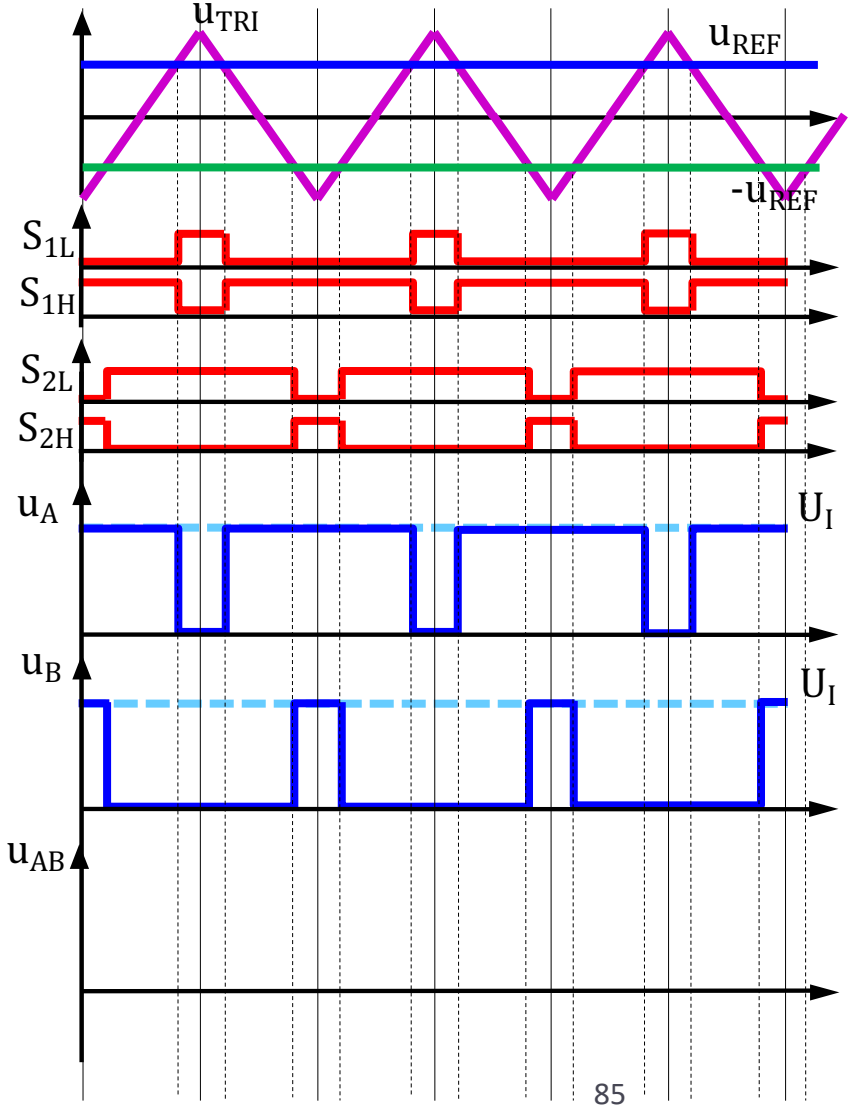


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

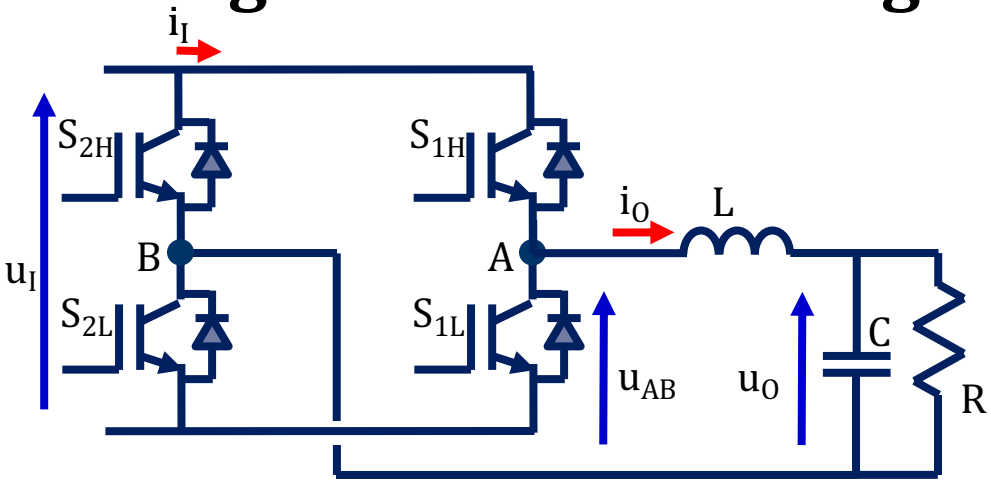


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

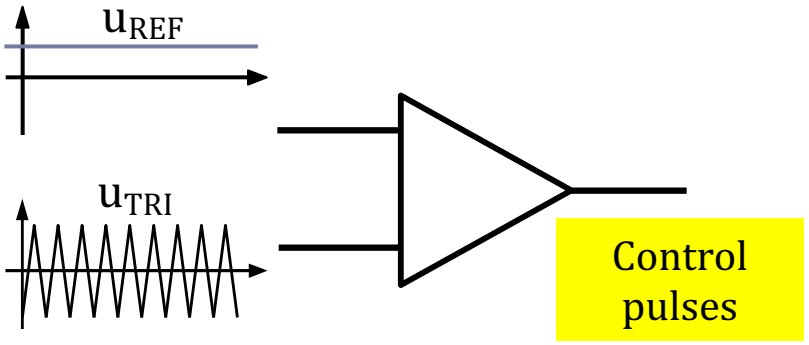


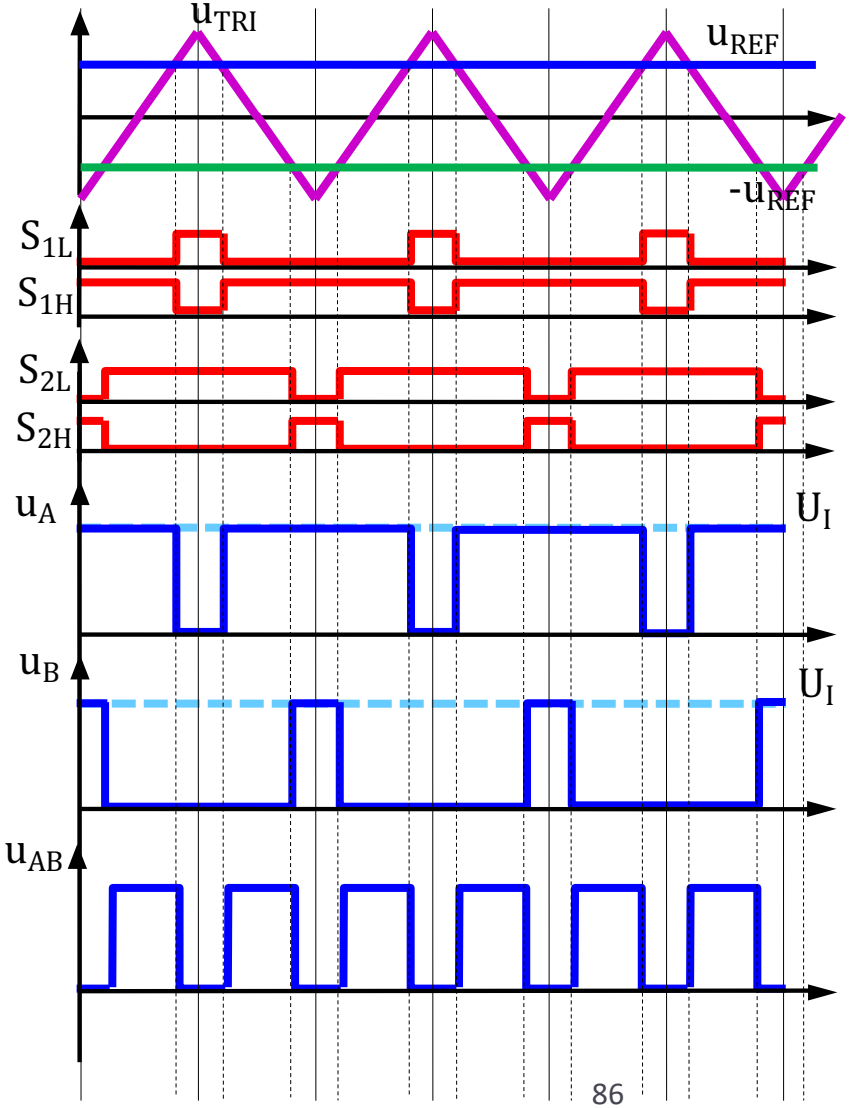


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

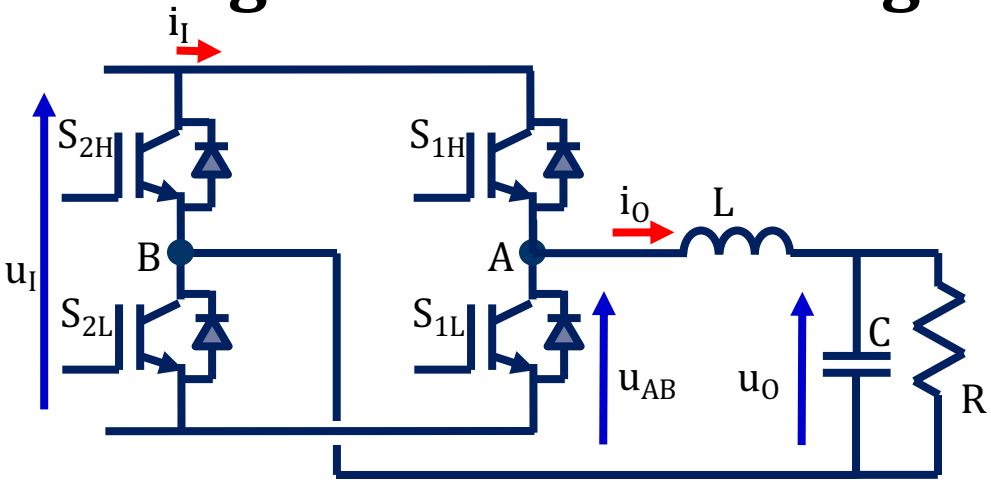


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

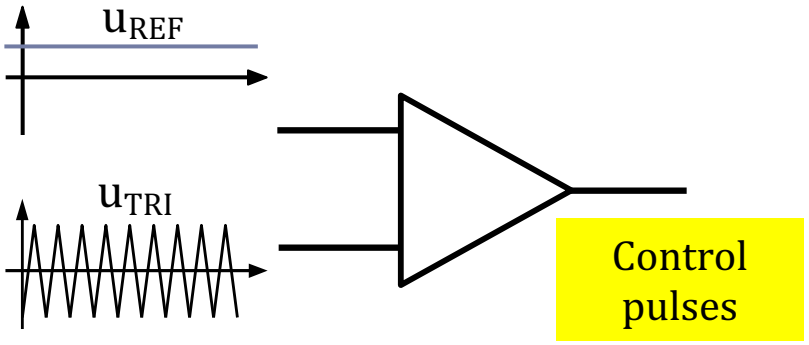


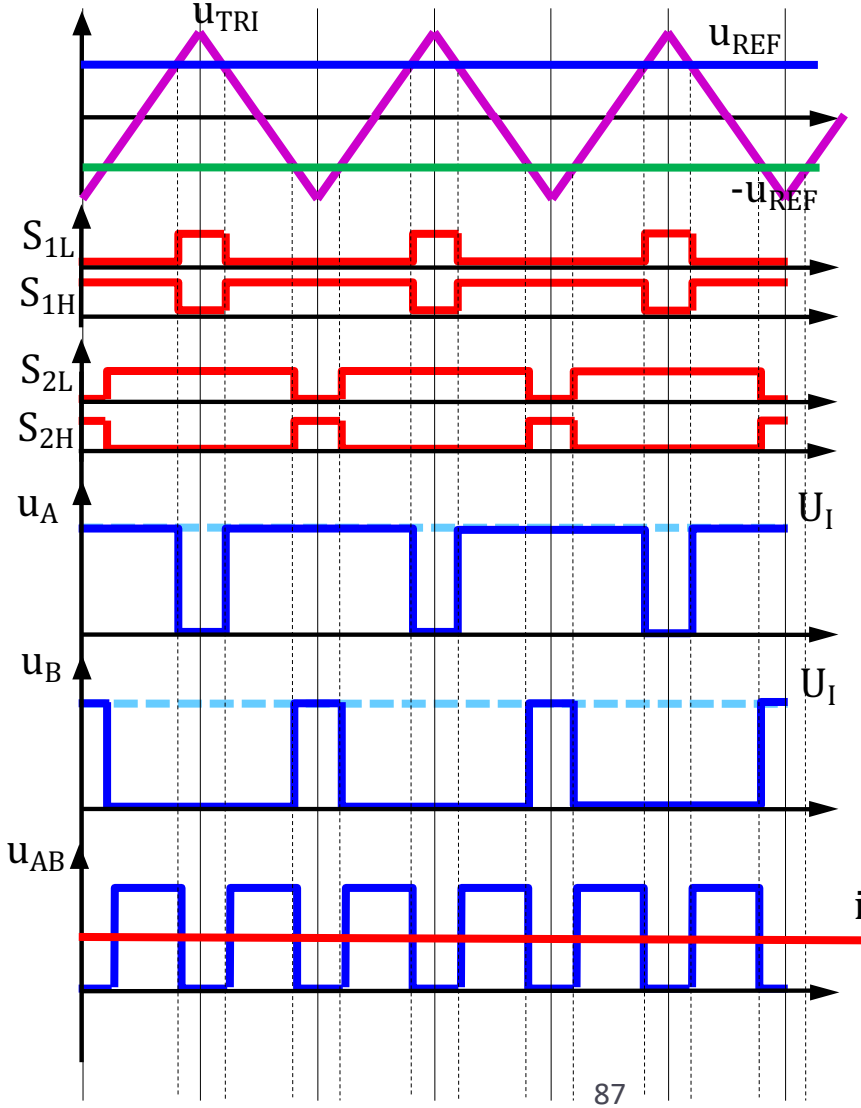


Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

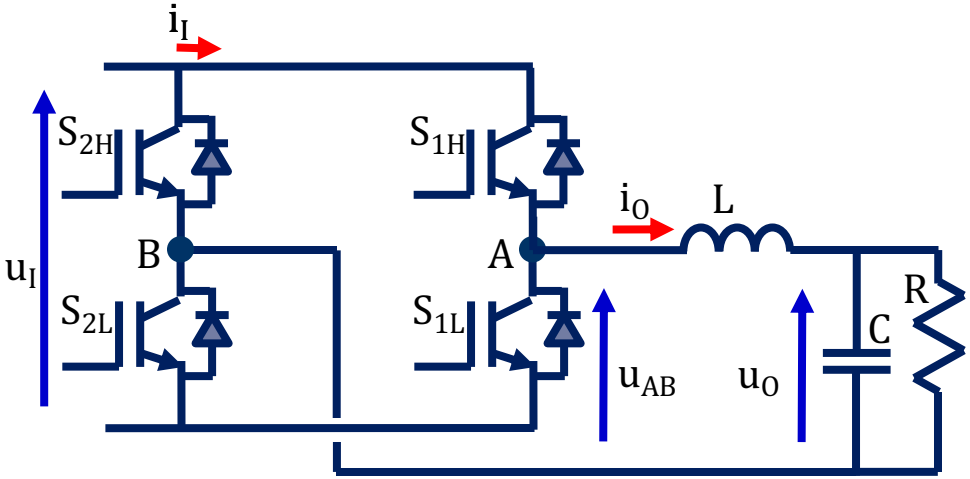


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

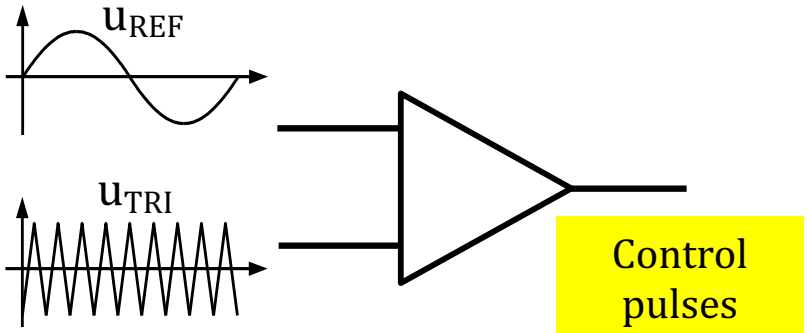


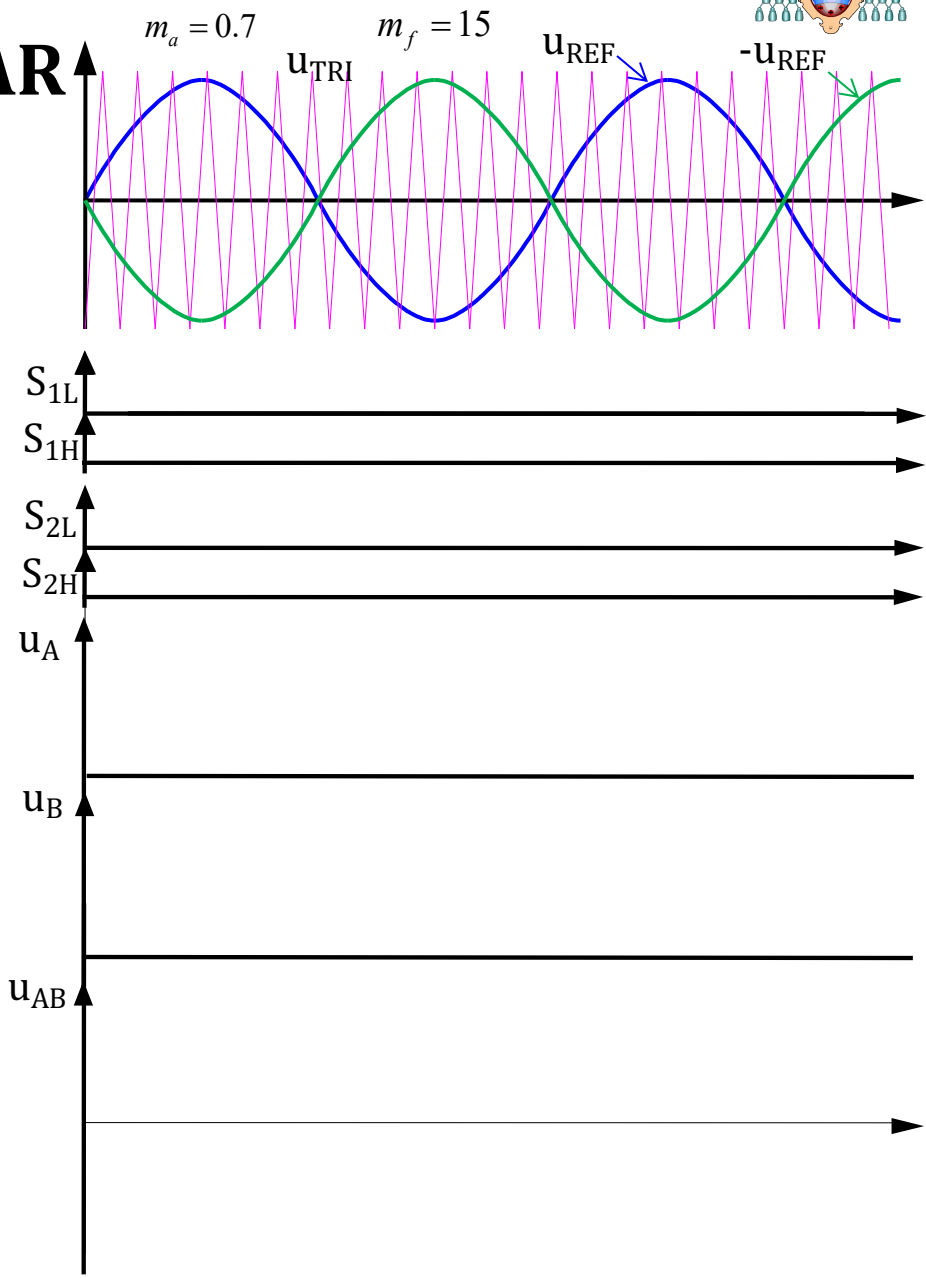


1-Φ FB PWM Inv: UNIPOLAR

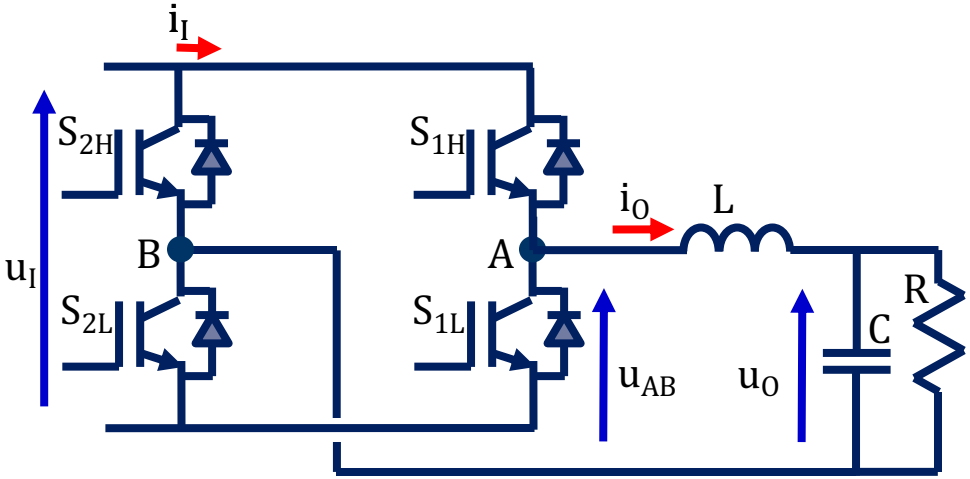


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

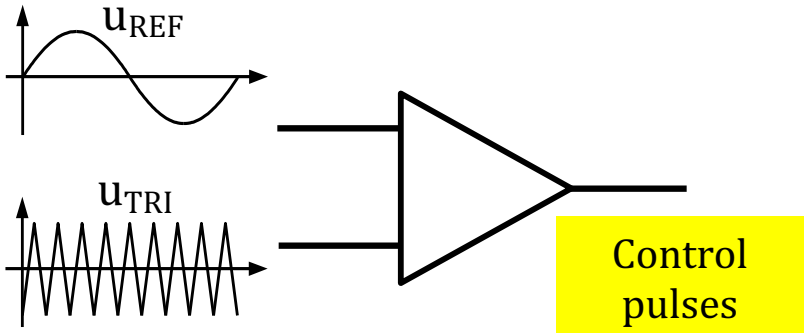


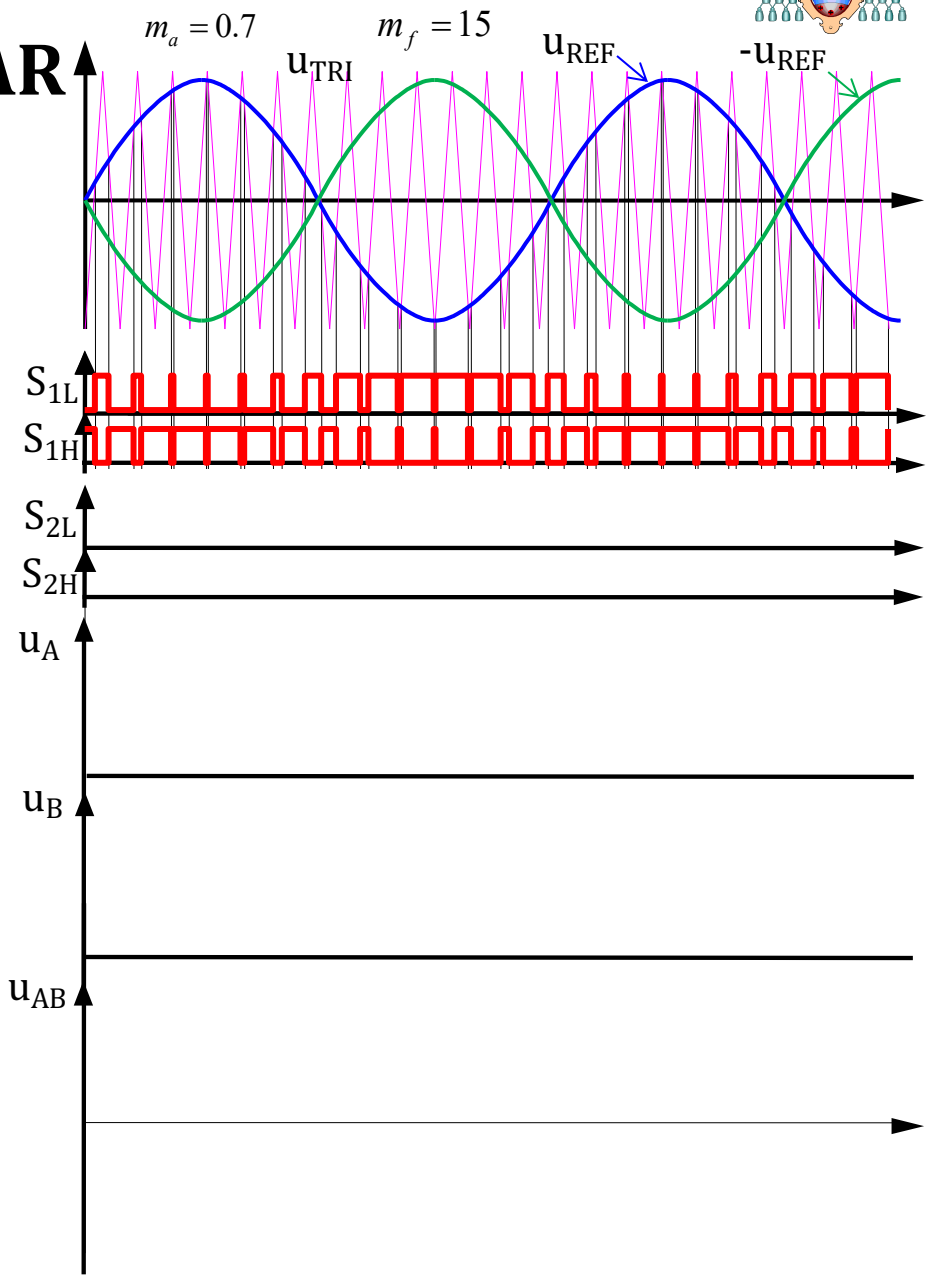


1-Φ FB PWM Inv: UNIPOLAR

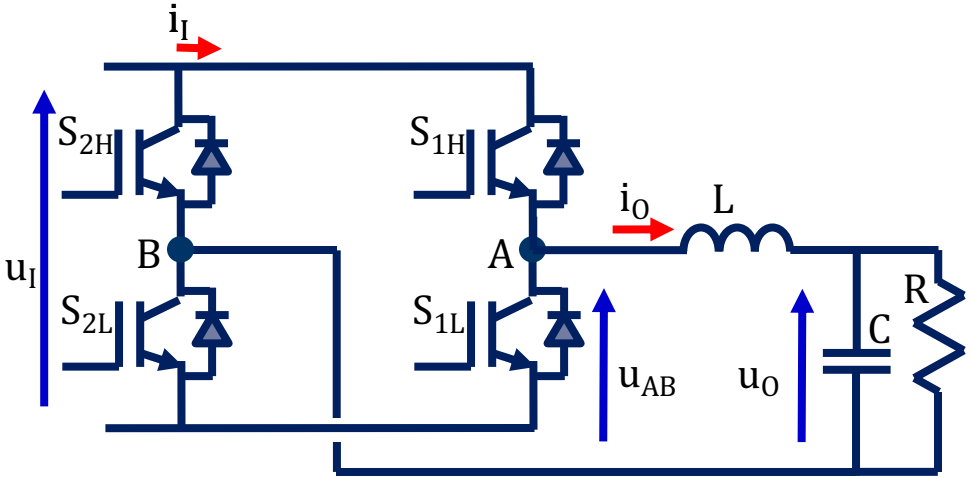


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

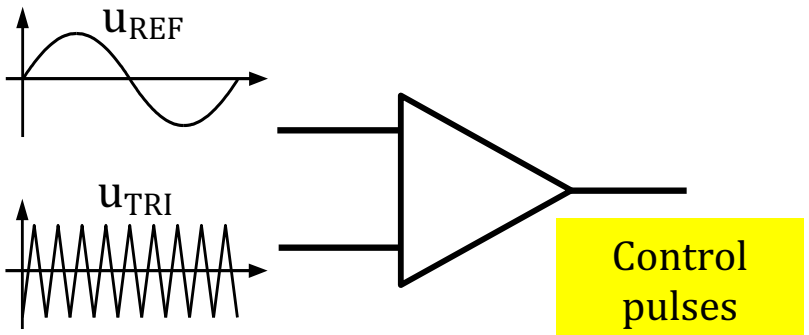


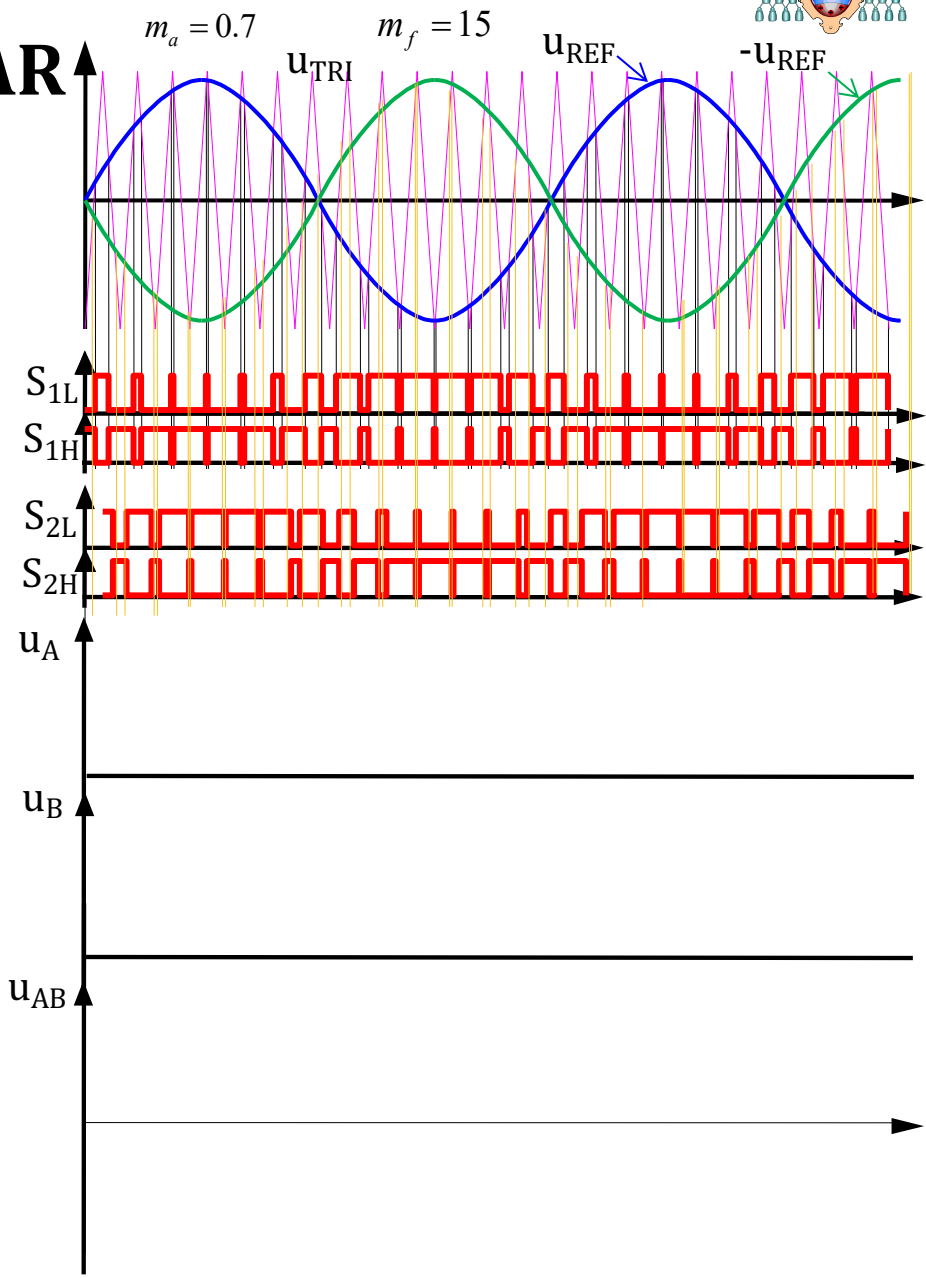


1-Φ FB PWM Inv: UNIPOLAR

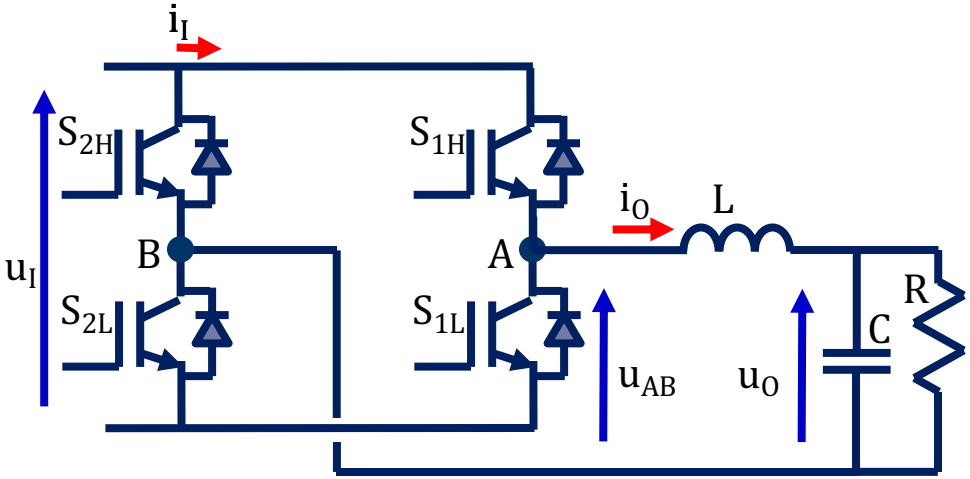


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

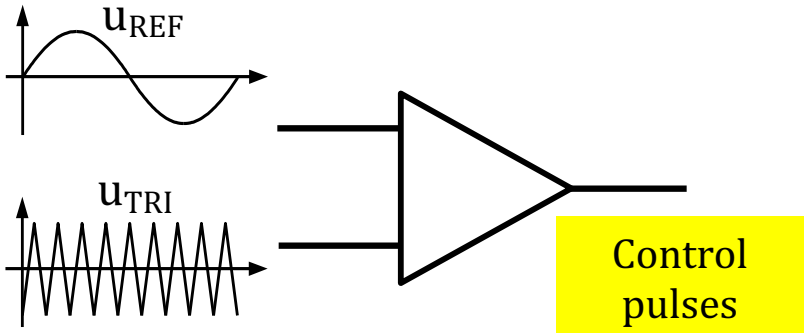


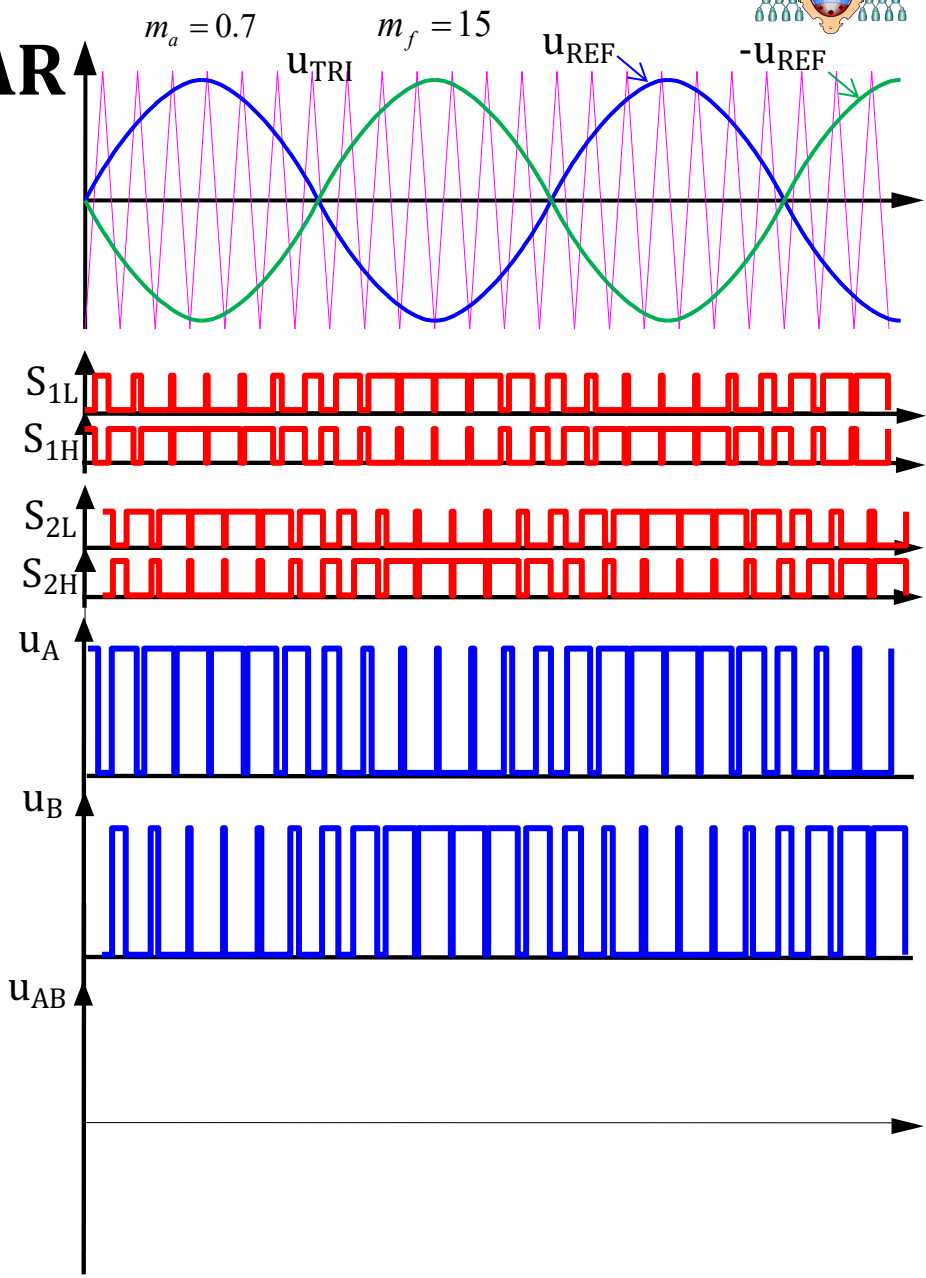


1-Φ FB PWM Inv: UNIPOLAR

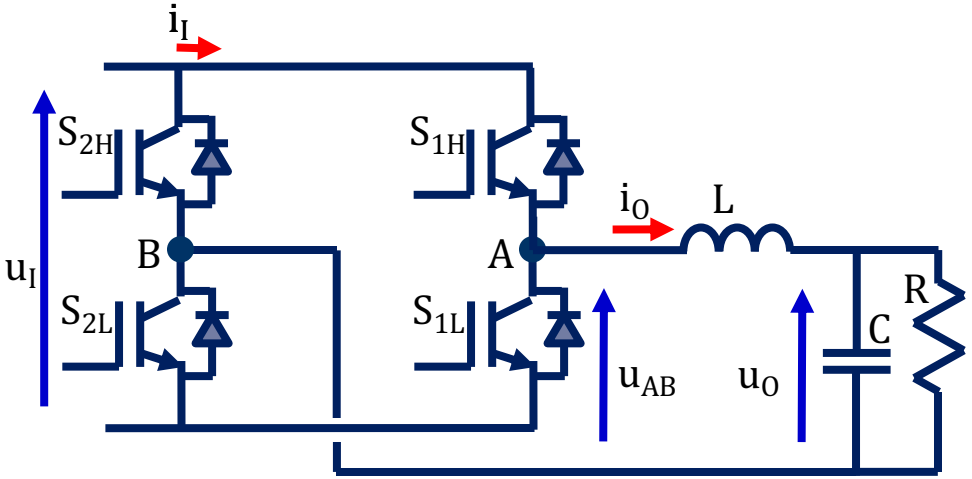


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

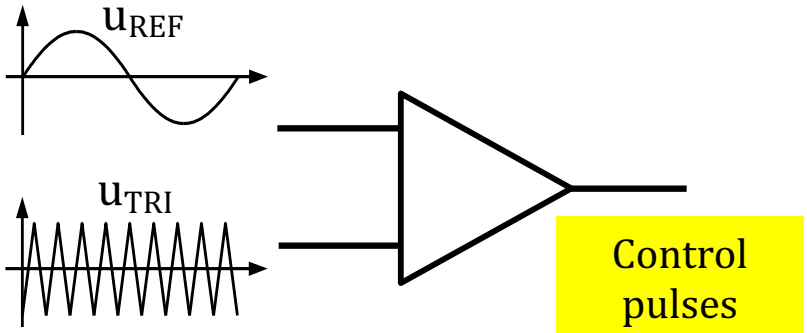


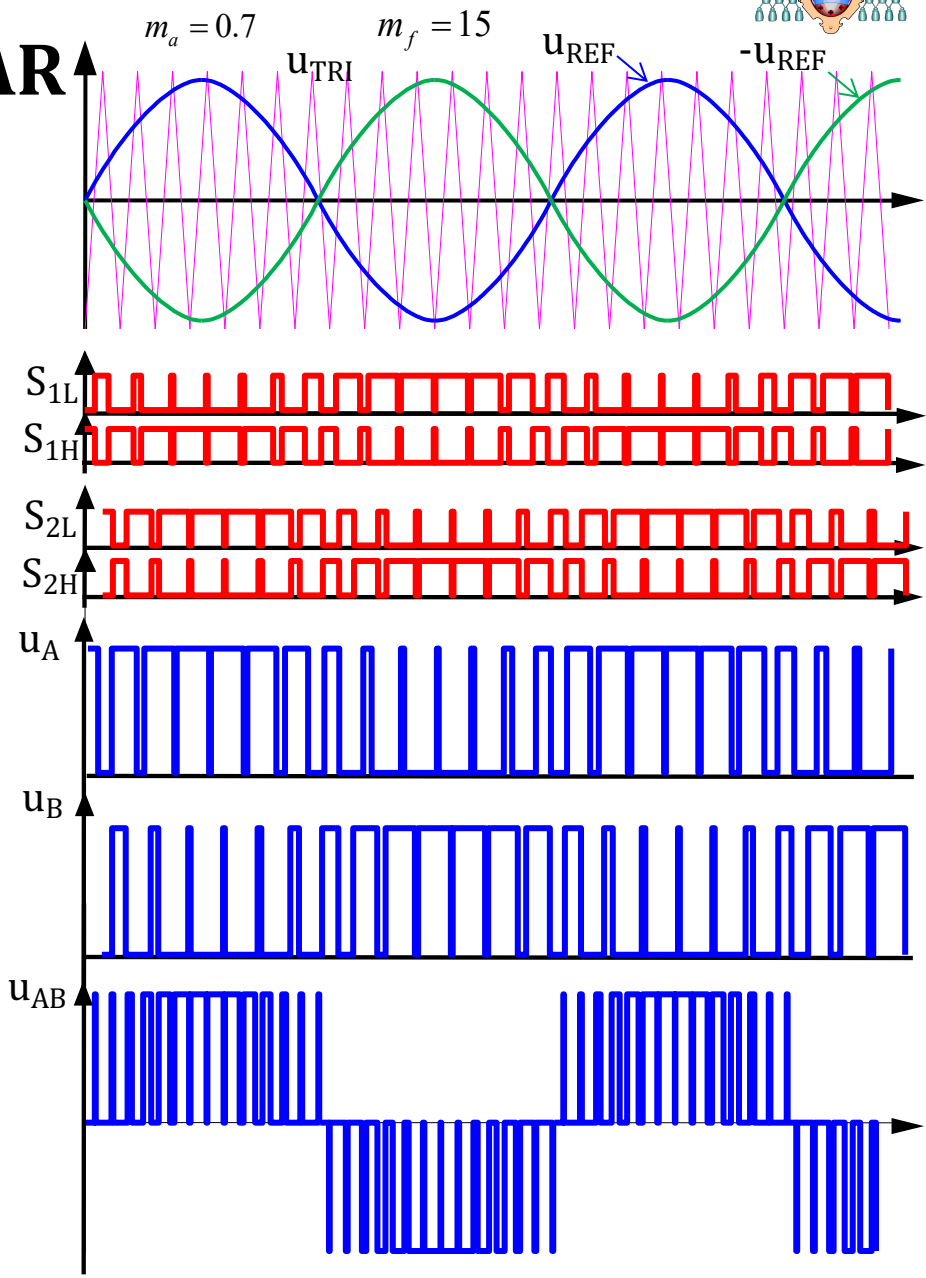


1-Φ FB PWM Inv: UNIPOLAR

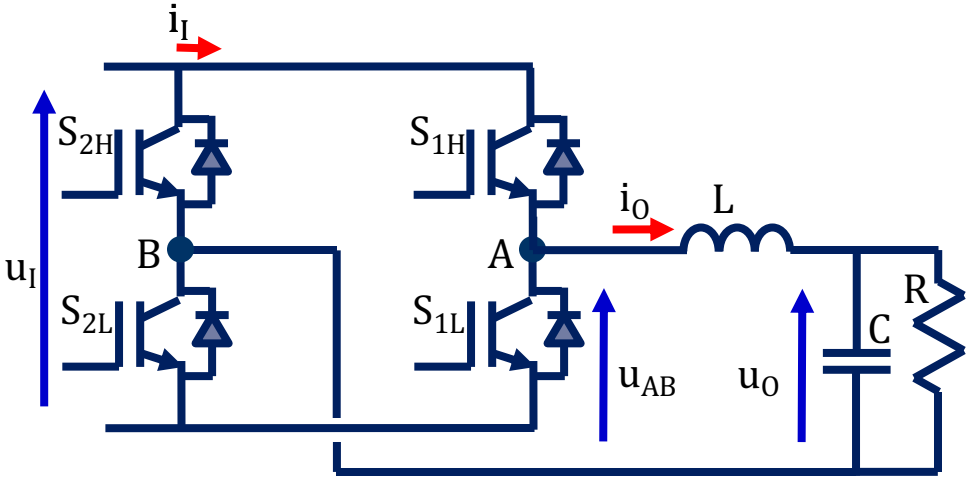


This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

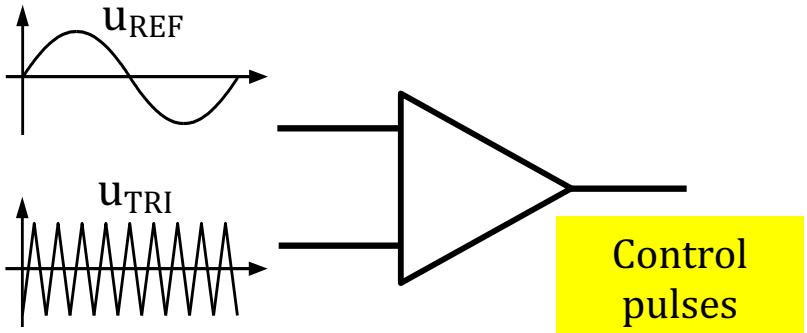




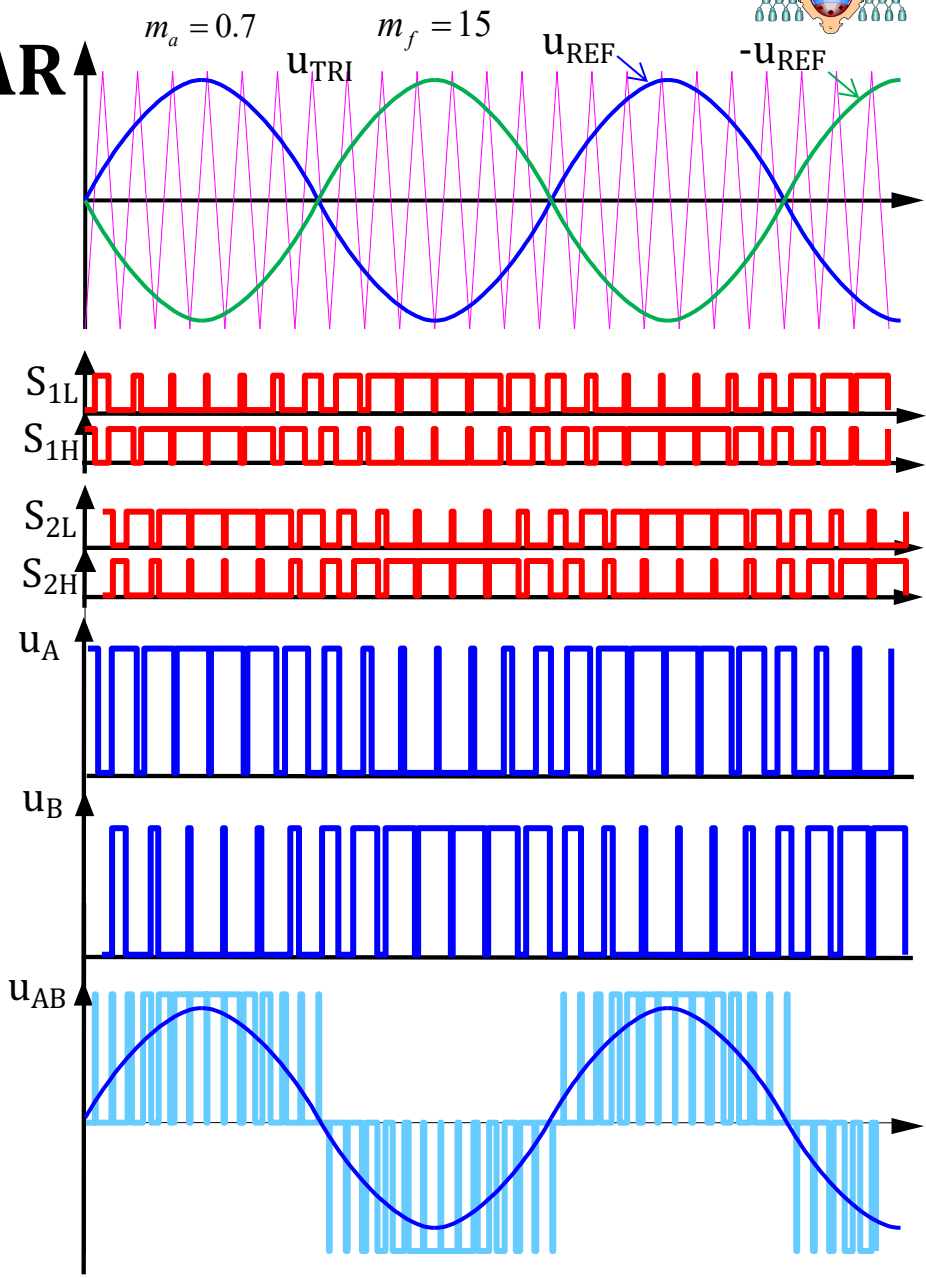
1-Φ FB PWM Inv: UNIPOLAR



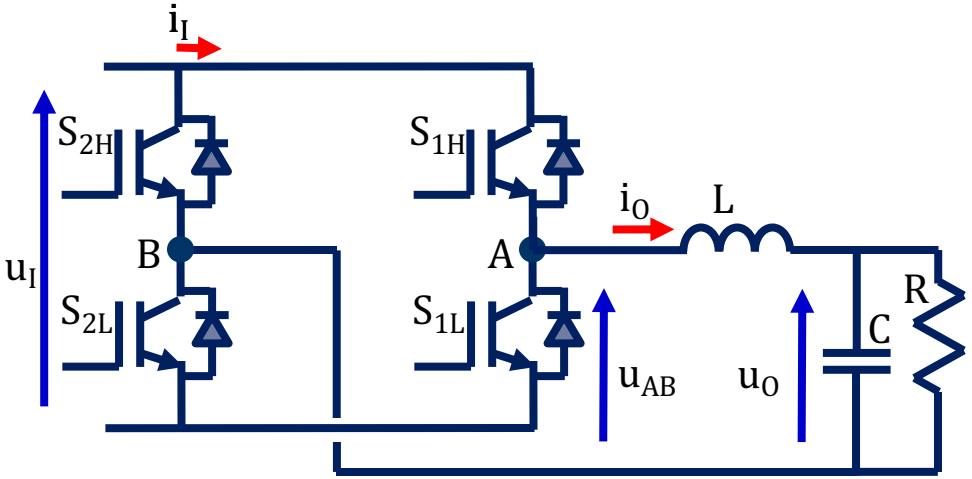
This pattern can be obtaining calculating with a triangular waveform, u_{TRI}



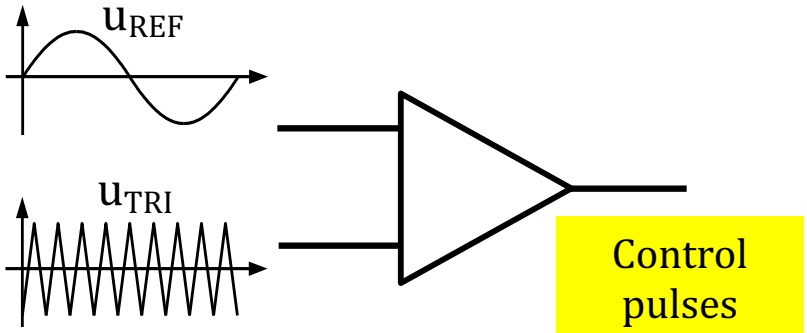
UNIPOLAR CONTROL (u_{AB} changes from 0 to $+u_I$, or from 0 to $-u_I$)



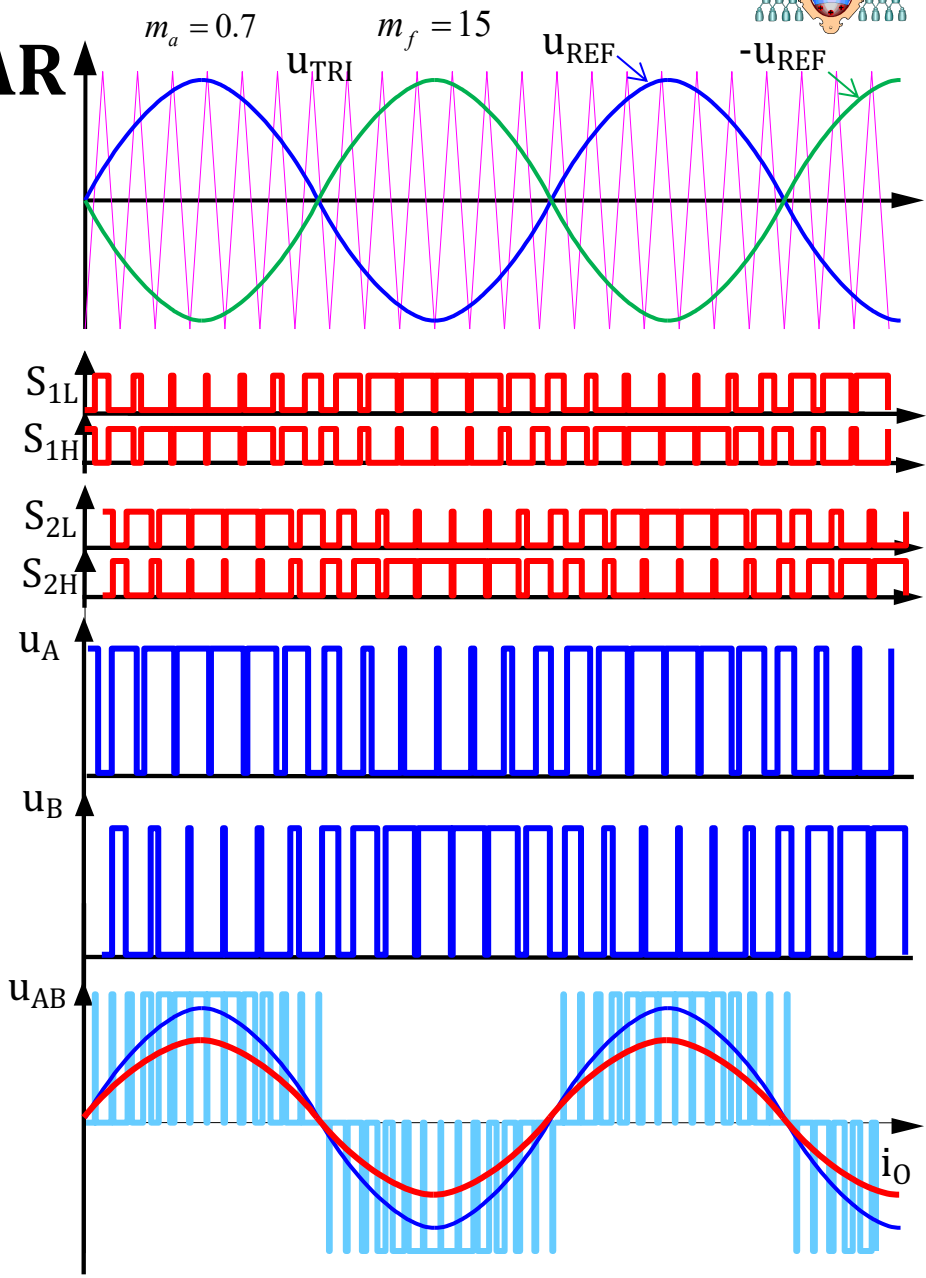
1-Φ FB PWM Inv: UNIPOLAR



This pattern can be obtaining calculating with a triangular waveform, u_{TRI}

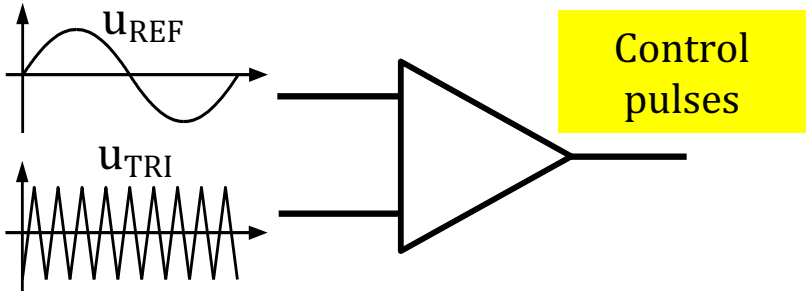


UNIPOLAR CONTROL (u_{AB} changes from 0 to $+u_I$, or from 0 to $-u_I$)



Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

UNIPOLAR CONTROL (u_{AB} changes from 0 to $+u_I$, or from 0 to $-u_I$)



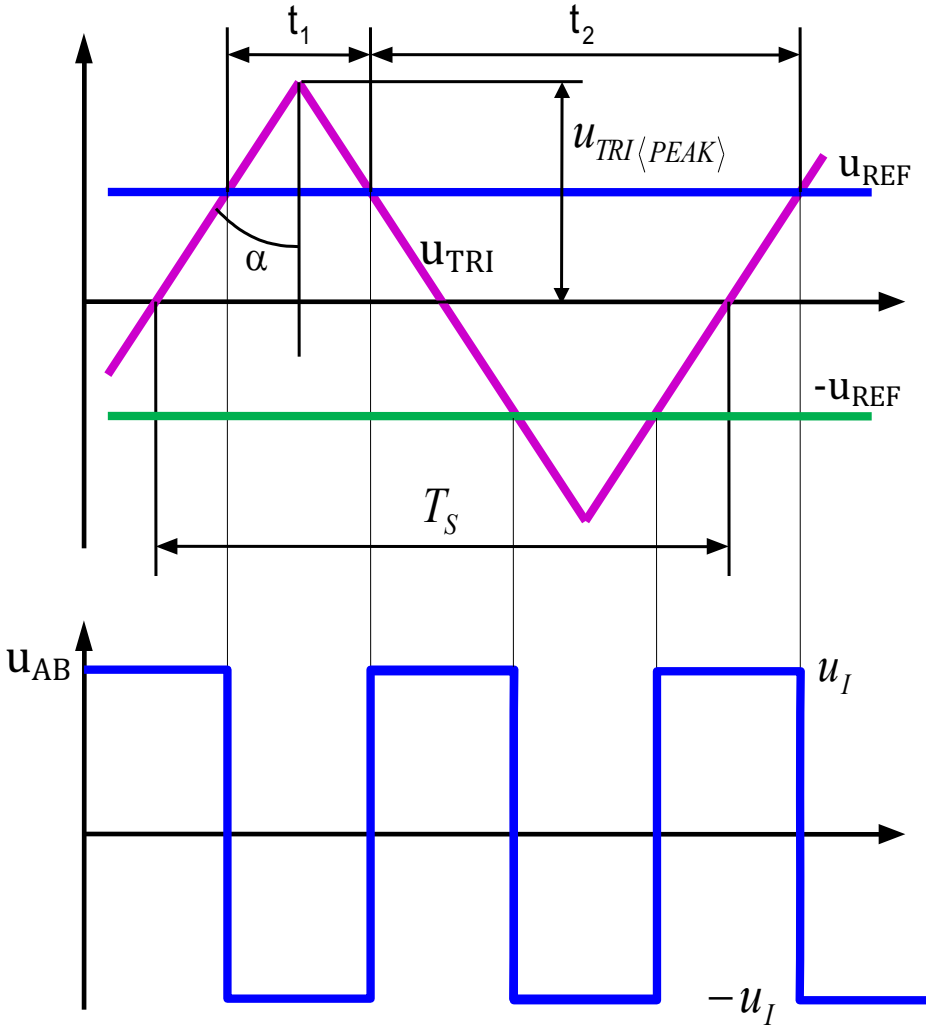
$$u_{REF}(t) = u_{REF\langle PEAK\rangle} \cdot \sin(2\cdot\pi\cdot f_R \cdot t)$$

$$u_{TRI}(t) = triangular\left(u_{TRI\langle PEAK\rangle}, T_S\right)$$

We define the amplitude and frequency **modulation ratios:**

$$m_f = \frac{f_S}{f_R} = \frac{T_R}{T_S} \qquad m_a = \frac{u_{REF\langle PEAK\rangle}}{u_{TRI\langle PEAK\rangle}}$$

$$\left\langle u_{AB}(t)\right\rangle_{T_S} = u_I \cdot m_a \cdot sen(2\cdot\pi\cdot f_R \cdot t)$$



Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

STUDY OF HARMONICS FOR UNIPOLAR CONTROL

(u_{AB} changes from 0 to $+u_I$, or from 0 to $-u_I$)

$$m_a = \frac{u_{REF\langle PEAK\rangle}}{u_{TRI\langle PEAK\rangle}} = 0.7 \qquad m_f = \frac{f_S}{f_R} = 41$$

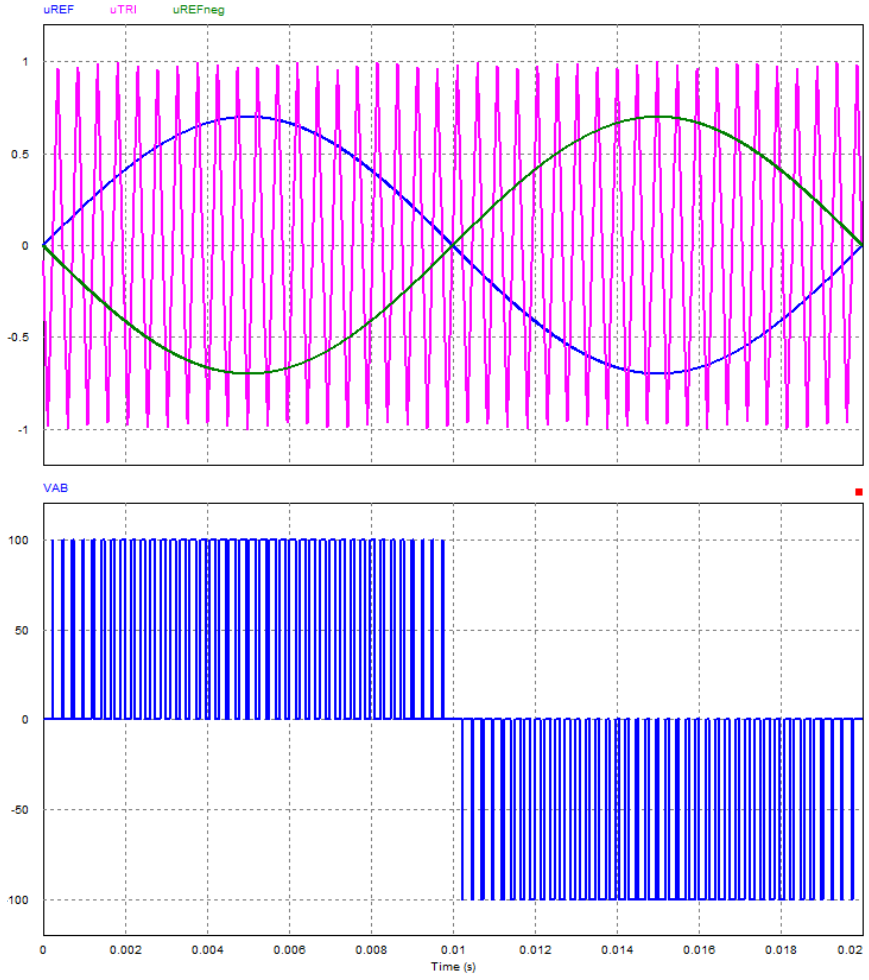
$$\langle u_{AB}(t)\rangle_{T_S} = u_I \cdot m_a \cdot sen(2\cdot\pi\cdot f_R \cdot t)$$

Table 8-1 Generalized Harmonics of v_{Ao} for a Large m_f.

h \ m_a	0.2	0.4	0.6	0.8	1.0
1 *Fundamental*	0.2	0.4	0.6	0.8	1.0
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$		0.024	0.071	0.139	0.212
$2m_f \pm 5$				0.013	0.033
$4m_f \pm 1$	0.163	0.157	0.008	0.105	0.068
$4m_f \pm 3$	0.012	0.070	0.132	0.115	0.009
$4m_f \pm 5$			0.034	0.084	0.119
$4m_f \pm 7$				0.017	0.050

AMPLITUDES

Note: $(\hat{V}_{Ao})_h/\frac{1}{2}V_d$ [= $(\hat{V}_{AN})_h/\frac{1}{2}V_d$] is tabulated as a function of m_a.



"Power Electronics Handbook, Second Edition: Devices, Circuits and Applications" *Editor: Muhammad H. Rashid, Ed. Elsevier Inc., 2007.*

Single-Phase Full-Bridge PWM Inverter: UNIPOLAR

STUDY OF HARMONICS FOR UNIPOLAR CONTROL

(u_{AB} changes from 0 to $+u_I$, or from 0 to $-u_I$)

$$m_a = \frac{u_{REF\langle PEAK\rangle}}{u_{TRI\langle PEAK\rangle}} = 0.7 \qquad m_f = \frac{f_S}{f_R} = 41$$

$$\langle u_{AB}(t)\rangle_{T_S} = u_I \cdot m_a \cdot sen(2\cdot\pi\cdot f_R \cdot t)$$

Table 8-1 Generalized Harmonics of v_{Ao} for a Large m_f.

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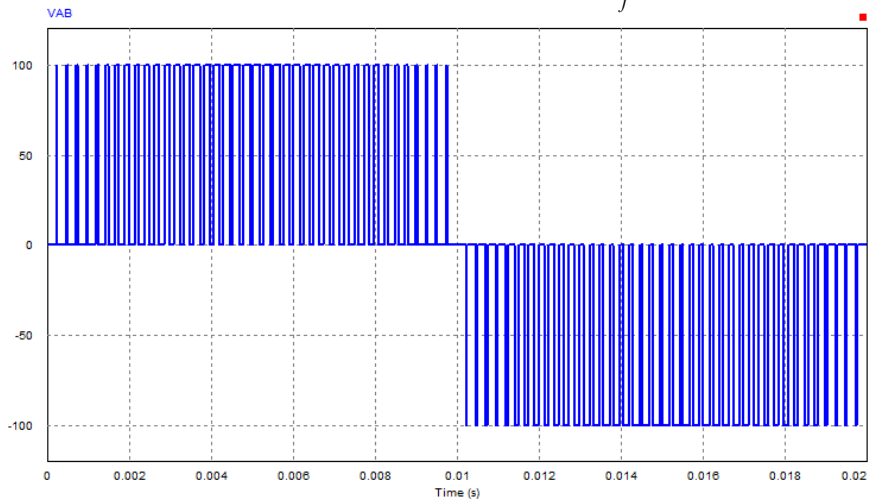
"Power Electronics Handbook, Second Edition: Devices, Circuits and Applications" *Editor: Muhammad H. Rashid, Ed. Elsevier Inc., 2007.*

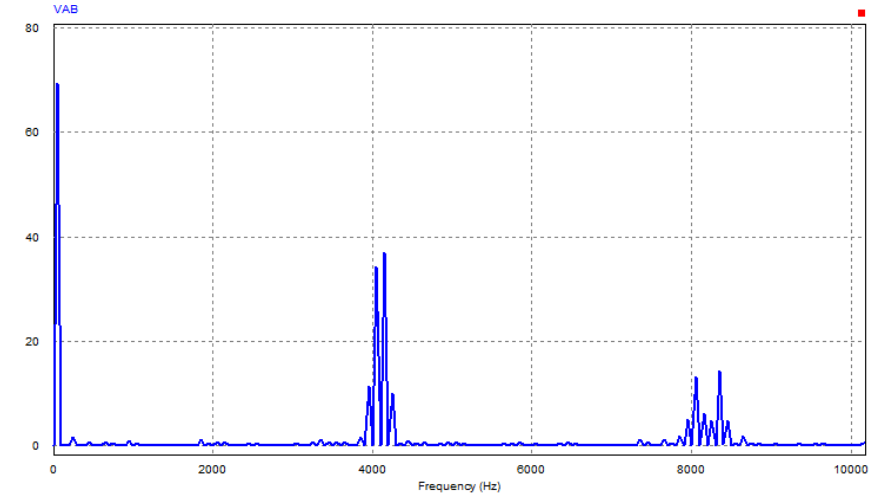
Single-Phase Full-Bridge PWM Inverter

UNIPOLAR CONTROL

$m_a = 0.7$

$m_f = 41$

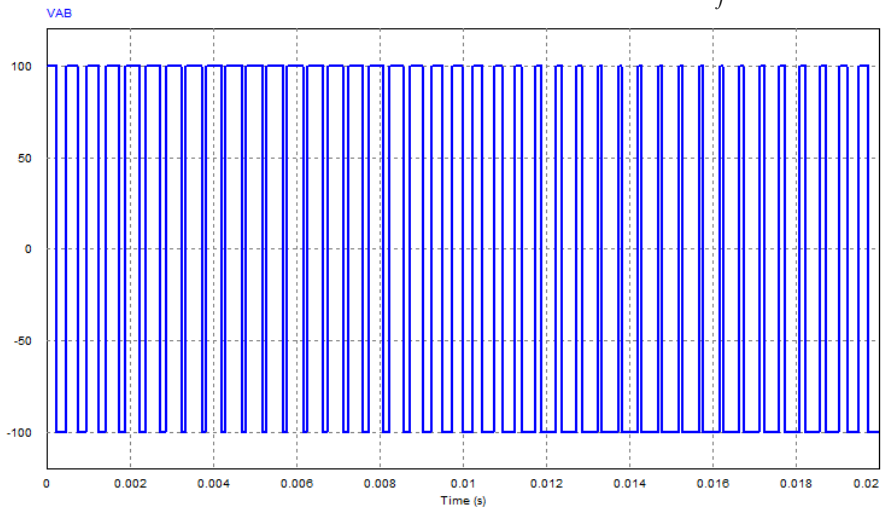


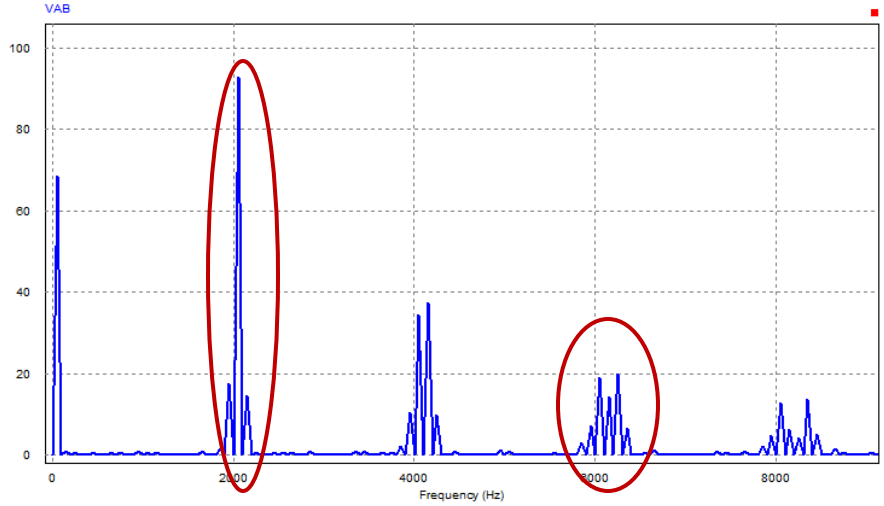


BIPOLAR CONTROL

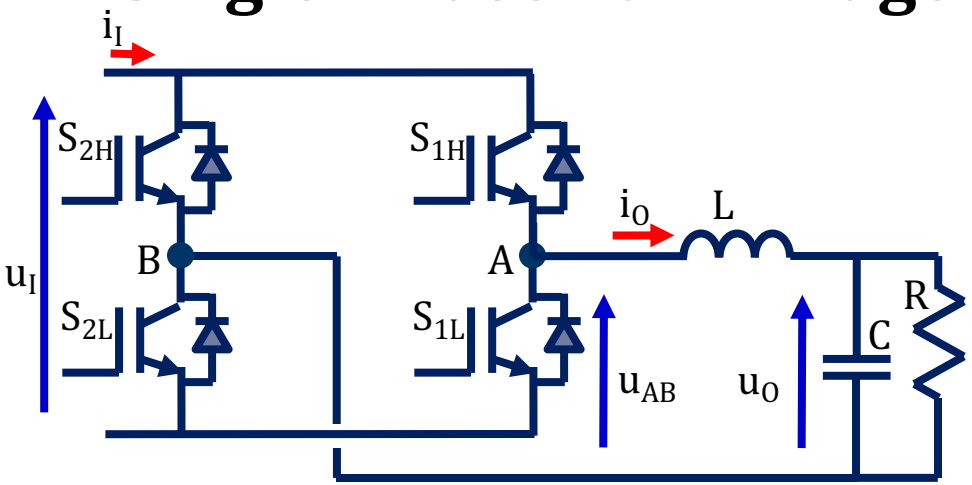
$m_a = 0.7$

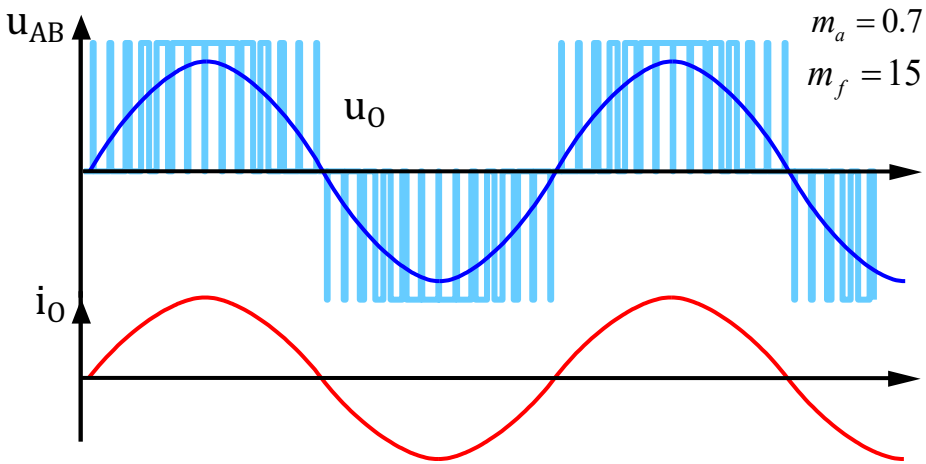
$m_f = 41$



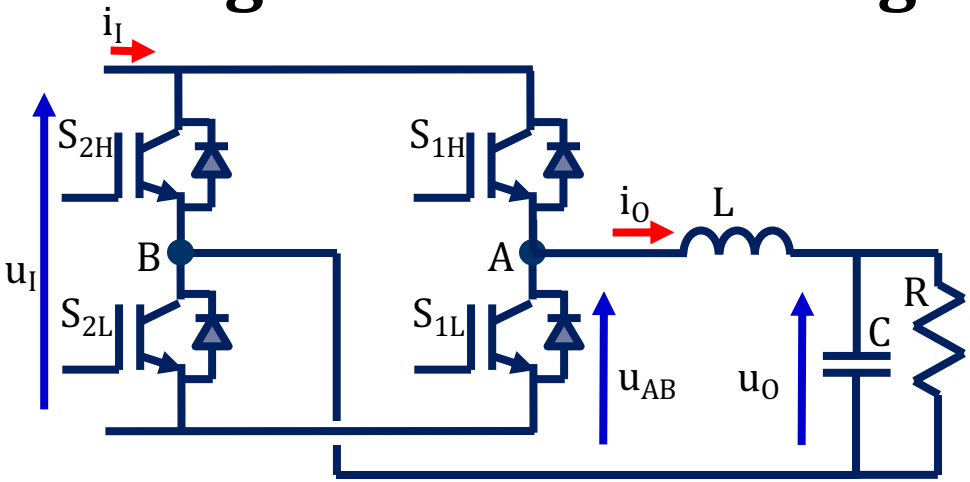


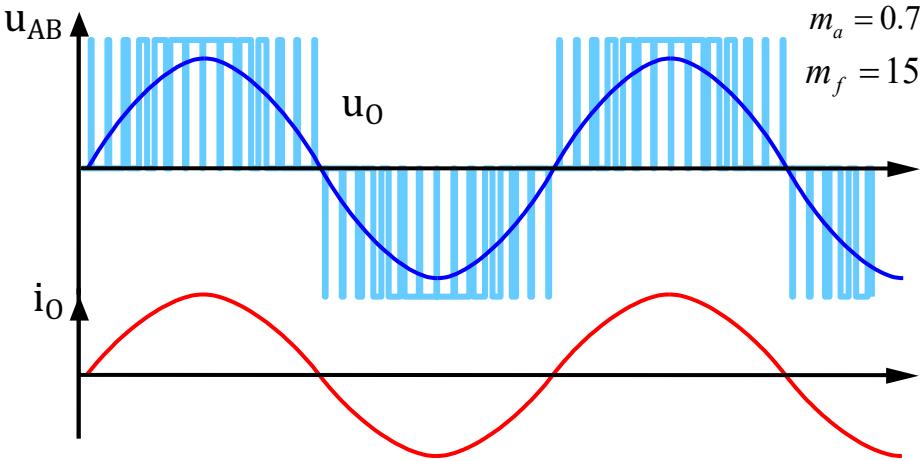
Single-Phase Full-Bridge PWM Inverter





Single-Phase Full-Bridge PWM Inverter



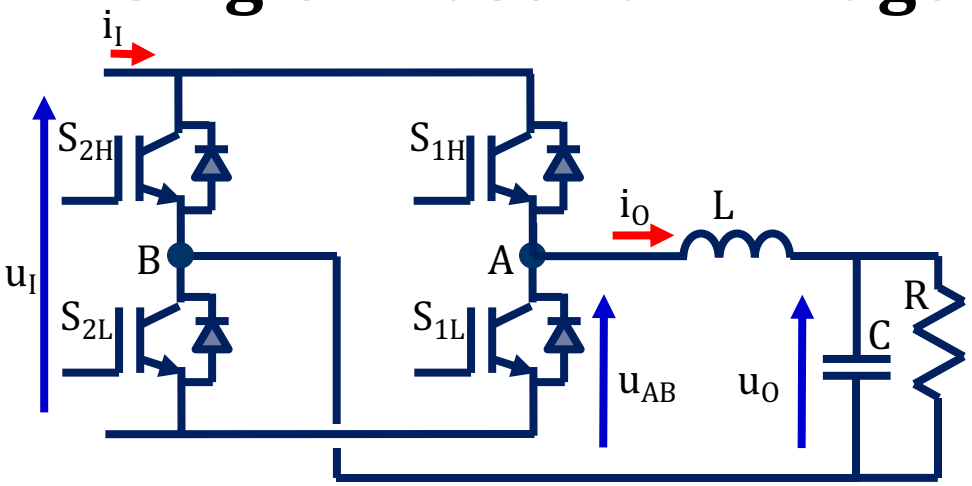


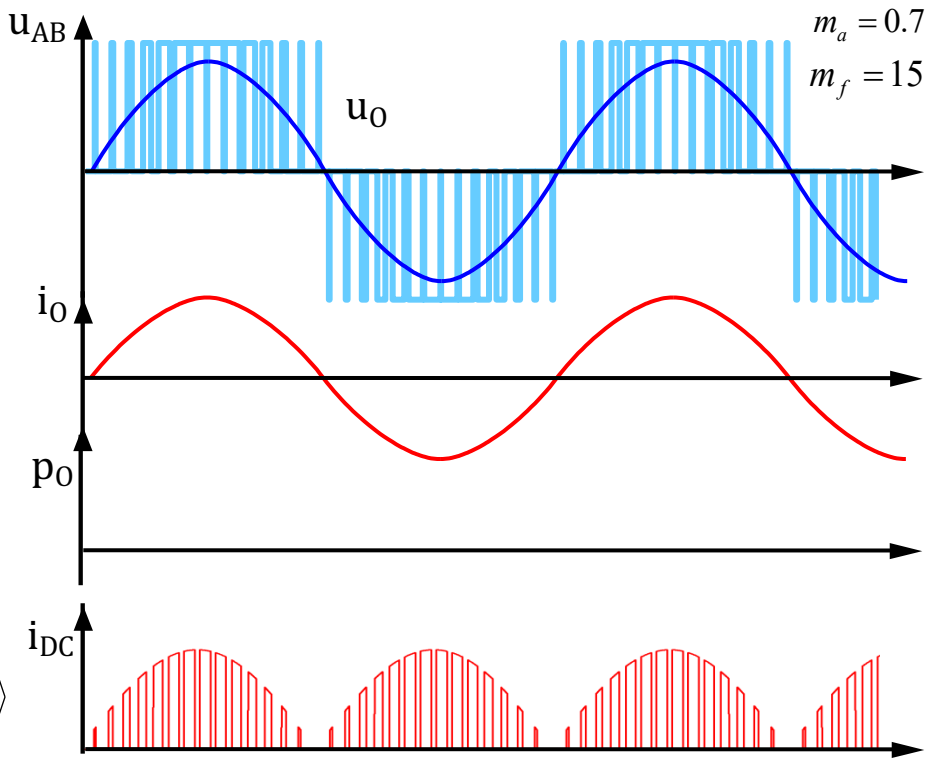
Size of components:

Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Single-Phase Full-Bridge PWM Inverter





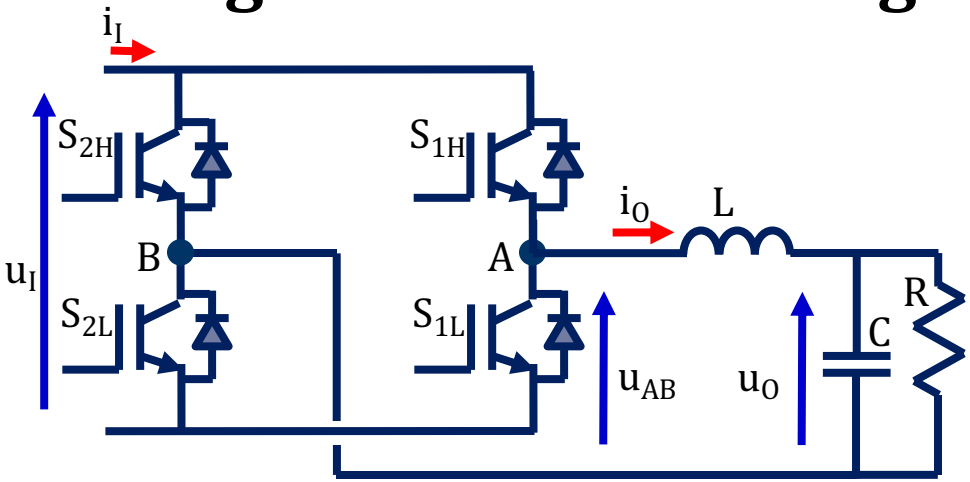
Size of components:

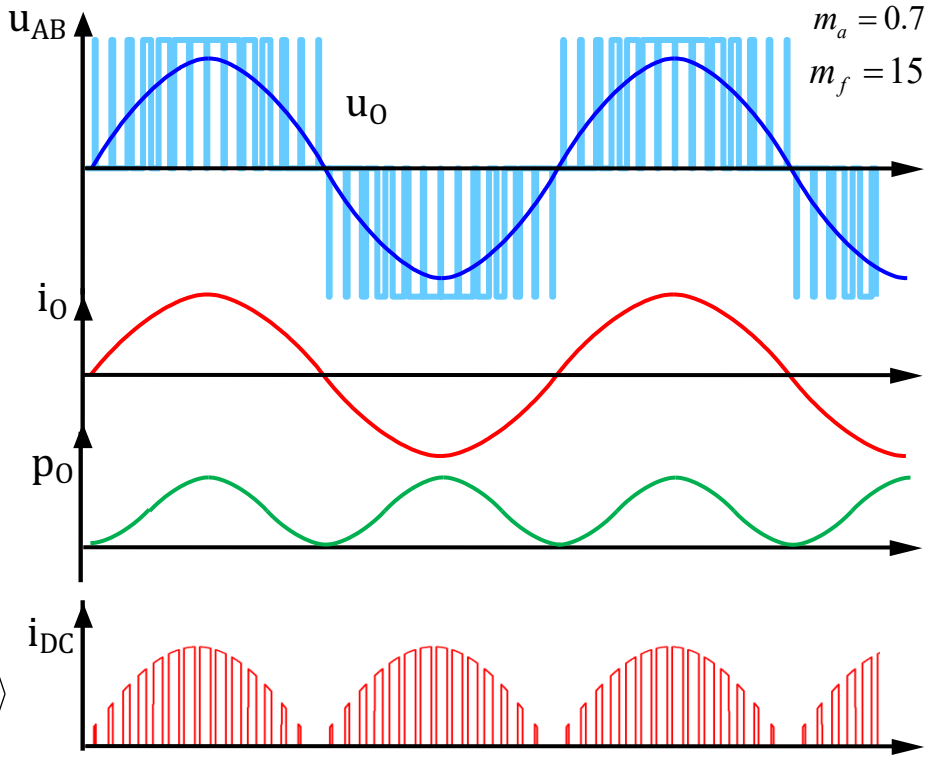
Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Input current:

Single-Phase Full-Bridge PWM Inverter





Size of components:

Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

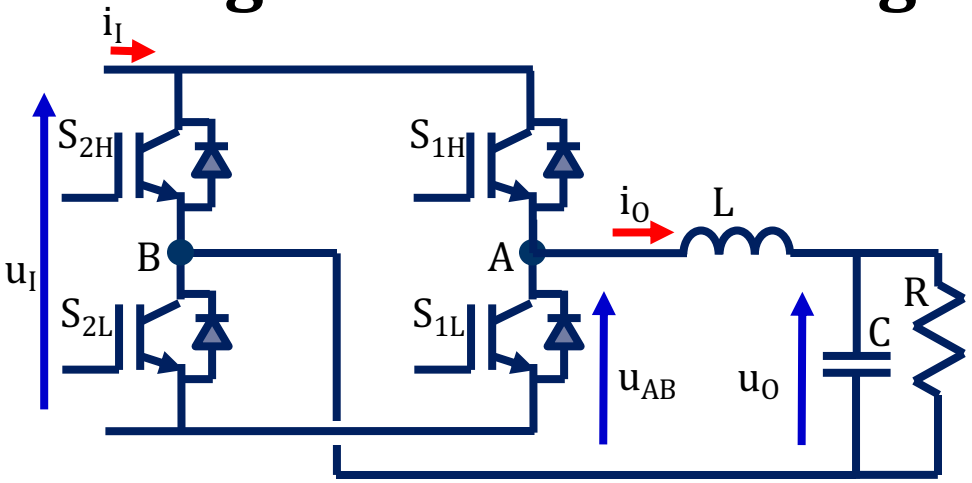
Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

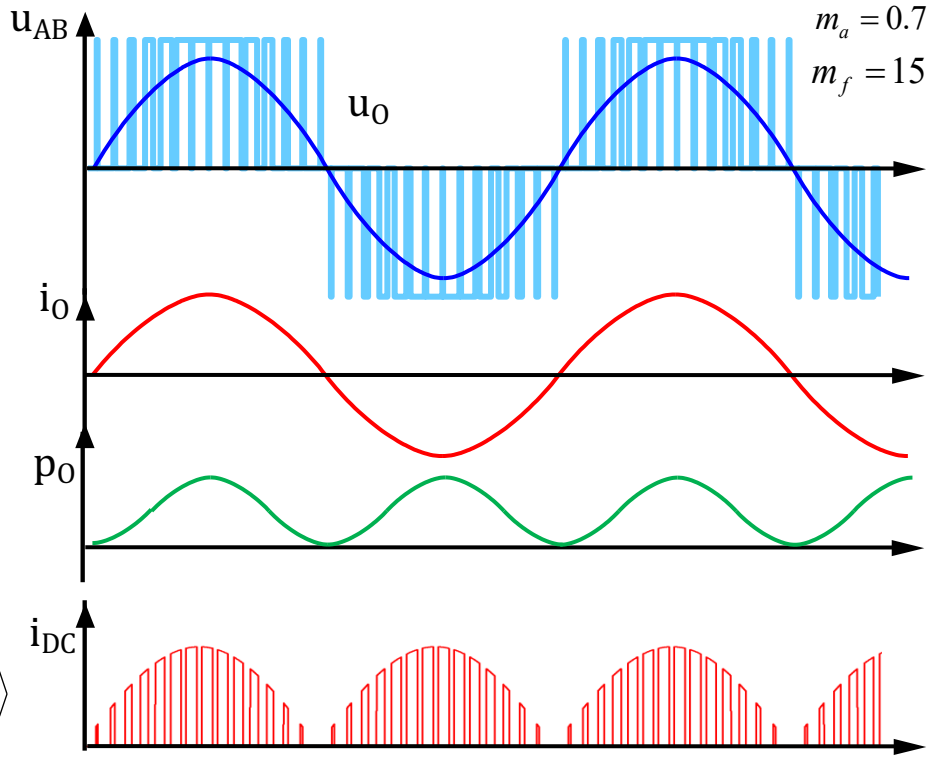
Input current:

$$u_O(t) = \langle u_{AB}(t)\rangle_{T_S} = U_I \cdot m_a \cdot \sin(2\cdot\pi\cdot f_R\cdot t) \qquad i_O(t) = \frac{U_I}{R}\cdot m_a \cdot \sin(2\cdot\pi\cdot f_R\cdot t)$$

$$p_O(t) = \frac{U_I^{\,2}}{R}\cdot m_a^{\,2}\cdot \sin^2(2\cdot\pi\cdot f_R\cdot t) \qquad p_O(t) = \frac{U_I^{\,2}}{R}\cdot m_a^{\,2}\,\frac{1-\cos(2\cdot\pi\cdot 2\cdot f_R\cdot t)}{2}$$

Single-Phase Full-Bridge PWM Inverter





Size of components:

Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Input current:

$$u_O(t) = \langle u_{AB}(t)\rangle_{T_S} = U_I \cdot m_a \cdot \sin(2\cdot\pi\cdot f_R \cdot t) \qquad i_O(t) = \frac{U_I}{R} \cdot m_a \cdot \sin(2\cdot\pi\cdot f_R \cdot t)$$

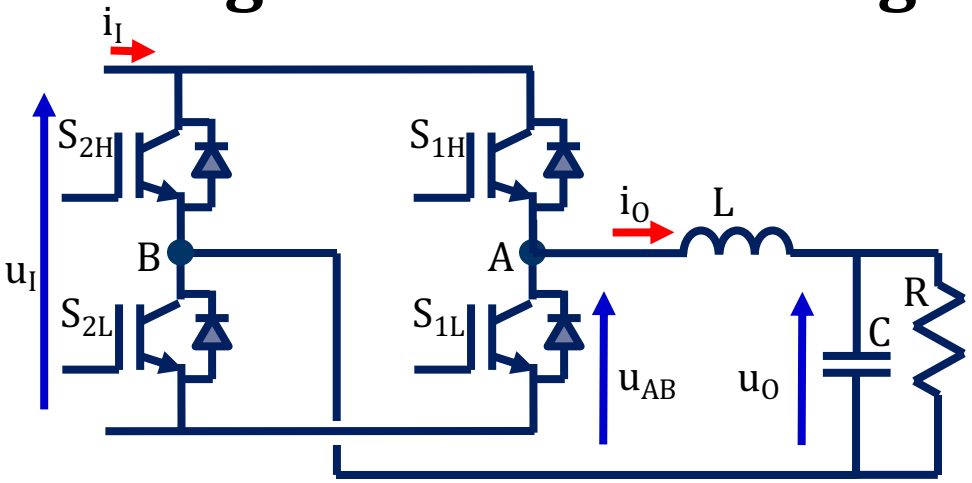
$$p_O(t) = \frac{U_I^{\,2}}{R} \cdot m_a^{\,2} \cdot \sin^2(2\cdot\pi\cdot f_R \cdot t) \qquad p_O(t) = \frac{U_I^{\,2}}{R} \cdot m_a^{\,2} \frac{1-\cos(2\cdot\pi\cdot 2\cdot f_R \cdot t)}{2}$$

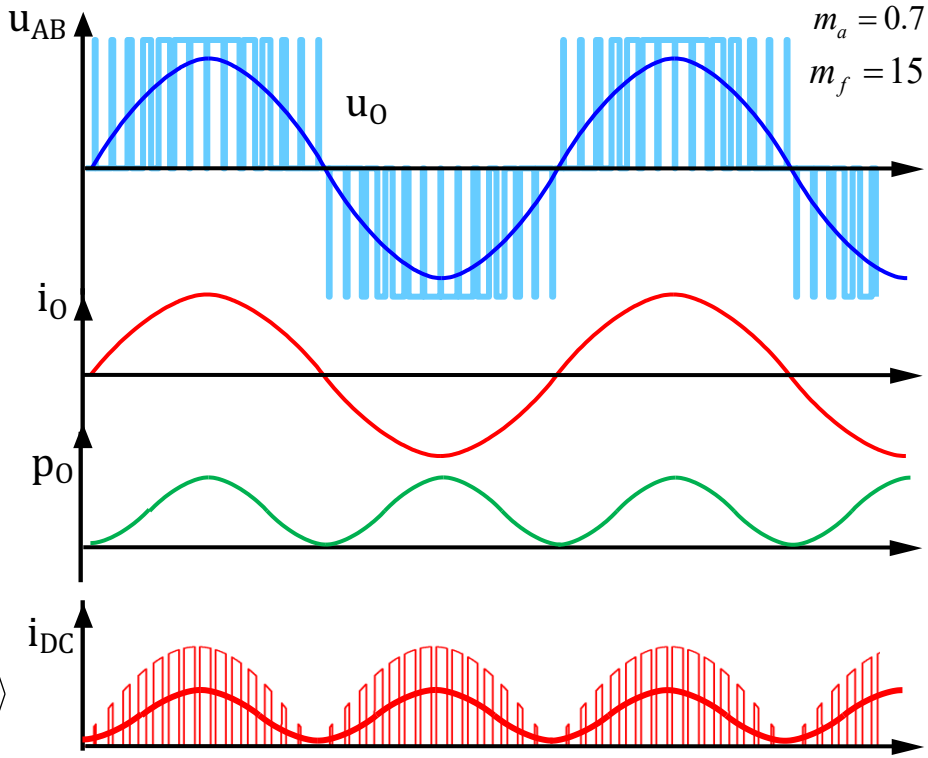
$$u_I(t) = U_I \qquad p_I(t) = u_I(t)\cdot i_I(t)$$

The filter has no effect at line frequencies!!!

$$p_I(t) = p_O(t)$$

Single-Phase Full-Bridge PWM Inverter





Size of components:

Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Input current:

$$u_O(t) = \langle u_{AB}(t)\rangle_{T_S} = U_I \cdot m_a \cdot \sin(2\cdot\pi\cdot f_R \cdot t) \qquad i_O(t) = \frac{U_I}{R} \cdot m_a \cdot \sin(2\cdot\pi\cdot f_R \cdot t)$$

$$p_O(t) = \frac{U_I^{\,2}}{R} \cdot m_a^{\,2} \cdot \sin^2(2\cdot\pi\cdot f_R \cdot t) \qquad p_O(t) = \frac{U_I^{\,2}}{R} \cdot m_a^{\,2} \frac{1-\cos(2\cdot\pi\cdot 2\cdot f_R \cdot t)}{2}$$

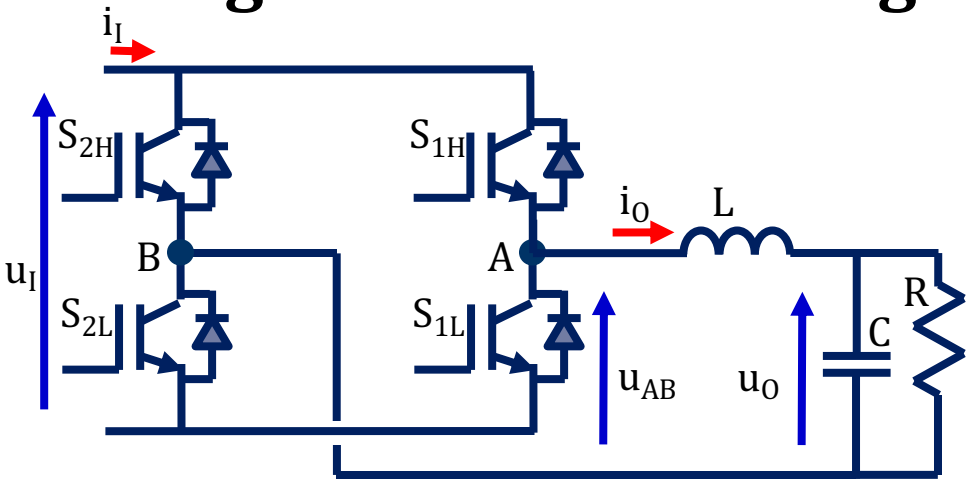
$$u_I(t) = U_I \qquad p_I(t) = u_I(t)\cdot i_I(t)$$

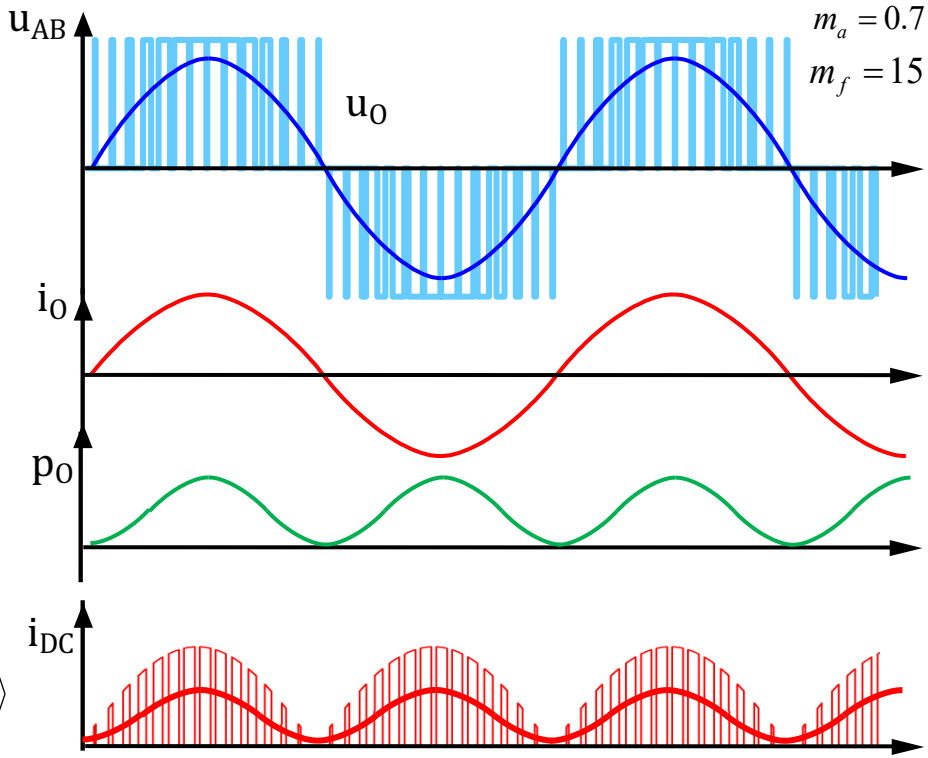
The filter has no effect at line frequencies!!!

$$p_I(t) = p_O(t)$$

$$i_I(t) = \frac{U_I}{R} \cdot m_a^{\,2} \frac{1-\cos(2\cdot\pi\cdot 2\cdot f_R \cdot t)}{2}$$

Single-Phase Full-Bridge PWM Inverter





Size of components:

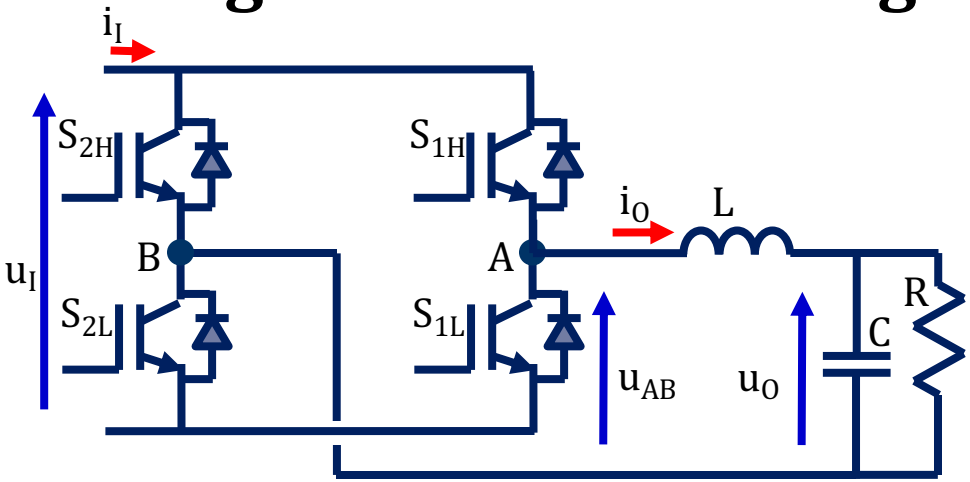
Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

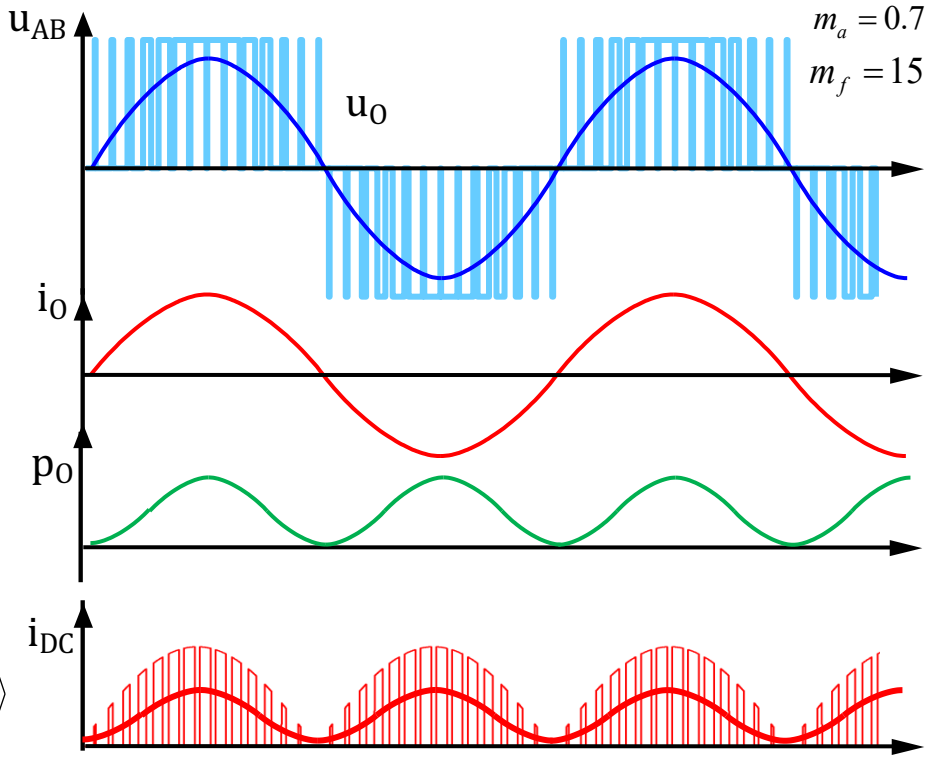
Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Input current:

$$i_I(t) = \frac{U_I}{R} \cdot m_a{}^2 \frac{1-\cos(2\cdot\pi\cdot 2\cdot f_R \cdot t)}{2}$$

Single-Phase Full-Bridge PWM Inverter





Size of components:

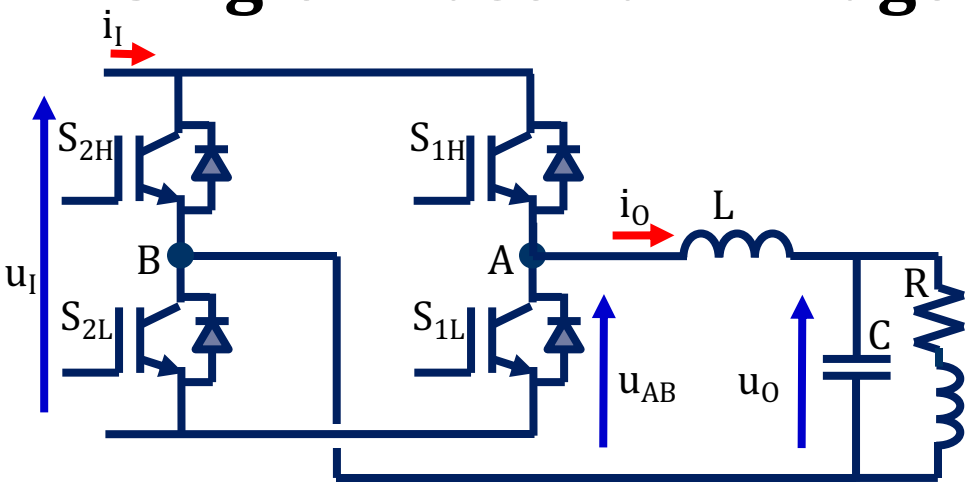
Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

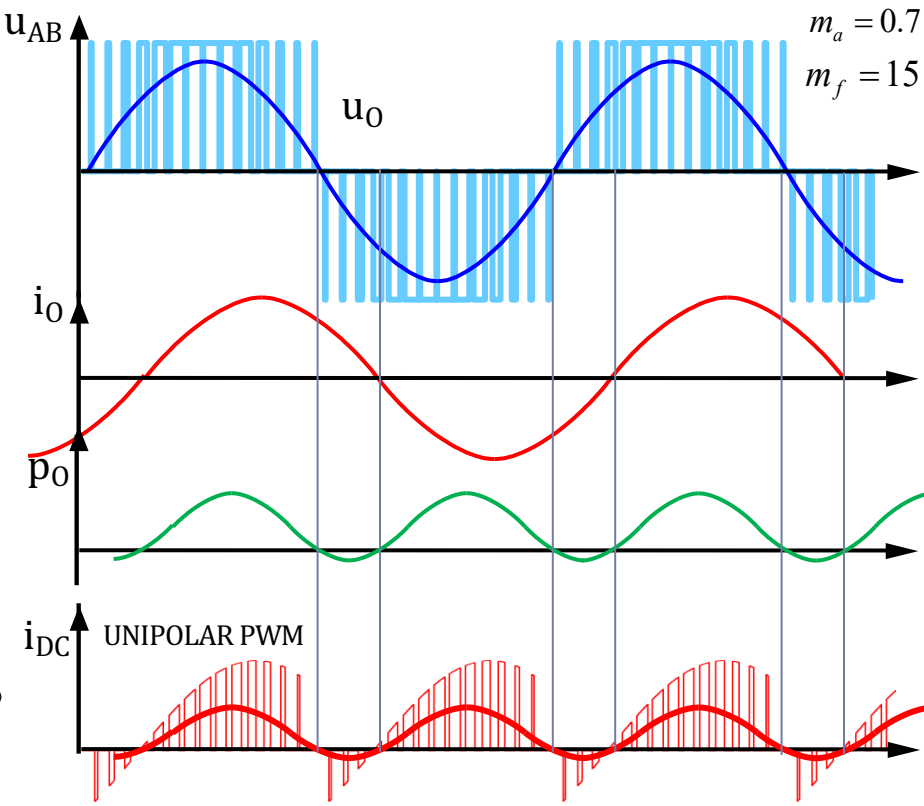
Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

Input current:

$$i_I(t) = i_{DC\langle AVG\rangle} + i_{RIPPLE} \cdot \sin(2 \cdot \pi \cdot 2 \cdot f_R t + \varphi)$$

Single-Phase Full-Bridge PWM Inverter





Size of components:

Voltage stress in Switches: $u_{CE\langle PEAK\rangle} = u_I$

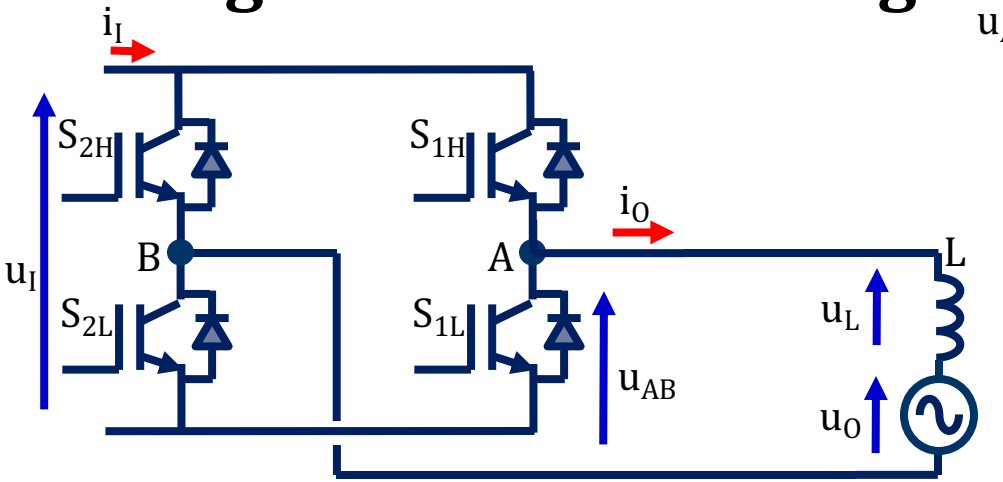
Current stress in Switches: $i_{C\langle PEAK\rangle} = i_{O\langle PEAK\rangle}$

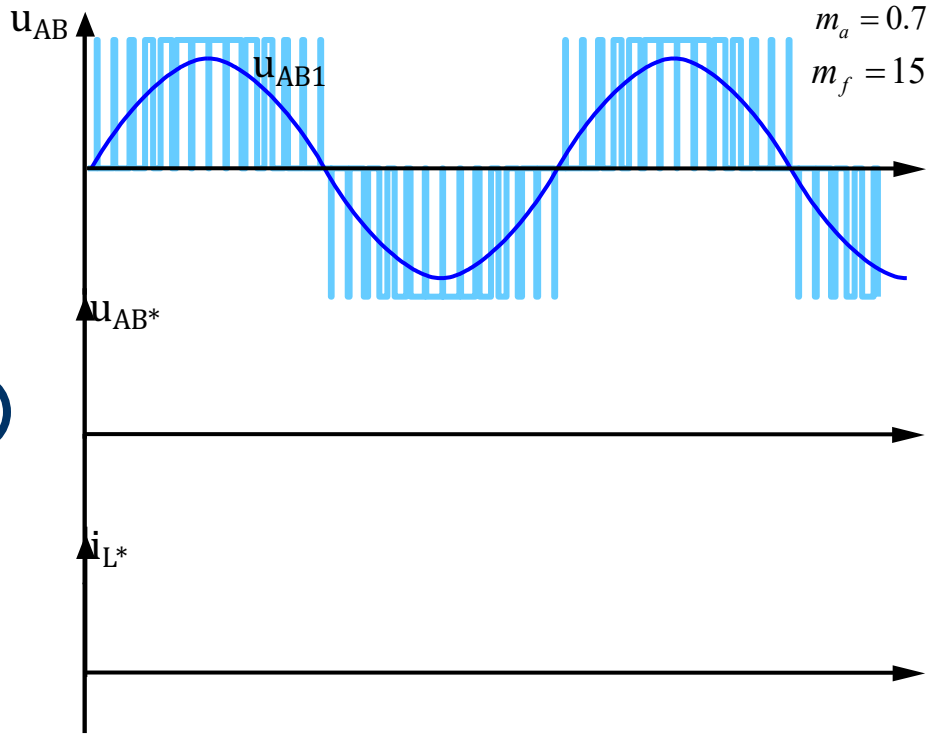
Input current:

$$i'_I(t) = i'_{DC\langle AVG\rangle} + i'_{RIPPLE} \cdot \sin(2\cdot\pi\cdot 2\cdot f_R t + \varphi')$$

The load can have reactive component

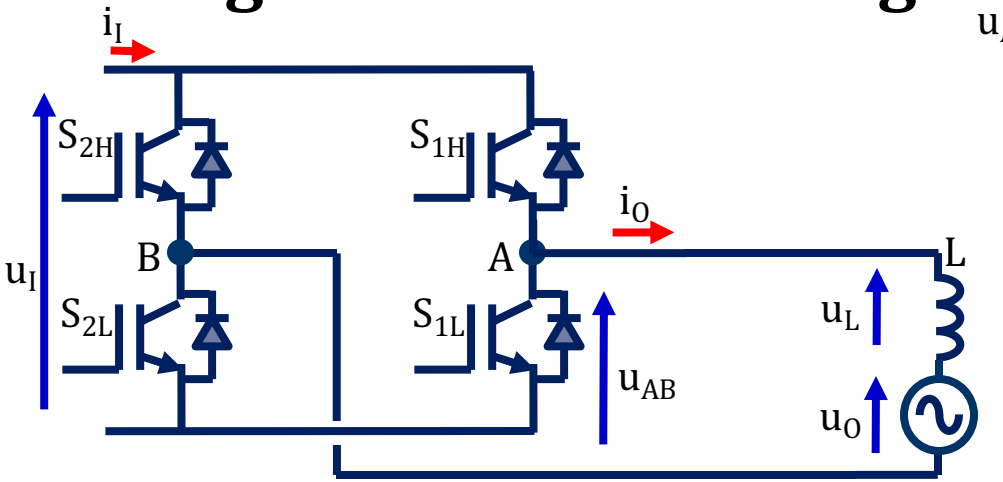
Single-Phase Full-Bridge PWM Inverter

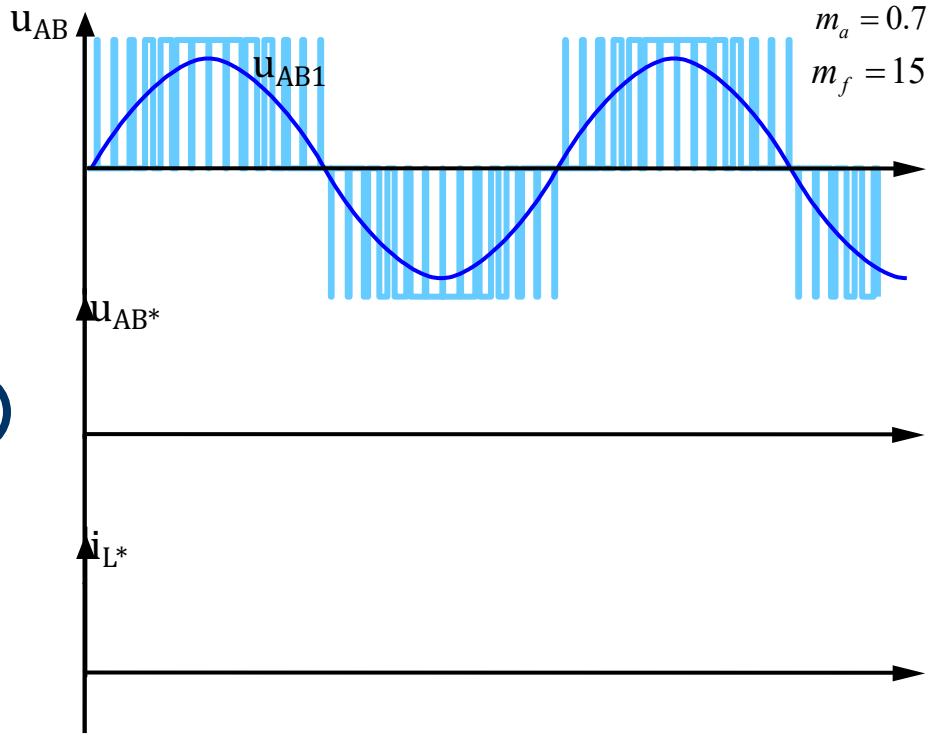




Output current ripple and switching frequency
Consider as a load an AC motor (or the grid line).

Single-Phase Full-Bridge PWM Inverter

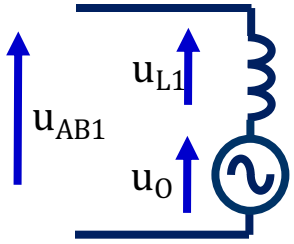




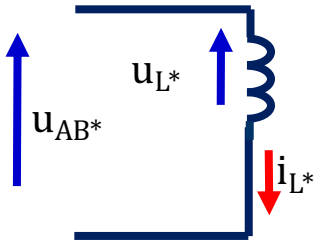
Output current ripple and switching frequency
Consider as a load an AC motor (or the grid line).

Superposition:

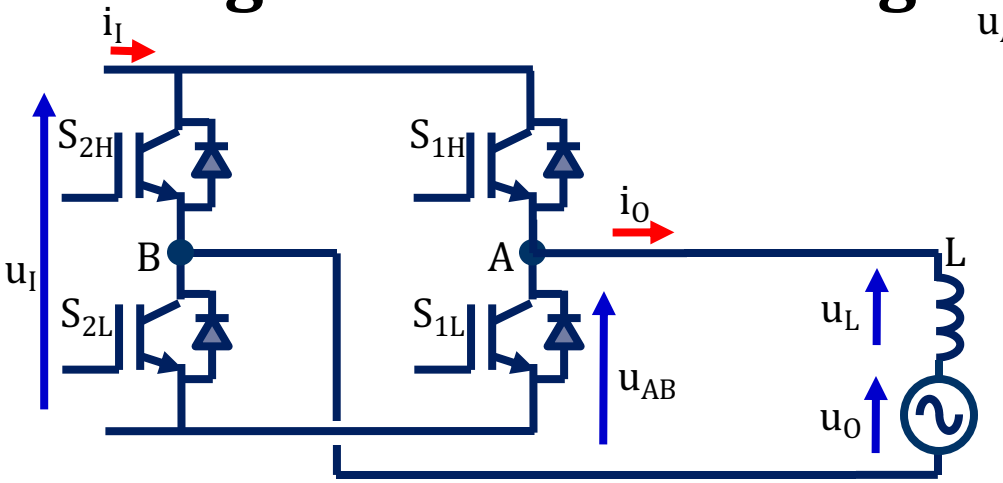
1st harmonic:

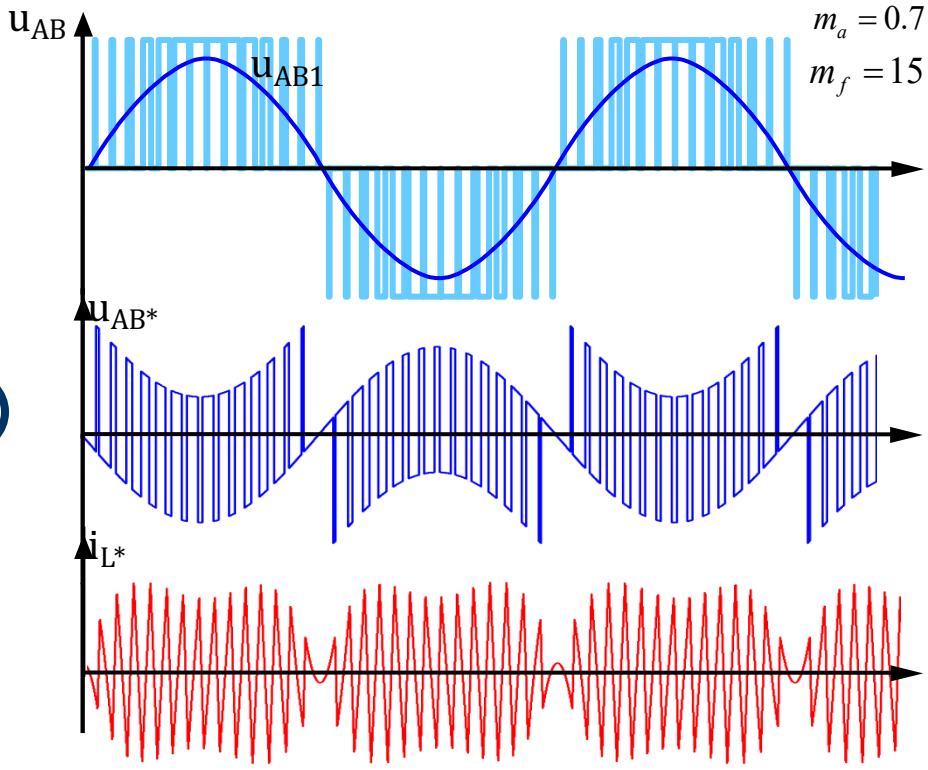


Rest of harmonics:



Single-Phase Full-Bridge PWM Inverter

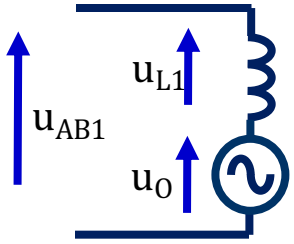




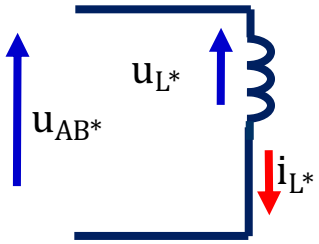
Output current ripple and switching frequency
Consider as a load an AC motor (or the grid line).

Superposition:

1st harmonic:



Rest of harmonics:

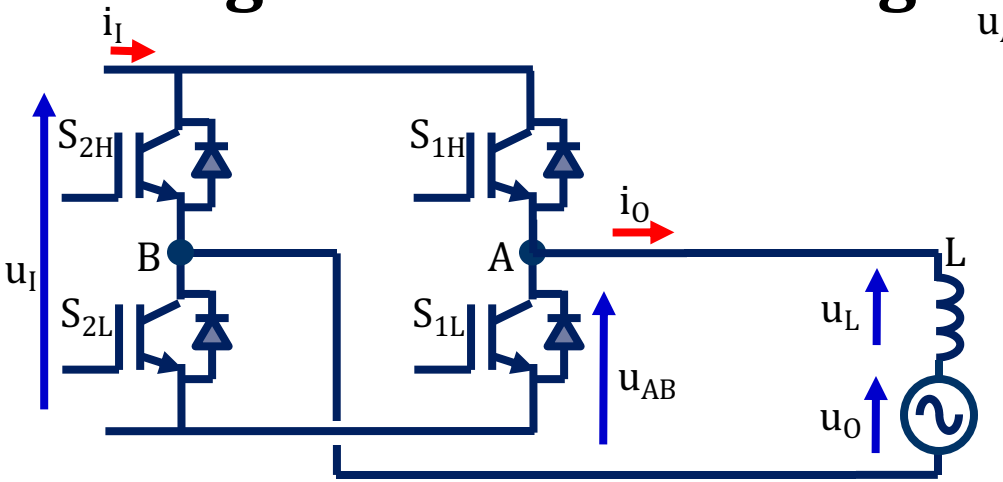


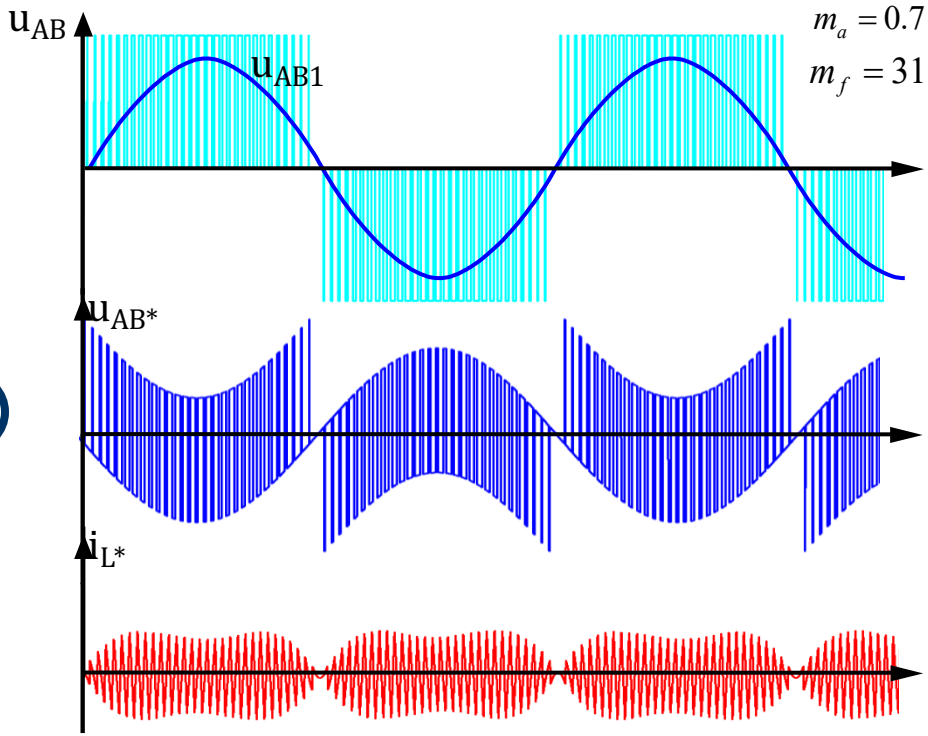
$$u_{AB*}(t) = u_{AB} - u_{AB1}$$

$$u_{AB*}(t) = L \cdot \frac{di_{L*}}{dt}$$

This current ripple implies losses (no active power delivered)

Single-Phase Full-Bridge PWM Inverter

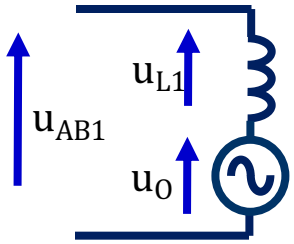




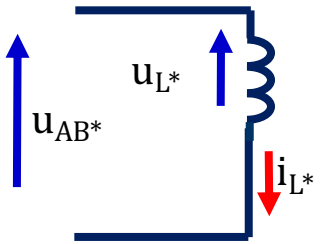
Output current ripple and switching frequency
Consider as a load an AC motor (or the grid line).

Superposition:

1st harmonic:



Rest of harmonics:



$$u_{AB*}(t) = u_{AB} - u_{AB1}$$

$$u_{AB*}(t) = L \cdot \frac{di_{L*}}{dt}$$

This current ripple implies losses (no active power delivered)

Increasing the modulation frequency decreases the current ripple and the losses.
Increasing the modulation frequency increases switching losses.

References:

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